Local Field Theory

***EYNTKA* Series**

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October 27, 2026

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0.0.1 Reading the numbering A star marks *bonus material*: a chapter, section or subsection that belongs in the book but that nothing later depends on. Skipping it costs nothing downstream. The star sits on the number, as in §2.7⋆; in divisions written before this convention it sits at the front of the title instead, as in **Uniform Spaces*. The two mean the same thing.

Introduction

This book is about doing arithmetic one prime at a time. Completing a number field at a single prime concentrates the information carried by that prime into one analytically tractable object, a local field. Questions about ramification, about whether an equation has a solution, about how a Galois group acts near the prime, all sharpen and often become decidable once localized. The technology for that localization, and the dictionary translating between a global field and the family of its completions, is what these pages develop.

The reader is assumed to have met algebraic number theory: rings of integers, Dedekind domains, the splitting of primes, and the ideal class group. From there the treatment is self-contained. It builds the local fields from the ground up, first as completions and as abstract complete discrete valuation rings, then through the fine structure of their extensions, and finally reassembles the local pictures into a global one in the language of places.

The journey runs through five chapters and an appendix. Chapter 1 constructs the objects: the p-adic numbers, discrete valuation rings and their completions, the definition of a local field, and the structure of its multiplicative group. Chapter 2 develops the two instruments for locating roots over a complete field, Hensel's lemma and the Newton polygon. Chapter 3 is the technical heart, the extension theory of local fields and the full anatomy of ramification, from the fundamental identity through the higher ramification filtration, the different, and the Hilbert symbol, closing with Witt vectors as the canonical source of unramified extensions. Chapter 4 completes the development with the local-global correspondence and its consequences, the Kronecker-Weber theorem among them. Chapter 5 recasts everything in the language of places, the uniform treatment of finite and infinite primes that opens onto adelic number theory, and appendix A proves the Hasse-Arf theorem.

The book stops deliberately short of local class field theory itself, the reciprocity map and the classification of abelian extensions by norm groups. That is the subject of the sequel [10], and the first place to go next.

1

p-adic Numbers, Valuations, and Local Fields

In this chapter, we shall explore special fields which focus into a single prime of a number field. For a prime $p \in \mathbb{Z}$, the fundamental object corresponding to this prime is the p-adic integers $\mathbb{Z}_p$. Intuitively, solving an equation over $\mathbb{Z}_p$ corresponds to solving an equation over $\mathbb{Z}$ mod p^n for each n (which can be thought of as solving the equation over all multiplicities). Each $\mathbb{Z}_p$ shall correspond to a field $\mathbb{Q}_p$, and finite extensions of $\mathbb{Q}_p$, $K_\mathfrak{p}/\mathbb{Q}_p$, shall only contain primes that are over p. The galois group of these extensions shall then be isomorphic to the decomposition group. The field extensions $K_\mathfrak{p}$ shall be called *local number fields*. Finite extensions $K/\mathbb{Q}$ will be called *global number fields*. A large part of the early sections is dedicated to building up the theory of local fields. This work culminates in the local-global correspondence (section 4.2), which readily translates many results between global fields and local fields.

In the process of developing the generalization, we shall see that the local fields $\mathbb{Q}_p$ correspond very naturally to all possible distinct absolute values that can be defined on $\mathbb{Q}$. In fact, the collection of all possible absolute absolute values that can be defined on $\mathbb{Q}$ shall create the following completions:

$$\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_5, \mathbb{Q}_7, \mathbb{Q}_{11}, ..., \mathbb{Q}_\infty = \mathbb{R}$$

where there is one distinct completion for each prime, as well as one completion that can in fact be naturally interpreted as the completion at the "prime at infinity". We shall make this idea precise in chapter 5. The idea of places being an extension of primes shall become important when developing class field theory, as well as linking it to the development of extensions of $\mathbb{F}_p(t)$.

Later, in chapter 4, we shall prove the *Kronecker-Weber Theorem*, which shows that every finite abelian extension of $\mathbb{Q}$ is contained in a cyclotomic field. The sequel [10] describes the abelian extensions of every number field K. One consequence is that K has an extension K_H whose galois group $\mathrm{Gal}_K(K_H)$ is isomorphic to the ideal class group of K.

1.1 p-adic Integers and p-adic Numbers

Let us start with giving a detailed description of the "new integers" where we only focus on a single prime. Let R be a commutative ring, $\mathfrak{p} \subseteq R$ a prime. Then $R_{\mathfrak{p}}$ will represent the localization of R at $\mathfrak{p}$ (i.e. at the set $R \setminus \mathfrak{p}$). Recall that the completion of R at a prime P is:

$$\varprojlim_n R/\mathfrak{p}^n R = \widehat{R}$$

where the $\mathfrak{p}$ is usually understood from context on the right hand side. For number theory, we are interested in the case where $R = \mathbb{Z}$. This gives us the localizations $\mathbb{Z}_{(p)}$ and the completions $\widehat{\mathbb{Z}}_p$ which we shall denote $\mathbb{Z}_p$. Elements in the completion can be represented as the formal power series:

$$\sum_i^{\infty} a_i p^i = a_0 + a_1 p + a_2 p^2 + \cdots$$

where $a_i \in \{0, ..., p-1\}$. More succinctly, as $\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$, as a compatible sequence of residues

$$(a_0 \bmod p,\ a_0 + a_1 p \bmod p^2,\ a_0 + a_1 p + a_2 p^2 \bmod p^3, \dots)$$

It is not immediately clear that these two are the same: in the first we have multiplication via convolution, while in the second we have point-wise multiplication. The equivalence in these two representation can be given by the following: any element can be found by seeing that when we are given:

$$s = \sum_i^{\infty} a_i p^i$$

it has to satisfy:

$$s_n = \sum_i^{n-1} a_i p^i$$

hence, for any $[r] \in \mathbb{Z}/p^n\mathbb{Z}$, there exists a representative of the form:

$$[r] = [a_0 + a_1 p + \cdots + a_{n-1} p^{n-1}]$$

Such an expansion can be written down for any $r \in \mathbb{Z}_{(p)}$, that is for any a/b where $p \nmid b$. From this, if we consider $\mathbb{Z}_p$ to be the formal power series with addition/multiplication extending naturally. From this:

Proposition 1.1.1: Representation Of p-addic

Let $\mathbb{Z}_p$ represent the formal p-adic integers and $\varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$ the completion definition. Then:

$$\mathbb{Z}_p \cong \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$$

***Proof*:**
Write a formal p-adic integer as $s = \sum_{i\geq 0} a_i p^i$ with digits $a_i \in \{0, \dots, p-1\}$, and let $s_n = \sum_{i=0}^{n-1} a_i p^i$ be its truncation, an integer with $0 \leq s_n < p^n$. Since $s_{n+1} - s_n = a_n p^n$, the classes $s_n \bmod p^n$ are compatible with the projections $\mathbb{Z}/p^{n+1}\mathbb{Z} \to \mathbb{Z}/p^n\mathbb{Z}$, so

$$\Phi(s) = (s_n \bmod p^n)_{n\geq 1} \in \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$$

defines a map Φ from the formal p-adic integers to the limit.

Bijectivity. Let $(c_n)_n$ be a compatible family, let r_n be the representative of c_n with $0 \leq r_n < p^n$, and put $a_0 = r_1 \in \{0, \dots, p-1\}$. Compatibility says $p^n \big| r_{n+1} - r_n$, and $-p^n < r_{n+1} - r_n < p^{n+1}$, so $r_{n+1} = r_n + a_n p^n$ with $0 \leq a_n \leq p-1$. By induction on n, starting from $r_1 = a_0$, we get $r_n = \sum_{i=0}^{n-1} a_i p^i$ for every n, so the family is $\Phi(s)$ for $s = \sum_i a_i p^i$. Each digit a_n is determined by $r_{n+1} - r_n$, so s is the only formal p-adic integer with $\Phi(s) = (c_n)_n$.

Ring structure. Formal p-adic integers are added and multiplied digit by digit with carries, and a carry only moves into higher digits. Hence the first n digits of $s+t$ and of st are the base-p digits of the least nonnegative residues of $s_n + t_n$ and of $s_n t_n$ modulo p^n, which says that $\Phi(s+t) = \Phi(s) + \Phi(t)$ and $\Phi(st) = \Phi(s)\Phi(t)$. Also $\Phi(1) = (1 \bmod p^n)_n$ is the identity of the limit. So Φ is a bijective ring homomorphism, hence a ring isomorphism, as we sought to show.

A motivation for defining the p-adic integers is in the following "universal property": the solutions to Diophantine equations (elements of $\mathbb{Z}[x_1, ..., x_n]$) mod p^n for all $n \in \mathbb{N}_{>0}$ equivalent to finding solution to Diophantine equations over the p-adics. Note that if we ask for solutions over some collection mod m, by the Chinese Remainder Theorem (CRT) we may reduce the case to solution over powers of primes.

Theorem 1.1.2: Diophantine Equations in p-adic

Let $f \in \mathbb{Z}[x_1, ..., x_m]$. Then the congruence

$$f \equiv 0 \bmod p^k$$

is solvable in integers for every $k \geq 1$ if and only if f has a zero in $\mathbb{Z}_p^m$.

***Proof*:**
By proposition 1.1.1, $\mathbb{Z}_p = \varprojlim_j \mathbb{Z}/p^j\mathbb{Z}$, and the projections $\pi_j \colon \mathbb{Z}_p \to \mathbb{Z}/p^j\mathbb{Z}$ are ring homomorphisms, applied componentwise to $\mathbb{Z}_p^m$. As f has integer coefficients, $\pi_j(f(x)) = f(\pi_j(x))$ for $x \in \mathbb{Z}_p^m$. Suppose first that $x \in \mathbb{Z}_p^m$ has $f(x) = 0$, and let $a \in \mathbb{Z}^m$ represent $\pi_k(x)$. Then $f(a) \bmod p^k = f(\pi_k(x)) = \pi_k(f(x)) = 0$, so a solves the congruence modulo p^k.

Conversely, for each $k \geq 1$ choose $a^{(k)} \in \mathbb{Z}^m$ with $f(a^{(k)}) \equiv 0 \bmod p^k$. There are finitely many residue classes in $(\mathbb{Z}/p\mathbb{Z})^m$, so infinitely many $a^{(k)}$ lie in one class $b_1 \bmod p$. Among these, infinitely many lie in one class $b_2 \bmod p^2$, with $b_2 \equiv b_1 \bmod p$. Continuing gives $b_j \in \mathbb{Z}^m$ with $b_{j+1} \equiv b_j \bmod p^j$, such that infinitely many $a^{(k)}$ satisfy $a^{(k)} \equiv b_j \bmod p^j$. The compatible

residues $b_j \bmod p^j$ define $x \in \mathbb{Z}_p^m$ with $\pi_j(x) = b_j \bmod p^j$ for every j. Given j, choose $k \geq j$ with $a^{(k)} \equiv b_j \bmod p^j$. Then $\pi_j(f(x)) = f(b_j) \bmod p^j = f(a^{(k)}) \bmod p^j = 0$. So every component of $f(x) \in \varprojlim_j \mathbb{Z}/p^j\mathbb{Z}$ is 0, and $f(x) = 0$, as we sought to show.

Example 1.1.3: p-adic Integers

1. The integer 10 has the expansions $10 = 20_5 = 1010_2 = 101_3$. For an integer n with at least two distinct prime factors, the n-adic completion $\mathbb{Z}_n = \varprojlim_k \mathbb{Z}/n^k\mathbb{Z}$ still exists, but it has zero divisors. For example,
$$\mathbb{Z}_{10} \cong \mathbb{Z}_2 \oplus \mathbb{Z}_5$$
obtained by passing to the limit in the Chinese remainder isomorphisms $\mathbb{Z}/10^k\mathbb{Z} \cong \mathbb{Z}/2^k\mathbb{Z} \times \mathbb{Z}/5^k\mathbb{Z}$.

2. Let $p = 3$ and consider -1_3, the expansion $[\cdots a_3a_2a_1]_3$ with
$$0001_3 + -1_3 = 0$$
The expansion $\cdots 222_3$ works, since adding 1 carries at every digit:
$$\cdots 222_3 + \cdots 0001_3 = \cdots 000_3$$
The same carrying gives the expansion of every negative integer.

3. Let $p = 3$ and consider $1/2$. Its expansion $[\cdots a_3a_2a_1]_3$ must satisfy
$$(1/2)_3 \cdot 2_3 = \cdots 0001_3$$
To construct such a number, first:
$$a_1 \cdot 2 \equiv 1 \bmod 3$$
hence $a_1 = 2$. We then need:
$$2 \cdot a_2 + 1 \equiv 0 \bmod 3$$
which gives $a_2 = 1$. Continuing we get:
$$(\cdots 1112_3) \cdot (\cdots 0002)_3 = (\cdots 0001)_3$$
This can be done for any prime $p \neq 2$.

4. Let $a/b \in \mathbb{Q}$ be in lowest terms with $p \mid b$. Then a/b has no expansion as a p-adic integer. For example, $1/5$ is not a 5-adic integer: 5 times any 5-adic integer has last digit 0, never 1. Such numbers need the negative powers of p introduced below.

5. Using the second example, find the p-adic representation of $\frac{1}{1-p}$.

By the above example, the elements of $\mathbb{Z}_{(p)} \subseteq \mathbb{Q}$ have expansions as p-adic integers, which gives a natural map:
$$\mathbb{Z}_{(p)} \to \mathbb{Z}_p$$

This map is not surjective, as the right hand side is uncountable and $\mathbb{Z}_{(p)}$ is countable. The ring $\mathbb{Z}_p$ is local with maximal ideal $p\mathbb{Z}_p$: an element u whose first digit is nonzero is a unit, since a v with $uv \equiv 1 \bmod p$ makes $1-uv \in p\mathbb{Z}_p$, and the geometric series inverts $uv = 1-(1-uv)$. More generally, the completion of a ring at a maximal ideal is a local ring [4, Completion At Maximal Ideals].

Every nonzero element of $\mathbb{Z}_p$ is a power of p times a unit, so inverting p gives the fraction field of $\mathbb{Z}_p$, denoted $\mathbb{Q}_p$. It consists of all "Laurent series", that is all elements of the form:

$$\sum_{-m}^{\infty} a_i p^i$$

for all $m \in \mathbb{Z}$, $0 \leq a_i < p$. This allows us to represent *any* rational number, for example if $a/b \in \mathbb{Q}$, then represent it as:

$$\frac{a}{b} = \frac{g}{h} p^{-m} \qquad g, h \in \mathbb{Z} \qquad \gcd(gh, p) = 1$$

We can then do the p-adic expansion of $g/h = p^m f$:

$$a_0 + a_1 p + a_2 p^2 + \cdots$$

we then shift everything down by m to get the p-adic expansion of f:

$$a_0 p^{-m} + a_1 p^{-m+1} + \cdots + a_{m-1} p^{-1} + a_m + a_{m+1} p + a_{m+2} p^2 + \cdots$$

this is called a p-adic number.

1.1.4 Relation to Complex Analysis These expansions resemble the Laurent expansions of meromorphic functions in complex analysis, which inspired the p-adic integers and numbers. Many results for number fields have exact parallels for function fields, and section 1.3 makes this precise.

Hence, we have the more general embedding:

$$\mathbb{Q} \to \mathbb{Q}_p = \mathbb{Z}_p[1/p]$$

where $\mathbb{Z}$ injects into $\mathbb{Z}_p$ (if a, b have the same p-adic expansion, $a-b$ is divisible by p^n for every n, hence $a = b$ in $\mathbb{Z}$). Given this, identify $\mathbb{Q} \subseteq \mathbb{Q}_p$ and $\mathbb{Z} \subseteq \mathbb{Z}_p$.

Let us now take another perspective on $\mathbb{Q}_p$ that is more analytic. Recall that in [5, Classification Of Absolute Values On Rationals] we classified all absolute values on $\mathbb{Q}$, namely the p-valuations and the usual absolute value. Recall the p-valuation is defined on $\mathbb{Q}$ as:

$$|a|_p = \left|\frac{b}{c}\right|_p = \left|p^m \frac{b'}{c'}\right|_p = p^{-m} \qquad \gcd(b'c', p) = 1$$

Where in the p-adic representation of a, we may think of ν_p as counting how many 0's there are before hitting the first non-zero digit, for example in $625 = 1000_5$ we have:

$$|625|_5 = \frac{1}{5^4}$$

Or more algebraically, we take the maximal n where $a \equiv 0 \bmod p^n$, $a \not\equiv 0 \bmod p^{n+1}$. Hence, the larger the order of the prime p in a/b, the smaller the valuation (intuitively, the "less" detectable

the number is when taking $\bmod p^n$). This is perhaps intuitive if we compare p-adic expansion to decimal expansion, in which we have:

$$a_0 + a_1\left(\frac{1}{10}\right) + a_2\left(\frac{1}{10}\right)^2 + \cdots$$

with $1 \leq a_0 < 10$. Then the more a_i's that are zero, the smaller the valuation. The above representation of a number in decimal-expansion is good for visualizing a valuation called the ∞-valuation, $|-|_\infty$ which is the unique Archimedean valuation on $\mathbb{Q}$ up to equivalence (as all p-valuations induce an ultra-metric, they are all non-Archimedean). In this construction, what is important is the exponent, and not the p. The intuition from this comes from the induced p-adic topology, namely from $|-|_p$ we can define the [ultra]-metric

$$d(x, y) = |x - y|_p$$

from which we can construct $\mathbb{Q}_p$ from $\mathbb{Q}$ by taking it's completion using the above metric. Then choosing another constant c will produce equivalent topologies; what matter's for the definition is the factoring of the p and not the "concrete representation".

To continue creating intuition's, let us motivate the $|-|_\infty$ symbol. This comes from considering $k(t)$ over a finite field k, where the analog of $\mathbb{Z}$ is $k[x]$. In $k[x]$, there are monic irreducible polynomials which are in one-to-one correspondence with the primes $\mathfrak{p}$ of $k[t]$ (recall $k[t]$ is a PID). Hence, define $|-|_\mathfrak{p} : k(t) \to \mathbb{R}$ as:

$$f(t) = \frac{g(t)}{h(t)} = p(t)^m \frac{g'(t)}{h'(t)} \qquad (g'h', p) = 1$$

Then define $\nu_\mathfrak{p}(f) = m$. Then topologically the choice of what to exponentiate does not matter, however an informed choice will give us more information, namely let $q_\mathfrak{p} = q^{d_\mathfrak{p}}$ where q is any real number > 1 and d_p is the degree of the residue class field of $\mathfrak{p}$ over k (i.e. of $\kappa(f)$). Then:

$$\nu_\mathfrak{p}(f) = m \qquad |f|_p = q_\mathfrak{p}^{-\nu_\mathfrak{p}(f)}$$

Notice that $\nu_\mathfrak{p}(0) = \infty$ and $|0|_\mathfrak{p} = 0$, giving us that these are indeed the discrete valuations. When $\mathfrak{p} = (t - a)$, the valuation $\nu_\mathfrak{p}(f)$ gives the order of the zero/poles of the function $f(t)$ at $t = a$. Then for the infinite case, we have:

$$\nu_\infty : k(t) \to \mathbb{Z} \cup \{\infty\} \qquad \nu_\infty(f) = \deg(h) - \deg(g)$$

for $f = g/h \neq 0$ and $g, h \in k[t]$. This can be thought of describing the zero and poles at the point at infinity, i.e. the zeros and pole of the function $f(1/t)$ at the point 0. It is in fact associated to a prime ideal, namely $\mathfrak{p} = (1/t)$ in $k[1/t] \subseteq k(t)$. Hence, when considering the absolute value $|-|$ on $\mathbb{Q}$, we may think of this as being the evaluation at the point at infinity, and hence the $|-|_\infty$ symbol. Notice too how this motivates that the size of $|-|_\infty$ diminishes the more zeros' there are in the decimal expansion and taking the reciprocal of integers. As the concrete value of "t" does not matter, we can consider any integer and get an expansion that mimics the properties of $|-|_\infty \to 0$, for example

$$a_0 + a_1\left(\frac{1}{2}\right) + a_2\left(\frac{1}{2}\right)^2 + \cdots$$

Finally, for $k(t)$, there is also a 0-valuation given by:

$$|f/g|_0 = \operatorname{ord} g - \operatorname{ord} f$$

where the order represents the first degree of the smallest non-trivial monomial, for example $\text{ord}\,(ax^3 + bx) = 1$. Then in $\mathbb{Q}$, $|-|_0$ is the trivial valuation.

Coming back to absolute values over $\mathbb{Q}$, as what is important is the exponent and not the choice of constant to be exponentiated, it is often better to define a new functional, namely a discrete valuation, which only captures the exponent information. The reader should recall from [4, Discrete Valuation] the following the function $\nu_p : \mathbb{Q} \to \mathbb{Z} \cup \{\infty\}$ where $x + \infty = \infty + x = \infty + \infty = \infty$:

1. $\nu_p(a) = \infty$ if and only if $a = 0$ (this mirror's the idea that p divides 0 infinitely many times
2. $\nu_p(ab) = \nu_p(a) + \nu_p(b)$
3. $\nu_p(a + b) \geq \min(\nu_p(a), \nu_p(b))$

The last condition can be thought of as the ultra-metric condition[1]. From this, we may define p-valuations as

$$|-|_p : \mathbb{Q} \to \mathbb{R} \qquad a \mapsto |a|_p = p^{-\nu_p(a)}$$

and we may choose to change p to any other constant $c > 1$.

Now, with the p-adic metric properly defined on $\mathbb{Q}$, we may take the completion of $\mathbb{Q}$ under this metric by taking all Cauchy-sequences of $\mathbb{Q}$ and taking the quotient of all null-sequences (sequences that approach 0). Note that null-sequences form a maximal ideal in the ring of Cauchy-sequences. If R is the ring of Cauchy-sequences and $\mathfrak{m}$, then:

$$\mathbb{Q}_p \cong R/\mathfrak{m}$$

and hence, is an equivalent definition. The field $\mathbb{Q}_p$ is complete, and the same construction with $|-|_\infty$ gives $\mathbb{Q}_\infty = \mathbb{R}$. The field $\mathbb{Q}$ embeds in $\mathbb{Q}_p$ via constant sequences, and the p-adic absolute value extends to $\mathbb{Q}_p$ by

$$|x|_p := \lim_{n\to\infty} |x_n|_p \in \mathbb{R}$$

where (x_n) is a Cauchy sequence representing x[2]. The extended absolute value still takes values in $p^{\mathbb{Z}} \cup \{0\}$. Let $x \neq 0$. As (x_n) is not a null sequence, there is a $\delta > 0$ with $|x_m|_p \geq \delta$ for infinitely many m. Choose N with $|x_n - x_m|_p < \delta$ for all $n, m \geq N$, and fix $m \geq N$ with $|x_m|_p \geq \delta$. For every $n \geq N$ we then have $|x_n - x_m|_p < |x_m|_p$, and the ultrametric inequality gives $|x_n|_p = |x_m|_p$. So $\nu_p(x_n)$ is constant for $n \geq N$, and defining $\nu_p(x)$ to be this constant value gives

$$\nu_p(x) = \nu_p(x_n) \quad \text{for } n \geq N, \qquad |x|_p = p^{-\nu_p(x)}$$

We now have a collection of complete fields with different absolute values:

$$\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_5, \mathbb{Q}_7, ..., \mathbb{Q}_\infty = \mathbb{R}$$

From each of these, we may extract $\mathbb{Z}_p$ in the following way:

[1] For $c > 1$, a discrete valuation ν gives the non-Archimedean absolute value $c^{-\nu}$, and its third axiom becomes the ultra-metric inequality of [5, Characterizing Non-Archimedean Valuation].

[2] The limit exists and does not depend on the representative, as the proof of the next proposition checks.

Proposition 1.1.5: Characterizing p-adic Integers

Considering $\mathbb{Q}_p$ as the completion as given above, show that:

$$\mathbb{Z}_p = \{x \in \mathbb{Q}_p \mid |x|_p \leq 1\}$$

is a well-defined ring. Then $\mathbb{Z}_p$ is the closure of $\mathbb{Z}$ with respect to $|-|_p$

***Proof*:**
Write $S = \{x \in \mathbb{Q}_p \mid |x|_p \leq 1\}$. The set S is well-defined because $|x|_p$ depends only on the class of the Cauchy-sequence (x_n): if (x_n) and (y_n) differ by a null-sequence, then the triangle inequality on $\mathbb{Q}$ gives

$$\big||x_n|_p - |y_n|_p\big| \leq |x_n - y_n|_p \to 0$$

and so $\lim_n |x_n|_p = \lim_n |y_n|_p$. Next, passing the relations $|x_n y_n|_p = |x_n|_p |y_n|_p$ and $|x_n + y_n|_p \leq \max(|x_n|_p, |y_n|_p)$ to the limit shows that $|-|_p$ is multiplicative and ultrametric on all of $\mathbb{Q}_p$. The set S contains 0 and 1 since $|0|_p = 0$ and $|1|_p = 1$, and for $x, y \in S$:

$$|xy|_p = |x|_p |y|_p \leq 1 \qquad |x+y|_p \leq \max(|x|_p, |y|_p) \leq 1 \qquad |-x|_p = |x|_p \leq 1$$

Hence S is a subring of $\mathbb{Q}_p$.

It remains to show that S is the closure of $\mathbb{Z}$ in $\mathbb{Q}_p$. For one inclusion, $\mathbb{Z} \subseteq S$ since $\nu_p(a) \geq 0$ for $a \in \mathbb{Z}$, and S is closed: if $|x|_p > 1$ and $|y - x|_p < |x|_p$, then the ultrametric inequality gives $|y|_p \leq \max(|y-x|_p, |x|_p) = |x|_p$ and $|x|_p \leq \max(|x-y|_p, |y|_p)$, where the latter maximum must be $|y|_p$ since $|x-y|_p < |x|_p$, so $|y|_p = |x|_p > 1$ and the complement of S is open. The closure of $\mathbb{Z}$ is therefore contained in S. Conversely, let $x \in S$ be nonzero (the case $x = 0$ is trivial) and let (x_n) be a representative Cauchy-sequence with each $x_n \neq 0$. As shown above, $\nu_p(x_n)$ is eventually constant equal to $\nu_p(x) \geq 0$, so after discarding finitely many terms each x_n lies in $\mathbb{Z}_{(p)}$, say $x_n = a_n/b_n$ with $a_n, b_n \in \mathbb{Z}$ and $p \nmid b_n$. Since b_n is invertible mod p^n, there is a $c_n \in \mathbb{Z}$ with $b_n c_n \equiv a_n \bmod p^n$, and then

$$|x_n - c_n|_p = \frac{|a_n - b_n c_n|_p}{|b_n|_p} \leq p^{-n}$$

Hence $|x - c_n|_p \leq \max(|x - x_n|_p, |x_n - c_n|_p) \to 0$, where $|x - x_n|_p = \lim_m |x_m - x_n|_p \to 0$ as $n \to \infty$ since (x_n) is Cauchy. Thus x is a limit of integers, and S is the closure of $\mathbb{Z}$, as we sought to show.

Using the p-adic valuation, we may characterize the units of $\mathbb{Z}_p$ as:

$$\mathbb{Z}_p^* = \{x \in \mathbb{Z}_p \mid |x|_p = 1\}$$

Using this, we may represent every $x \in \mathbb{Q}_p^*$ as :

$$x = p^m u \qquad m \in \mathbb{Z},\ u \in \mathbb{Z}_p^*$$

that is, every element of $\mathbb{Q}_p^*$ is some unit times some power of prime. For if:

$$\nu_p(x) = m \in \mathbb{Z} \quad \Rightarrow \quad \nu_p(xp^{-m}) = 0 \quad \Rightarrow \quad |xp^{-m}|_p = 1$$

thus $u = xp^{-m} \in \mathbb{Z}_p^*$. In the localization $\mathbb{Z}_{(p)}$ every prime other than p is a unit, so a prime factorization in $\mathbb{Z}_{(p)}$ has only the prime p left, and the above argument shows the same in the completion. From this, we can characterize the ideals of $\mathbb{Z}_p$:

Proposition 1.1.6: Classifying Ideals Of p-adic Integers

Let $\mathbb{Z}_p$ be the p-adic integers. Then the nonzero ideals of $\mathbb{Z}_p$ are the principal ideals:

$$p^n\mathbb{Z}_p = \{x \in \mathbb{Q}_p \mid \nu_p(x) \geq n\}$$

with $n \geq 0$. Furthermore:

$$\mathbb{Z}_p/p^n\mathbb{Z}_p \cong \mathbb{Z}/p^n\mathbb{Z}$$

***Proof*:**

By proposition 1.1.5, $\mathbb{Z}_p = \{x \in \mathbb{Q}_p \mid \nu_p(x) \geq 0\}$, and $|-|_p = p^{-\nu_p(-)}$ is multiplicative on $\mathbb{Q}_p$, so $\nu_p(xy) = \nu_p(x) + \nu_p(y)$. An element $u \in \mathbb{Z}_p$ is a unit of $\mathbb{Z}_p$ if and only if $\nu_p(u) = 0$: if $uv = 1$ with $v \in \mathbb{Z}_p$, then $\nu_p(u) + \nu_p(v) = 0$ with both terms nonnegative, and if $\nu_p(u) = 0$, then $\nu_p(u^{-1}) = 0$ and $u^{-1} \in \mathbb{Z}_p$. Hence every nonzero $x \in \mathbb{Z}_p$ is $x = p^{\nu_p(x)}u$ with $u = xp^{-\nu_p(x)}$ a unit. For $n \geq 0$ this gives $p^n\mathbb{Z}_p = \{x \in \mathbb{Q}_p \mid \nu_p(x) \geq n\}$, since $\nu_p(p^n z) = n + \nu_p(z) \geq n$ for $z \in \mathbb{Z}_p$, and an x with $\nu_p(x) \geq n$ is $p^n(xp^{-n})$ with $\nu_p(xp^{-n}) \geq 0$. Now let $I \neq 0$ be an ideal of $\mathbb{Z}_p$, and let n be the least value of ν_p on $I \setminus \{0\}$, attained at $x = p^n u \in I$ with u a unit. Then $p^n = xu^{-1} \in I$, so $p^n\mathbb{Z}_p \subseteq I$, and every element of I has valuation at least n, so $I \subseteq p^n\mathbb{Z}_p$. Thus $I = p^n\mathbb{Z}_p$.

To show the isomorphism, we show that the map $\mathbb{Z} \to \mathbb{Z}_p/p^n\mathbb{Z}_p$, $a \mapsto a \bmod p^n\mathbb{Z}_p$, is surjective. As $\mathbb{Z}$ is dense in $\mathbb{Z}_p$ (proposition 1.1.5), for any $x \in \mathbb{Z}_p$ there exists an $a \in \mathbb{Z}$ such that

$$|x - a|_p \leq \frac{1}{p^n}$$

Thus $\nu_p(x - a) \geq n$, so $x - a \in p^n\mathbb{Z}_p$ by the first part, i.e. $x \equiv a \bmod p^n\mathbb{Z}_p$. Hence the map is surjective, and its kernel is $\mathbb{Z} \cap p^n\mathbb{Z}_p = \{a \in \mathbb{Z} \mid \nu_p(a) \geq n\} = p^n\mathbb{Z}$, hence:

$$\mathbb{Z}_p/p^n\mathbb{Z}_p \cong \mathbb{Z}/p^n\mathbb{Z}$$

as we sought to show.

Using this, we have that:

Proposition 1.1.7: Next Equivalence Of p-adic Integers

Show that with $\mathbb{Z}_p$ as defined from above we have:

$$\mathbb{Z}_p \cong \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$$

***Proof*:**
By proposition 1.1.6, every $x \in \mathbb{Z}_p$ satisfies $x - a_n \in p^n\mathbb{Z}_p$ for an integer a_n whose class modulo p^n is determined by x, and $x \mapsto a_n \bmod p^n$ is a surjective ring homomorphism $\mathbb{Z}_p \to \mathbb{Z}/p^n\mathbb{Z}$. An integer that works for $n+1$ also works for n, so these maps are compatible with the projections $\mathbb{Z}/p^{n+1}\mathbb{Z} \to \mathbb{Z}/p^n\mathbb{Z}$ and define a ring homomorphism

$$\varphi\colon \mathbb{Z}_p \to \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$$

If $\varphi(x) = 0$, then $x \in p^n\mathbb{Z}_p$, that is $\nu_p(x) \geq n$, for every n, so $x = 0$. For surjectivity, let $(c_n)_n$ be a compatible family and choose integers a_n representing c_n. Compatibility gives $a_{n+1} - a_n \in p^n\mathbb{Z}$, so $|a_m - a_n|_p \leq p^{-n}$ for $m \geq n$ by the ultrametric inequality, and (a_n) is Cauchy. Its limit x lies in the complete field $\mathbb{Q}_p$, and in $\mathbb{Z}_p$, which is closed as the closure of $\mathbb{Z}$ (proposition 1.1.5). Letting $m \to \infty$ gives $|x - a_n|_p \leq p^{-n}$, so $x - a_n \in p^n\mathbb{Z}_p$ and $\varphi(x) = (c_n)_n$. Hence φ is an isomorphism, as we sought to show.

Finally, let us characterize p-adic integers in one more way and quickly address a possible misconception of the fields $\mathbb{Q}_p$ with respect to $\mathbb{R}$. For the new characterization:

Proposition 1.1.8: Equivalence Of p-adic Via Power-series Quotient

$$\mathbb{Z}_p \cong \mathbb{Z}[[x]]/(x - p)$$

***Proof*:**
For each $n \geq 1$ define $\varphi_n : \mathbb{Z}[[x]] \to \mathbb{Z}/p^n\mathbb{Z}$ by

$$\varphi_n\left(\sum_{i=0}^{\infty} a_i x^i\right) = \sum_{i=0}^{n-1} a_i p^i \bmod p^n$$

which is a finite sum. Each φ_n is additive, and it is multiplicative because in the product $\left(\sum_{i<n} a_i p^i\right)\left(\sum_{j<n} b_j p^j\right)$ every term with $i + j \geq n$ lies in $p^n\mathbb{Z}$, so both $\varphi_n(fg)$ and $\varphi_n(f)\varphi_n(g)$ are equal to $\sum_{k<n}\left(\sum_{i+j=k} a_i b_j\right) p^k \bmod p^n$. The φ_n are compatible with the projections $\mathbb{Z}/p^{n+1}\mathbb{Z} \to \mathbb{Z}/p^n\mathbb{Z}$, and hence induce a ring homomorphism

$$\varphi : \mathbb{Z}[[x]] \to \varprojlim_n \mathbb{Z}/p^n\mathbb{Z} \cong \mathbb{Z}_p$$

where the identification is proposition 1.1.7.

For surjectivity, let $(\overline{r}_n)_n \in \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$ and choose integers r_n representing $\overline{r}_n$, so that $r_{n+1} \equiv r_n \bmod p^n$, say $r_{n+1} - r_n = p^n t_n$ with $t_n \in \mathbb{Z}$. Set $f = r_1 + \sum_{n\geq 1} t_n x^n$. Then for each $m \geq 1$, telescoping gives:

$$\varphi_m(f) = r_1 + \sum_{n=1}^{m-1} t_n p^n = r_1 + \sum_{n=1}^{m-1} (r_{n+1} - r_n) = r_m \bmod p^m$$

and so $\varphi(f) = (\overline{r}_n)_n$.

For the kernel, first $\varphi_n(x-p) = p - p = 0$ for $n \geq 2$ and $\varphi_1(x-p) = -p \equiv 0 \bmod p$, so $(x-p) \subseteq \ker\varphi$. Conversely, let $f = \sum_i a_i x^i \in \ker\varphi$, which means that the partial sums $s_n := \sum_{i=0}^{n} a_i p^i$ satisfy $p^{n+1} \big| s_n$ in $\mathbb{Z}$ for every $n \geq 0$. Then $b_n := -s_n/p^{n+1} \in \mathbb{Z}$, and $g := \sum_n b_n x^n \in \mathbb{Z}[[x]]$ satisfies $(x-p)g = f$: the constant coefficient of $(x-p)g$ is $-pb_0 = s_0 = a_0$, and for $n \geq 1$ the coefficient of x^n is

$$b_{n-1} - pb_n = -\frac{s_{n-1}}{p^n} + \frac{s_n}{p^n} = \frac{a_n p^n}{p^n} = a_n$$

Hence $\ker\varphi = (x-p)$, and the first isomorphism theorem gives $\mathbb{Z}[[x]]/(x-p) \cong \mathbb{Z}_p$, as we sought to show.

A common mistake a reader may do is think that $\mathbb{Q}_p \cong \mathbb{R}$ algebraically. We proved this is not the case in [4], however a concrete example should help.

Example 1.1.9: Cautions for p-adic Numbers

1. Recall that $1/p \notin \mathbb{Z}_p$ (review why if this is fuzzy). As $1/p \in \mathbb{Q}_p$, we cannot use it to construct our counter-example. Instead, we shall show that $\sqrt{p} \notin \mathbb{Q}_p$. To see this, notice that $\sqrt{p}$ by definition satisfies the equation $x^2 - p$. If $\sqrt{p} \in \mathbb{Q}_p$, then $|\sqrt{p}|_p^2 = |p|_p < 1$, so $\sqrt{p} \in \mathbb{Z}_p$, and by theorem 1.1.2 the congruence $x^2 \equiv p \bmod p^2$ would have an integer solution. It has none, as $x^2 \equiv p \bmod p$ forces $p \big| x$ and then $p^2 \big| x^2$. Hence $\sqrt{p} \notin \mathbb{Q}_p$.

   On the other hand, for odd p we have $\sqrt{1-p^3} \in \mathbb{Q}_p$. Indeed, $x_1 = 1$ solves $x^2 \equiv 1 - p^3 \bmod p$, and if $x_k \equiv 1 \bmod p$ solves it modulo p^k, with $1 - p^3 - x_k^2 = p^k c$, then $x_k + tp^k$ solves it modulo p^{k+1} for an integer t with $2x_k t \equiv c \bmod p$, which exists as $2x_k$ is invertible modulo p. By theorem 1.1.2, $x^2 - (1-p^3)$ has a root in $\mathbb{Z}_p$. However, $\sqrt{1-p^3} \notin \mathbb{R}$, as $1 - p^3 < 0$.

2. By field theory, we know that $\mathbb{Q}_p$ embeds into $\mathbb{C}$. This is an example where we really require the axiom of choice. The embeddings is far away from preserving the topological structure of $\mathbb{Q}_p$ or being represented in $\mathbb{C}$. To see this, first notice that $\pm\sqrt{-1} = \pm i \in \mathbb{Q}_5$. It may seem like we should map i to either $\pm i$. However, note that i is closer to 2 while $-i$ is closer to 3. In fact, i is even closer to 57 while the other is much closer to 68. In fact, without the axiom of choice, this embedding would not be possible. Algebraically, you may also try to check that $\mathbb{Q}_5$ has no automorphisms that swap i and $-i$ (in fact it has *no* non-trivial automorphisms, mirroring how $\mathbb{R}$ has no non-trivial automorphisms).

3. Another interesting distinction between $\mathbb{Q}_p$ and $\mathbb{R}$ the degree of $\overline{\mathbb{Q}_p}/\mathbb{Q}_p$ vs $\overline{\mathbb{R}}/\mathbb{R}$. In the latter, we have $\overline{\mathbb{R}} \cong \mathbb{C}$, and we have a degree two extension. After covering Hensel's lemma, we shall see that $x^n - p$ is irreducible for each n in $\mathbb{Q}_p$. It can be shown that there are finitely many extensions for each degree n, and so $\overline{\mathbb{Q}_p}/\mathbb{Q}_p$ is a countable extension.

Finally, we introduce the following generalization of prime decompositions to $\mathbb{Q}$[3]:

[3]Proposition 5.2.2 generalizes this result to number fields.

Proposition 1.1.10: Artin-Whaples Product Formula

Let $0 \neq r \in \mathbb{Q}$. Then

$$\prod_{p,\infty} |r|_p = 1$$

where p varies over all primes number and ∞

***Proof*:**
Any nonzero $r \in \mathbb{Q}$ factors as

$$r = \pm \prod_{p \neq \infty} p^{\nu_p(r)}$$

with only finitely many $\nu_p(r) \neq 0$. Hence $|r|_p = p^{-\nu_p(r)}$ for each prime p, while $|r|_\infty = \prod_{p\neq\infty} p^{\nu_p(r)}$, and the product of all of these is 1, as we sought to show.

The same product formula holds for nonzero $f \in k(t)$, with the absolute values $|f|_\mathfrak{p} = q_\mathfrak{p}^{-\nu_\mathfrak{p}(f)}$ and $|f|_\infty = q^{-\nu_\infty(f)}$ from the discussion of $k(t)$ above:

$$\prod_{\mathfrak{p},\infty} |f|_\mathfrak{p} = 1$$

Indeed, writing $f = c\prod_\mathfrak{p} p_\mathfrak{p}^{\nu_\mathfrak{p}(f)}$ with $c \in k^\times$ and monic irreducible $p_\mathfrak{p}$ of degree $d_\mathfrak{p}$, the product is $q^{-\sum_\mathfrak{p} d_\mathfrak{p}\nu_\mathfrak{p}(f) - \nu_\infty(f)}$, and $\sum_\mathfrak{p} d_\mathfrak{p}\nu_\mathfrak{p}(f) = \deg f = -\nu_\infty(f)$.
Rearranged, the product formula expresses the archimedean absolute value of a nonzero rational number through all the p-adic ones:

$$|r|_\infty = \prod_p |r|_p^{-1}$$

Exercise 1.1.1

1. Let $f_1, \ldots, f_m \in \mathbb{Q}[x_1, ..., x_n]$, and for a field $F \supseteq \mathbb{Q}$ let $V(F)$ be the set of common zeros of the f_i in F^n. Show that if $V(\mathbb{Q})$ is nonempty, then $V(\mathbb{Q}_p)$ is nonempty for each prime p, and $V(\mathbb{R})$ is nonempty.

   The converse fails: Selmer showed that the equation $3x^3 + 4y^3 + 5z^3 = 0$ has solutions $(x, y, z) \neq 0$ over every $\mathbb{Q}_p$ and over $\mathbb{R}$, but none over $\mathbb{Q}$. So the *Hasse principle*, or *local-to-global principle*, does not hold for this cubic.

1.2 Discrete Valuation Rings and Completions

This construction started with the integers, i.e. the "fundamental" ring of integers. We now need to have a way to generalize this notion for rings of integers more generally. Recall that Dedekind domains are locally DVRs (see [2, Dedekind Domain As DVR]). The intuition is that the localization of a Dedekind domain to the DVR gives information of the multiplicity of a prime. Topologically, we

may take advantage of the fact that ideals in $\mathbb{Z}_p$ form a chain:

$$\mathbb{Z}_p \supseteq (p) \supseteq (p)^2 \supseteq ...$$

More generally, as any discrete valuation ring $\mathcal{O}$ has a chain of ideals, we have:

$$\mathcal{O} \supseteq \mathfrak{p} \supseteq \mathfrak{p}^2 \supseteq ...$$

Then these ideals form a neighborhood-basis:

Definition 1.2.1: Complete DVR

let $\mathcal{O}$ be a DVR with discrete valuation ν. Then $\hat{\mathcal{O}}$ is its completion with respect to ν.

If ν is a discrete valuation [4, Discrete Valuation] normalized so that $\nu(K^\times) = \mathbb{Z}$, and $|-| = q^{-\nu}$ for $q > 1$, we have:

$$\mathfrak{p}^n = \left\{ x \in K \;\middle|\; |x| < \frac{1}{q^{n-1}} \right\}$$

and we can define a topology on K^* through the sets:

$$U^{(n)} = 1 + \mathfrak{p}^n = \left\{ x \in K^* \;\middle|\; |1 - x| < \frac{1}{q^{n-1}} \right\} \qquad n > 0$$

so that we have the descending chain:

$$\mathcal{O}^* = U^{(0)} \supseteq U^{(1)} \supseteq U^{(2)} \supseteq \cdots$$

Notice these are all subgroups under multiplication (use that the valuation is a homomorphism and that $1 + \mathfrak{p}^n = \left\{ x \in K^* \;\middle|\; |1 - x|_p < \frac{1}{q^{n-1}} \right\}$ for some choice of q to define the p-absolute value). These subgroups are called the *higher unit groups* and $U^{(1)}$ is called the group of *principal units*. These groups have the following property:

Proposition 1.2.2: Higher Unit Group

For $n \geq 1$,

$$\mathcal{O}^*/U^{(n)} \cong (\mathcal{O}/\mathfrak{p}^n)^* \qquad \text{and} \qquad U^{(n)}/U^{(n+1)} \cong \mathcal{O}/\mathfrak{p}$$

***Proof*:**
Take the surjection $\mathcal{O}^* \to (\mathcal{O}/\mathfrak{p}^n)^*$. The kernel then is certainly $U^{(n)}$. Similarly for the 2nd isomorphism, where $U^{(n)} = 1 + \pi^n\mathcal{O}$ which maps $1 + \pi^n a \mapsto a \bmod \mathfrak{p}$.

Naturally, a complete DVR is best characterized as a limit of the rings $\mathcal{O}/\mathfrak{p}^n$:

Proposition 1.2.3: Characterizing Complete DVR Via Limit

Let $\mathcal{O}$ be a complete DVR. Then:

$$\mathcal{O} \cong \varprojlim_n \mathcal{O}/\mathfrak{p}^n \qquad \mathcal{O}^* \cong \varprojlim_n \mathcal{O}^*/U^{(n)}$$

showing that each complete valuation ring is naturally characterized as a limit given by its unique maximal ideal.

***Proof*:**
Write ν for the normalized valuation of $\mathcal{O}$, π for a uniformizer, $\mathfrak{p} = (\pi)$ for the maximal ideal, and $|x| = q^{-\nu(x)}$ for the associated absolute value, so that $\mathfrak{p}^n = (\pi^n) = \{x \in \mathcal{O} \mid \nu(x) \geq n\}$ as in the display above. The quotient maps $\mathcal{O}/\mathfrak{p}^{n+1} \to \mathcal{O}/\mathfrak{p}^n$ make $(\mathcal{O}/\mathfrak{p}^n)_n$ an inverse system, and $\varprojlim_n \mathcal{O}/\mathfrak{p}^n$ is the subring of $\prod_n \mathcal{O}/\mathfrak{p}^n$ consisting of the compatible families. Reduction modulo the powers of $\mathfrak{p}$ is a ring homomorphism:

$$\varphi : \mathcal{O} \to \varprojlim_n \mathcal{O}/\mathfrak{p}^n \qquad \varphi(x) = (x \bmod \mathfrak{p}^n)_n$$

Injectivity. If $\varphi(x) = 0$ then $x \in \bigcap_n \mathfrak{p}^n$. A nonzero x has $\nu(x) = m < \infty$ and so $x \notin \mathfrak{p}^{m+1}$. Hence $\bigcap_n \mathfrak{p}^n = 0$ and $\ker \varphi = 0$.

Surjectivity. Let $(x_n)_n$ be a compatible family and choose $a_n \in \mathcal{O}$ representing x_n. Compatibility says $a_{n+1} - a_n \in \mathfrak{p}^n$, that is $\nu(a_{n+1} - a_n) \geq n$, and for $m > n$ the ultrametric inequality gives:

$$\nu(a_m - a_n) \geq \min_{n \leq i < m} \nu(a_{i+1} - a_i) \geq n$$

So $|a_m - a_n| \leq q^{-n}$ and (a_n) is Cauchy. Completeness of $\mathcal{O}$ gives an $a \in \mathcal{O}$ with $a_n \to a$. Fix n. For m large enough $\nu(a - a_m) \geq n$, while $\nu(a_m - a_n) \geq n$, so $\nu(a - a_n) \geq n$ and $a \equiv a_n \equiv x_n \bmod \mathfrak{p}^n$. Hence $\varphi(a) = (x_n)_n$ and φ is a ring isomorphism.

Topologies. The ideals $\mathfrak{p}^n$ are a basis of neighbourhoods of 0 in $\mathcal{O}$, and the kernels of the projections $\varprojlim_m \mathcal{O}/\mathfrak{p}^m \to \mathcal{O}/\mathfrak{p}^n$ are a basis of neighbourhoods of 0 in the limit. The preimage under φ of the nth such kernel is $\{x \in \mathcal{O} \mid x \equiv 0 \bmod \mathfrak{p}^n\} = \mathfrak{p}^n$, so φ matches the two bases and is a homeomorphism as well.

The unit groups. For $n \geq 1$ the inclusions $U^{(n+1)} \subseteq U^{(n)}$ make $(\mathcal{O}^*/U^{(n)})_n$ an inverse system of groups, and reduction gives a group homomorphism $\psi : \mathcal{O}^* \to \varprojlim_n \mathcal{O}^*/U^{(n)}$. Its kernel is $\bigcap_n U^{(n)}$, and $u \in \bigcap_n (1 + \mathfrak{p}^n)$ has $u - 1 \in \bigcap_n \mathfrak{p}^n = 0$, so ψ is injective. For surjectivity let $\left(a_n U^{(n)}\right)_n$ be a compatible family with $a_n \in \mathcal{O}^*$. Compatibility says $a_{n+1} a_n^{-1} \in U^{(n)}$, hence $a_{n+1} - a_n \in a_n \mathfrak{p}^n = \mathfrak{p}^n$ because a_n is a unit, and the estimate of the previous paragraphs again makes (a_n) Cauchy with a limit $a \in \mathcal{O}$ satisfying $a \equiv a_n \bmod \mathfrak{p}^n$ for every n. Taking $n = 1$ gives $a \equiv a_1 \bmod \mathfrak{p}$ with $a_1 \notin \mathfrak{p}$, so $\nu(a) = 0$ and $a \in \mathcal{O}^*$. Then $a a_n^{-1} - 1 = (a - a_n) a_n^{-1} \in \mathfrak{p}^n$, so $a a_n^{-1} \in U^{(n)}$ for every n and $\psi(a)$ is the given family.

Completion of Discrete Valuation Ring

Let k be a field with a discrete valuation defined on it. This discrete valuation can be extended to $\widehat{k}$ and denoted $\widehat{\nu}$, the completion of k by setting:

$$\widehat{\nu}(a) = \lim_{n \to \infty} \nu(a_n)$$

Now, for any arbitrarily small ϵ, there exists an a_n such that $|a - a_n| < \epsilon$. Letting $\epsilon = p^{-N}$ for some large n ,we see that at some point it must be that:

$$\widehat{\nu}(a - a_n) \geq \widehat{\nu}(a)$$

and so

$$\nu(a_n) = \widehat{\nu}(a_n - a + a) = \min\{\widehat{\nu}(a_n - a), \widehat{\nu}(a)\} = \widehat{\nu}(a)$$

showing that the valuation extends to the completion:

$$\nu(k^*) = \widehat{\nu}(\widehat{k}^*)$$

Proposition 1.2.4: Valuation Ring Quotient And Completion

Let K be a field with valuation ν. Let $\mathcal{O} \subseteq K$ be the valuation ring of K and $\widehat{\mathcal{O}} \subseteq \widehat{K}$ with completed valuation $\widehat{\nu}$. Then:

$$\widehat{\mathcal{O}}/\widehat{\mathfrak{p}} \cong \mathcal{O}/\mathfrak{p}$$

If ν is discrete, then $\widehat{\nu}(\widehat{K}^\times) = \nu(K^\times)$ and, for every $n \geq 1$,

$$\widehat{\mathcal{O}}/\widehat{\mathfrak{p}}^n \cong \mathcal{O}/\mathfrak{p}^n$$

In particular, $\#\widehat{\mathcal{O}}/\widehat{\mathfrak{p}} = \#\mathcal{O}/\mathfrak{p}$, so the residue field of $\widehat{K}$ is finite exactly when that of K is.

***Proof*:**

The valuation ν is real-valued, and $\widehat{K}$ is the completion of K for $|x| = c^{\nu(x)}$ with a fixed $0 < c < 1$. Passing the relations $\nu(x_n y_n) = \nu(x_n) + \nu(y_n)$ and $\nu(x_n + y_n) \geq \min(\nu(x_n), \nu(y_n))$ to the limit shows that $\widehat{\nu}$ is again a valuation, with $\widehat{\mathcal{O}} = \left\{ x \in \widehat{K} \;\middle|\; \widehat{\nu}(x) \geq 0 \right\}$ and $\widehat{\mathfrak{p}} = \left\{ x \in \widehat{K} \;\middle|\; \widehat{\nu}(x) > 0 \right\}$. On constant sequences $\widehat{\nu}$ restricts to ν, so $\widehat{\mathcal{O}} \cap K = \mathcal{O}$ and $\widehat{\mathfrak{p}} \cap K = \mathfrak{p}$, and the inclusion $\mathcal{O} \to \widehat{\mathcal{O}}$ induces an injective ring homomorphism

$$\mathcal{O}/\mathfrak{p} \to \widehat{\mathcal{O}}/\widehat{\mathfrak{p}}$$

For surjectivity, let $x \in \widehat{\mathcal{O}}$. As K is dense in $\widehat{K}$, there is a $y \in K$ with $|x - y| < 1$, that is $\widehat{\nu}(x - y) > 0$. Then $\nu(y) = \widehat{\nu}(y) \geq \min(\widehat{\nu}(x), \widehat{\nu}(y - x)) \geq 0$, so $y \in \mathcal{O}$ and $x \equiv y \bmod \widehat{\mathfrak{p}}$. This proves the first isomorphism, and with it the equality of the sizes of the two residue fields.

Now let ν be discrete. Replacing ν by a positive multiple changes neither the valuation rings, their ideals, nor the completion, and it multiplies $\nu(K^\times)$ and $\widehat{\nu}(\widehat{K}^\times)$ by the same factor, so we may assume $\nu(K^\times) = \mathbb{Z}$. Let $\pi \in K$ have $\nu(\pi) = 1$. The extended valuation takes the same values as ν: for $x \in \widehat{K}^\times$ choose $y \in K$ with $|x - y| < |x|$, and the ultrametric inequality gives $|y| = |x|$, so $\widehat{\nu}(x) = \nu(y) \in \mathbb{Z}$. Hence $\widehat{\nu}(\widehat{K}^\times) = \mathbb{Z} = \nu(K^\times)$. Every $x \in \widehat{K}^\times$ is then $\pi^{\widehat{\nu}(x)} u$ with $\widehat{\nu}(u) = 0$, so $\widehat{\mathfrak{p}}^n = \pi^n \widehat{\mathcal{O}} = \left\{ x \in \widehat{K} \;\middle|\; \widehat{\nu}(x) \geq n \right\}$, and in the same way $\mathfrak{p}^n = \{x \in K \mid \nu(x) \geq n\}$. The kernel of the reduction $\mathcal{O} \to \widehat{\mathcal{O}}/\widehat{\mathfrak{p}}^n$ is $\mathcal{O} \cap \widehat{\mathfrak{p}}^n = \mathfrak{p}^n$. For surjectivity, given $x \in \widehat{\mathcal{O}}$, density gives $y \in K$ with $|x - y| \leq c^n$, that is $\widehat{\nu}(x - y) \geq n$. As before $y \in \mathcal{O}$, and $x \equiv y \bmod \widehat{\mathfrak{p}}^n$. Hence $\widehat{\mathcal{O}}/\widehat{\mathfrak{p}}^n \cong \mathcal{O}/\mathfrak{p}^n$, completing the proof.

Using this, the p-adic expansion extends to the completion of any field K with a discrete valuation. Let $\pi \in K$ be a uniformizer and let $R \subseteq \widehat{\mathcal{O}}$ be a set of representatives of $\widehat{\mathcal{O}}/\widehat{\mathfrak{p}} \cong \mathcal{O}/\mathfrak{p}$ containing 0. For $x \in \widehat{K}^\times$ with $\widehat{\nu}(x) = m$, the unit $x\pi^{-m}$ is congruent modulo $\widehat{\mathfrak{p}}$ to a unique $a_m \in R$, and $x - a_m\pi^m$ has valuation greater than m. Repeating this with $x - a_m\pi^m$ in place of x gives digits $a_i \in R$ with $\widehat{\nu}\big(x - \sum_{i=m}^{N} a_i \pi^i\big) > N$ for every N, so that

$$x = \sum_{i \geq m} a_i \pi^i$$

The digits are unique: at the first index j where two such expansions of x differ, the difference of the two digits is a unit, so the difference of the two series has valuation j and is not 0. For the t-adic valuation of $\mathbb{F}_q(t)$, with $\pi = t$ and $R = \mathbb{F}_q$, every element of the completion is thus a Laurent series $\sum_{i \geq m} a_i t^i$ with $a_i \in \mathbb{F}_q$. As $\mathbb{F}_q$ is a subfield, sending a formal Laurent series to the sum of its series respects sums and products, since a product of partial sums differs from the corresponding partial sum of the formal product by terms of arbitrarily large valuation. So the completion of $\mathbb{F}_q(t)$ is the field $\mathbb{F}_q((t))$ of formal Laurent series.

Exercise 1.2.1

1. Let p be an odd prime. Show that $\sqrt{2} \in \mathbb{Q}_p$ if and only if $x^2 - 2$ factors into linear factors modulo p.

1.3 Local and Global Fields

An overarching theme in this entire book is to understand the relation between a field (such as a number field) and its localizations. It was already noted in box 1.1.4 that function field and number fields share many of the same characteristics when localizing. Let us box these notions to study this further:

Definition 1.3.1: Global And Local Fields

1. A *global field* if a finite extension of either $\mathbb{Q}$ or $\mathbb{F}_p(t)$.
2. A *local field* is a field which is complete with respect to a discrete valuation ν and its residue field is finite.

Note that the valuation on the completion of global fields is discrete and has a finite residue class field. If we do not specify the finite residue field, we shall get some constructions we are not looking for. We should immediately re-characterize local fields to make them easier to work with:

Proposition 1.3.2: Characterizing Local Fields

Let K be a local field, with normalized valuation ν and residue field $\mathbb{F}_q$ of characteristic p. Then K has characteristic 0 or p.

1. If K has characteristic 0, then the closure of $\mathbb{Q}$ in K is a copy of $\mathbb{Q}_p$, on which ν restricts to $e\,\nu_p$ with $e = \nu(p)$, and K is a finite extension of it.
2. If K has characteristic p, then $K \cong \mathbb{F}_q((t))$, which is a finite extension of $\mathbb{F}_p((t))$ spanned over it by $\mathbb{F}_q$.

Some authors will define local fields to be completion of fields with respect to an absolute value to allow for $\mathbb{R}$ to be a local field, but this will not be done here.

***Proof*:**
Let ν be the discrete valuation of K, normalized so that $\nu(K^\times) = \mathbb{Z}$, let $\mathcal{O}$ be its valuation ring, π a uniformizer and $\mathfrak{p} = \pi\mathcal{O}$, and let $\kappa = \mathcal{O}/\mathfrak{p}$ be the finite residue field, of characteristic p with $q = p^f$ elements. If ℓ is a prime with $\ell \cdot 1 = 0$ in K, then $\ell \cdot 1 = 0$ in κ as well, so $\ell = p$. Hence K has characteristic 0 or p, and the two cases give the two items.

1. Suppose K has characteristic 0, so that $\mathbb{Q} \subseteq K$. Every integer is a sum of copies of 1 or of -1, so $\nu(n) \geq 0$ for $n \in \mathbb{Z}$ by the ultrametric inequality. The image of p in κ is 0, so $e := \nu(p) \geq 1$, while an integer n with $p \nmid n$ has nonzero image in κ, so $\nu(n) = 0$. Writing a nonzero rational number as $p^k a/b$ with $p \nmid ab$ gives $\nu(p^k a/b) = ke$, that is, $\nu = e\,\nu_p$ on $\mathbb{Q}$. So a sequence in $\mathbb{Q}$ is Cauchy, or null, for $|-|$ if and only if it is so for the absolute value of K. As K is complete, sending the class of a Cauchy sequence (x_n) in $\mathbb{Q}$ to $\lim_n x_n \in K$ is a well-defined ring homomorphism $\iota\colon \mathbb{Q}_p \to K$, injective because $\mathbb{Q}_p$ is a field, and $\nu(\iota(x)) = e\,\nu_p(x)$ because $\nu_p(x_n)$ is eventually constant equal to $\nu_p(x)$. The image of ι contains $\mathbb{Q}$, and it is complete, as $\nu \circ \iota = e\,\nu_p$, hence closed in K. As $\mathbb{Q}$ is dense in $\mathbb{Q}_p$, this image is the closure of $\mathbb{Q}$ in K. We identify $\mathbb{Q}_p$ with its image, so that $\mathbb{Z}_p \subseteq \mathcal{O}$ and $p^N\mathbb{Z}_p \subseteq p^N\mathcal{O}$ for every $N \geq 0$.

   Choose $\omega_1, \ldots, \omega_f \in \mathcal{O}$ whose residues form an $\mathbb{F}_p$-basis of κ, and let S be the set of sums $\sum_{i,j} c_{ij}\omega_i\pi^j$ with $c_{ij} \in \mathbb{Z}$, $1 \leq i \leq f$ and $0 \leq j < e$. Every $y \in \mathcal{O}$ lies in $S + p\mathcal{O}$: the residue of y is an $\mathbb{F}_p$-combination of the residues of the ω_i, so $y - \sum_i c_{i0}\omega_i \in \pi\mathcal{O}$ for suitable integers c_{i0}, and repeating this with $(y - \sum_i c_{i0}\omega_i)/\pi$ in place of y, e times in all, gives $y \in S + \pi^e\mathcal{O}$, where $\pi^e\mathcal{O} = p\mathcal{O}$ because $\nu(p) = e$. Now let $x \in \mathcal{O}$. Applying this to x, then to the resulting element of $\mathcal{O}$, and so on, gives $s_k \in S$ with $x - \sum_{k<N} p^k s_k \in p^N\mathcal{O}$ for every N. Write $s_k = \sum_{i,j} c_{ij}^{(k)}\omega_i\pi^j$. For each (i,j) the series $a_{ij} = \sum_k c_{ij}^{(k)}p^k$ converges in $\mathbb{Z}_p$, with $a_{ij} - \sum_{k<N} c_{ij}^{(k)}p^k \in p^N\mathbb{Z}_p$, and

$$x - \sum_{i,j} a_{ij}\omega_i\pi^j = \Big(x - \sum_{k<N} p^k s_k\Big) - \sum_{i,j}\Big(a_{ij} - \sum_{k<N} c_{ij}^{(k)}p^k\Big)\omega_i\pi^j \in p^N\mathcal{O}$$

   for every N. As $\bigcap_N p^N\mathcal{O} = 0$, the ef elements $\omega_i\pi^j$ generate $\mathcal{O}$ as a $\mathbb{Z}_p$-module. Every element of K lies in $p^{-N}\mathcal{O}$ for N large, so these elements span K over $\mathbb{Q}_p$, and $[K : \mathbb{Q}_p] \leq ef$.

2. Suppose K has characteristic p. For $\bar{a} \in \kappa$ choose a lift $a \in \mathcal{O}$. Since $\bar{a}^q = \bar{a}$, the element $a^q - a$ lies in $\mathfrak{p}$, and as raising to a power of p is additive in characteristic p,

$$a^{q^{n+1}} - a^{q^n} = (a^q - a)^{q^n} \in \mathfrak{p}^{q^n}$$

   So $(a^{q^n})_n$ is Cauchy, and its limit $\omega(\bar{a}) \in \mathcal{O}$ satisfies $\omega(\bar{a}) \equiv a \bmod \mathfrak{p}$, as every term does and $\mathfrak{p}$ is closed. Another lift $a + t$ with $t \in \mathfrak{p}$ has $(a+t)^{q^n} - a^{q^n} = t^{q^n} \to 0$, so $\omega(\bar{a})$ does not depend on the lift. As $a + b$ and ab lift $\bar{a} + \bar{b}$ and $\bar{a}\bar{b}$, and $(a+b)^{q^n} = a^{q^n} + b^{q^n}$, the map $\omega\colon \kappa \to \mathcal{O}$ is a ring homomorphism whose composite with reduction is the identity of κ.

   Identify $\kappa = \mathbb{F}_q$ and define $\Phi\colon \mathbb{F}_q[[t]] \to \mathcal{O}$ by $\Phi\big(\sum_i c_i t^i\big) = \sum_i \omega(c_i)\pi^i$, which converges as $\nu(\omega(c_i)\pi^i) \geq i$. It is additive, and for every N, $\Phi(g) \equiv \sum_{i<N} \omega(c_i)\pi^i \bmod \pi^N\mathcal{O}$, so, since ω is a ring homomorphism, $\Phi(gh) \equiv \Phi(g)\Phi(h) \bmod \pi^N\mathcal{O}$ for all N, whence $\Phi(gh) = \Phi(g)\Phi(h)$.

It is injective: if $g \neq 0$ has first nonzero coefficient c_j, then $\Phi(g) = \pi^j(\omega(c_j) + \pi y)$ with $y \in \mathcal{O}$ and $\omega(c_j)$ a unit, so $\nu(\Phi(g)) = j$. It is surjective: for $x \in \mathcal{O}$ put $c_0 = x \bmod \mathfrak{p}$, write $x - \omega(c_0) = \pi x_1$, put $c_1 = x_1 \bmod \mathfrak{p}$, and continue, so that $x - \Phi\big(\sum_{i<N} c_i t^i\big) \in \pi^N \mathcal{O}$ for every N and $x = \Phi\big(\sum_i c_i t^i\big)$. Hence $\mathcal{O} \cong \mathbb{F}_q[[t]]$, and as every element of $K^\times$ is $\pi^m u$ with $m \in \mathbb{Z}$ and $u \in \mathcal{O}$, Φ extends to an isomorphism $\mathbb{F}_q((t)) \to K$ sending t^{-1} to π^{-1}. Splitting each coefficient of a Laurent series along an $\mathbb{F}_p$-basis of $\mathbb{F}_q$ shows that such a basis spans $\mathbb{F}_q((t))$ over $\mathbb{F}_p((t))$, so K is an extension of $\mathbb{F}_p((t))$ of degree at most f.

In either case K is a finite extension of $\mathbb{Q}_p$ or of $\mathbb{F}_p((t))$, completing the proof.

Hence, the local fields of characteristic $p \neq 0$ are of the form $\mathbb{F}_q((t))$ where $q = p^n$, and local fields of characteristic 0 are finite extensions of $\mathbb{Q}_p$. The former will be called *power series fields*, and the latter will be called *p-adic number fields*.

Proposition 1.3.3: Compactness Of Local Fields

Let K be a local field. Then K is locally compact, and its valuation ring $\mathcal{O}$ is compact

***Proof*:**
Let π be a uniformizer, $\mathfrak{p} = \pi\mathcal{O}$, and $q = \#\mathcal{O}/\mathfrak{p}$. For $n \geq 0$, multiplication by π^n induces a bijection $\mathcal{O}/\mathfrak{p} \to \mathfrak{p}^n/\mathfrak{p}^{n+1}$, so every coset $b + \mathfrak{p}^n$ is the disjoint union of q cosets of $\mathfrak{p}^{n+1}$. The cosets $b + \mathfrak{p}^n$ are the open balls around b, as the absolute value takes discrete values.

Suppose that an open cover of $\mathcal{O}$ has no finite subcover. Then one of the q cosets of $\mathfrak{p}$ in $\mathcal{O}$, say $a_0 + \mathfrak{p}$, has no finite subcover, and one of the q cosets of $\mathfrak{p}^2$ in it, say $a_1 + \mathfrak{p}^2$, has none either. Continuing gives $a_n \in \mathcal{O}$ with $a_{n+1} \equiv a_n \bmod \mathfrak{p}^{n+1}$ such that no $a_n + \mathfrak{p}^{n+1}$ is covered by finitely many members of the cover. The sequence (a_n) is Cauchy, so it converges to some $a \in \mathcal{O}$, and $a \equiv a_n \bmod \mathfrak{p}^{n+1}$ for every n, as $\mathfrak{p}^{n+1}$ is closed. Some member U of the cover contains a, and as U is open, $a + \mathfrak{p}^N \subseteq U$ for some $N \geq 1$. But $a + \mathfrak{p}^N = a_{N-1} + \mathfrak{p}^N$, which is then covered by U alone, a contradiction. Hence $\mathcal{O}$ is compact. For every $a \in K$, the set $a + \mathcal{O}$ is an open neighbourhood of a, and it is compact, being a translate of $\mathcal{O}$. So K is locally compact, completing the proof.

Conversely, a field that is locally compact for a nondiscrete topology is isomorphic to $\mathbb{R}$ or $\mathbb{C}$, or to a local field with its valuation topology [12, Chapter II, §5].

Definition 1.3.4: Local Number Field

Let $K_\mathfrak{p}/\mathbb{Q}_p$ be a finite extension of $\mathbb{Q}_p$. Then $K_\mathfrak{p}$ is called a *local number field*

Notice that the extension has a sub-script $\mathfrak{p}$ attached to it. Each local number field shall be associated to a prime, as theorem 3.1.2 suggests (proved in section 3.1).

Proposition 1.3.5: Units Of Local Field

Let K be a local field. Then:

$$K^* = (\pi) \times \mu_{q-1} \times U^{(1)}$$

where π is a prime element, $q = \#\kappa(\mathfrak{p}) = \#(\mathcal{O}/\mathfrak{p})$, and $U^{(1)} = 1 + \mathfrak{p}$ is the group of principal units

***Proof*:**

Let ν be the normalized valuation and $\kappa = \mathcal{O}/\mathfrak{p}$, of characteristic p. Every $x \in K^*$ is uniquely $\pi^n u$ with $n = \nu(x) \in \mathbb{Z}$ and $u = x\pi^{-n} \in \mathcal{O}^*$, so $K^* = (\pi) \times \mathcal{O}^*$.

For $a, b \in \mathcal{O}$ with $a \equiv b \bmod \mathfrak{p}^k$ and $k \geq 1$, we have $a^p \equiv b^p \bmod \mathfrak{p}^{k+1}$: writing $a = b + c$ with $c \in \mathfrak{p}^k$, each term $\binom{p}{i} b^{p-i} c^i$ with $0 < i < p$ is divisible by $p \in \mathfrak{p}$, hence lies in $\mathfrak{p}^{k+1}$, and $c^p \in \mathfrak{p}^{pk} \subseteq \mathfrak{p}^{k+1}$. As q is a power of p, iterating gives $a^q \equiv b^q \bmod \mathfrak{p}^{k+1}$. Now let $\bar{a} \in \kappa$ with lift $a \in \mathcal{O}$. Since $\bar{a}^q = \bar{a}$, we have $a^q \equiv a \bmod \mathfrak{p}$, and induction gives $a^{q^{n+1}} \equiv a^{q^n} \bmod \mathfrak{p}^{n+1}$. So $(a^{q^n})_n$ is Cauchy, and its limit $\omega(\bar{a}) \in \mathcal{O}$ satisfies $\omega(\bar{a})^q = \omega(\bar{a})$ and $\omega(\bar{a}) \equiv a \bmod \mathfrak{p}$, as every term does and $\mathfrak{p}$ is closed. For $\bar{a} \neq 0$, the element $\omega(\bar{a})$ is a unit with $\omega(\bar{a})^{q-1} = 1$. These are $q - 1$ roots of $x^{q-1} - 1$ with distinct residues, and this polynomial has at most $q - 1$ roots in the field K, so they form the group μ_{q-1} of $(q-1)$-th roots of unity in K, and reduction maps μ_{q-1} bijectively onto κ^*.

The reduction homomorphism $\mathcal{O}^* \to \kappa^*$ is surjective with kernel $U^{(1)}$ (proposition 1.2.2), and its restriction to μ_{q-1} is an isomorphism. Hence $\mu_{q-1} \cap U^{(1)} = 1$, and every $u \in \mathcal{O}^*$ is $\zeta \cdot (\zeta^{-1} u)$ with $\zeta \in \mu_{q-1}$ the element of the same residue and $\zeta^{-1} u \in U^{(1)}$. So $\mathcal{O}^* = \mu_{q-1} \times U^{(1)}$ and $K^* = (\pi) \times \mu_{q-1} \times U^{(1)}$, as we sought to show.

This decomposition is only the coarse structure. The principal units $U^{(1)}$ carry a much finer description through the p-adic logarithm and exponential, which turn the multiplicative filtration into an additive one and pin down $K^\times$ as a topological group; this is developed in full in section 1.4.

1.4 The Multiplicative Group of a Local Field

The additive group of a local field is easy: it is a finite-dimensional vector space over $\mathbb{Q}_p$ or $\mathbb{F}_q((t))$. The multiplicative group carries the arithmetic, and it is the object local class field theory attaches to abelian extensions. This section pins it down completely. The coarse decomposition $K^\times = \pi^{\mathbb{Z}} \times \mu_{q-1} \times U^{(1)}$ of proposition 1.3.5 splits off a copy of $\mathbb{Z}$ (the valuation) and the prime-to-p roots of unity. The work is to understand the principal units $U^{(1)}$, and the tool is the p-adic logarithm, which trades the multiplicative filtration for an additive one. Throughout, K is a p-adic field of residue field $\mathbb{F}_q$ and absolute ramification index $e = \nu(p)$, with ν normalized so that $\nu(K^\times) = \mathbb{Z}$.

The residue field has canonical multiplicative representatives, which for a local field are the prime-to-p roots of unity. Their construction needs only completeness and a perfect residue field.

Proposition 1.4.1: Teichmüller Representatives

Let A be a complete DVR with maximal ideal $\mathfrak{p}$ whose residue field κ is perfect of characteristic $p > 0$, and write $A^{p^n} = \left\{a^{p^n} \mid a \in A\right\}$.

1. For $c \in \kappa$, choose lifts $x_n \in A$ of the elements $c^{p^{-n}}$. The limit $\omega(c) = \lim_{n\to\infty} x_n^{p^n}$ exists and does not depend on the choice of lifts. The map $\omega\colon \kappa \to A$ is the unique multiplicative map whose composite with the reduction $A \to \kappa$ is the identity, and $\omega(c)$ is the unique element of $\bigcap_{n\geq 0} A^{p^n}$ that reduces to c.
2. If A has characteristic p, then ω is a ring homomorphism.
3. If $\kappa = \mathbb{F}_q$, then $\omega(c)$ is the unique root of $X^q - X$ in A that reduces to c, and ω restricts to an isomorphism $\mathbb{F}_q^\times \to \mu_{q-1}$ onto the group of $(q-1)$-th roots of unity of $\mathrm{Frac}(A)$.

***Proof*:**
If $a \equiv b \pmod{\mathfrak{p}^m}$ with $m \geq 1$, then $a^p \equiv b^p \pmod{\mathfrak{p}^{m+1}}$: writing $a = b + y$ with $y \in \mathfrak{p}^m$, the terms $\binom{p}{i} b^{p-i} y^i$ of $a^p - b^p$ with $0 < i < p$ lie in $\mathfrak{p}^{m+1}$, as $p \in \mathfrak{p}$ divides $\binom{p}{i}$, and $y^p \in \mathfrak{p}^{pm} \subseteq \mathfrak{p}^{m+1}$. By induction, $a \equiv b \pmod{\mathfrak{p}}$ implies $a^{p^n} \equiv b^{p^n} \pmod{\mathfrak{p}^{n+1}}$.

1. As κ is perfect, c has a unique p^n-th root $c^{p^{-n}}$ for every n. Both x_{n+1}^p and x_n reduce to $c^{p^{-n}}$, so $x_{n+1}^{p^{n+1}} \equiv x_n^{p^n} \pmod{\mathfrak{p}^{n+1}}$. The sequence $(x_n^{p^n})$ is therefore Cauchy, and it converges as A is complete. For other lifts y_n, the congruence $x_n \equiv y_n \pmod{\mathfrak{p}}$ gives $x_n^{p^n} \equiv y_n^{p^n} \pmod{\mathfrak{p}^{n+1}}$, so the limit is the same. Every $x_n^{p^n}$ reduces to c, hence so does $\omega(c)$, as $\mathfrak{p}$ is closed. If y_n lifts $d^{p^{-n}}$, then $x_n y_n$ lifts $(cd)^{p^{-n}}$, so $\omega(cd) = \lim (x_n y_n)^{p^n} = \omega(c)\omega(d)$.

   Multiplicativity gives $\omega(c) = \omega(c^{p^{-n}})^{p^n} \in A^{p^n}$ for every n. Conversely, let $z \in \bigcap_n A^{p^n}$ reduce to c, and write $z = z_n^{p^n}$ with $z_n \in A$. The residue of z_n is a p^n-th root of c, so it is $c^{p^{-n}}$, and the z_n are lifts as in the construction. Hence $\omega(c) = \lim_n z_n^{p^n} = z$. Finally, a multiplicative map $s\colon \kappa \to A$ whose composite with reduction is the identity has $s(c) = s(c^{p^{-n}})^{p^n} \in A^{p^n}$ for every n, and $s(c)$ reduces to c, so $s(c) = \omega(c)$.
2. In characteristic p, $(x+y)^{p^n} = x^{p^n} + y^{p^n}$ for all $x, y \in A$. With x_n, y_n lifts of $c^{p^{-n}}, d^{p^{-n}}$ as above, $x_n + y_n$ lifts $c^{p^{-n}} + d^{p^{-n}} = (c+d)^{p^{-n}}$, so $\omega(c+d) = \lim (x_n + y_n)^{p^n} = \omega(c) + \omega(d)$. The lifts $x_n = 1$ of 1 give $\omega(1) = 1$, so ω is a ring homomorphism.
3. Let $c \in \mathbb{F}_q$. Multiplicativity gives $\omega(c)^q = \omega(c^q) = \omega(c)$, so the q values of ω, which are distinct as their residues are, are roots of $X^q - X$. This polynomial has at most q roots in the field $\mathrm{Frac}(A)$, so these are all of its roots, and $\omega(c)$ is the only root reducing to c. The nonzero values are the $q-1$ roots of $X^{q-1} - 1$, which form μ_{q-1}, and the restriction of ω to $\mathbb{F}_q^\times$ is an injective homomorphism onto μ_{q-1}, completing the proof.

With the roots of unity understood, only the principal units remain. Their structure is governed by two power series. Recall that the higher unit groups $U^{(n)} = 1 + \mathfrak{p}^n$ filter $U^{(1)}$, with $U^{(n)}/U^{(n+1)} \cong \mathbb{F}_q$ (proposition 1.2.2).

Theorem 1.4.2: The p-adic Logarithm and Exponential

The series

$$\log(1+x) = \sum_{n\geq 1} \frac{(-1)^{n-1}}{n} x^n, \qquad \exp(x) = \sum_{n\geq 0} \frac{x^n}{n!}$$

converge for $\nu(x) > 0$ and for $\nu(x) > \frac{e}{p-1}$ respectively. The logarithm defines a homomorphism $\log\colon U^{(1)} \to K$, and for every integer $n > \frac{e}{p-1}$ the maps

$$\log\colon U^{(n)} \to \mathfrak{p}^n, \qquad \exp\colon \mathfrak{p}^n \to U^{(n)}$$

are mutually inverse isomorphisms of topological groups, the multiplicative $U^{(n)}$ with the additive $\mathfrak{p}^n$.

***Proof*:**

Convergence. For log, $\nu(x^n/n) = n\nu(x) - e\, v_p(n) \geq n\nu(x) - e\log_p n \to \infty$ when $\nu(x) > 0$, so the series converges for $x \in \mathfrak{p}$. For exp, Legendre's formula gives $v_p(n!) = \frac{n - s_p(n)}{p-1} < \frac{n}{p-1}$, so $\nu(x^n/n!) > n\big(\nu(x) - \frac{e}{p-1}\big) \to \infty$ exactly when $\nu(x) > \frac{e}{p-1}$.

Homomorphism. The identity $\log\big((1+x)(1+y)\big) = \log(1+x) + \log(1+y)$ holds as an identity of formal power series over $\mathbb{Q}$, and both sides converge for $x, y \in \mathfrak{p}$, so it holds in K; thus log is a homomorphism $U^{(1)} \to K$.

Isomorphism. Fix $n > \frac{e}{p-1}$ and x with $\nu(x) \geq n$. For $k \geq 2$,

$$\nu\left(\frac{x^k}{k}\right) - \nu(x) = (k-1)\nu(x) - e\, v_p(k) \geq (k-1)\tfrac{e}{p-1} - e\log_p k > 0$$

the last inequality because $(k-1)/(p-1) \geq \log_p k$ with equality only at $k = p$, where $\nu(x) > \frac{e}{p-1}$ makes it strict. Hence the leading term dominates and $\nu(\log(1+x)) = \nu(x)$, so log carries $U^{(n)}$ bijectively onto $\mathfrak{p}^n$ preserving valuation; the same estimate for exp gives $\nu(\exp(x) - 1) = \nu(x)$, carrying $\mathfrak{p}^n$ into $U^{(n)}$. The formal identities $\exp(\log(1+x)) = 1 + x$ and $\log(\exp(x)) = x$ converge in this range, so the two maps are mutually inverse, completing the proof.

The logarithm converts the finer structure of $U^{(1)}$ into a question about a $\mathbb{Z}_p$-lattice, which the next result resolves.

Theorem 1.4.3: Structure of the Principal Units

Let $K/\mathbb{Q}_p$ have degree $d = [K : \mathbb{Q}_p]$. Then

$$U^{(1)} \cong \mu_{p^\infty}(K) \times \mathbb{Z}_p^d$$

where $\mu_{p^\infty}(K)$ is the finite group of p-power roots of unity in K. Consequently

$$K^\times \cong \mathbb{Z} \times \mu(K) \times \mathbb{Z}_p^d$$

where $\mu(K) = \mu_{q-1} \times \mu_{p^\infty}(K)$ is the finite group of all roots of unity in K.

***Proof*:**

Choose $n > \frac{e}{p-1}$. By theorem 1.4.2, log identifies $U^{(n)}$ with the additive group $\mathfrak{p}^n$, which is a free $\mathbb{Z}_p$-module of rank d (it is an open $\mathbb{Z}_p$-lattice in the d-dimensional $\mathbb{Q}_p$-vector space K). Thus $U^{(n)} \cong \mathbb{Z}_p^d$, in particular torsion-free. The quotient $U^{(1)}/U^{(n)}$ is finite (each $U^{(k)}/U^{(k+1)} \cong \mathbb{F}_q$), so $U^{(1)}$ is a finitely generated $\mathbb{Z}_p$-module with free part of rank d; its torsion subgroup is exactly the p-power roots of unity $\mu_{p^\infty}(K)$, finite since $K^\times$ has only finitely many roots of unity of p-power order. The structure theorem for modules over the principal ideal domain $\mathbb{Z}_p$ ([5, Invariant Factor Form of Modules over PID's]) then splits $U^{(1)} \cong \mu_{p^\infty}(K) \times \mathbb{Z}_p^d$. Feeding this into $K^\times = \pi^{\mathbb{Z}} \times \mu_{q-1} \times U^{(1)}$ (proposition 1.3.5) and combining the prime-to-p and p-power roots of unity into $\mu(K)$ gives the second isomorphism, completing the proof.

Example 1.4.4: The Multiplicative Group of $\mathbb{Q}_p$

For $K = \mathbb{Q}_p$ with p odd, $e = 1$ and $\frac{e}{p-1} < 1$, so already $\log\colon U^{(1)} \xrightarrow{\sim} \mathfrak{p} = p\mathbb{Z}_p \cong \mathbb{Z}_p$; there are no nontrivial p-power roots of unity, and

$$\mathbb{Q}_p^\times \cong \mathbb{Z} \times \mathbb{Z}/(p-1)\mathbb{Z} \times \mathbb{Z}_p$$

the factors being the valuation, the Teichmüller units μ_{p-1}, and the principal units. For $p = 2$, $\frac{e}{p-1} = 1$ so the logarithm is an isomorphism only from $U^{(2)}$ onward; the extra torsion is $\mu_{2^\infty}(\mathbb{Q}_2) = \{\pm 1\}$, and $\mathbb{Q}_2^\times \cong \mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}_2$.

Exercise 1.4.1

1. Compute the Teichmüller representative $\omega(2) \in \mathbb{Z}_5$ of $2 \in \mathbb{F}_5^\times$ modulo 25, using $\omega(2)^4 = 1$.
2. Show that each unit $x \in \mathcal{O}^\times$ factors uniquely as $x = \omega(\bar{x})\,\langle x \rangle$ with $\langle x \rangle \in U^{(1)}$ a principal unit.
3. Using theorem 1.4.2, show that for p odd the group $U^{(1)}$ of $\mathbb{Q}_p$ has no element of order p, and hence is torsion-free.
4. Let $K/\mathbb{Q}_p$ be a finite extension of degree d. Show that $[U^{(1)} : (U^{(1)})^p] = p^d \cdot |\mu_p(K)|$, and interpret the factor $|\mu_p(K)|$.

2

Hensel's Lemma and Newton Polygons

Having constructed the local fields and their valuations, we turn to the tool that makes them so useful for arithmetic: the ability to lift solutions from the residue field back to the ring itself. Hensel's lemma gives the exact condition under which a factorization modulo the maximal ideal lifts, and Newton polygons refine it by reading the valuations of the roots of a polynomial directly off its coefficients. Together they turn questions about roots over a local field into finite, combinatorial computations.

2.1 Hensel's Lemma

Hensel's lemma can be thought as the foundational result on why we want to work over local fields. Recall that given a polynomial $f(x) \in \mathbb{Z}[x]$, it is possible that $f(x)$ has a root a root mod each prime, but f need not have a root. For example recall that $x^4 + 1 \in \mathbb{Z}[x]$ is irreducible (use the fact that $\sqrt{2}$ is irrational in the proof). However, this polynomial is reducible mod every prime (see [5, Galois Fields]). When working over local fields, it was shown in theorem 1.1.2 that a polynomial has a solution if it has a solution mod each p^n (or equivalently for $\varprojlim_n R/\mathfrak{m}^n$, for each $\mathfrak{m}^m$). Hensel's lemma says that under mild conditions, it suffices to check that a polynomial has a root mod p (resp. mod $\mathfrak{m}$) for it to have a root mod for each p^n (resp $\mathfrak{m}^n$)! In particular, Hensel's lemma give a way of lifting roots.

Let K be complete with respect to a discrete valuation, with absolute value $|-|$, valuation ring $\mathcal{O}$, maximal ideal $\mathfrak{p}$ and residue field $\kappa = \mathcal{O}/\mathfrak{p}$. For example, $K = \mathbb{Q}_p$ and $\mathcal{O} = \mathbb{Z}_p$ with ideal (p) and $\kappa = \mathbb{Z}_p/p\mathbb{Z}_p \cong \mathbb{Z}/p\mathbb{Z}$. A polynomial $f(x) = a_0 + \cdots + a_n x^n \in \mathcal{O}[x]$ is *primitive* if

$$f(x) \not\equiv 0 \bmod \mathfrak{p}$$

which means

$$|f| = \max\{|a_0|, ..., |a_n|\} = 1$$

A primitive polynomial need not be monic. Its factorizations modulo $\mathfrak{p}$ into relatively prime factors lift.

Theorem 2.1.1: Hensel's Lemma

Let K be complete with respect to a discrete valuation, with valuation ring $\mathcal{O}$, maximal ideal $\mathfrak{p}$ and residue field κ. Let $f(x) \in \mathcal{O}[x]$ be a primitive polynomial. If it admits a factorization modulo $\mathfrak{p}$

$$f(x) \equiv \overline{g}(x)\overline{h}(x) \bmod \mathfrak{p}$$

into relatively prime polynomials $\overline{g}, \overline{h} \in \kappa[x]$, then $f(x)$ admits a factorization

$$f(x) = g(x)h(x)$$

with $g, h \in \mathcal{O}[x]$, $\deg g = \deg \overline{g}$, and

$$g(x) \equiv \overline{g}(x) \bmod \mathfrak{p} \qquad \text{and} \qquad h(x) \equiv \overline{h}(x) \bmod \mathfrak{p}$$

***Proof*:**

Let π be a uniformizer, $d = \deg f$ and $m = \deg \overline{g}$. As f is primitive, $\overline{f} = \overline{g}\overline{h}$ is nonzero, so $\deg \overline{h} = \deg \overline{f} - m \leq d - m$. Choose lifts $g_0, h_0 \in \mathcal{O}[x]$ of $\overline{g}, \overline{h}$ with $\deg g_0 = m$ and $\deg h_0 \leq d - m$. The leading coefficient of g_0 reduces to that of $\overline{g}$, so it is a unit. As $\overline{g}$ and $\overline{h}$ are relatively prime, there are $\overline{a}, \overline{b} \in \kappa[x]$ with $\overline{a}\overline{g} + \overline{b}\overline{h} = 1$, and lifts $a, b \in \mathcal{O}[x]$ satisfy

$$ag_0 + bh_0 \equiv 1 \bmod \mathfrak{p}$$

Step 1: the induction. We construct $g_n, h_n \in \mathcal{O}[x]$ for $n \geq 0$ with

$$f \equiv g_n h_n \bmod \mathfrak{p}^{n+1} \qquad g_n \equiv g_0,\ h_n \equiv h_0 \bmod \mathfrak{p} \qquad \deg(g_n - g_0) < m,\ \deg h_n \leq d - m$$

where a congruence of polynomials modulo $\mathfrak{p}^k$ means that every coefficient of the difference lies in $\mathfrak{p}^k$. For $n = 0$ these hold by the choice of g_0, h_0. Given g_n, h_n, write $f - g_n h_n = \pi^{n+1} r$ with $r \in \mathcal{O}[x]$ of degree at most d. Division by g_0, whose leading coefficient is a unit, gives $br = qg_0 + u$ with $q, u \in \mathcal{O}[x]$ and $\deg u < m$. Put $v_0 = ar + qh_0$. Then

$$g_0 v_0 + h_0 u = (ag_0 + bh_0)r \equiv r \bmod \mathfrak{p}$$

Reducing modulo $\mathfrak{p}$ gives $\overline{g}\,\overline{v}_0 = \overline{r} - \overline{h}\,\overline{u}$, a polynomial of degree at most d, as $\deg(\overline{h}\,\overline{u}) < (d - m) + m$. So $\deg \overline{v}_0 \leq d - m$, and the coefficients of v_0 in degrees above $d - m$ lie in $\mathfrak{p}$. Let v be v_0 with these coefficients removed, so that $\deg v \leq d - m$ and $g_0 v + h_0 u \equiv r \bmod \mathfrak{p}$. Put $g_{n+1} = g_n + \pi^{n+1} u$ and $h_{n+1} = h_n + \pi^{n+1} v$. As $g_n \equiv g_0$ and $h_n \equiv h_0$ modulo $\mathfrak{p}$,

$$f - g_{n+1}h_{n+1} = \pi^{n+1}\big(r - g_n v - h_n u\big) - \pi^{2n+2} uv \equiv 0 \bmod \mathfrak{p}^{n+2}$$

and the degree conditions hold, as $\deg u < m$ and $\deg v \leq d - m$.

Step 2: the limit. For each k, the coefficients of x^k in g_n and in h_n form Cauchy sequences, as

$g_{n+1} - g_n$ and $h_{n+1} - h_n$ have coefficients in $\mathfrak{p}^{n+1}$. Since $\mathcal{O}$ is complete, they converge, defining $g, h \in \mathcal{O}[x]$ with $g \equiv g_n$ and $h \equiv h_n$ modulo $\mathfrak{p}^{n+1}$ for every n. Then $f \equiv gh \bmod \mathfrak{p}^{n+1}$ for every n, and as $\bigcap_n \mathfrak{p}^{n+1} = 0$, $f = gh$. Finally $\deg(g - g_0) < m$ and $\deg g_0 = m$ give $\deg g = m = \deg \overline{g}$, and $g \equiv g_0 \equiv \overline{g}$, $h \equiv h_0 \equiv \overline{h}$ modulo $\mathfrak{p}$, completing the proof.

Corollary 2.1.2: Hensel's Lemma – Lifting Roots

Let R be a complete DVR with maximal ideal $\mathfrak{p}$ and residue field $k = R/\mathfrak{p}$. Then if f is a monic polynomial whose reduction to $k[x]$ has a *simple* root $\overline{a} \in k$, then $\overline{a}$ can be lifted to a root of f in R.

***Proof*:**
Since $\overline{a}$ is a simple root of the reduction $\overline{f}$, we can write $\overline{f} = (x - \overline{a})\overline{h}$ with $\overline{h}(\overline{a}) \neq 0$, so $x - \overline{a}$ and $\overline{h}$ are relatively prime. The monic polynomial f is primitive, so theorem 2.1.1 gives $f = gh$ with $g \in R[x]$ of degree 1 and $g \equiv x - \overline{a} \bmod \mathfrak{p}$. Writing $g = cx - d$, the coefficient $c \equiv 1 \bmod \mathfrak{p}$ is a unit and $d \equiv \overline{a} \bmod \mathfrak{p}$, so $a = d/c \in R$ is a root of f lifting $\overline{a}$, as we sought to show.

Example 2.1.3: Hensel's Lemma

1. (roots of unity) Consider $x^{p-1} - 1 \in \mathbb{Z}_p[x]$. Over $\mathbb{F}_p \cong \mathbb{Z}_p/p\mathbb{Z}_p$, this polynomials splits completely into linear factors. Hence through repeated use of Hensel's lemma it splits in $\mathbb{Z}_p$. As such, the field $\mathbb{Q}_p$ of p-adic number contains the $(p-1)$ roots of unity!

   These along with 0 can even from a system of representatives for the residue class field (the usual representatives are $0, 1, ..., p-1$).

2. Consider $x^2 + 1$. Modulo 2, $x^2 + 1 \equiv (x+1)^2$ has a double root, so Hensel's lemma does not apply. A root of $x^2 + 1$ in $\mathbb{Q}_2$ would be a unit, hence in $\mathbb{Z}_2$, and by theorem 1.1.2 the congruence $x^2 + 1 \equiv 0 \bmod 4$ would then be solvable. It is not, as squares are 0 or 1 modulo 4, so $x^2 + 1$ has no root in $\mathbb{Q}_2$.

3. Hensel's lemma is a sufficient, but not necessary condition. Consider $x^2 - p^2$ over $\mathbb{Q}_p$. Then $x^2 - p^2 \equiv x^2 \bmod p$ has multiple roots, hence Hensel's lemma does not apply. However, $x^2 - p^2$ has two roots in $\mathbb{Q}_p$, namely $\pm p$. The problem can be seen as there being two possible liftings of 0 mod p.

4. Show that $x^2 - 98 = x^2 - 2 \cdot 7^2$ in $\mathbb{Q}_7$ has two roots (hint: quadratic reciprocity). However again it does not satisfy the condition of Hensel's lemma.

Corollary 2.1.4: Hensel's Lemma Giving Newtons Method

Let R be a complete DVR, $f \in R[x]$, and $a_0 \in R$ satisfies:

$$|f(a_0)| < |f'(a_0)|^2$$

implying $f'(a_0)$ divides $f(a_0)$. For any $n \geq 0$ define

$$a_{n+1} := a_n - \frac{f(a_n)}{f'(a_n)}$$

Then the sequence (a_n) is well-defined and converges to a root $a \in R$ of f for which

$$|a - a_0| \leq \epsilon := \frac{|f(a_0)|}{|f'(a_0)|^2}$$

and a is the only root $b \in R$ of f with $|b - a_0| < |f'(a_0)|$. Furthermore, $|f(a_n)| \leq \epsilon^{2^n} |f'(a_0)|^2$ for all $n \geq 0$.

***Proof*:**
Let $|-|$ be the absolute value of the fraction field of R, so that $|r| \leq 1$ for $r \in R$. Expanding each power $(x+y)^k$ by the binomial theorem gives polynomials $G, H \in R[x,y]$ with

$$f(x+y) = f(x) + f'(x)\,y + y^2 G(x,y) \qquad f'(x+y) = f'(x) + y\,H(x,y)$$

so for $a, h \in R$ the bounds $|f(a+h) - f(a) - f'(a)h| \leq |h|^2$ and $|f'(a+h) - f'(a)| \leq |h|$ hold. As $|f'(a_0)|^2 > |f(a_0)| \geq 0$, also $f'(a_0) \neq 0$ and $\epsilon < 1$.

Step 1: the iteration. By induction on n, $a_n \in R$ is defined, $|f'(a_n)| = |f'(a_0)|$, and $|f(a_n)| \leq \epsilon^{2^n} |f'(a_0)|^2$. For $n = 0$ the last inequality is the equality $|f(a_0)| = \epsilon |f'(a_0)|^2$. Assume the claims for n. Then $f'(a_n) \neq 0$, and $h_n := -f(a_n)/f'(a_n)$ satisfies

$$|h_n| = \frac{|f(a_n)|}{|f'(a_0)|} \leq \epsilon^{2^n} |f'(a_0)| < |f'(a_0)| \leq 1$$

so $h_n \in R$ and $a_{n+1} = a_n + h_n \in R$. The expansion gives $f(a_{n+1}) = f(a_n) + f'(a_n)h_n + h_n^2 G(a_n, h_n) = h_n^2 G(a_n, h_n)$, so $|f(a_{n+1})| \leq |h_n|^2 \leq \epsilon^{2^{n+1}} |f'(a_0)|^2$. Also $|f'(a_{n+1}) - f'(a_n)| \leq |h_n| < |f'(a_0)| = |f'(a_n)|$, so the ultrametric inequality gives $|f'(a_{n+1})| = |f'(a_n)| = |f'(a_0)|$.

Step 2: convergence. By Step 1, $|a_{k+1} - a_k| = |h_k| \leq \epsilon^{2^k} |f'(a_0)|$, so for $m > n$ the ultrametric inequality gives $|a_m - a_n| \leq \max_{n \leq k < m} |h_k| \leq \epsilon^{2^n} |f'(a_0)|$. As $\epsilon < 1$ the sequence is Cauchy, and since R is complete it converges to some $a \in R$. Polynomials are continuous, so $f(a) = \lim_n f(a_n) = 0$. Taking $n = 0$ and letting $m \to \infty$ gives $|a - a_0| \leq \epsilon |f'(a_0)| \leq \epsilon$, and in particular $|a - a_0| < |f'(a_0)|$.

Step 3: uniqueness. The expansion gives $|f'(a) - f'(a_0)| \leq |a - a_0| < |f'(a_0)|$, so $|f'(a)| = |f'(a_0)|$. Let $b \in R$ be a root of f with $|b - a_0| < |f'(a_0)|$, and put $h = b - a$, so that

$|h| \leq \max(|b-a_0|, |a_0-a|) < |f'(a_0)|$. Then

$$0 = f(b) = f(a) + f'(a)h + h^2G(a,h) = h\big(f'(a) + hG(a,h)\big)$$

and $|hG(a,h)| \leq |h| < |f'(a)|$, so $f'(a) + hG(a,h) \neq 0$. Hence $h = 0$ and $b = a$, completing the proof.

Corollary 2.1.5: Hensel-Kürschák Lemma

Let K be the fraction field of a complete DVR $\mathcal{O}$, with absolute value $|-|$. Then every irreducible polynomial $f(x) = a_0 + \cdots + a_n x^n \in K[x]$ with $a_0 a_n \neq 0$ satisfies

$$|f| = \max\{|a_0|, |a_n|\}$$

In particular, if $a_n = 1$ and $a_0 \in \mathcal{O}$, then $f \in \mathcal{O}[x]$.

***Proof*:**
By multiplying by an appropriate element of K, assume that $f \in \mathcal{O}[x]$ and $|f| = 1$. Let a_r be the first coefficient where $|a_r| = 1$ so that:

$$f(x) \equiv x^r(a_r + a_{r+1}x + \cdots + a_n x^{n-r}) \bmod \mathfrak{p}$$

If $|f| > \max\{|a_0|, |a_n|\}$, then $\max\{|a_0|, |a_n|\} < 1$, so $0 < r < n$. The factors x^r and $\overline{a}_r + \cdots + \overline{a}_n x^{n-r}$ of the reduction are relatively prime, as the second has nonzero constant term, so theorem 2.1.1 gives $f = gh$ with $\deg g = r$, a nontrivial factorization contradicting the irreducibility of f. Hence $|f| = \max\{|a_0|, |a_n|\}$. If $a_n = 1$ and $a_0 \in \mathcal{O}$, this gives $|f| = 1$, so every coefficient of f lies in $\mathcal{O}$, as we sought to show.

2.1.6 Henselian Fields The Hensel-Kürschák lemma uses completeness only through Hensel's lemma, and the same is true of much of the theory of complete fields. A field with a nonarchimedean absolute value whose valuation ring satisfies Hensel's lemma is called *henselian*. For a field K with a nonarchimedean absolute value and completion $\widehat{K}$, the separable closure of K in $\widehat{K}$ is henselian, and it is called the *henselization* of K [12, Chapter II, §6]. Unlike $\widehat{K}$, the henselization is algebraic over K. The same contrast appears in algebraic geometry. For the local ring $k[x]_{(x)}$ of the affine line at the origin, over a field k of characteristic 0, the completion $k[[x]]$ contains $e^x = \sum_{n\geq 0} x^n/n!$, which is transcendental over $k(x)$, while every element of the henselization is algebraic over $k(x)$.

Lemma 2.1.7: Detecting Integral Elements, Complete DVR

Let A be a complete DVR with fraction field K, and L/K be a finite extension of degree n. Then $\alpha \in L$ is integral over A if and only if $N_{L/K}(\alpha) \in A$

Recall that in the case for number fields, this was *not* a sufficient condition [2, Trace And Norm Insufficient To Detect Integral Elements].

***Proof*:**

For $\alpha = 0$ both conditions hold, so let $\alpha \neq 0$, let $f = \sum_{i=0}^{d} a_i x^i \in K[x]$ be its minimal polynomial, of degree d, and let $e = [L : K(\alpha)]$, so that $n = de$ and $N_{L/K}(\alpha) = \big((-1)^d a_0\big)^e = (-1)^n a_0^e$. If α is integral over A, then so are its conjugates, the roots of f in a splitting field, and the coefficients of f, being symmetric functions of these roots, are elements of K integral over A. As the DVR A is integrally closed, $f \in A[x]$, so $a_0 \in A$ and $N_{L/K}(\alpha) \in A$. Conversely, if $N_{L/K}(\alpha) \in A$, then $a_0 \in K$ is a root of $x^e - (-1)^n N_{L/K}(\alpha) \in A[x]$, so $a_0 \in A$ as A is integrally closed. The polynomial f is monic and irreducible with $a_0 \neq 0$ and $a_0 \in A$, so corollary 2.1.5 gives $f \in A[x]$, and α is integral over A, completing the proof.

A *norm* on a vector space V over a field K with absolute value $|-|$ is a function $\|-\| \colon V \to \mathbb{R}_{\geq 0}$ with $\|v\| = 0$ only for $v = 0$, $\|\lambda v\| = |\lambda|\,\|v\|$ and $\|v + w\| \leq \|v\| + \|w\|$.

Lemma 2.1.8: Norms over a Complete Field

Let K be complete with respect to an absolute value $|-|$, let V be a K-vector space with basis $b_1, \ldots, b_n$, and let $\|-\|$ be a norm on V. Then there are constants $c, C > 0$ with

$$c \max_i |x_i| \leq \Big\| \sum_{i=1}^{n} x_i b_i \Big\| \leq C \max_i |x_i|$$

for all $x_1, \ldots, x_n \in K$. In particular, any two norms on V define the same topology, and V is complete with respect to every norm.

***Proof*:**

For $x = \sum_i x_i b_i$ write $\|x\|_\infty = \max_i |x_i|$. The triangle inequality gives $\|x\| \leq C\|x\|_\infty$ with $C = \sum_i \|b_i\|$. The lower bound is proved by induction on n, the case $n = 1$ being $\|x_1 b_1\| = |x_1|\,\|b_1\|$. Let $n \geq 2$, and suppose that no $c > 0$ works. Then there are $x^{(m)} \in V$ with $\|x^{(m)}\|_\infty = 1$ and $\|x^{(m)}\| \to 0$. Some index i has $|x_i^{(m)}| = 1$ for infinitely many m, so after renumbering the basis and passing to a subsequence, $|x_n^{(m)}| = 1$ for all m. Then $y^{(m)} = x^{(m)}/x_n^{(m)}$ satisfies $\|y^{(m)}\| = \|x^{(m)}\| \to 0$ and $y^{(m)} = w^{(m)} + b_n$ with $w^{(m)}$ in the span W of $b_1, \ldots, b_{n-1}$. So $w^{(m)} \to -b_n$ for $\|-\|$, and $(w^{(m)})$ is Cauchy for $\|-\|$. By induction the restriction of $\|-\|$ to W satisfies a lower bound $c'\|w\|_\infty \leq \|w\|$, so the coordinates of the $w^{(m)}$ are Cauchy in K. They converge, as K is complete, to the coordinates of some $w \in W$, and $\|w^{(m)} - w\| \leq C\|w^{(m)} - w\|_\infty \to 0$. Limits in a metric space are unique, so $-b_n = w \in W$, which is impossible. Hence a lower bound $c\|x\|_\infty \leq \|x\|$ holds.

Two norms on V are each bounded above and below by positive multiples of $\|-\|_\infty$, hence by positive multiples of each other, so every ball for one contains a ball for the other with the same centre, and the two topologies agree. A sequence that is Cauchy for $\|-\|$ is Cauchy for $\|-\|_\infty$, so its coordinates are Cauchy in K and converge, and the sequence converges for $\|-\|_\infty$ and hence for $\|-\|$, completing the proof.

Theorem 2.1.9: Extension of DVR's, Completion

Let A be a complete DVR with maximal ideal $\mathfrak{p}$, $K = \mathrm{Frac}(A)$, and $\nu_\mathfrak{p}$ a discrete valuation which gives the absolute value $|x|_\mathfrak{p} := c^{\nu_\mathfrak{p}(x)}$ where $0 < c < 1$. Let L/K be a finite extension of degree n. Then:

1. There is a unique absolute value $|x| := |N_{L/K}(x)|_\mathfrak{p}^{1/n}$ on L that extends $|-|_\mathfrak{p}$
2. The field L is complete with respect to $|-|$, and its valuation ring $\{x \in L \mid |x| \leq 1\}$ is equal to the integral closure B of A in L

Furthermore, B is a complete DVR whose maximal ideal $\mathfrak{P}$ induces:

$$|x| = |x|_\mathfrak{P} := c^{\frac{1}{e_\mathfrak{P}}\nu_\mathfrak{P}(x)}$$

where $e_\mathfrak{P}$ is the ramification index of $\mathfrak{P}$ (recall that $\mathfrak{p}B = \mathfrak{P}^{e_\mathfrak{P}}$).

***Proof*:**

1. The function $|x| = |N_{L/K}(x)|_\mathfrak{p}^{1/n}$ vanishes only at $x = 0$, and it is multiplicative because $N_{L/K}$ is. For $x \in L$,

   $$|x| \leq 1 \iff |N_{L/K}(x)|_\mathfrak{p} \leq 1 \iff N_{L/K}(x) \in A \iff x \in B$$

   the last equivalence by lemma 2.1.7. As B is a ring containing 1, $|x| \leq 1$ implies $|x+1| \leq 1$. For $y, z \in L$ with $0 < |y| \leq |z|$, applying this to $x = y/z$ gives $|y+z| = |z|\,|y/z+1| \leq |z|$, so $|y+z| \leq \max(|y|, |z|)$, which also holds when $y = 0$. Hence $|-|$ is an absolute value. For $x \in K$, $N_{L/K}(x) = x^n$, so $|x| = |x^n|_\mathfrak{p}^{1/n} = |x|_\mathfrak{p}$, and $|-|$ extends $|-|_\mathfrak{p}$. For uniqueness, let $|-|'$ be an absolute value on L extending $|-|_\mathfrak{p}$. Both $|-|$ and $|-|'$ are norms on the K-vector space L, as $|\lambda x|' = |\lambda|_\mathfrak{p}|x|'$ for $\lambda \in K$, so they define the same topology by lemma 2.1.8. By [5, Equivalent Topoloies] there is an $s > 0$ with $|x|' = |x|^s$ for all $x \in L$, and evaluating at an $x \in K$ with $|x|_\mathfrak{p} \neq 1$ gives $s = 1$. So $|-|' = |-|$.

2. The field K is complete and $|-|$ is a norm on the K-vector space L, so L is complete by lemma 2.1.8. The equivalences in the previous item show that $\{x \in L \mid |x| \leq 1\} = B$.

For the last assertion, the values $|x| = |N_{L/K}(x)|_\mathfrak{p}^{1/n}$ with $x \neq 0$ lie in $c^{\frac{1}{n}\mathbb{Z}}$, so $w(x) := \log_c |x|$ is a valuation on L with values in $\frac{1}{n}\mathbb{Z}$ that agrees with $\nu_\mathfrak{p}$ on K. Its valuation ring B is therefore a DVR. Let $\mathfrak{P}$ be its maximal ideal, $\nu_\mathfrak{P}$ its normalized valuation and Π a uniformizer. Every $x \in L^\times$ is $\Pi^{\nu_\mathfrak{P}(x)}u$ with $w(u) = 0$, so $w = w(\Pi)\,\nu_\mathfrak{P}$. A uniformizer π_K of A generates $\mathfrak{p}B = \mathfrak{P}^{e_\mathfrak{P}}$, so $\nu_\mathfrak{P}(\pi_K) = e_\mathfrak{P}$, while $w(\pi_K) = \nu_\mathfrak{p}(\pi_K) = 1$. Hence $w(\Pi) = 1/e_\mathfrak{P}$ and $|x| = c^{w(x)} = c^{\nu_\mathfrak{P}(x)/e_\mathfrak{P}}$. Finally, $B = \{x \mid |x| \leq 1\}$ is closed in L: if $|x| > 1$ and $|y - x| < |x|$, then $|y| = |x| > 1$ by the ultrametric inequality. So B is complete, as L is, completing the proof.

For finite extensions $K \subseteq L \subseteq L'$, the restriction to L of the absolute value of L' extends $|-|_\mathfrak{p}$, so it is the absolute value of L by uniqueness. Hence the absolute value of K extends uniquely to an algebraic closure $\overline{K}$, by giving $x \in \overline{K}$ its absolute value in any finite extension of K containing it. Every $\sigma \in \mathrm{Aut}_K(\overline{K})$ preserves this absolute value: on each finite extension L of K in $\overline{K}$, the function

$x \mapsto |\sigma(x)|$ is an absolute value extending $|-|_{\mathfrak{p}}$, so it equals $|-|$.

Definition 2.1.10: Ramification Index and Inertia Degree

Let A be a complete DVR with fraction field K, maximal ideal $\mathfrak{p}$ and residue field κ, and let L/K be a finite extension, with valuation ring B, maximal ideal $\mathfrak{P}$ and residue field $\lambda = B/\mathfrak{P}$ (theorem 2.1.9). The *ramification index* of L/K is the exponent $e(L:K) = e_{\mathfrak{P}}$ in $\mathfrak{p}B = \mathfrak{P}^{e_{\mathfrak{P}}}$, and the *inertia degree* of L/K is $f(L:K) = [\lambda : \kappa]$. These are also written $e_{L/K}$ and $f_{L/K}$. The extension L/K is *totally ramified* if $e(L:K) = [L:K]$.

Let $w = \log_c |-|$ be the valuation of L from theorem 2.1.9, which restricts to $\nu_{\mathfrak{p}}$ on K. That theorem gives $w = \nu_{\mathfrak{P}}/e(L:K)$ and $w = \frac{1}{n}\,\nu_{\mathfrak{p}} \circ N_{L/K}$ with $n = [L:K]$. So the value group $w(L^\times) = \frac{1}{e(L:K)}\mathbb{Z}$ contains $\nu_{\mathfrak{p}}(K^\times) = \mathbb{Z}$ with index $e(L:K)$, and it lies in $\frac{1}{n}\mathbb{Z}$, whence $e(L:K)$ divides $[L:K]$.

2.2 Newton Polygons

Hensel's lemma lifts a root once its reduction is known, but it says nothing about where the roots of a polynomial sit when no root is visible modulo $\mathfrak{p}$. The Newton polygon fills this gap: from the valuations of the coefficients alone it reads off the valuations of all the roots, and it sorts the roots into groups of equal valuation, each group cut out by a genuine factor of the polynomial over K. It is the standard combinatorial instrument for p-adic factorization, and downstream it determines reduction types of elliptic curves and the slopes of L-functions. Throughout, K is complete with respect to a discrete valuation ν, extended uniquely to the algebraic closure $\overline{K}$ (theorem 2.1.9), and $f(T) = \sum_{i=0}^{n} a_i T^i \in K[T]$ has $a_0 a_n \neq 0$.

Definition 2.2.1: Newton Polygon

The *Newton polygon* of f is the lower convex hull of the set of points

$$\{(i, \nu(a_i)) \mid 0 \le i \le n,\ a_i \neq 0\} \subseteq \mathbb{R}^2$$

Its *slopes* are the slopes of the successive segments, read left to right, and the *horizontal length* of a segment is the difference of the x-coordinates of its endpoints.

Because the hull is convex, the slopes strictly increase from one segment to the next. The content of the theory is that these slopes are precisely the negatives of the valuations of the roots.

Theorem 2.2.2: Newton Polygon and Root Valuations

If the Newton polygon of f has a segment of slope $-\lambda$ and horizontal length ℓ, then f has exactly ℓ roots (counted with multiplicity) in $\overline{K}$ of valuation λ.

***Proof*:**

Let $\alpha_1, \dots, \alpha_n \in \overline{K}$ be the roots of f, indexed so that $\nu(\alpha_1) \le \cdots \le \nu(\alpha_n)$, and write $f(T) = a_n \prod_j (T - \alpha_j)$. The coefficients are the elementary symmetric functions: $a_{n-k}/a_n = (-1)^k e_k(\alpha_1, \dots, \alpha_n)$. Each monomial of e_k is a product of k distinct roots, so by the ultrametric

inequality $\nu(e_k) \geq \nu(\alpha_1) + \cdots + \nu(\alpha_k)$, the right side being the smallest possible sum of k root valuations. Moreover, whenever $\nu(\alpha_k) < \nu(\alpha_{k+1})$ the monomial $\alpha_1 \cdots \alpha_k$ has strictly smaller valuation than every other monomial of e_k, so no cancellation occurs and

$$\nu(a_{n-k}) - \nu(a_n) = \nu(e_k) = \nu(\alpha_1) + \cdots + \nu(\alpha_k)$$

Thus at every index k where the root valuation jumps, the point $(n-k, \nu(a_{n-k}))$ lies on the lower convex hull, and between two consecutive jumps the points lie on or above the segment joining them. The segment of the hull spanning x-coordinates from $n-k'$ to $n-k$ (for a maximal run $\nu(\alpha_{k+1}) = \cdots = \nu(\alpha_{k'}) =: \lambda$) therefore has slope

$$\frac{\nu(a_{n-k'}) - \nu(a_{n-k})}{(n-k') - (n-k)} = \frac{(\nu(\alpha_1) + \cdots + \nu(\alpha_{k'})) - (\nu(\alpha_1) + \cdots + \nu(\alpha_k))}{k - k'} = \frac{(k'-k)\lambda}{-(k'-k)} = -\lambda$$

and horizontal length $k' - k$, the number of roots of valuation λ. Hence each segment of slope $-\lambda$ has horizontal length equal to the number of roots of valuation λ, completing the proof.

The theorem separates the roots by valuation, and this separation is realized by a factorization over K: distinct slopes cannot mix within an irreducible factor.

Corollary 2.2.3: Factorization by Slopes

Write the distinct slopes of the Newton polygon of f as $-\lambda_1 < \cdots < -\lambda_r$ with horizontal lengths $\ell_1, \ldots, \ell_r$. Then $f = a_n\, g_1 \cdots g_r$ with each $g_s \in K[T]$ monic of degree ℓ_s, all of whose roots have valuation λ_s. In particular, if the Newton polygon is a single segment then all roots of f share one valuation.

***Proof*:**
Group the roots by valuation: for each s, let $g_s = \prod_{\nu(\alpha_j)=\lambda_s} (T - \alpha_j)$, a factor of degree ℓ_s by theorem 2.2.2. The Galois group $\mathrm{Gal}_K(\overline{K})$ preserves the valuation, since for $\sigma \in \mathrm{Gal}_K(\overline{K})$ the map $\nu \circ \sigma$ is again an extension to $\overline{K}$ of the valuation of K, and such an extension is unique (theorem 2.1.9); so σ permutes the roots of each fixed valuation among themselves, each g_s is $\mathrm{Gal}_K(\overline{K})$-invariant, and hence $g_s \in K[T]$. The product is f/a_n, giving the factorization, as we sought to show.

A single segment of slope $-1/n$ recovers the Eisenstein criterion and more.

Corollary 2.2.4: Eisenstein via the Newton Polygon

Suppose f is monic of degree n with Newton polygon a single segment from $(0,1)$ to $(n,0)$ passing through no lattice point strictly between its endpoints, that is $\nu(a_0) = 1$ and $\nu(a_i) \geq 1$ for $0 < i < n$. Then f is irreducible over K, and each root generates a totally ramified extension of degree n.

***Proof*:**
By theorem 2.2.2 every root α has valuation $1/n$. Let $m = [K(\alpha) : K]$ and $e = e(K(\alpha) : K)$. The value group of $K(\alpha)$ is $\frac{1}{e}\mathbb{Z}$, it contains $1/n$, and e divides m (definition 2.1.10 and the

computation after it). So $n \mid e$ and $e \mid m$, while $m \leq n$ as α is a root of f. Hence $e = m = n$, and $K(\alpha)/K$ is totally ramified of degree n. Then f, monic of degree $n = [K(\alpha) : K]$ with α as a root, is the minimal polynomial of α and is irreducible. The hypothesis $\nu(a_0) = 1$, $\nu(a_i) \geq 1$ $(0 < i < n)$, $\nu(a_n) = 0$ is exactly the Eisenstein condition, completing the proof.

Example 2.2.5: Slopes of $T^n - p$ and a Two-Slope Polynomial

1. For $f(T) = T^n - p$ over $\mathbb{Q}_p$ the points are $(0, 1)$ and $(n, 0)$, a single segment of slope $-1/n$. Every root has valuation $1/n$, the polynomial is Eisenstein, and $\mathbb{Q}_p(p^{1/n})/\mathbb{Q}_p$ is totally ramified of degree n.
2. For $f(T) = T^3 + pT^2 + pT + p^3$ over $\mathbb{Q}_p$ the points are $(0, 3), (1, 1), (2, 1), (3, 0)$. The lower hull runs from $(0, 3)$ to $(1, 1)$ (slope -2) and from $(1, 1)$ to $(3, 0)$ (slope $-1/2$). So f has one root of valuation 2 and two roots of valuation $1/2$, and factors over $\mathbb{Q}_p$ as a linear times a quadratic. The roots of the quadratic have valuation $1/2$, so each generates an extension of ramification index divisible by 2, hence a totally ramified quadratic extension.

Exercise 2.2.1

1. Draw the Newton polygon of $f(T) = T^4 + pT^3 + p^2T + p^5$ over $\mathbb{Q}_p$ and list the valuations of its roots.
2. Show that if the Newton polygon of a monic f has all integer slopes, then the valuations of the roots are integers, and conversely.
3. Use the Newton polygon to prove that $T^p - T - 1/p$ over $\mathbb{Q}_p$ is irreducible (its roots have negative valuation).
4. Let f be monic of degree n with a single Newton-polygon slope $-a/n$, where $\gcd(a, n) = 1$. Show that f is irreducible over K.

Lemma 2.2.6: Cyclotomic Extensions of p-Power Level

Let $n \geq 1$ and let ζ be a primitive p^n-th root of unity. Then $\mathbb{Q}_p(\zeta)/\mathbb{Q}_p$ is totally ramified of degree $\varphi(p^n) = p^{n-1}(p-1)$, the element $\zeta - 1$ is a uniformizer of $\mathbb{Q}_p(\zeta)$, and $\mathbb{Q}_p(\zeta)/\mathbb{Q}_p$ is Galois with

$$\mathrm{Gal}_{\mathbb{Q}_p}(\mathbb{Q}_p(\zeta)) \xrightarrow{\sim} (\mathbb{Z}/p^n\mathbb{Z})^\times \qquad \sigma \mapsto a_\sigma$$

where $\sigma(\zeta) = \zeta^{a_\sigma}$.

***Proof*:**

The primitive p^n-th roots of unity are the roots of $\Phi(x) = (x^{p^n} - 1)/(x^{p^{n-1}} - 1) = \sum_{j=0}^{p-1} x^{jp^{n-1}}$. Put $g(x) = \Phi(x + 1) \in \mathbb{Z}[x]$, a monic polynomial of degree $(p-1)p^{n-1}$ with $g(\zeta - 1) = 0$ and constant term $g(0) = \Phi(1) = p$. Modulo p, $(x+1)^{p^k} \equiv x^{p^k} + 1$, so reducing the identity $g(x)\big((x+1)^{p^{n-1}} - 1\big) = (x+1)^{p^n} - 1$ gives $\bar{g}(x)\, x^{p^{n-1}} = x^{p^n}$ in $\mathbb{F}_p[x]$, that is, $\bar{g}(x) = x^{(p-1)p^{n-1}}$. So every non-leading coefficient of g is divisible by p and $\nu_p(g(0)) = 1$, which is the hypothesis of corollary 2.2.4. Hence g is irreducible over $\mathbb{Q}_p$, the extension $\mathbb{Q}_p(\zeta) = \mathbb{Q}_p(\zeta - 1)$ is totally

ramified of degree $\varphi(p^n)$, and $\zeta - 1$ has valuation $1/\varphi(p^n)$ by theorem 2.2.2, so it is a uniformizer.

Every p^n-th root of unity is a power of ζ, so $\mathbb{Q}_p(\zeta)$ is the splitting field of $x^{p^n} - 1$ and is Galois over $\mathbb{Q}_p$. Each σ sends ζ to a primitive p^n-th root of unity ζ^{a_σ} with $a_\sigma \in (\mathbb{Z}/p^n\mathbb{Z})^\times$ determined by σ. As $\sigma\tau(\zeta) = \sigma(\zeta^{a_\tau}) = \zeta^{a_\sigma a_\tau}$, the map $\sigma \mapsto a_\sigma$ is a homomorphism, injective as σ is determined by $\sigma(\zeta)$. Both groups have order $\varphi(p^n)$, so it is an isomorphism, completing the proof.

3

Extensions of Local Fields and Ramification

We now study how a local field sits inside its finite extensions. Because a single prime of the extension lies above the base prime, the whole arithmetic is governed by one ramification index and one residue degree, tied together by the fundamental identity $n = ef$. This chapter develops that theory: the extension of valuations, the local Dedekind-Kummer theorem, and the decomposition of an extension into its unramified and ramified parts.

3.1 Extension of Valuations and the Fundamental Identity

With the definition of the local fields, and the ground work for how to complete any discrete valuation, we now look at how valuations change under field extensions. Throughout this section we consider the following setup:

Let A be an integral domain, K the field of fractions of A, L a finite separable extension of K, and B the integral closure of A in L. This setup shall be called the *AKLB setup*.

We shall put some different conditions on A depending on what we need. In number theory, we usually have A be a Dedekind domain.

Recall that each nonzero prime $\mathfrak{p} \subseteq A$ where A is a Dedekind domain determines a discrete valuation $\nu_\mathfrak{p} : \mathcal{I}_A \to \mathbb{Z}$ by giving the exponent $n_\mathfrak{p}$ from the unique factorization (i.e. $\nu_\mathfrak{p}(I) = n$ if and only if $IA_\mathfrak{p} = \mathfrak{p}^n A_\mathfrak{p}$). Thus for each nonzero prime $\mathfrak{p} \subseteq A$, we have the discrete valuation:

$$\nu_\mathfrak{p} : K^\times \to \mathbb{Z} \qquad \nu_\mathfrak{p}(x) = \nu_\mathfrak{p}(xA)$$

and hence an absolute value $|x|_{\mathfrak{p}} := c^{\nu_{\mathfrak{p}}(x)}$ for $0 < c < 1$. Each prime $\mathfrak{P}$ of B lying over $\mathfrak{p}$ gives a discrete valuation $\nu_{\mathfrak{P}}$ on L, whose restriction to K need not equal $\nu_{\mathfrak{p}}$.

Definition 3.1.1: Extending Valuations

Let L/K be a finite extension, and let ν and w be discrete valuations on K and L. If $w|_K = e\nu$ for some integer $e \geq 1$, then w is said to *extend* ν *with index* e.

Theorem 3.1.2: Extension of Valuation

Let A be a Dedekind domain with fraction field K, let L/K be a finite extension, let B be the integral closure of A in L, and let $\mathfrak{p} \subseteq A$ be a nonzero prime. Then at least one prime of B lies over $\mathfrak{p}$. For a prime $\mathfrak{P}$ of B lying over $\mathfrak{p}$, let $e_{\mathfrak{P}}$ be the exponent of $\mathfrak{P}$ in the factorization of $\mathfrak{p}B$. Then the $\mathfrak{P}$-adic valuation $\nu_{\mathfrak{P}}$ of L extends $\nu_{\mathfrak{p}}$ with index $e_{\mathfrak{P}}$, and every discrete valuation w on L with $w(L^\times) = \mathbb{Z}$ that extends $\nu_{\mathfrak{p}}$ is $\nu_{\mathfrak{P}}$ for exactly one such $\mathfrak{P}$. Thus $\mathfrak{P} \mapsto \nu_{\mathfrak{P}}$ is a bijection from the set of primes of B lying over $\mathfrak{p}$ to the set of discrete valuations on L with value group $\mathbb{Z}$ that extend $\nu_{\mathfrak{p}}$.

***Proof*:**
By [2, Extending Dedekind Domains], B is a Dedekind domain. As L/K is algebraic, every element of L is b/a with $b \in B$ and $a \in A \setminus \{0\}$, so L is the fraction field of B. The nonzero primes of B are maximal, and for each of them, $\mathfrak{P}$, the localization $B_{\mathfrak{P}}$ is a DVR with fraction field L [2, Dedekind Domain As DVR]. Its normalized valuation is the $\mathfrak{P}$-adic valuation $\nu_{\mathfrak{P}}$: for $x \in L^\times$, $\nu_{\mathfrak{P}}(x)$ is the integer k with $xB_{\mathfrak{P}} = \mathfrak{P}^k B_{\mathfrak{P}}$.

Step 1: primes over $\mathfrak{p}$. By [2, Dedekind Domain, Lying Over And Lying Under], $\mathfrak{p}B \neq B$, so $\mathfrak{p}B$ lies in a maximal ideal $\mathfrak{P}$ of B. Then $\mathfrak{P} \cap A$ is a proper ideal of A containing the maximal ideal $\mathfrak{p}$, so $\mathfrak{P} \cap A = \mathfrak{p}$, and $\mathfrak{P}$ lies over $\mathfrak{p}$. Every prime $\mathfrak{P}$ lying over $\mathfrak{p}$ contains $\mathfrak{p} \neq 0$, so it is nonzero, hence maximal. It also satisfies $A \setminus \mathfrak{p} \subseteq B \setminus \mathfrak{P}$, which gives $A_{\mathfrak{p}} \subseteq B_{\mathfrak{P}}$.

Step 2: the index. Let $\mathfrak{P}$ lie over $\mathfrak{p}$. The ideal $\mathfrak{p}B$ is nonzero and proper, so $\mathfrak{p}B = \mathfrak{Q}_1^{a_1} \cdots \mathfrak{Q}_r^{a_r}$ with distinct primes $\mathfrak{Q}_j$ and exponents $a_j \geq 1$ [2, Unique Factorization Of Ideals In Dedekind Domains]. Localizing at $\mathfrak{P}$ gives $\mathfrak{p}B_{\mathfrak{P}} = \prod_j (\mathfrak{Q}_j B_{\mathfrak{P}})^{a_j}$. A prime $\mathfrak{Q}_j \neq \mathfrak{P}$ is a maximal ideal not contained in $\mathfrak{P}$, so it meets $B \setminus \mathfrak{P}$ and $\mathfrak{Q}_j B_{\mathfrak{P}} = B_{\mathfrak{P}}$. As $\mathfrak{p}B_{\mathfrak{P}} \subseteq \mathfrak{P}B_{\mathfrak{P}} \neq B_{\mathfrak{P}}$, the prime $\mathfrak{P}$ is one of the $\mathfrak{Q}_j$, its exponent $e_{\mathfrak{P}}$ is at least 1, and $\mathfrak{p}B_{\mathfrak{P}} = \mathfrak{P}^{e_{\mathfrak{P}}} B_{\mathfrak{P}}$. Now let $a \in A \setminus \{0\}$ and $n = \nu_{\mathfrak{p}}(a) \geq 0$, so that $aA_{\mathfrak{p}} = \mathfrak{p}^n A_{\mathfrak{p}}$. Extending to $B_{\mathfrak{P}} \supseteq A_{\mathfrak{p}}$ gives $aB_{\mathfrak{P}} = \mathfrak{p}^n B_{\mathfrak{P}} = \mathfrak{P}^{e_{\mathfrak{P}} n} B_{\mathfrak{P}}$, that is, $\nu_{\mathfrak{P}}(a) = e_{\mathfrak{P}}\, \nu_{\mathfrak{p}}(a)$. Every element of $K^\times$ is a quotient of two such elements a, and both valuations are homomorphisms on $K^\times$, so $\nu_{\mathfrak{P}}|_K = e_{\mathfrak{P}}\, \nu_{\mathfrak{p}}$. Thus $\nu_{\mathfrak{P}}$ extends $\nu_{\mathfrak{p}}$ with index $e_{\mathfrak{P}}$, and $\nu_{\mathfrak{P}}(L^\times) = \mathbb{Z}$.

Step 3: injectivity. Let $\mathfrak{P} \neq \mathfrak{P}'$ lie over $\mathfrak{p}$. Both are maximal, so there is an $x \in \mathfrak{P} \setminus \mathfrak{P}'$. Then $\nu_{\mathfrak{P}}(x) > 0$, while x is a unit of $B_{\mathfrak{P}'}$, so $\nu_{\mathfrak{P}'}(x) = 0$. Hence $\nu_{\mathfrak{P}} \neq \nu_{\mathfrak{P}'}$.

Step 4: surjectivity. Let w be a discrete valuation on L with $w(L^\times) = \mathbb{Z}$ that extends $\nu_{\mathfrak{p}}$ with index e, and let $W = \{x \in L \mid w(x) \geq 0\}$ and $\mathfrak{m} = \{x \in L \mid w(x) > 0\}$. For $a \in A$, $w(a) = e\, \nu_{\mathfrak{p}}(a) \geq 0$, so $A \subseteq W$. The ring W is integrally closed in L: if $x \in L \setminus W$ satisfied $x^m + c_{m-1}x^{m-1} + \cdots + c_0 = 0$

with $c_i \in W$, then $w(x) < 0$ would give $w(c_i x^i) \geq i\, w(x) > m\, w(x) = w(x^m)$ for $i < m$, and the term x^m, of strictly smallest valuation, could not cancel. Hence $B \subseteq W$. Put $\mathfrak{P} = \mathfrak{m} \cap B$, a prime of B. An element $a \in A$ lies in $\mathfrak{m}$ exactly when $\nu_\mathfrak{p}(a) > 0$, that is, when $a \in \mathfrak{p}$, as the elements of $A \setminus \mathfrak{p}$ are units of $A_\mathfrak{p}$. So $\mathfrak{P} \cap A = \mathfrak{p}$, and $\mathfrak{P}$ lies over $\mathfrak{p}$. Elements of $B \setminus \mathfrak{P}$ have valuation 0, so they are units of W, and $B_\mathfrak{P} \subseteq W$.

Let Π be a uniformizer of $B_\mathfrak{P}$, and write $\Pi = b/s$ with $b \in \mathfrak{P}$ and $s \in B \setminus \mathfrak{P}$. Then $w(\Pi) = w(b) - w(s) = w(b) > 0$. Every $x \in L^\times$ is $u\Pi^k$ with $u \in B_\mathfrak{P}^\times$ and $k = \nu_\mathfrak{P}(x)$, and $w(u) = 0$ as u and u^{-1} lie in $B_\mathfrak{P} \subseteq W$. So $w = w(\Pi)\,\nu_\mathfrak{P}$. As $w(L^\times) = \mathbb{Z}$, $w(\Pi) = 1$, and $w = \nu_\mathfrak{P}$. By Step 3 this $\mathfrak{P}$ is unique. Comparing $w|_K = e\,\nu_\mathfrak{p}$ with $\nu_\mathfrak{P}|_K = e_\mathfrak{P}\,\nu_\mathfrak{p}$ at a nonzero element of $\mathfrak{p}$ gives $e = e_\mathfrak{P}$, completing the proof.

Theorem 3.1.3: Extension of DVR's

Assume $AKLB$ and let A be a complete DVR with maximal ideal $\mathfrak{p}$. Then B is a DVR whose maximal ideal $\mathfrak{P}$ is necessarily the unique prime ideal above $\mathfrak{p}$.

***Proof*:**
By [2, Extending Dedekind Domains], B is a Dedekind domain, as the DVR A is one. At least one prime $\mathfrak{P}$ of B lies over $\mathfrak{p}$ by theorem 3.1.2. Suppose, for the sake of contradiction, that $\mathfrak{P} \neq \mathfrak{P}'$ both lie over $\mathfrak{p}$. As distinct nonzero primes of the Dedekind domain B, they are maximal and neither contains the other, so we may choose $b \in \mathfrak{P} \setminus \mathfrak{P}'$. Let $f \in A[x]$ be the minimal polynomial of b over K, which has coefficients in A as b is integral over the integrally closed ring A. As f is monic, division by f shows that $A[b] \cong A[x]/(f)$, so

$$\frac{A[b]}{\mathfrak{p}A[b]} \cong \frac{A[x]}{(\mathfrak{p}, f)} \cong \frac{(A/\mathfrak{p})[x]}{(\overline{f})}$$

The ideals $\mathfrak{P} \cap A[b]$ and $\mathfrak{P}' \cap A[b]$ contain $\mathfrak{p}A[b]$, and they are distinct, as b lies in the first but not the second. Each is maximal: $A[b]/(\mathfrak{P} \cap A[b])$ is a domain and a quotient of $(A/\mathfrak{p})[x]/(\overline{f})$, hence a finite-dimensional domain over the field $A/\mathfrak{p}$, hence a field. So $(A/\mathfrak{p})[x]/(\overline{f})$ has at least two maximal ideals, and as $(A/\mathfrak{p})[x]$ is a PID, $\overline{f}$ has at least two distinct monic irreducible factors. Let $\overline{g}$ be the highest power of one of them dividing $\overline{f}$, and $\overline{h} = \overline{f}/\overline{g}$. Then $\overline{g}$ and $\overline{h}$ are relatively prime and of positive degree, and theorem 2.1.1, applied to the monic, hence primitive, polynomial f, gives $f = gh$ with $\deg g = \deg \overline{g}$, strictly between 0 and $\deg f$. This contradicts the irreducibility of f. So $\mathfrak{P}$ is the only prime of B over $\mathfrak{p}$.

Every nonzero prime $\mathfrak{Q}$ of B lies over $\mathfrak{p}$: a nonzero $b \in \mathfrak{Q}$ has a minimal polynomial over K with nonzero constant term $a_0 \in A$, and a_0 is b times an element of B, so $\mathfrak{Q} \cap A$ is a nonzero prime of the DVR A, which is $\mathfrak{p}$. Hence $\mathfrak{P}$ is the only nonzero prime of B, so $B = B_\mathfrak{P}$, which is a DVR by [2, Dedekind Domain As DVR], completing the proof.

Note that completion is necessary for this lemma to work:

Example 3.1.4: Failure Without Completion

Take $A = \mathbb{Z}_{(5)}$ which is a DVR with maximal ideal (5) but is not complete. Let $K = \mathbb{Q}$ and $L = \mathbb{Q}(i)$. Then $\mathcal{O}_L = \mathbb{Z}_{(5)}[i]$, which is a PID, but is *not* a DVR, as the ideals $(1+2i)$ and $(1-2i)$ are both maximal and not equal. However, if we take $A = \mathbb{Z}_5$ and $K = \mathbb{Q}_5$ with $L = \mathbb{Q}_5(i) = \mathbb{Q}_5$, and $B = \mathbb{Z}_5[i] = \mathbb{Z}_5$, it is a DVR, since $x^2 + 1$ has roots in $\mathbb{F}_5$ and we can lift roots in $\mathbb{Z}_5$ via Hensel's lemma.

Notice that the factorization $5 = (1+2i)(1-2i)$ in $\mathbb{Z}_{(5)}[i]$ when replaced with $\mathbb{Z}_5$ we will get that (5) is the product of the maximal ideal and a unit. In fact, not matter what prime we localize at we get this result!

Corollary 3.1.5: The Fundamental Identity For Local Fields

Let A be a complete DVR with fraction field K, maximal ideal $\mathfrak{p}$ and residue field κ, and let L/K be a finite extension of degree n, with valuation ring B, maximal ideal $\mathfrak{P}$ and residue field λ (theorem 2.1.9). Let $e = e(L:K)$ and $f = f(L:K)$. Then

$$[L:K] = n = ef$$

More precisely, if Π is a uniformizer of B and $\omega_1, \dots, \omega_f \in B$ have residues forming a κ-basis of λ, then the elements $\omega_i \Pi^j$ with $1 \le i \le f$ and $0 \le j < e$ form an A-basis of B. In particular, this holds when K is a local field.

***Proof*:**

Let $\nu_{\mathfrak{P}}$ be the normalized valuation of B, so that $\nu_{\mathfrak{P}} = e\,\nu_{\mathfrak{p}}$ on K, as $\mathfrak{p}B = \mathfrak{P}^e$. By definition 2.1.10 and the remark after it, e divides n.

Step 1: f is finite. Let $\overline{y}_1, \dots, \overline{y}_r \in \lambda$ be linearly independent over κ, with lifts $y_i \in B$. In a relation $\sum_i c_i y_i = 0$ with $c_i \in K$ not all zero, dividing by a c_k of largest absolute value gives coefficients in A, one of them equal to 1, and reducing modulo $\mathfrak{P}$ gives a nontrivial relation among the $\overline{y}_i$, which is impossible. So the y_i are linearly independent over K, $r \le n$, and $f = [\lambda : \kappa] \le n$.

Step 2: independence. Let $c_{ij} \in K$, not all zero, and for each j put $s_j = \sum_i c_{ij}\omega_i$. If the c_{ij} with a given j are not all zero, let c be one of largest absolute value. Then $s_j/c = \sum_i (c_{ij}/c)\,\omega_i$ has coefficients in A, one of them equal to 1, so its residue is nonzero and s_j/c is a unit of B. Hence $\nu_{\mathfrak{P}}(s_j) = \nu_{\mathfrak{P}}(c) = e\,\nu_{\mathfrak{p}}(c) \in e\mathbb{Z}$, and $\nu_{\mathfrak{P}}(s_j\Pi^j) \equiv j \pmod{e}$. So the nonzero terms $s_j\Pi^j$ with $0 \le j < e$ have pairwise distinct valuations, and by the ultrametric inequality their sum $\sum_{i,j} c_{ij}\omega_i\Pi^j$ is nonzero. Hence the $\omega_i\Pi^j$ are linearly independent over K.

Step 3: spanning. Let $M = \sum_{i,j} A\omega_i\Pi^j$. As the residues of the ω_i span λ, every $b \in B$ satisfies $b - \sum_i a_i\omega_i \in \Pi B$ for suitable $a_i \in A$, so $B = \sum_i A\omega_i + \Pi B$. Substituting this equality into itself e times gives $B = M + \Pi^e B = M + \mathfrak{p}B$. Let π_A be a uniformizer of A and $b \in B$. Applying $B = M + \pi_A B$ to b, then to the resulting element of B, and so on, gives $m_k \in M$ with $b - \sum_{k<N} \pi_A^k m_k \in \pi_A^N B$ for every N. Write $m_k = \sum_{i,j} a_{ij}^{(k)}\omega_i\Pi^j$ with $a_{ij}^{(k)} \in A$. The series $a_{ij} = \sum_k \pi_A^k a_{ij}^{(k)}$ converge in the complete ring A, with $a_{ij} - \sum_{k<N} \pi_A^k a_{ij}^{(k)} \in \pi_A^N A$, so

$b - \sum_{i,j} a_{ij}\omega_i\Pi^j \in \pi_A^N B$ for every N. As $\bigcap_N \pi_A^N B = \bigcap_N \mathfrak{P}^{eN} = 0$, $b = \sum_{i,j} a_{ij}\omega_i\Pi^j \in M$.

Hence the $\omega_i\Pi^j$ form an A-basis of B. Every element of L becomes integral over A after multiplication by a nonzero element of A, so it is b/a with $b \in B$ and $a \in A \setminus \{0\}$, and the $\omega_i\Pi^j$ also form a K-basis of L. So $n = ef$, completing the proof.

Corollary 3.1.6: DVR Completion And Number Rings

Assume $AKLB$ and let A be a complete DVR with maximal ideal $\mathfrak{p}$. Let $\mathfrak{P}/\mathfrak{p}$. Then

$$\nu_{\mathfrak{P}}(x) = \frac{1}{f_{\mathfrak{P}}}\nu_{\mathfrak{p}}(N_{L/K}(x))$$

for all $x \in L$

***Proof*:**
Let $n = [L : K]$, $e = e_{\mathfrak{P}}$ and $f = f_{\mathfrak{P}}$, and let $x \in L^\times$. By theorem 2.1.9, the absolute value of L is $|x| = |N_{L/K}(x)|_{\mathfrak{p}}^{1/n}$ and also $|x| = c^{\nu_{\mathfrak{P}}(x)/e}$, while $|N_{L/K}(x)|_{\mathfrak{p}} = c^{\nu_{\mathfrak{p}}(N_{L/K}(x))}$. Comparing exponents gives $\nu_{\mathfrak{p}}(N_{L/K}(x)) = (n/e)\,\nu_{\mathfrak{P}}(x)$, and $n/e = f$ by corollary 3.1.5, as we sought to show.

3.2 Dedekind-Kummer for Local Rings

Recall the Dedekind-Kummer Theorem ([2, Dedekind-Kummer Theorem]) where a prime $\mathfrak{P}/\mathfrak{p}$ factors like the minimal polynomial of the finite separable extension L/K (i.e. $AKLB$) if the ring of integers of L is monogenic or the prime does not contain the conductor. When A is a complete DVR and the residue field extension is separable, for instance when the residue fields are finite, B is always monogenic.

Lemma 3.2.1: Nakayama's Lemma

Let A be a local ring with maximal ideal $\mathfrak{p}$ and let M be a finitely generated A-module. If the image of $x_1, ..., x_n \in M$ generate $M/\mathfrak{p}M$ as an $(A/\mathfrak{p})$-vector space, then $x_1, ..., x_n$ generate M as an A-module.

***Proof*:**
see [4, Nakayama's Lemma III]

The following local version of the Dedekind-Kummer theorem does not need A or B to be Dedekind domains.

Corollary 3.2.2: Local Ring Maximal Ideal Extension

Let A be a local Noetherian ring with maximal ideal $\mathfrak{p}$, $g \in A[x]$ be a monic polynomial with image $\overline{g} \in (A/\mathfrak{p})[x]$, and let α be the image of x in the ring $B := A[x]/(g(x))$. Then the maximal ideals of B are $(\mathfrak{p}, g_i(\alpha))$ where $g_1, ..., g_m \in A[x]$ are lifts of the distinct irreducible polynomials $\overline{g}_i \in (A/\mathfrak{p})[x]$ that divide $\overline{g}$.

***Proof*:**
As g is monic, B is a finitely generated A-module. Let $\mathfrak{M}$ be a maximal ideal of B. If $\mathfrak{p}B \not\subseteq \mathfrak{M}$, then $\mathfrak{M} + \mathfrak{p}B = B$, so the finitely generated A-module $B/\mathfrak{M}$ satisfies $\mathfrak{p}(B/\mathfrak{M}) = B/\mathfrak{M}$, and lemma 3.2.1, applied with no generators, gives $B/\mathfrak{M} = 0$, which is impossible. So every maximal ideal of B contains $\mathfrak{p}B$, and the maximal ideals of B correspond to those of

$$B/\mathfrak{p}B \cong A[x]/(\mathfrak{p}, g(x)) \cong (A/\mathfrak{p})[x]/(\overline{g}(x))$$

As $(A/\mathfrak{p})[x]$ is a PID, the maximal ideals of the right hand side are generated by the irreducible divisors $\overline{g}_i$ of $\overline{g}$, and their preimages in B are the ideals $(\mathfrak{p}, g_i(\alpha))$, as we sought to show.

Theorem 3.2.3: Local Dedekind-Kummer Theorem

Assume $AKLB$ with A a complete DVR, and let $\kappa = A/\mathfrak{p}$ and $\lambda = B/\mathfrak{P}$ be the residue fields of A and of the DVR B. If λ/κ is separable, then

$$B = A[\alpha]$$

for some $\alpha \in B$. If L/K is unramified, this holds for every lift $\alpha \in B$ of every generator $\overline{\alpha}$ of $\lambda = \kappa(\overline{\alpha})$.

***Proof*:**
Let e and $f = [\lambda : \kappa]$ be the ramification index and inertia degree, so that $ef = n := [L : K]$ (corollary 3.1.5). As the extension λ/κ is separable, there exists a primitive element

$$\lambda = \kappa(\overline{\alpha}_0)$$

for some $\overline{\alpha}_0 \in \lambda$ whose minimal polynomial $\overline{g}$ is separable of degree f. Let $g \in A[x]$ be a monic lift of $\overline{g}$ and let α_0 be any lift of $\overline{\alpha}_0$ to B. Then $g(\alpha_0) \in \mathfrak{P}$, as $\overline{g}(\overline{\alpha}_0) = 0$, and $g'(\alpha_0)$ is a unit, as $\overline{g}$ is separable. If $\nu_{\mathfrak{P}}(g(\alpha_0)) = 1$, let $\alpha := \alpha_0$. Otherwise, let π_0 be any uniformizer of B and let $\alpha := \alpha_0 + \pi_0$, so that $\alpha \equiv \alpha_0 \bmod \mathfrak{P}$. Expanding by the binomial theorem gives

$$g(\alpha_0 + \pi_0) = g(\alpha_0) + \pi_0 g'(\alpha_0) + \pi_0^2 h$$

for some $h \in B$. Here $\nu_{\mathfrak{P}}(\pi_0 g'(\alpha_0)) = 1$, while the other two terms have valuation at least 2, so $\nu_{\mathfrak{P}}(g(\alpha)) = 1$. In either case $\Pi := g(\alpha)$ is a uniformizer of B, and the residues of $1, \alpha, \ldots, \alpha^{f-1}$ form a κ-basis of $\lambda = \kappa(\overline{\alpha})$.

By corollary 3.1.5, the elements $\alpha^i \Pi^j = \alpha^i g(\alpha)^j$ with $0 \le i < f$ and $0 \le j < e$ form an A-basis of B. Each of them lies in $A[\alpha]$, so $B = A[\alpha]$. If L/K is unramified, then $e = 1$, and the same

corollary, applied with any uniformizer of B and $j = 0$ only, shows that $1, \alpha_0, \ldots, \alpha_0^{f-1}$ is an A-basis of B for every lift α_0 of a generator of λ, so $B = A[\alpha_0]$, as we sought to show.

3.3 Unramified, Tame, and Wild Extensions

Let L/K be a finite extension of local fields, with ramification index $e = e(L : K)$ and inertia degree $f = f(L : K)$ (definition 2.1.10). When L/K is separable, it decomposes into an unramified extension followed by a totally ramified one (theorem 3.3.16). If π and Π are uniformizers of K and L, then $\mathfrak{p}\mathcal{O}_L = \mathfrak{P}^e$ gives

$$\pi = u\Pi^e$$

for a unit $u \in \mathcal{O}_L^\times$, as for the ramification index of a prime in an extension of global fields.

3.3.1 Unramified Extensions

With the fundamental identity property re-defined, let us upgrade the definition of unramified extensions from [2, Ramified and Unramified Extensions]:

Definition 3.3.1: Unramified Extension

Let A be a complete DVR with fraction field K and residue field κ, and let L/K be a finite extension with residue field λ. Then L/K is *unramified* if $[L : K] = [\lambda : \kappa]$ and λ/κ is separable.

When κ is finite, as for local fields, the separability condition holds automatically, and L/K is unramified exactly when $f(L : K) = [L : K]$.

Lemma 3.3.2: Separable Reduction Gives Unramified Extensions

Let A be a complete DVR with fraction field K and residue field κ, and let α be a root, in an algebraic closure of K, of a monic polynomial $h \in A[x]$ whose reduction $\overline{h} \in \kappa[x]$ is separable. Then the residue field λ of $K(\alpha)$ is $\kappa(\overline{\alpha})$, it is separable over κ, and $[K(\alpha) : K] = [\lambda : \kappa]$. In particular $K(\alpha)/K$ is unramified.

***Proof*:**
Let $L = K(\alpha)$, let B be the integral closure of A in L, which is the valuation ring of L, with maximal ideal $\mathfrak{P}$ and residue field $\lambda = B/\mathfrak{P}$ (theorem 2.1.9), and let g be the minimal polynomial of α over K. As α is integral over A, so are its conjugates, and the coefficients of g, symmetric functions of these conjugates, are elements of K integral over A, hence lie in A. Since g is monic and divides h in $K[x]$, the quotient h/g lies in $A[x]$, so $\overline{g}$ divides $\overline{h}$ and is separable. It is irreducible: otherwise $\overline{g} = \overline{g}_1\overline{g}_2$ with $\overline{g}_1$ irreducible and $\deg \overline{g}_1 < \deg g$, the factors being relatively prime as $\overline{g}$ is separable, and theorem 2.1.1 would give a factorization $g = g_1 g_2$ of the monic polynomial g with $0 < \deg g_1 < \deg g$. The residue $\overline{\alpha} \in \lambda$ of $\alpha \in B$ is a root of $\overline{g}$, so $[\kappa(\overline{\alpha}) : \kappa] = \deg g = [L : K]$.

Next, let $\overline{y}_1, \ldots, \overline{y}_r \in \lambda$ be linearly independent over κ, with lifts $y_i \in B$. In a relation $\sum_i c_i y_i = 0$

with $c_i \in K$ not all zero, dividing by a c_j of largest absolute value gives coefficients in A, one of them equal to 1, and reducing modulo $\mathfrak{P}$ gives a nontrivial relation among the $\overline{y}_i$, which is impossible. So the y_i are linearly independent over K and $r \leq [L:K]$, that is, $[\lambda:\kappa] \leq [L:K]$. Hence $\kappa(\overline{\alpha}) \subseteq \lambda$ with $[\lambda:\kappa] \leq [L:K] = [\kappa(\overline{\alpha}):\kappa]$, so $\lambda = \kappa(\overline{\alpha})$, which is separable over κ as $\overline{\alpha}$ is a root of the separable polynomial $\overline{g}$, and $[L:K] = [\lambda:\kappa]$, completing the proof.

We may thus ask what is the relation of L/K and λ/κ. In particular, if we fix K with residue field κ, what is the relationship between the finite unramified extensions L/K of degree n and the finite separable extensions λ/κ of degree n. We already know that each L/K uniquely determines a corresponding λ/κ, but is the converse true too? In fact, it is almost as nice as it gets:

Theorem 3.3.3: Unramified Extension And Residue Field Category Equivalence

Let A be a complete DVR with fraction field K and residue field κ. Then the category $\mathcal{C}_K^{nr}$ and $\mathcal{C}_\kappa^{\text{sep}}$ are equivalent; the functor $\mathcal{F} : \mathcal{C}_K^{nr} \to \mathcal{C}_\kappa^{\text{sep}}$ sends each unramified extension L/K to its residue field λ, and each K-algebra homomorphism $\varphi : L_1 \to L_2$ to the κ-algebra homomorphism $\overline{\varphi} : \lambda_1 \to \lambda_2$ given by $\overline{\varphi}(\overline{\alpha}) := \overline{\varphi(\alpha)}$ where α is any lift of $\overline{\alpha} \in \lambda_1 = B_1/\mathfrak{P}_1$ to B_1 and $\overline{\varphi}$ is the reduction of $\varphi(\alpha) \in B_2$ to $\lambda_2 := B_2/\mathfrak{P}_2$.

This means that $\mathcal{F}$ gives a bijection between the isomorphism classes in $\mathcal{C}_K^{nr}$, and $\mathcal{C}_\kappa^{\text{sep}}$ and if L_1, L_2 have residue fields λ_1, λ_2, then $\mathcal{F}$ induces a bijection of finite sets:

$$\text{Hom}_K(L_1, L_2) \xrightarrow{\sim} \text{Hom}_\kappa(\lambda_1, \lambda_2)$$

The most interesting part of the proof is $\text{Hom}_K(L_1, L_2) \xrightarrow{\sim} \text{Hom}_\kappa(\lambda_1, \lambda_2)$, which will ultimately come down to Hensel's lemma after all the appropriate build-up.

***Proof*:**
Let us start with showing that $\mathcal{F}$ is well-defined. Objects are sent to objects, so we need to show that the morphism map is independent of lifts. Let $\varphi : L_1 \to L_2$ be a K-algebra homomorphism, and for $\overline{\alpha} \in \lambda_1$, let α, α' be two lifts to B_1. Then $\alpha - \alpha' \in \mathfrak{P}_1$, implying $\varphi(\alpha - \alpha') \in \varphi(\mathfrak{P}_1) \overset{!}{=} \varphi(B_1) \cap \mathfrak{P}_2 \subseteq \mathfrak{P}_2$, and hence $\overline{\varphi(\alpha)} = \overline{\varphi(\alpha')}$, showing it is well-defined. The $\overset{!}{=}$ equality comes from φ restricts to the ring homomorphism $B_1 \to B_2$ and $B_2/\varphi(B_1)$ is a finite extension of DVRs in which $\mathfrak{P}_2$ lies over the prime $\varphi(\mathfrak{P}_1)$ of $\varphi(B_1)$. Checking that the identity is set to the identity and composition is well-defined now follows quickly.

Let us now show that $\mathcal{F}$ gives an equivalence of categories. Recall from [3, Characterization of Equivalences] that this means that:

1. $\mathcal{F}$ is essentially surjective: each separable extension λ/κ is isomorphic to the residue field of some unramified L/K
2. $\mathcal{F}$ is full and faithful, i.e. the hom-map is a bijection.

Starting with essential surjectivity, given a finite separable extension λ/κ, we may apply the primitive element theorem and get:

$$\lambda \cong \kappa(\overline{\alpha}) \cong \frac{\kappa[x]}{(\overline{g}(x))}$$

for some $\overline{\alpha} \in \lambda$ whose minimal polynomial $\overline{g}$ is monic, irreducible, separable, and of degree $n := [\lambda : \kappa]$. Take any monic lift $g \in A[x]$ of $\overline{g}$, a root α of g in an algebraic closure of K, and $L := K(\alpha)$. By lemma 3.3.2, L/K is unramified, and its residue field is generated over κ by the residue of α, a root of the irreducible polynomial $\overline{g}$, so it is isomorphic to $\kappa[x]/(\overline{g}(x)) \cong \lambda$. Also $[L : K] = [\lambda : \kappa] = n = \deg g$, so g is the minimal polynomial of α and $L \cong K[x]/(g(x))$. The extension L/K is separable, as the discriminant of g, a polynomial in its coefficients with integer coefficients, reduces to the discriminant of $\overline{g}$, which is nonzero.

Next, to show that $\mathcal{F}$ is full and faithful, let us start with the induced maps:

$$\mathrm{Hom}_K(L_1, L_2) \xrightarrow{\sim} \mathrm{Hom}_A(B_1, B_2) \to \mathrm{Hom}_\kappa(\lambda_1, \lambda_2)$$

The first map is a bijection. A K-algebra map $L_1 \to L_2$ carries elements integral over A to elements integral over A, so it restricts to an A-algebra map $B_1 \to B_2$. Every element of L_1 becomes integral over A after multiplication by a nonzero element of A, so every element of L_1 is b/a with $b \in B_1$ and $a \in A \setminus \{0\}$. An A-algebra map $\psi\colon B_1 \to B_2$ therefore extends uniquely to a K-algebra map $L_1 \to L_2$, by $b/a \mapsto \psi(b)/a$, since a is a unit in L_2. The two constructions are inverse to each other. It must be shown that the same applies for the second map.

First, notice that:

$$\mathrm{Hom}_A(B_1, B_2) = \mathrm{Hom}_A\left(\frac{A[x]}{(g(x))}, B_2\right) = \mathrm{Hom}_A(A[\alpha], B_2)$$

which comes from B_1 always being a monogenic extension of A (Theorem 3.2.3) and that each A-algebra homomorphism is uniquely determined by the image of x in B_2. This gives a bijection between $\mathrm{Hom}_A(B_1, B_2)$ and the roots of g in B_2. Now, with the right choice of α, we can give the map. By the primitive element theorem, $\lambda_1 = \kappa(\overline{\alpha})$. Lift $\overline{\alpha}$ to $\alpha \in B_1$, and let g be the minimal polynomial of α over K, which lies in $A[x]$ because α is integral over the integrally closed ring A. Reducing $g(\alpha) = 0$ gives $\overline{g}(\overline{\alpha}) = 0$, so

$$[\lambda_1 : \kappa] \leq \deg \overline{g} = \deg g = [K(\alpha) : K] \leq [L_1 : K] = [\lambda_1 : \kappa]$$

with the last equality because L_1/K is unramified. Equality holds throughout, so $L_1 = K(\alpha)$ and $\overline{g}$ is the minimal polynomial of $\overline{\alpha}$. Then we saw in theorem 3.2.3 that $B_1 = A[\alpha]$. Now, consider:

$$\mathrm{Hom}_\kappa(\lambda_1, \lambda_2) = \mathrm{Hom}_\kappa\left(\frac{\kappa[x]}{(\overline{g}(x))}, \lambda_2\right) = \mathrm{Hom}_\kappa(\kappa(\overline{\alpha}), \lambda_2)$$

where the equality again comes from the uniquely determined image of x in λ_2, and there is a bijection between $\mathrm{Hom}_\kappa(\lambda_1, \lambda_2)$ and the roots of $\overline{g}$ in λ_2. Now, $\overline{g}$ is separable, so by Hensel's lemma every root of $\overline{g}$ in $\lambda_2 = B_2/\mathfrak{P}_2$ lifts to unique roots of g in B_2. But then the map is a bijection, as we sought to show.

3.3.4 Scope of the Hom Bijection For the bijection it was only necessary that L_1/K be unramified: the proof used the unramified hypothesis only to write $B_1 = A[\alpha]$ with separable reduced minimal polynomial, and on the target side it used only Hensel's lemma in B_2, which is a complete DVR for every finite separable extension L_2/K by theorem 2.1.9. The bijection therefore holds whether or not L_2/K is unramified.

Using this bijection we can compute the galois group of automorphisms of a field by looking at the automorphisms of the residue field.

Corollary 3.3.5: Characterizing Unramified Extensions through residue field

Assume $AKLB$ where A is a complete DVR with residue field κ. Then L/K is unramified if and only if $B = A[\alpha]$ for some $\alpha \in L$ whose minimal polynomial has separable image in $\kappa[x]$.

***Proof*:**
For the "if" direction, let g be the minimal polynomial of α, whose image $\overline{g}$ is separable. Then $\overline{g}$ must be irreducible, otherwise Hensel's lemma would lift a non-trivial factorization of $\overline{g}$ to one of g. Since $B = A[\alpha] \cong A[x]/(g)$, corollary 3.2.2 identifies λ with $\kappa[x]/(\overline{g})$, so the residue field extension is separable of degree $\deg \overline{g} = \deg g = [L : K]$, and hence L/K is unramified as $[L : K] = f$.

Conversely, assume L/K is unramified, so that $[L : K] = [\lambda : \kappa]$ and the residue extension λ/κ is separable (definition 3.3.1). By theorem 3.1.3 the ring B is a DVR, so theorem 3.2.3 applies: writing $\lambda = \kappa(\overline{\alpha})$ by the primitive element theorem, any lift $\alpha \in B$ of $\overline{\alpha}$ satisfies $B = A[\alpha]$. The minimal polynomial g of α is then monic of degree $[L : K] = [\lambda : \kappa]$, and its image $\overline{g}$ is monic of the same degree with $\overline{g}(\overline{\alpha}) = 0$, so $\overline{g}$ is the minimal polynomial of $\overline{\alpha}$ over κ. As $\overline{\alpha}$ generates the separable extension λ/κ, the image $\overline{g}$ is separable, completing the proof.

We shall now build-up to show that all unramified extensions where the residue field κ is finite will be isomorphic to a Cyclotomic extension. Starting with the following:

Lemma 3.3.6: Unramified Cyclotomic Extensions

Let A be a complete DVR with fraction field K and residue field κ, and let ζ_n be a primitive nth root of unity in some algebraic closure of K with $(n, p) = 1$ where $\operatorname{char} \kappa = p$. Then $K(\zeta_n)/K$ is unramified

***Proof*:**
The element ζ_n is a root of $x^n - 1 \in A[x]$, whose reduction $x^n - 1 \in \kappa[x]$ is separable: its derivative nx^{n-1} is nonzero as $p \nmid n$, and vanishes only at 0, which is not a root. By lemma 3.3.2, $K(\zeta_n)/K$ is unramified, as we sought to show.

Adding on the extra condition that the residue field is finite (which is always the case for local fields), we get to characterize unramified extnesions:

Proposition 3.3.7: Characterizing Unramified Extension Of Local Fields

Let A be a complete DVR with fraction field K and finite residue field $\mathbb{F}_q$. Let L/K be a degree n extension. Then L/K is unramified iff

$$L \cong K(\zeta_{q^n-1})$$

Furthermore, $A[\zeta_{q^n-1}]$ is the integral closure of A in L, and L/K is a galois extension with

$$\mathrm{Gal}_K(L) \cong \mathbb{Z}/n\mathbb{Z}$$

***Proof*:**
Let p be the characteristic of $\mathbb{F}_q$. For the "if" direction, if $L \cong K(\zeta_{q^n-1})$, then L/K is unramified by lemma 3.3.6, as $p \nmid q^n - 1$.

For the "only if" direction, assume L/K is unramified, let B be the integral closure of A in L, a complete DVR with maximal ideal $\mathfrak{P}$ (theorem 2.1.9), and let $\lambda = B/\mathfrak{P}$. Then $[\lambda : \mathbb{F}_q] = [L : K] = n$, so $\lambda = \mathbb{F}_{q^n}$, whose multiplicative group is cyclic of order $q^n - 1$, generated by some $\overline{\alpha}$. The polynomial $x^{q^n-1} - 1 \in B[x]$ is monic, and $\overline{\alpha}$ is a simple root of its reduction, as the derivative $(q^n - 1)x^{q^n-2}$ does not vanish at $\overline{\alpha} \neq 0$. By corollary 2.1.2 there is a root $\alpha \in B$ of $x^{q^n-1} - 1$ lifting $\overline{\alpha}$. Its powers α^j with $0 \leq j < q^n - 1$ have the distinct residues $\overline{\alpha}^j$, so α is a primitive $(q^n - 1)$-th root of unity. The residue field of $K(\alpha) \subseteq L$ contains $\overline{\alpha}$, which generates λ, so its residue degree over K is n and $[K(\alpha) : K] \geq n$ by corollary 3.1.5. Hence $L = K(\alpha)$, which gives $L \cong K(\zeta_{q^n-1})$.

As α lifts the generator $\overline{\alpha}$ of λ over $\mathbb{F}_q$ and L/K is unramified, theorem 3.2.3 gives $B = A[\alpha] = A[\zeta_{q^n-1}]$. Every $(q^n - 1)$-th root of unity is a power of α, so L is the splitting field of $x^{q^n-1} - 1$ over K and L/K is Galois, of degree n. Each $\sigma \in \mathrm{Gal}_K(L)$ maps B to B and $\mathfrak{P}$ to $\mathfrak{P}$, so it induces an automorphism $\overline{\sigma}$ of λ over $\mathbb{F}_q$. If $\overline{\sigma}$ is the identity, then $\sigma(\alpha) = \alpha^j$ has residue $\overline{\alpha}^j = \overline{\alpha}$, forcing $\alpha^j = \alpha$, so $\sigma = 1$. Thus $\sigma \mapsto \overline{\sigma}$ embeds $\mathrm{Gal}_K(L)$, of order n, into the cyclic group $\mathrm{Gal}_{\mathbb{F}_q}(\mathbb{F}_{q^n})$ of order n, and $\mathrm{Gal}_K(L) \cong \mathbb{Z}/n\mathbb{Z}$, as we sought to show.

The fact that an extension is unramified if and only if $L \cong K(\zeta_{q^n-1})$ implies that we can also deduce when a cyclic extension *is* ramified:

Corollary 3.3.8: Ramification Of Cyclic Extension

Let A be a complete DVR with field of fraction K of characteristic 0 and finite residue field of characteristic p, and let $e = \nu(p)$ be its absolute ramification index, for the normalized valuation ν of K. Suppose that $p - 1$ does not divide e. Then the extension $K(\zeta_m)/K$ is ramified if and only if $p \mid m$.

***Proof*:**
If $p \nmid m$, by lemma 3.3.6 $K(\zeta_m)/K$ is unramified. Let $p \mid m$, so that $\zeta_p = \zeta_m^{m/p}$ is a primitive p-th root of unity in $K(\zeta_m)$. The field K is a local field, so it contains a copy of $\mathbb{Q}_p$ on which ν restricts to $e\,\nu_p$ (proposition 1.3.2(1)). Let w be the extension of ν to $K(\zeta_m)$. On the subfield $\mathbb{Q}_p(\zeta_p)$,

the valuation w/e extends ν_p, so by uniqueness of extended absolute values (theorem 2.1.9) it is the extension of ν_p to $\mathbb{Q}_p(\zeta_p)$. By lemma 2.2.6, $\zeta_p - 1$ is a uniformizer of the totally ramified extension $\mathbb{Q}_p(\zeta_p)$ of degree $p-1$, at which the extension of ν_p takes the value $1/(p-1)$. Hence $w(\zeta_p - 1) = e/(p-1)$, which is not an integer. So $w(K(\zeta_m)^\times)$ strictly contains $\nu(K^\times) = \mathbb{Z}$, the ramification index of $K(\zeta_m)/K$ is greater than 1, and $K(\zeta_m)/K$ is ramified, as we sought to show.

Now with a choice of complete DVR, we can fully understand the unramified extensions

Example 3.3.9: Local Number Fields

Let $A = \mathbb{Z}_p$ so that $K = \mathbb{Q}_p$ and $\kappa = \mathbb{F}_p$. Then for each n, the galois field $\mathbb{F}_{p^n}$ is the unique extension of degree n. Thus, by the above results, there is a unique unramified extension of degree n, $L/\mathbb{Q}_p$. This can be explicitly constructed by adjoining a primitive root of unity ζ_{p^n-1} to $\mathbb{Q}_p$.

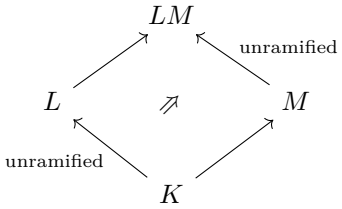
Maximal Unramified Extension

Let us build-up to the notion of maximal unramified extension, which is an important object for the Artin map. First, the "3rd isomorphism theorem" property applies for unramified extensions

Proposition 3.3.10: Unramified Square

Let K be the fraction field of a complete DVR, and let L/K and M/K be finite extensions contained in a common field. If L/K is unramified, then LM/M is unramified.

Visually,



***Proof*:**

Let κ, λ and μ be the residue fields of K, L and M, let A be the valuation ring of K, and let A_M be that of M, a complete DVR by theorem 2.1.9. As L/K is unramified, λ/κ is finite and separable, so $\lambda = \kappa(\overline{\alpha})$ for some $\overline{\alpha}$, which we lift to an element α of the valuation ring of L. The minimal polynomial f of α over K lies in $A[x]$, as α is integral over A, and its reduction $\overline{f}$ has the root $\overline{\alpha}$, so

$$[\lambda : \kappa] \leq \deg(\overline{f}) = \deg(f) = [K(\alpha) : K] \leq [L : K] = [\lambda : \kappa]$$

Hence $L = K(\alpha)$, and $\overline{f}$ is the minimal polynomial of $\overline{\alpha}$ over κ, which is separable. Then $LM = M(\alpha)$, where α is a root of the monic polynomial $f \in A[x] \subseteq A_M[x]$, whose reduction in $\mu[x]$ is the separable polynomial $\overline{f}$. By lemma 3.3.2 applied over M, the extension LM/M is unramified, as we sought to show.

Corollary 3.3.11: Composite Of Unramified Extensions

The composite of two unramified extensions of K is again unramified

***Proof*:**
Let L/K and M/K be unramified extensions contained in a common field, and let κ, μ and ρ be the residue fields of K, M and LM, so that $\kappa \subseteq \mu \subseteq \rho$. By proposition 3.3.10, LM/M is unramified, so $[LM : M] = [\rho : \mu]$ with ρ/μ separable, while $[M : K] = [\mu : \kappa]$ with μ/κ separable. The tower law for $LM \supseteq M \supseteq K$ and for $\rho \supseteq \mu \supseteq \kappa$ gives $[LM : K] = [\rho : \kappa]$, and ρ/κ is separable by transitivity of separability. So LM/K is unramified, as we sought to show.

By corollary 3.3.11, the finite unramified extensions of K inside a fixed algebraic extension of K are closed under composites, so their union is a field.

Definition 3.3.12: Maximal Unramified Extension

Let K be the fraction field of a complete DVR, and let L/K be an algebraic extension. The union of all finite unramified extensions of K contained in L is the *maximal unramified extension of K in L*. For $L = \overline{K}$ an algebraic closure of K, it is written K_{nr}, or K^{nr}, and called the *maximal unramified extension* of K.

The letters nr abbreviate the French *non ramifiée*.

Example 3.3.13: Maximal Unramified Extension For Local Fields
Take $K = \mathbb{Q}_p$. Then $\mathbb{Q}_p^{nr}$ is an infinite extension of $\mathbb{Q}_p$, with galois group:

$$\mathrm{Gal}_{\mathbb{Q}_p}(\mathbb{Q}_p^{nr}) \cong \mathrm{Gal}_{\mathbb{F}_p}(\overline{\mathbb{F}}_p) = \varprojlim_n \mathrm{Gal}_{\mathbb{F}_p}(\mathbb{F}_{p^n}) \cong \varprojlim_n \mathbb{Z}/n\mathbb{Z} =: \hat{\mathbb{Z}}$$

where the last ring is called the *profinite completion* of $\mathbb{Z}$. The field has value group $\mathbb{Z}$ (its image under the valuation) and residue field $\overline{\mathbb{F}}_p$.

Corollary 3.3.14: Root Of Unity In Maximally Unramified Extension

Let K_{nr}/K be the maximal unramified extension. Then K_{nr} contains all the roots of unity of order m not divisible by the characteristic of κ

If κ is a finite field, then the extension K_{nr}/K is generated by the roots of unity of order prime to the characteristic of κ.

***Proof*:**
Let p be the characteristic of κ and let $p \nmid m$. A primitive m-th root of unity $\zeta_m \in \overline{K}$ generates an unramified extension $K(\zeta_m)/K$ by lemma 3.3.6, so $\zeta_m \in K_{nr}$, and with it every m-th root of unity, each being a power of ζ_m.

Now let $\kappa = \mathbb{F}_q$, and let $L \subseteq \overline{K}$ be a finite unramified extension of K, of degree n. By propo-

sition 3.3.7, L is isomorphic over K to the splitting field $N \subseteq \overline{K}$ of $x^{q^n-1} - 1$. As N/K is finite and normal, every K-embedding of N into $\overline{K}$ has image N [5, Finite Normal Extension Equivalences], and the isomorphism $N \to L$ is such an embedding, so $L = N = K(\zeta_{q^n-1})$. Hence K_{nr}, the composite of all such L, is generated over K by roots of unity of order prime to p, and by the first part it contains all of them, as we sought to show.

This also shows then that K_{nr} is a Cyclotomic extension of K.

Proposition 3.3.15: Residue Field Of Maximally Unramified Extension

Let K be the fraction field of a complete DVR with finite residue field κ, and let M/K be the maximal unramified extension of K. Then the residue field of M is an algebraic closure of κ, which is its separable closure κ_s since κ is perfect.

***Proof*:**
Write $\kappa = \mathbb{F}_q$. Every element of M lies in the composite of finitely many finite unramified extensions of K, which is again a finite unramified extension of K by corollary 3.3.11. The valuation of K extends uniquely to each finite extension of K (theorem 2.1.9), so these extensions agree on overlaps and define a valuation on M, whose valuation ring $\mathcal{O}_M$ and maximal ideal $\mathfrak{m}_M$ are the unions of the valuation rings $\mathcal{O}_L$ and maximal ideals $\mathfrak{m}_L$ over the finite unramified $L \subseteq M$. Hence the residue field $\lambda_M = \mathcal{O}_M/\mathfrak{m}_M$ is the union of the images of the residue fields $\mathcal{O}_L/\mathfrak{m}_L$, each a finite extension of κ. So λ_M is algebraic over κ, and we fix an algebraic closure $\overline{\kappa}$ of κ containing λ_M.

For every $n \geq 1$, theorem 3.3.3 gives an unramified extension of K with residue field $\mathbb{F}_{q^n}$, and embedding it into $\overline{K}$ over K places it inside M. So λ_M contains a subfield with q^n elements, which is the unique such subfield $\mathbb{F}_{q^n}$ of $\overline{\kappa}$, the set of roots of $x^{q^n} - x$. Every element of $\overline{\kappa}$ generates a finite extension of $\mathbb{F}_q$, which is some $\mathbb{F}_{q^n}$, and so lies in λ_M. Thus $\lambda_M = \overline{\kappa}$ is an algebraic closure of κ, which is the separable closure of κ because the finite field κ is perfect, as we sought to show.

To wrap up this section, let us now look at decomposing an extension L/K into the ramified and unramified portion, and in the case where L/K is galois to find what it is:

Theorem 3.3.16: Tower Extension And Galois Group

Assume $AKLB$ with A a complete DVR and separable residue field extension λ/κ. Let e, f be the ramification index and residue field degree, and let $\mathfrak{P}$ be the unique prime of B. Then:

1. There is a unique intermediate field extension E/K that contains every unramified extension of K in L and has degree $[E : K] = f$
2. The extension L/E is totally ramified and has degree $[L : E] = e$
3. If L/K is galois, then the decomposition group $D_{\mathfrak{P}}$ is all of $\mathrm{Gal}_K(L)$, the inertia group $I_{\mathfrak{P}}$ is $\mathrm{Gal}_E(L)$, and E/K is galois with $\mathrm{Gal}_K(E) \cong D_{\mathfrak{P}}/I_{\mathfrak{P}} \cong \mathrm{Gal}_\kappa(\lambda)$

***Proof*:**

1. Since λ/κ is a finite separable extension of degree f, the equivalence of categories of theorem 3.3.3 (whose hypotheses hold as A is a complete DVR) provides an unramified extension E/K with residue field λ, and $[E:K] = [\lambda:\kappa] = f$ because E/K is unramified. By box 3.3.4, the bijection

$$\mathrm{Hom}_K(E, L) \xrightarrow{\sim} \mathrm{Hom}_\kappa(\lambda, \lambda)$$

holds here: its source-side input is supplied by corollary 3.3.5 as E/K is unramified, and its target-side input by B being a complete DVR by theorem 2.1.9. The identity $\lambda \to \lambda$ thus corresponds to a K-embedding $E \to L$. Replacing E by its image, E is an intermediate field of L/K of degree f whose residue field is λ, as the embedding induces the identity on residue fields.

Now let M/K be an unramified extension of K contained in L. The composite EM inside L is unramified over K by corollary 3.3.11. The ring $B \cap EM$ is the integral closure of A in EM, and is therefore the valuation ring of EM by theorem 2.1.9. The inclusion $B \cap EM \to B$ descends to an embedding of its residue field into $B/\mathfrak{P} = \lambda$, hence:

$$[EM : K] = f_{EM/K} \leq [\lambda : \kappa] = f$$

As $E \subseteq EM$ and $[E:K] = f$, this forces $EM = E$, and so $M \subseteq E$. Finally, if E' is another intermediate field that contains every unramified extension of K in L and has $[E':K] = f$, then $E \subseteq E'$ because E is itself an unramified extension of K in L, and the equality of degrees gives $E' = E$, establishing uniqueness.

2. Let $B_E := B \cap E$ be the valuation ring of E and $\mathfrak{P}_E := \mathfrak{P} \cap B_E$ its maximal ideal. The extension E/K is separable, since the construction of theorem 3.3.3 presents E as $K[x]/(g)$ with g separable. Hence theorem 2.1.9 applied to E/K makes B_E a complete DVR, B is the integral closure of B_E in L by transitivity of integral closure, and the $AKLB$ hypotheses hold for B_E, E, L, B. The residue field of E is λ by the previous item, so the residue field extension of L/E is trivial and $f_{L/E} = 1$. The fundamental identity (corollary 3.1.5) for the complete DVR B_E and the extension L/E gives:

$$[L : E] = e_{L/E} f_{L/E} = e_{L/E}$$

so L/E is totally ramified. The same identity applied to L/K gives $[L:K] = ef$, and multiplicativity of degrees in towers gives:

$$[L : E] = \frac{[L:K]}{[E:K]} = \frac{ef}{f} = e$$

3. Let $G = \mathrm{Gal}_K(L)$. Each $\sigma \in G$ maps B onto B, as it preserves integrality over A, and maps $\mathfrak{P}$ onto a maximal ideal of B, which is $\mathfrak{P}$ as B is a DVR (theorem 3.1.3). So the decomposition group $D_\mathfrak{P} = \{\sigma \in G \mid \sigma(\mathfrak{P}) = \mathfrak{P}\}$ is all of G. Each $\sigma \in G$ therefore induces an automorphism $\overline{\sigma}$ of $\lambda = B/\mathfrak{P}$ over κ, and $\sigma \mapsto \overline{\sigma}$ is a homomorphism $G \to \mathrm{Aut}_\kappa(\lambda)$ whose kernel is the inertia group $I_\mathfrak{P} = \{\sigma \in G \mid \sigma(x) \equiv x \bmod \mathfrak{P} \text{ for all } x \in B\}$.

First, $\mathrm{Gal}_E(L) \subseteq I_\mathfrak{P}$. Let $\sigma \in \mathrm{Gal}_E(L)$ and $x \in B$. The residue field of E is λ by the

first item, so $x \equiv y \bmod \mathfrak{P}$ for some $y \in B_E$, and $\sigma(x) - x = \sigma(x-y) - (x-y) \in \mathfrak{P}$, as $\sigma(y) = y$ and $\sigma(\mathfrak{P}) = \mathfrak{P}$.

Next, $I_{\mathfrak{P}} \subseteq \mathrm{Gal}_E(L)$. The extension E/K is unramified and separable, so by corollary 3.3.5 $B_E = A[\beta]$ for some β whose minimal polynomial g over K has separable reduction $\overline{g}$, and $E = K(\beta)$. As L/K is normal, $g = \prod_k (x - \beta_k)$ with all β_k in B, so $\overline{g} = \prod_k (x - \overline{\beta}_k)$, and the residues of distinct roots of g are distinct, as $\overline{g}$ is separable. For $\sigma \in I_{\mathfrak{P}}$, the root $\sigma(\beta)$ of g has the same residue as β, so $\sigma(\beta) = \beta$ and σ fixes $E = K(\beta)$.

Hence $I_{\mathfrak{P}} = \mathrm{Gal}_E(L)$. It is normal in G, being a kernel, so E/K is galois and $\mathrm{Gal}_K(E) \cong G/I_{\mathfrak{P}} = D_{\mathfrak{P}}/I_{\mathfrak{P}}$ [5, Fundamental Theorem Of Galois Theory III]. The map $\sigma \mapsto \overline{\sigma}$ induces an injective homomorphism $G/I_{\mathfrak{P}} \to \mathrm{Aut}_\kappa(\lambda)$, whose source has order $[E:K] = f = [\lambda:\kappa]$ by the first item. As λ/κ is separable, $|\mathrm{Aut}_\kappa(\lambda)| \leq [\lambda:\kappa]$ [5, Order of $\mathrm{Aut}_k(F)$ Of Finite Extension]. So the map is bijective, equality holds, λ/κ is galois by the same corollary, and $G/I_{\mathfrak{P}} \cong \mathrm{Gal}_\kappa(\lambda)$, completing the proof.

In the complete case the decomposition group is all of $G = \mathrm{Gal}_K(L)$, so the tower of fixed fields loses the bottom step of the global tower (fig. 3.1). The global degrees are [2, Inertia Group Properties], and the bottom one, $[G : D_{\mathfrak{P}}] = g$, is the number of primes above $\mathfrak{p}$, since G permutes them transitively [2, Galois Group Transitive Action On Ideals] and $D_{\mathfrak{P}}$ is the stabilizer of $\mathfrak{P}$.

Global: L/K number fields

$$\begin{array}{c} L = L^{\{1\}} \\ | e \\ L^{I_{\mathfrak{P}}} \\ | f \\ L^{D_{\mathfrak{P}}} \\ | g \\ K = L^G \end{array} \quad \Big\} n = efg$$

Local: A a complete DVR

$$\begin{array}{c} L = L^{\{1\}} \\ | e \\ E = L^{I_{\mathfrak{P}}} \\ | f \\ K = L^{D_{\mathfrak{P}}} = L^G \end{array} \quad \Big\} n = ef$$

Figure 3.1: The fixed fields of the inertia and decomposition groups of a prime $\mathfrak{P}$ above $\mathfrak{p}$, for a galois extension L/K of degree n

Exercise 3.3.1

1. Assume $AKLB$, with A a complete DVR with finite residue field. Then L/K is unramified if and only if $N_{L/K}(B^*) = A^*$. Is this the case for global fields?

3.3.2 Ramified Extensions

We are now in a position where if we have any finite separable extension L/K where A is a complete DVR, then we have a tower $L/E/K$ where E/K is unramified and L/E is totally ramified. We just classified the unramified extensions (or in particular the unramified extensions of field of fractions of complete DVRs), so let us now focus on the ramified case.

Totally ramified extensions of complete DVR

Recall that a monic polynomial $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0 \in R[x]$ is *Eisenstein* at a prime $\mathfrak{p} \subseteq R$ if $a_i \in \mathfrak{p}$ for all $i < n$ and $a_0 \notin \mathfrak{p}^2$. When R is a DVR with maximal ideal $\mathfrak{p}$, this says $\nu_\mathfrak{p}(a_i) \geq 1$ for $i < n$ and $\nu_\mathfrak{p}(a_0) = 1$, so a_0 is a uniformizer. As a DVR is a PID and hence a UFD, Gauss's lemma and Eisenstein's criterion show that such an f is irreducible over the field of fractions.

Lemma 3.3.17: DVR Extension Using Eisenstein Polynomial

Let A be a DVR and let $f \in A[x]$ be an Eisenstein polynomial. Then $B = A[\pi] = A[x]/(f)$ is a DVR with uniformizer π, where π is the image of x in $A[x]/(f)$.

***Proof*:**
Let $\mathfrak{p}$ be the maximal ideal of A and $n = \deg f$. Then $f \equiv x^n \bmod \mathfrak{p}$, and so by corollary 3.2.2 the ideal $\mathfrak{P} = (\mathfrak{p}, \pi)$ is the only maximal ideal of B. Let $f = \sum_i a_i x^i$. Then as f is Eisenstein, $\mathfrak{p} = (a_0)$, $\mathfrak{P} = (a_0, \pi)$ and $a_0 = -a_1\pi - a_2\pi^2 - \cdots - \pi^n \in (\pi)$, hence $\mathfrak{P} = (\pi)$.

The ring B is a domain: f is irreducible over the fraction field K of A by Eisenstein's criterion, and as f is monic, division by f shows $A[x] \cap fK[x] = fA[x]$, so B embeds in the field $K[x]/(f)$. As f is monic, B is a free A-module with basis $1, \pi, \ldots, \pi^{n-1}$, so it is Noetherian, being finitely generated over the Noetherian ring A. It is not a field, as π is nonzero and lies in $\mathfrak{P}$. From $f(\pi) = 0$ and $a_i \in \mathfrak{p}$, $\pi^n \in \mathfrak{p}B$, so $\pi^{nk}B \subseteq \mathfrak{p}^k B$ for every k, and $\bigcap_k \mathfrak{p}^k B = 0$ as $B \cong A^n$. So a nonzero $b \in B$ lies in a largest $\pi^k B$, and $b = \pi^k u$ with $u \notin \pi B = \mathfrak{P}$, a unit as $\mathfrak{P}$ is the only maximal ideal. Hence every nonzero element of B is a unit times a power of the non-unit π, and B is a DVR with uniformizer π, as we sought to show.

Theorem 3.3.18: Characterizing Totally Ramified Extensions

Assume $AKLB$ with A a complete DVR and π a uniformizer for B. Then the extension L/K is totally ramified if and only if $B = A[\pi]$ and the minimal polynomial of π is Eisenstein

***Proof*:**
Let $n = [L : K]$, $e = e(L : K)$, and let $\nu_\mathfrak{P}$ be the normalized valuation of B, so that $\nu_\mathfrak{P}(\pi) = 1$ and $\nu_\mathfrak{P} = e\,\nu_\mathfrak{p}$ on K. In both directions we use that if finitely many elements of L sum to 0, the least valuation among them is attained at least twice, by the ultrametric inequality.

Suppose that $B = A[\pi]$ and that the minimal polynomial $f = x^m + a_{m-1}x^{m-1} + \cdots + a_0$ of π over K is Eisenstein. As B spans L over K, $L = K(\pi)$ and $m = n$. In $f(\pi) = 0$, the term π^n

has valuation n, the term a_0 has valuation e, and for $0 < i < n$ the term $a_i\pi^i$ has valuation at least $e + i > e$. If $n < e$, then π^n alone has the least valuation, and if $n > e$, then a_0 alone does. Hence $n = e$, and L/K is totally ramified.

Conversely, suppose that L/K is totally ramified, so that $e = n$ and the residue field of L is that of K. By corollary 3.1.5, applied with $\omega_1 = 1$, the elements $1, \pi, \dots, \pi^{n-1}$ form an A-basis of B, so $B = A[\pi]$ and $L = K(\pi)$. Let $f = x^n + a_{n-1}x^{n-1} + \cdots + a_0$ be the minimal polynomial of π, which lies in $A[x]$ as π is integral over the integrally closed ring A. For $0 \le i < n$ with $a_i \neq 0$, $\nu_{\mathfrak{P}}(a_i\pi^i) = n\,\nu_{\mathfrak{p}}(a_i) + i \equiv i \pmod{n}$, so these terms of $f(\pi) = 0$ have pairwise distinct valuations, while $\nu_{\mathfrak{P}}(\pi^n) = n$. The least valuation is therefore attained by π^n together with exactly one other term, whose valuation is congruent to n modulo n, which is a_0. So $\nu_{\mathfrak{P}}(a_0) = n$, that is $\nu_{\mathfrak{p}}(a_0) = 1$, and for $0 < i < n$ with $a_i \neq 0$, $n\,\nu_{\mathfrak{p}}(a_i) + i > n$, which forces $\nu_{\mathfrak{p}}(a_i) \ge 1$. Hence f is Eisenstein, completing the proof.

Example 3.3.19: Composite of Totally Ramified Extensions

Let p be odd, and let $u \in \mathbb{Z}_p^\times$ have residue $\bar{u}$ a non-square in $\mathbb{F}_p$, which exists as squaring is not surjective on the cyclic group $\mathbb{F}_p^\times$ of even order. Then u is not a square in $\mathbb{Q}_p$, since a square root of u would be a unit whose residue squares to $\bar{u}$. The polynomials $x^2 - p$ and $x^2 - up$ are Eisenstein, so $\mathbb{Q}_p(\sqrt{p})$ and $\mathbb{Q}_p(\sqrt{up})$ are totally ramified of degree 2 (corollary 2.2.4). Their composite L contains the square root $\sqrt{up}/\sqrt{p}$ of u, and $\mathbb{Q}_p(\sqrt{u})/\mathbb{Q}_p$ is unramified of degree 2 by lemma 3.3.2, as $x^2 - \bar{u}$ is separable over $\mathbb{F}_p$. So the residue field of L contains $\mathbb{F}_{p^2}$, and the value group of L contains that of $\mathbb{Q}_p(\sqrt{p})$, so $e(L/\mathbb{Q}_p) \ge 2$ and $f(L/\mathbb{Q}_p) \ge 2$, while $e(L/\mathbb{Q}_p)f(L/\mathbb{Q}_p) = [L : \mathbb{Q}_p] \le 4$ (corollary 3.1.5). Hence $e(L/\mathbb{Q}_p) = f(L/\mathbb{Q}_p) = 2$ and $[L : \mathbb{Q}_p] = 4$, so the composite of the two totally ramified extensions is not totally ramified.

Definition 3.3.20: Tamely Ramified And Wildly Ramified

Assume $AKLB$ with A a complete DVR and a separable field extension of characteristic $p \ge 0$. Then:

1. The extension L/K is *tamely ramified* if $p \nmid e_{L/K}$ (which is always true if $p = 0$)
2. The extension L/K is *wildly ramified* if $p \mid e_{L/K}$

All unramified extensions are tamely ramified. A totally ramified extension L/K is totally tamely ramified if $p \nmid e_{L/K}$ and is *totally wildly ramified* if $e_{L/K}$ is a power of p. Note that a totally ramified extension that is wildly ramified need not be totally wildly ramified.

Proposition 3.3.21: Tower of Ramified Extensions

Let $M/L/K$ be a tower of finite separable extensions of fraction fields of complete DVR's with separable residue field extensions. Then

$$e_{M/K} = e_{M/L}e_{L/K} \qquad f_{M/K} = f_{M/L}f_{L/K}$$

and each type of ramified extension is transitive: if M/L and L/K are both unramified, both totally ramified, both tamely ramified, both wildly ramified, both totally tamely ramified, or both totally wildly ramified, then so is M/K.

***Proof*:**
Let $M/L/K$ be a tower of finite separable extensions of fraction fields of complete DVRs with separable residue field extensions, say with valuation rings $A \subseteq B \subseteq C$ and maximal ideals $\mathfrak{p}, \mathfrak{P}_B, \mathfrak{P}_C$, so that B is the integral closure of A in L and C is the integral closure of B in M (each again a complete DVR by theorem 2.1.9). By transitivity of integral closure, C is also the integral closure of A in M, and by theorem 3.1.3 the ideal $\mathfrak{P}_C$ is the unique prime of C above $\mathfrak{p}$.

Both e and f are multiplicative in the tower. For f, the inclusions $A \subseteq B \subseteq C$ induce inclusions of residue fields $\kappa \subseteq \lambda \subseteq \mu$, and the tower law for field degrees gives:

$$f_{M/K} = [\mu : \kappa] = [\mu : \lambda][\lambda : \kappa] = f_{M/L}f_{L/K}$$

For e, theorem 3.1.2 applied to M/L and to L/K gives $\nu_{\mathfrak{P}_C}|_L = e_{M/L}\nu_{\mathfrak{P}_B}$ and $\nu_{\mathfrak{P}_B}|_K = e_{L/K}\nu_{\mathfrak{p}}$, while applied to M/K it gives $\nu_{\mathfrak{P}_C}|_K = e_{M/K}\nu_{\mathfrak{p}}$. Evaluating both expressions for $\nu_{\mathfrak{P}_C}|_K$ at a uniformizer of A yields:

$$e_{M/K} = e_{M/L}e_{L/K}$$

Note also that κ, λ, μ all have the same characteristic p, that μ/κ is separable by transitivity of separability applied to the tower $\mu/\lambda/\kappa$, and that $[M : K] = [M : L][L : K]$ by the tower law.

Now suppose M/L and L/K are both of one of the named types. If both are unramified, then $f_{M/K} = f_{M/L}f_{L/K} = [M : L][L : K] = [M : K]$, so M/K is unramified. If both are totally ramified, the same computation with e in place of f shows $e_{M/K} = [M : K]$. If both are tamely ramified and $p = 0$, every extension under consideration is tamely ramified. If both are tamely ramified and $p > 0$, then p divides neither $e_{M/L}$ nor $e_{L/K}$, and as p is prime, $p \nmid e_{M/L}e_{L/K} = e_{M/K}$. If both are wildly ramified, then $p|e_{L/K}|e_{M/K}$. The totally tamely ramified case combines the totally ramified and tamely ramified cases, and if both extensions are totally wildly ramified, then M/K is totally ramified by the second case while $e_{M/K}$ is a product of two powers of p and hence a power of p. In each case the tower M/K is of the same type, as we sought to show.

The same is not the case for compositum's, for example the compositum of totally ramified extensions need not be totally ramified (see example 3.3.19).

Theorem 3.3.22: Characterizing Totally Tamely Ramified Extensions

Assume $AKLB$ with A a complete DVR and separable residue field extension of characteristic $p \geq 0$ not dividing $n := [L : K]$. Then the extension L/K is *totally tamely ramified* if and only if $L = K(\pi_A^{1/n})$ for some uniformizer π_A of A.

The main place the tame assumption shall come is when we need to use Hensel's lemma to lift a certain solution. The equivalent of this result in the global case required that the extension be a *Kummer Extension*, which is an extension with enough roots of unity and that there are certain properties satisfied by the galois group of K.

***Proof*:**

If $L = K(\pi_A^{1/n})$, then $\pi = \pi_A^{1/n}$ has minimal polynomial $x^n - \pi_A$, which his Eisenstein and so $A[\pi]$ is a DVR by lemma 3.3.17. By maximality of DVR's, $B = A[\pi]$, and by theorem 3.3.18 L/K is totally tamely ramified, as $p \nmid n$.

Conversely, assume L/K is totally tamely ramified and $\mathfrak{p}, \mathfrak{P}$ are the maximal ideals of A, B with uniformizer π_A, π_B respectfully. Then $\nu_{\mathfrak{P}}$ extends $\nu_{\mathfrak{p}}$ with index $e_{\mathfrak{P}} = n$ and $\nu_{\mathfrak{P}}(\pi_B^n) = n = \nu_{\mathfrak{P}}(\pi_A)$. Then $\pi_B^n = u\pi_A$ for some unit $u \in B^*$. As we are totally ramified, $f_{\mathfrak{P}} = 1$, so B and A have the *same* residue field, and so if we lift the image of $u \in B/\mathfrak{P} \cong A/\mathfrak{p}$ to a unit $u_A \in A$, and replace π_A with $u_A\pi_A$, the unit u is replaced by uu_A^{-1}, so we may assume $u \equiv 1 \bmod \mathfrak{P}$. From this, define

$$g(x) = x^n - u \in B[x] \qquad \Rightarrow \qquad \overline{g}(x) = x^n - 1 \in (B/\mathfrak{P})[x]$$

Then $\overline{g}'(1) = n \neq 0$ as $p \nmid n$, showing it is a primitive polynomial and so by Hensel's lemma we can lift the root of 1 to r for $g(x)$ in B. Say $\pi = \pi_B/r$. Then π is a uniformizer for B, hence by theorem 3.3.18 $B = A[\pi]$, so $L = K(\pi)$. As $\pi^n = \pi_B^n/r^n = \pi_B^n/u = \pi_A$, we have:

$$L = K(\pi_A^{1/n})$$

we completed the other direction, as we sought to show.

Proposition 3.3.23: Tower of Ramification

Let K be the field of fractions of a complete DVR and L/K a totally ramified extension. Then there is a unique intermediate field E/K that is totally tamely ramified and L/E is totally wildly ramified

***Proof*:**

Let A be the valuation ring of K with maximal ideal $\mathfrak{p}$ and uniformizer π_A, and let B be the valuation ring of L with maximal ideal $\mathfrak{P}$ and uniformizer π_B, again a complete DVR by theorem 2.1.9. Put $n = [L : K]$, let $e = e_{L/K}$, and let $p \geq 0$ be the characteristic of the residue field $\kappa = A/\mathfrak{p}$. That L/K is totally ramified says $n = e$ and $f_{L/K} = 1$, so $B/\mathfrak{P} = \kappa$. Ramification indices and inertia degrees are multiplicative in towers (proposition 3.3.21).

The tame case. If $p \nmid e$ then L/K is already totally tamely ramified and $E = L$ works, since L/L has degree $1 = p^0$. For uniqueness, if L/E is totally wildly ramified then $e_{L/E}$ is a power of p dividing $e_{L/K} = e$, hence $e_{L/E} = 1$, and L/E being totally ramified forces $[L : E] = e_{L/E} = 1$.

So assume from now on that $p \mid e$, and write $e = mp^a$ with $p \nmid m$ and $a \geq 1$.

A convenient pair of uniformizers. By theorem 3.1.2 the valuation $\nu_{\mathfrak{P}}$ extends $\nu_{\mathfrak{p}}$ with index $e = n$, so $\nu_{\mathfrak{P}}(\pi_A) = n = \nu_{\mathfrak{P}}(\pi_B^n)$ and $\pi_B^n = u\pi_A$ for some $u \in B^*$. As $B/\mathfrak{P} = \kappa$, the image of u in $B/\mathfrak{P}$ is the image of some $u_A \in A^*$. Replacing π_A by the uniformizer $u_A\pi_A$ replaces u by uu_A^{-1}, so we may assume $u \equiv 1 \bmod \mathfrak{P}$.

Existence. Let $g(x) = x^m - u \in B[x]$, whose reduction $\overline{g} = x^m - 1 \in (B/\mathfrak{P})[x]$ has 1 as a root with $\overline{g}'(1) = m \neq 0$, the last because $p \nmid m$. The root is therefore simple and by corollary 2.1.2 it lifts to a root $r \in B$ of g, so that $r^m = u$ and $r \equiv 1 \bmod \mathfrak{P}$ is a unit. Put:

$$\pi = \frac{\pi_B^{p^a}}{r} \qquad E = K(\pi)$$

Then $\pi^m = \pi_B^{mp^a}/r^m = \pi_B^n/u = \pi_A$, so π is a root of $x^m - \pi_A$, which satisfies the hypothesis of corollary 2.2.4 with $\nu_{\mathfrak{p}}(\pi_A) = 1$ and all intermediate coefficients zero. That corollary makes $x^m - \pi_A$ irreducible over K and E/K totally ramified of degree m. In particular the residue field extension of E/K is trivial and hence separable, and $p \nmid m$, so $E = K(\pi_A^{1/m})$ is totally tamely ramified by theorem 3.3.22. Finally $[L : E] = n/m = p^a$, and $e_{L/E}e_{E/K} = e$ with $e_{E/K} = m$ gives $e_{L/E} = p^a = [L : E]$. So L/E is totally ramified with ramification index a power of p, which is to say totally wildly ramified.

Uniqueness. Let E' be an intermediate field with E'/K totally tamely ramified and L/E' totally wildly ramified. Then $[E' : K] = e_{E'/K}$ with $p \nmid e_{E'/K}$, the degree $[L : E'] = e_{L/E'}$ is a power of p, and their product is $e = mp^a$. Comparing the p-parts gives $[E' : K] = m$ and $[L : E'] = p^a$. The residue field extension of E'/K is again trivial, so theorem 3.3.22 applies and $E' = K(\sigma)$ for some σ with $\sigma^m = \tau$ a uniformizer of A.

Rewrite this in terms of the uniformizer π_A fixed above. Write $\tau = w\pi_A$ with $w \in A^*$ and put $\theta = \sigma/\pi \in L$, so that $\theta^m = \tau/\pi_A = w$ and $\nu_{\mathfrak{P}}(\theta) = 0$. Reducing modulo $\mathfrak{P}$ and using $B/\mathfrak{P} = \kappa$, the element $\overline{\theta} \in \kappa$ is a root of $\overline{h}$ where $h(x) = x^m - w \in A[x]$, and $\overline{h}'(\overline{\theta}) = m\overline{\theta}^{\,m-1} \neq 0$ because $p \nmid m$ and $\overline{\theta} \neq 0$. So corollary 2.1.2 applied inside the complete DVR A lifts $\overline{\theta}$ to a root $\theta_A \in A^*$ of h, and $\rho = \sigma/\theta_A$ lies in E' with $\rho^m = \tau/w = \pi_A$ and $E' = K(\sigma) = K(\rho)$.

Both E and E' are therefore generated by roots of $x^m - \pi_A$ lying in L, and $\zeta = \rho/\pi \in L$ satisfies $\zeta^m = 1$. Thus ζ is a primitive m'th root of unity for some $m' \mid m$, and $p \nmid m'$, so $K(\zeta)/K$ is unramified by lemma 3.3.6. But $K(\zeta) \subseteq L$ with L/K totally ramified, so $f_{K(\zeta)/K}$ divides $f_{L/K} = 1$, while unramifiedness says $[K(\zeta) : K] = f_{K(\zeta)/K}$. Hence $\zeta \in K$, giving $\rho = \zeta\pi \in E$ and $E' \subseteq E$. The two fields have the same degree m over K, so $E' = E$, as we sought to show.

Corollary 3.3.24: Tower of Ramification in Separable Case

Let K be the field of fractions of a complete DVR with separable residue field and L/K be a finite separable extension with separable residue field extensions. Then there is a unique intermediate field E such that E/K is tamely ramified and L/E is totally wildly ramified.

***Proof*:**
Let A be the valuation ring of K with maximal ideal $\mathfrak{p}$ and residue field κ of characteristic p, let B be the integral closure of A in L with maximal ideal $\mathfrak{P}$, and let $e = e_{L/K}$ and $f = f_{L/K}$. The hypotheses are those of theorem 3.3.16, which gives a unique intermediate field F/K containing

every unramified extension of K inside L, with $[F:K]=f$ and L/F totally ramified of degree e. The field F is the fraction field of the complete DVR $B\cap F$ by theorem 2.1.9, and B is the integral closure of $B\cap F$ in L by transitivity of integral closure, so the $AKLB$ hypotheses hold for $B\cap F, F, L, B$ as well. Ramification indices and inertia degrees are multiplicative in towers (proposition 3.3.21).

Existence. Apply proposition 3.3.23 to the totally ramified extension L/F: there is an intermediate field E with E/F totally tamely ramified and L/E totally wildly ramified. The extension F/K is unramified, hence tamely ramified, and E/F is tamely ramified, so E/K is tamely ramified by proposition 3.3.21.

Uniqueness. Let E' be an intermediate field with E'/K tamely ramified and L/E' totally wildly ramified. First, $F\subseteq E'$. Indeed, F/K is unramified, so the compositum $E'F$ taken inside L is unramified over E' by proposition 3.3.10, which gives $[E'F:E']=f_{E'F/E'}$. But L/E' is totally ramified, so $f_{L/E'}=1$ and the factor $f_{E'F/E'}$ of it equals 1 as well. Hence $E'F=E'$ and $F\subseteq E'$.

Now E' is an intermediate field of L/F. Since L/F is totally ramified, $f_{E'/F}$ divides $f_{L/F}=1$, and the fundamental identity (corollary 3.1.5), applied to the extension E'/F of fraction fields of complete DVRs, gives $[E':F]=e_{E'/F}f_{E'/F}=e_{E'/F}$. So E'/F is totally ramified. Its ramification index $e_{E'/F}$ divides $e_{E'/K}$, which is prime to p because E'/K is tamely ramified, so E'/F is totally tamely ramified. With L/E' totally wildly ramified, the uniqueness clause of proposition 3.3.23 applied to L/F forces $E'=E$, as we sought to show.

Exercise 3.3.2

1. Let L/K be a finite extension. Then L/K is a tamely ramified extension (not necessarily totally tamely ramified) if and only if $L=K(\sqrt[m_1]{a_1},\ldots,\sqrt[m_r]{a_r})$ where $\gcd(m_i,p)=1$ (generalize theorem 3.3.22).
2. Generalize proposition 3.3.23 to tamely ramified extensions. Use this to show there exists a maximally tamely ramified extension.

3.4 Higher Ramification Groups

The trichotomy of the previous section sorts every extension into unramified, tamely ramified, or wildly ramified. The first two are transparent: an unramified extension is governed entirely by its residue field, and a totally tamely ramified one by a single root of a uniformizer. Wild ramification is subtler. When $p\mid e$ the inertia group no longer injects into the residue field, and the single integer e no longer records how the Galois group acts near the prime. What is needed is a *filtration* of the inertia group, each step isolating the automorphisms that act trivially to higher and higher order on the ring of integers. This filtration, the higher ramification groups, is the finest invariant of ramification: it computes the different that measures wildness quantitatively, and through the conductor it controls the local constants attached to Galois representations.

Throughout this section L/K is a finite Galois extension of local fields with group $G=\mathrm{Gal}_K(L)$, and ν is the valuation on L normalized so that $\nu(L^\times)=\mathbb{Z}$. We write $\mathfrak{P}$ for the maximal ideal of $\mathcal{O}_L$ and $\lambda=\mathcal{O}_L/\mathfrak{P}$ for its residue field, of characteristic p.

Definition 3.4.1: Ramification Groups in the Lower Numbering

For an integer $i \geq -1$, the i-th *ramification group* of L/K is

$$G_i = \{\sigma \in G \mid \nu(\sigma(x) - x) \geq i + 1 \text{ for all } x \in \mathcal{O}_L\}$$

The condition tightens as i grows, so the groups form a descending chain

$$G = G_{-1} \supseteq G_0 \supseteq G_1 \supseteq \cdots$$

The first two steps are already familiar. For $i = -1$ the condition $\nu(\sigma(x) - x) \geq 0$ is automatic, so $G_{-1} = G$. For $i = 0$ the condition says $\sigma(x) \equiv x \bmod \mathfrak{P}$ for every $x \in \mathcal{O}_L$, that is, σ acts trivially on the residue field λ; thus $G_0 = I$ is precisely the inertia group of theorem 3.3.16. The higher groups $G_1, G_2, \ldots$ dissect the wild part of the inertia, and it is their behaviour that the rest of the section describes.

Verifying membership in G_i against every $x \in \mathcal{O}_L$ would be laborious; a single generator suffices.

Lemma 3.4.2: Ramification Groups from a Generator

Suppose $\mathcal{O}_L = \mathcal{O}_K[\alpha]$. Then for every $\sigma \in G$ and every integer $i \geq -1$,

$$\sigma \in G_i \quad \Longleftrightarrow \quad \nu(\sigma(\alpha) - \alpha) \geq i + 1$$

In particular the integer $i_G(\sigma) := \nu(\sigma(\alpha) - \alpha)$ satisfies $G_i = \{\sigma \in G \mid i_G(\sigma) \geq i + 1\}$, and the resulting groups do not depend on the choice of generator α.

***Proof*:**
The forward implication is immediate, since $\alpha \in \mathcal{O}_L$. For the converse, suppose $\nu(\sigma(\alpha) - \alpha) \geq i + 1$ and let $x \in \mathcal{O}_L$ be arbitrary. As $\mathcal{O}_L = \mathcal{O}_K[\alpha]$ we may write $x = f(\alpha)$ for some $f \in \mathcal{O}_K[t]$. Then

$$\sigma(x) - x = f(\sigma(\alpha)) - f(\alpha) = (\sigma(\alpha) - \alpha)\, g(\sigma(\alpha), \alpha)$$

for a polynomial g with coefficients in $\mathcal{O}_K$, because $t - s$ divides $f(t) - f(s)$ in $\mathcal{O}_K[t, s]$. Both factors lie in $\mathcal{O}_L$, so $\nu(\sigma(x) - x) \geq \nu(\sigma(\alpha) - \alpha) \geq i + 1$. Thus $\sigma \in G_i$. Since the groups G_i were defined without reference to α, the characterization $G_i = \{\sigma \mid i_G(\sigma) \geq i + 1\}$ shows i_G records the same data for any generator, as we sought to show.

Proposition 3.4.3: The Ramification Filtration

Each G_i is a normal subgroup of G, the chain $G_0 \supseteq G_1 \supseteq \cdots$ is decreasing, and $G_i = 1$ for all sufficiently large i.

***Proof*:**
Recall that the valuation ν on L is preserved by every $\tau \in G$, since $\nu \circ \tau$ is another valuation extending that of K and the extension is unique (theorem 2.1.9).

Subgroup. Let $\sigma, \tau \in G_i$ and $x \in \mathcal{O}_L$. Then $\tau(x) \in \mathcal{O}_L$, so

$$\nu(\sigma\tau(x) - x) \geq \min\big(\nu(\sigma(\tau x) - \tau x),\, \nu(\tau x - x)\big) \geq i + 1$$

giving $\sigma\tau \in G_i$. For inverses, $\nu(\sigma^{-1}(x) - x) = \nu\big(\sigma(\sigma^{-1}x - x)\big) = \nu(x - \sigma(x)) \geq i + 1$, so $\sigma^{-1} \in G_i$.

Normality. For $\tau \in G$ and $\sigma \in G_i$, using $\nu \circ \tau = \nu$ and $\tau(x) \in \mathcal{O}_L$,

$$\nu\big(\tau^{-1}\sigma\tau(x) - x\big) = \nu\big(\sigma\tau(x) - \tau(x)\big) \geq i + 1$$

so $\tau^{-1}\sigma\tau \in G_i$.

Eventual triviality. Fix a generator $\mathcal{O}_L = \mathcal{O}_K[\alpha]$, which exists because the residue extension is separable (theorem 3.2.3). For $\sigma \neq 1$ we have $\sigma(\alpha) \neq \alpha$, since α generates L over K, so $i_G(\sigma) = \nu(\sigma(\alpha) - \alpha)$ is finite. As G is finite, $m := \max_{\sigma \neq 1} i_G(\sigma)$ is finite, and lemma 3.4.2 gives $G_i = 1$ for all $i \geq m$, completing the proof.

An integer i with $G_i \neq G_{i+1}$ is called a *ramification break* or *jump* of L/K; the breaks are the finitely many indices at which the filtration strictly decreases. The next theorem is the structural heart of the subject: it identifies each successive quotient with a piece of the residue field, and so pins down what kind of group the inertia can be.

Theorem 3.4.4: Ramification Quotients

Let π be a uniformizer of L and let $U^{(0)} = \mathcal{O}_L^\times$, $U^{(n)} = 1 + \mathfrak{P}^n$ for $n \geq 1$ denote the unit filtration of L. For each $i \geq 0$ the map

$$\theta_i \colon G_i/G_{i+1} \to U^{(i)}/U^{(i+1)}, \qquad \sigma \mapsto \frac{\sigma(\pi)}{\pi} \bmod U^{(i+1)}$$

is a well-defined injective homomorphism, independent of the choice of π. Consequently:

1. G_0/G_1 embeds into $\lambda^\times$; it is cyclic of order prime to p.
2. For $i \geq 1$, G_i/G_{i+1} embeds into the additive group λ^+; it is an elementary abelian p-group.

***Proof*:**
The groups G_i for $i \geq 0$ lie in $G_0 = I$, and their defining condition refers only to the action of σ on $\mathcal{O}_L$; it is therefore unchanged if K is replaced by the inertia field $E = L^{G_0}$ (theorem 3.3.16). Over E the extension L/E is totally ramified, so $\mathcal{O}_L = \mathcal{O}_E[\pi]$ for any uniformizer π (theorem 3.3.18), and we may compute with $\alpha = \pi$ in lemma 3.4.2: for $i \geq 0$,

$$\sigma \in G_i \iff \nu(\sigma(\pi) - \pi) \geq i + 1 \iff \nu\left(\frac{\sigma(\pi)}{\pi} - 1\right) \geq i \iff \frac{\sigma(\pi)}{\pi} \in U^{(i)}$$

and likewise $\sigma \in G_{i+1} \iff \sigma(\pi)/\pi \in U^{(i+1)}$. Thus θ_i carries G_i into $U^{(i)}$ and has kernel exactly G_{i+1}; injectivity of the induced map on G_i/G_{i+1} follows once we check it is a homomorphism.

Homomorphism. Let $\sigma, \tau \in G_i$ and write $\tau(\pi)/\pi = 1 + b$ with $\nu(b) \geq i$. Applying σ,

$$\frac{\sigma\tau(\pi)}{\pi} = \frac{\sigma(\pi)}{\pi} \cdot \sigma\left(\frac{\tau(\pi)}{\pi}\right) = \frac{\sigma(\pi)}{\pi}\,(1 + \sigma(b))$$

Now $b \in \mathcal{O}_L$ and $\sigma \in G_i$, so $\nu(\sigma(b) - b) \geq i + 1$, whence $(1 + \sigma(b))/(1 + b) = 1 + (\sigma(b) - b)/(1 + b) \in$

$U^{(i+1)}$. Therefore

$$\theta_i(\sigma\tau) = \frac{\sigma(\pi)}{\pi} \cdot \frac{\tau(\pi)}{\pi} \bmod U^{(i+1)} = \theta_i(\sigma)\,\theta_i(\tau)$$

Independence of π. Any other uniformizer is $\pi' = u\pi$ with $u \in \mathcal{O}_L^\times$, and $\sigma(\pi')/\pi' = (\sigma(u)/u)(\sigma(\pi)/\pi)$. For $\sigma \in G_i$ we have $\nu(\sigma(u) - u) \geq i + 1$, so $\sigma(u)/u \in U^{(i+1)}$ and θ_i is unchanged modulo $U^{(i+1)}$.

Finally, the unit filtration was computed in proposition 1.2.2: reduction modulo $\mathfrak{P}$ gives $U^{(0)}/U^{(1)} \cong \lambda^\times$, and $1 + a\pi^i \mapsto a \bmod \mathfrak{P}$ gives $U^{(i)}/U^{(i+1)} \cong \lambda^+$ for $i \geq 1$. Composing with θ_i embeds G_0/G_1 into $\lambda^\times$, a finite subgroup of the multiplicative group of a field and hence cyclic, of order prime to $p = \operatorname{char}\lambda$; and it embeds G_i/G_{i+1} into λ^+, all of whose nonzero elements have order p, so G_i/G_{i+1} is an elementary abelian p-group, as we sought to show.

The consequences for the structure of the inertia group are immediate and decisive.

Corollary 3.4.5: Structure of the Inertia Group

Let $G_0 = I$ be the inertia group of L/K, of order e. Then:

1. G_1 is a p-group, normal in G_0, and is the unique Sylow p-subgroup of G_0; it is called the *wild inertia group*.
2. G_0/G_1 is cyclic of order prime to p, and $G_0 = G_1 \rtimes C$ for a cyclic subgroup C of order prime to p.
3. L/K is tamely ramified if and only if $G_1 = 1$.
4. G is solvable.

***Proof*:**
By theorem 3.4.4 the chain $G_1 \supseteq G_2 \supseteq \cdots \supseteq 1$ has successive quotients that are elementary abelian p-groups, so G_1 is a p-group; it is normal in G_0 by proposition 3.4.3. Since G_0/G_1 has order prime to p, the order of G_1 is the full power of p dividing $|G_0|$, so G_1 is the unique Sylow p-subgroup. This proves (1), and (2) follows: the quotient G_0/G_1 is cyclic of order m prime to p by theorem 3.4.4. Choose $g \in G_0$ mapping to a generator of G_0/G_1. Then $g^m \in G_1$, a p-group, so g^m has order p^k for some k, and $h = g^{p^k}$ has order m. Its image generates G_0/G_1, since p^k is prime to m. As m is prime to p, the subgroup $C = \langle h \rangle$ meets G_1 trivially, so C maps isomorphically onto G_0/G_1 and is a complement.

For (3), L/K is tamely ramified exactly when $p \nmid e = |G_0|$, which by (1) holds if and only if the Sylow p-subgroup G_1 is trivial. For (4), G_1 is a p-group, hence solvable; G_0/G_1 is abelian; and $G/G_0 \cong \mathrm{Gal}_\kappa(\lambda)$ is cyclic (theorem 3.3.16). A group with a solvable normal subgroup and solvable quotient is solvable, applied twice, so G is solvable, completing the proof.

The classical fact that Galois groups of local fields are solvable, opaque from the definitions, is thus a direct reading of the ramification filtration. The two examples below show the filtration: the first is the wild cyclotomic tower, the reference example for the whole theory; the second is its positive-characteristic analogue, where a single Artin-Schreier equation produces a single wild break.

Example 3.4.6: Ramification of the Cyclotomic Tower $\mathbb{Q}_p(\zeta_{p^n})$

Let $L=\mathbb{Q}_p(\zeta_{p^n})$ with $n\geq 1$, so $L/\mathbb{Q}_p$ is totally ramified of degree $e=\varphi(p^n)=p^{n-1}(p-1)$, with uniformizer $\pi=\zeta-1$ where $\zeta=\zeta_{p^n}$, and $G=\mathrm{Gal}_{\mathbb{Q}_p}(L)\cong(\mathbb{Z}/p^n\mathbb{Z})^\times$ via $\sigma_a(\zeta)=\zeta^a$ (lemma 2.2.6). To compute the filtration, evaluate

$$i_G(\sigma_a)=\nu(\sigma_a(\pi)-\pi)=\nu(\zeta^a-\zeta)=\nu(\zeta^{a-1}-1)$$

using $\nu(\zeta)=0$. If $a\equiv 1 \bmod p^k$ but $a\not\equiv 1 \bmod p^{k+1}$, then ζ^{a-1} is a primitive p^{n-k}-th root of unity, and since $\zeta_{p^{n-k}}-1$ is a uniformizer of $\mathbb{Q}_p(\zeta_{p^{n-k}})$ (lemma 2.2.6) over which L has ramification index $\varphi(p^n)/\varphi(p^{n-k})=p^k$,

$$i_G(\sigma_a)=\nu(\zeta_{p^{n-k}}-1)=p^k$$

Therefore $\sigma_a\in G_i$ if and only if $p^{v_p(a-1)}\geq i+1$, that is, $a\equiv 1 \bmod p^k$ where $k=\lceil\log_p(i+1)\rceil$. Writing this out, for $p^{k-1}\leq i\leq p^k-1$,

$$G_i=\left\{\sigma_a \;\middle|\; a\equiv 1 \bmod p^k\right\}=\mathrm{Gal}_{\mathbb{Q}_p(\zeta_{p^k})}(L)$$

with $G_0=G$ and $G_i=1$ for $i\geq p^{n-1}$. The breaks occur at $i=p^k-1$ for $k=0,1,\ldots,n-1$. The quotients match theorem 3.4.4: $G_0/G_1\cong(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic of order $p-1$ prime to p, while for $k\geq 1$ each $G_{p^{k-1}}/G_{p^k}$ is cyclic of order p, an elementary abelian p-group.

Example 3.4.7: An Artin-Schreier Extension

Let $K=\mathbb{F}_p((t))$ and let $L=K(y)$ where $y^p-y=t^{-m}$ with $m\geq 1$ prime to p. This is a cyclic totally (wildly) ramified extension of degree p, with $G=\mathrm{Gal}_K(L)\cong\mathbb{Z}/p\mathbb{Z}$ generated by $\sigma\colon y\mapsto y+1$. From $y^p-y=t^{-m}$, whose right side has valuation $-mp$ while the left has valuation $p\,\nu_L(y)$, it follows that $\nu_L(y)=-m$ and $\nu_L(t)=p$. Since $\gcd(m,p)=1$, choose integers a,b with $-ma+pb=1$ and set $\pi=y^at^b$, a uniformizer of L; reducing $-ma+pb=1$ modulo p shows $p\nmid a$. For $1\leq j\leq p-1$,

$$\sigma^j(\pi)-\pi=t^b\big((y+j)^a-y^a\big)$$

and the lowest-valuation term of $(y+j)^a-y^a=aj\,y^{a-1}+\cdots$ is $aj\,y^{a-1}$, of valuation $-m(a-1)$ because $p\nmid aj$. Hence

$$i_G(\sigma^j)=\nu_L(\sigma^j(\pi)-\pi)=pb-m(a-1)=(pb-ma)+m=m+1$$

independent of j. Therefore

$$G_0=G_1=\cdots=G_m=G\cong\mathbb{Z}/p\mathbb{Z},\qquad G_{m+1}=1$$

so the extension has a single break, at $i=m$. There is no tame part ($G_0=G_1$), as befits a wildly ramified extension of degree p, and the lone quotient $G_m/G_{m+1}\cong\mathbb{Z}/p\mathbb{Z}$ sits inside λ^+ exactly as theorem 3.4.4 predicts.

Exercise 3.4.1

1. Compute the ramification filtration of $\mathbb{Q}_2(\sqrt{2})/\mathbb{Q}_2$: find $i_G(\sigma)$ for the nontrivial automorphism σ, and list the groups G_i.
2. Show that the integer $i_G(\sigma)=\nu(\sigma(\alpha)-\alpha)$ of lemma 3.4.2 is independent of the choice of

generator α with $\mathcal{O}_L = \mathcal{O}_K[\alpha]$.

3. Let L/K be totally and tamely ramified of degree e (so $p \nmid e$). Show directly from theorem 3.4.4 that $G_0 = G$ is cyclic of order e and $G_1 = 1$, so the only break is at $i = 0$.

4. Using corollary 3.4.5, show that if $[L : K]$ is a prime $\ell \neq p$ and L/K is totally ramified, then L/K is tame and cyclic.

3.5 The Different and Discriminant

The ramification groups of the previous section record how the Galois group acts near the prime, but they are a group-theoretic invariant of a Galois extension. The *different* repackages the same information as an ideal of $\mathcal{O}_L$, defined for any finite separable extension whether or not it is Galois, and its valuation counts ramification directly: it equals $e - 1$ exactly when the extension is tame, and the excess over $e - 1$ measures wildness. Taking norms down to K produces the *discriminant*, the invariant that detects which primes ramify in a global field. This section defines both, computes the different from the ramification filtration, and records the tower formula that makes the two multiplicative.

Throughout, L/K is a finite separable extension of local fields of degree n, with $\mathcal{O}_L = \mathcal{O}_K[\alpha]$ a monogenic ring of integers (theorem 3.2.3) and $g \in \mathcal{O}_K[T]$ the minimal polynomial of α. The trace form $(x, y) \mapsto \mathrm{tr}_{L/K}(xy)$ is nondegenerate because L/K is separable ([2]), which is what makes the following dual object a fractional ideal.

Definition 3.5.1: Different and Codifferent

The *codifferent* (or inverse different) of L/K is

$$\mathcal{C}_{L/K} = \left\{ x \in L \;\middle|\; \mathrm{tr}_{L/K}(x\mathcal{O}_L) \subseteq \mathcal{O}_K \right\}$$

a fractional ideal of $\mathcal{O}_L$ containing $\mathcal{O}_L$. Its inverse, the *different*, is the integral ideal

$$\mathfrak{d}_{L/K} = \mathcal{C}_{L/K}^{-1} \subseteq \mathcal{O}_L$$

The codifferent $\mathcal{C}_{L/K}$ contains $\mathcal{O}_L$, since $\mathrm{tr}_{L/K}(\mathcal{O}_L) \subseteq \mathcal{O}_K$. Thus $\mathfrak{d}_{L/K} \subseteq \mathcal{O}_L$ is a genuine ideal, and it is the unit ideal precisely when $\mathrm{tr}_{L/K}(\mathcal{O}_L) = \mathcal{O}_K$. In the monogenic case the codifferent has a closed form, which reduces every computation of the different to a derivative.

Proposition 3.5.2: The Different of a Monogenic Extension

With $\mathcal{O}_L = \mathcal{O}_K[\alpha]$ and g the minimal polynomial of α,

$$\mathcal{C}_{L/K} = \frac{1}{g'(\alpha)}\mathcal{O}_L, \qquad \mathfrak{d}_{L/K} = \big(g'(\alpha)\big)$$

***Proof*:**
Let $\alpha = \alpha_1, \dots, \alpha_n$ be the conjugates of α in a splitting field, so $g(T) = \prod_i (T - \alpha_i)$. Euler's interpolation identity states that

$$\sum_{i=1}^{n} \frac{g(T)}{(T-\alpha_i)\, g'(\alpha_i)}\, \alpha_i^k = T^k \qquad (0 \le k \le n-1)$$

both sides being polynomials of degree $\le n-1$ agreeing at the n points $T = \alpha_i$. Reading off the trace, the elements $\beta_0, \dots, \beta_{n-1} \in L$ defined by $\frac{g(T)}{(T-\alpha)g'(\alpha)} = \sum_j \beta_j T^j$ satisfy $\mathrm{tr}_{L/K}(\alpha^k \beta_j) = \delta_{jk}$. So $\{\beta_j\}$ is the trace dual basis of $\{\alpha^k\}$. Since $g(T)/(T-\alpha) = \sum_j \gamma_j T^j$ has coefficients $\gamma_j \in \mathcal{O}_K[\alpha] = \mathcal{O}_L$ (they are polynomials in α with coefficients built from those of g), we get $\beta_j = \gamma_j / g'(\alpha) \in \frac{1}{g'(\alpha)} \mathcal{O}_L$.

Now $x \in \mathcal{C}_{L/K}$ iff $\mathrm{tr}_{L/K}(x\mathcal{O}_L) \subseteq \mathcal{O}_K$, iff $\mathrm{tr}_{L/K}(x\alpha^k) \in \mathcal{O}_K$ for all k, iff x lies in the $\mathcal{O}_K$-span of the dual basis $\{\beta_j\}$. That span is $\bigoplus_j \mathcal{O}_K \beta_j = \frac{1}{g'(\alpha)} \bigoplus_j \mathcal{O}_K \gamma_j = \frac{1}{g'(\alpha)} \mathcal{O}_L$, the last equality because $\{\gamma_j\}$ is an $\mathcal{O}_K$-basis of $\mathcal{O}_L$ (the γ_j are unit-triangular in the $\{\alpha^k\}$, as $\gamma_{n-1} = 1$). Hence $\mathcal{C}_{L/K} = \frac{1}{g'(\alpha)} \mathcal{O}_L$ and $\mathfrak{d}_{L/K} = (g'(\alpha))$, as we sought to show.

For a Galois extension the derivative $g'(\alpha)$ is a product of differences of conjugates, and each difference is measured by the function i_G of the previous section. The valuation of the different is therefore exactly the sum that also computes the orders of the ramification groups.

Theorem 3.5.3: The Different and the Ramification Filtration

Let L/K be a finite Galois extension of local fields with group G and ramification groups G_i of orders $g_i = |G_i|$. Then

$$\nu_L\big(\mathfrak{d}_{L/K}\big) = \sum_{s \in G,\, s \neq 1} i_G(s) = \sum_{i \ge 0} (g_i - 1)$$

***Proof*:**
Take α with $\mathcal{O}_L = \mathcal{O}_K[\alpha]$. As L/K is Galois, the conjugates of α are the $s\alpha$ for $s \in G$, so $g(T) = \prod_{s \in G} (T - s\alpha)$ and

$$g'(\alpha) = \prod_{s \neq 1} (\alpha - s\alpha)$$

By proposition 3.5.2, $\mathfrak{d}_{L/K} = (g'(\alpha))$, so taking valuations and using $\nu_L(\alpha - s\alpha) = \nu_L(s\alpha - \alpha) = i_G(s)$ (lemma 3.4.2, since α generates $\mathcal{O}_L$ over $\mathcal{O}_K$),

$$\nu_L(\mathfrak{d}_{L/K}) = \sum_{s \neq 1} \nu_L(\alpha - s\alpha) = \sum_{s \neq 1} i_G(s)$$

For the second equality, count each $s \neq 1$ once for every $i \ge 0$ with $s \in G_i$. Since $s \in G_i \iff i_G(s) \ge i + 1 \iff i \le i_G(s) - 1$, there are exactly $i_G(s)$ such indices, and so

$$\sum_{s \neq 1} i_G(s) = \sum_{s \neq 1} \#\{i \ge 0 \mid s \in G_i\} = \sum_{i \ge 0} \#\{s \neq 1 \mid s \in G_i\} = \sum_{i \ge 0} (g_i - 1)$$

completing the proof.

Corollary 3.5.4: The Different Detects Tameness

Let L/K be finite Galois with ramification index e. Then $\nu_L(\mathfrak{d}_{L/K}) \geq e-1$, with equality if and only if L/K is tamely ramified. In general

$$\nu_L(\mathfrak{d}_{L/K}) = (e-1) + \sum_{i\geq 1}(g_i - 1)$$

and the wild term $\sum_{i\geq 1}(g_i-1)$ is positive exactly when $p \mid e$.

***Proof*:**
By theorem 3.5.3, $\nu_L(\mathfrak{d}_{L/K}) = (g_0-1) + \sum_{i\geq 1}(g_i-1)$, and $g_0 = |G_0| = e$ is the order of the inertia group. Each term $g_i - 1 \geq 0$, so $\nu_L(\mathfrak{d}_{L/K}) \geq e-1$, with equality iff $g_i = 1$ for all $i \geq 1$, that is, iff the wild inertia G_1 is trivial. By corollary 3.4.5 this happens exactly when L/K is tamely ramified, equivalently $p \nmid e$, as we sought to show.

Descending from L to K turns the different into an ideal of the base. Recall that the norm of ideals satisfies $N_{L/K}(\mathfrak{P}^m) = \mathfrak{p}^{fm}$, where f is the residue degree.

Definition 3.5.5: Discriminant

The *discriminant* of L/K is the ideal $\mathrm{disc}_{L/K} = N_{L/K}(\mathfrak{d}_{L/K})$ of $\mathcal{O}_K$.

Proposition 3.5.6: Valuation of the Discriminant

If L/K is Galois with a single prime $\mathfrak{P}$ of residue degree f, then $\nu_K(\mathrm{disc}_{L/K}) = f \cdot \nu_L(\mathfrak{d}_{L/K})$. In particular, for a totally ramified extension $\nu_K(\mathrm{disc}_{L/K}) = \nu_L(\mathfrak{d}_{L/K})$, and L/K is unramified if and only if $\mathrm{disc}_{L/K} = \mathcal{O}_K$.

***Proof*:**
Writing $\mathfrak{d}_{L/K} = \mathfrak{P}^d$ with $d = \nu_L(\mathfrak{d}_{L/K})$, the norm is $N_{L/K}(\mathfrak{P}^d) = (N_{L/K}\mathfrak{P})^d = \mathfrak{p}^{fd}$, so $\nu_K(\mathrm{disc}_{L/K}) = fd$. For totally ramified $f = 1$. Finally $\mathrm{disc}_{L/K} = \mathcal{O}_K$ iff $d = 0$ iff $\mathfrak{d}_{L/K} = \mathcal{O}_L$ iff every $g_i - 1 = 0$ including $g_0 = e = 1$, which is unramifiedness (an unramified extension has $G_0 = 1$), completing the proof.

Finally, both invariants are multiplicative in towers, the property that lets a global discriminant be assembled from local contributions and that will drive the quotient theory of the next section.

Theorem 3.5.7: Transitivity of the Different

For a tower $L/K'/K$ of finite separable extensions of local fields,

$$\mathfrak{d}_{L/K} = \mathfrak{d}_{L/K'} \cdot \left(\mathfrak{d}_{K'/K}\mathcal{O}_L\right)$$

and consequently $\nu_L(\mathfrak{d}_{L/K}) = \nu_L(\mathfrak{d}_{L/K'}) + e_{L/K'}\,\nu_{K'}(\mathfrak{d}_{K'/K})$.

***Proof*:**
Transitivity of the trace, $\mathrm{tr}_{L/K} = \mathrm{tr}_{K'/K} \circ \mathrm{tr}_{L/K'}$, gives the corresponding statement for codifferents: for $x \in L$,

$$x \in \mathcal{C}_{L/K} \iff \mathrm{tr}_{L/K}(x\mathcal{O}_L) \subseteq \mathcal{O}_K \iff \mathrm{tr}_{K'/K}\big(\mathrm{tr}_{L/K'}(x\mathcal{O}_L)\big) \subseteq \mathcal{O}_K$$

The inner trace $\mathrm{tr}_{L/K'}(x\mathcal{O}_L)$ is a fractional ideal of $\mathcal{O}_{K'}$, and it lands in $\mathcal{C}_{K'/K}$ exactly when the outer condition holds. Unwinding, $x \in \mathcal{C}_{L/K}$ iff $\mathrm{tr}_{L/K'}(x\mathcal{O}_L) \subseteq \mathcal{C}_{K'/K}$, iff $x \cdot \mathcal{C}_{K'/K}^{-1}\mathcal{O}_L \subseteq \mathcal{C}_{L/K'}$, that is $\mathcal{C}_{L/K} = \mathcal{C}_{L/K'} \cdot (\mathcal{C}_{K'/K}\mathcal{O}_L)$. Inverting gives $\mathfrak{d}_{L/K} = \mathfrak{d}_{L/K'} \cdot (\mathfrak{d}_{K'/K}\mathcal{O}_L)$. Applying ν_L and using $\nu_L|_{K'} = e_{L/K'}\nu_{K'}$ yields the valuation formula, completing the proof.

Example 3.5.8: Different of the Cyclotomic Fields
For $L = \mathbb{Q}_p(\zeta_p)$ the extension $L/\mathbb{Q}_p$ is totally and tamely ramified of degree $e = p-1$ (lemma 2.2.6), so corollary 3.5.4 gives $\nu_L(\mathfrak{d}_{L/\mathbb{Q}_p}) = e - 1 = p - 2$. Since it is totally ramified, $\nu_p(\mathrm{disc}_{L/\mathbb{Q}_p}) = p - 2$ as well, recovering $\mathrm{disc}(\mathbb{Q}(\zeta_p)) = \pm p^{p-2}$. For $L = \mathbb{Q}_p(\zeta_{p^2})$ the ramification groups computed in example 3.4.6 have $g_0 = p(p-1)$ and $g_1 = \cdots = g_{p-1} = p$, whence

$$\nu_L(\mathfrak{d}_{L/\mathbb{Q}_p}) = (g_0 - 1) + (p-1)(g_1 - 1) = \big(p(p-1) - 1\big) + (p-1)^2 = p(2p-3)$$

in agreement with the general formula $\nu_L(\mathfrak{d}_{\mathbb{Q}_p(\zeta_{p^n})/\mathbb{Q}_p}) = p^{n-1}(np - n - 1)$, which gives $p - 2$ at $n = 1$ and $p(2p-3)$ at $n = 2$.

Example 3.5.9: Different of an Eisenstein Extension
Let $L = K(\pi)$ with π a root of an Eisenstein polynomial $g(T) = T^e + a_{e-1}T^{e-1} + \cdots + a_0$, so L/K is totally ramified of degree e and $\mathcal{O}_L = \mathcal{O}_K[\pi]$ (theorem 3.3.18). Then $\mathfrak{d}_{L/K} = (g'(\pi))$ with $g'(\pi) = e\pi^{e-1} + (e-1)a_{e-1}\pi^{e-2} + \cdots + a_1$. When $p \nmid e$ the leading term $e\pi^{e-1}$ has the strictly smallest valuation $e - 1$ (the others have valuation $\geq e$, as $\nu_L(a_j) \geq e$ for $j \geq 1$ while $\nu_L(\pi^{j-1}) = j - 1$), so $\nu_L(\mathfrak{d}_{L/K}) = e - 1$, matching corollary 3.5.4. When $p \mid e$ the term $e\pi^{e-1}$ is no longer dominant and the different jumps above $e - 1$, detecting the wild ramification.

Exercise 3.5.1

1. Compute $\nu_L(\mathfrak{d}_{L/\mathbb{Q}_2})$ for $L = \mathbb{Q}_2(\sqrt{2})$ directly from the minimal polynomial, and check it against $\sum_{i \geq 0}(g_i - 1)$ using the ramification groups found in section 3.4.
2. Show that $\nu_L(\mathfrak{d}_{L/K}) = 0$ if and only if L/K is unramified, working directly from theorem 3.5.3.
3. Let L/K be totally tamely ramified of degree e (so $p \nmid e$). Using an Eisenstein uniformizer, prove $\mathfrak{d}_{L/K} = (\pi^{e-1})$ and $\mathrm{disc}_{L/K} = \mathfrak{p}^{e-1}$.
4. Using theorem 3.5.7, verify for the tower $\mathbb{Q}_p(\zeta_{p^2})/\mathbb{Q}_p(\zeta_p)/\mathbb{Q}_p$ that the middle and top differents add up (weighted by $e_{L/K'}$) to the different of $\mathbb{Q}_p(\zeta_{p^2})/\mathbb{Q}_p$.

3.6 The Herbrand Functions and Upper Numbering

The lower-numbering filtration has one defect: it does not pass to quotients. If $H \trianglelefteq G$ and $K' = L^H$, the ramification groups of $\mathrm{Gal}_K(K') = G/H$ are *not* the images G_iH/H of the groups of G. The wild ramification of L/K' distorts the indices. For a subgroup the lower numbering behaves perfectly, but a quotient is exactly what one needs to descend from a large extension to a small one, and ultimately to make sense of ramification in an infinite abelian extension. Herbrand's theorem repairs this by reindexing the filtration through a piecewise-linear change of variable, the function $\varphi_{L/K}$. In the resulting *upper numbering* the ramification groups pass to quotients exactly. This is the numbering that class field theory attaches to characters, and in which the Hasse-Arf theorem asserts that the jumps of an abelian extension are integers.

Throughout, L/K is a finite Galois extension with group $G = \mathrm{Gal}_K(L)$, lower-numbering ramification groups G_i of orders $g_i = |G_i|$, and ν the normalized valuation on L. The lower numbering restricts to subgroups without change.

Proposition 3.6.1: Subgroups and the Lower Numbering

Let $H \leq G$ and $K' = L^H$, so $H = \mathrm{Gal}_{K'}(L)$. Then the ramification groups of L/K' are $H_u = H \cap G_u$ for all $u \geq -1$.

***Proof*:**
The extensions L/K' and L/K share the ring $\mathcal{O}_L$ and the valuation ν, and the defining condition of a ramification group refers only to these. Thus

$$H_u = \{s \in H \mid \nu(s(x) - x) \geq u + 1 \text{ for all } x \in \mathcal{O}_L\} = H \cap G_u$$

as we sought to show.

Definition 3.6.2: The Herbrand Function

The *Herbrand function* of L/K is $\varphi_{L/K} \colon [-1, \infty) \to [-1, \infty)$,

$$\varphi_{L/K}(u) = \int_0^u \frac{dt}{(G_0 : G_t)}$$

where $G_t = G_{\lceil t \rceil}$ for $t > 0$ and $G_t = G_0$ for $-1 \leq t \leq 0$. Its inverse is $\psi_{L/K} = \varphi_{L/K}^{-1}$.

Proposition 3.6.3: Properties of the Herbrand Function

The function $\varphi = \varphi_{L/K}$ is continuous, piecewise linear, strictly increasing and concave, with $\varphi(u) = u$ on $[-1, 0]$. On $(m-1, m]$ for an integer $m \geq 1$ its slope is g_m/g_0, so for integer $m \geq 0$

$$\varphi(m) = \frac{1}{g_0}\big(g_1 + g_2 + \cdots + g_m\big)$$

It is a homeomorphism of $[-1, \infty)$ onto itself, and $\psi = \varphi^{-1}$ is continuous, piecewise linear, strictly increasing, convex, with $\psi(v) = v$ on $[-1, 0]$.

***Proof*:**
On $(m-1, m]$ the integrand $1/(G_0 : G_t) = g_m/g_0$ is constant, so φ is continuous and piecewise linear with the stated slopes. The slopes are positive and nonincreasing since the g_m decrease, giving strict monotonicity and concavity. Integrating the constant slopes yields $\varphi(m) = \sum_{i=1}^{m} g_i/g_0$, and on $[-1, 0]$ the integrand is 1, so $\varphi(u) = u$ there. As φ is a strictly increasing continuous surjection onto $[-1, \infty)$ (its slope is at least $1/g_0 > 0$, so it is unbounded), it is a homeomorphism, and $\psi = \varphi^{-1}$ carries the reciprocal slopes, hence is convex and equals v on $[-1, 0]$, as we sought to show.

Definition 3.6.4: Ramification Groups in the Upper Numbering

For $v \geq -1$, the *upper-numbering ramification group* is $G^v = G_{\psi_{L/K}(v)}$. Equivalently, $G^{\varphi_{L/K}(u)} = G_u$ for all $u \geq -1$.

Since $\varphi(-1) = -1$ and $\varphi(0) = 0$, the two numberings agree at the bottom: $G^{-1} = G$ and $G^0 = G_0$ is the inertia group. They encode the same filtration through the substitution $v = \varphi(u)$. A *jump* of the upper numbering (a v at which $G^v \neq G^{v+\epsilon}$ for every $\epsilon > 0$) is the φ-image of a lower ramification break.

We now turn to the quotient. Fix $H \trianglelefteq G$, set $K' = L^H$ and $\bar{G} = G/H = \mathrm{Gal}_K(K')$, and write $H_i = H \cap G_i$ (proposition 3.6.1), $h_i = |H_i|$, and $e' = e_{L/K'} = h_0$. For $\bar{\sigma} \in \bar{G}$ put $i_{\bar{G}}(\bar{\sigma}) = \nu_{K'}(\bar{\sigma}(y') - y')$, computed with a generator $\mathcal{O}_{K'} = \mathcal{O}_K[y']$. The bridge between the two filtrations is the following identity, whose proof leans on the transitivity of the different.

Lemma 3.6.5: The Quotient Identity

For $\bar{\sigma} \in \bar{G}$ with $\bar{\sigma} \neq 1$, and any lift $\sigma \in G$,

$$e'\, i_{\bar{G}}(\bar{\sigma}) = \sum_{s \in \sigma H} i_G(s)$$

***Proof*:**
Step 1: the inequality. Let $\mathcal{O}_L = \mathcal{O}_K[x]$ and $\mathcal{O}_{K'} = \mathcal{O}_K[y']$. Since L/K' is Galois with group H, the minimal polynomial of x over K' is $h(T) = \prod_{\tau \in H}(T - \tau x) \in \mathcal{O}_{K'}[T]$. Applying σ to its

coefficients gives $\sigma h(T) = \prod_{\tau\in H}(T - \sigma\tau x)$, and since $h(x) = 0$,

$$\prod_{\tau\in H}(x - \sigma\tau x) = \sigma h(x) - h(x) = \sum_k (\sigma c_k - c_k)\, x^k$$

where $c_k \in \mathcal{O}_{K'}$ are the coefficients of h. Writing $c_k = P_k(y')$ with $P_k \in \mathcal{O}_K[T]$, each $\sigma c_k - c_k = P_k(\sigma y') - P_k(y')$ is divisible by $\sigma y' - y'$, so $\sigma y' - y'$ divides the whole sum. Hence

$$\sum_{s\in\sigma H} i_G(s) = \sum_{\tau\in H}\nu(x - \sigma\tau x) = \nu\Big(\prod_{\tau\in H}(x - \sigma\tau x)\Big) \geq \nu(\sigma y' - y') = e'\, i_{\bar{G}}(\bar{\sigma})$$

the last equality because $y' \in K'$ and $\nu|_{K'} = e'\,\nu_{K'}$.

Step 2: equality by summation. Sum the inequality over all $\bar{\sigma} \neq 1$. The left total is $\sum_{s\in G\setminus H} i_G(s)$. By the ramification formula for the different (theorem 3.5.3) and its transitivity (theorem 3.5.7), together with $\nu(\mathfrak{d}_{L/K'}) = \sum_{i\geq 0}(h_i - 1) = \sum_{s\in H\setminus 1} i_G(s)$ (from proposition 3.6.1),

$$\sum_{s\in G\setminus 1} i_G(s) = \nu(\mathfrak{d}_{L/K}) = \nu(\mathfrak{d}_{L/K'}) + e'\,\nu_{K'}(\mathfrak{d}_{K'/K}) = \sum_{s\in H\setminus 1} i_G(s) + e'\sum_{\bar{\sigma}\neq 1} i_{\bar{G}}(\bar{\sigma})$$

Subtracting the sum over $H\setminus 1$ leaves $\sum_{s\in G\setminus H} i_G(s) = e'\sum_{\bar{\sigma}\neq 1} i_{\bar{G}}(\bar{\sigma})$. Thus the left totals equal the right totals of Step 1 while each left term is at least its right term. Equality of the totals therefore forces equality term by term, which is the claim, completing the proof.

To read off the groups, the coset sum is evaluated in terms of the filtration of H.

Theorem 3.6.6: Herbrand's Theorem

Let $H \trianglelefteq G$ and $K' = L^H$. Then for all $u \geq -1$,

$$(G/H)_{\varphi_{L/K'}(u)} = G_u\, H/H$$

***Proof*:**
Fix $\bar{\sigma} \neq 1$ and choose a lift $\sigma_0 \in \sigma H$ with $m := i_G(\sigma_0) = \max_{s\in\sigma H} i_G(s)$. The subgroup property of the G_j gives $i_G(ab) \geq \min(i_G(a), i_G(b))$ for all a, b, with equality when $i_G(a) \neq i_G(b)$: if $\min = i_G(a) < i_G(b)$ and $i_G(ab) > i_G(a)$ then $a = (ab)b^{-1}$ would lie in $G_{i_G(a)}$, a contradiction. Hence for $\tau \in H$, $i_G(\sigma_0\tau) = \min(m, i_H(\tau))$: when $i_H(\tau) \neq m$ this is the equality case, and when $i_H(\tau) = m$ we get $i_G(\sigma_0\tau) \geq m$ while maximality of m over the coset forces $i_G(\sigma_0\tau) \leq m$. Here $i_H(\tau) = i_G(\tau)$ for $\tau \in H$, both equal to $1 + \max\{j \mid \tau \in H_j\}$ by proposition 3.6.1. As τ runs over H, $\sigma_0\tau$ runs over σH, so

$$\sum_{s\in\sigma H} i_G(s) = \sum_{\tau\in H}\min(m, i_H(\tau)) = \sum_{\tau\in H}\#\{j \mid 0 \leq j \leq m-1,\ \tau \in H_j\} = \sum_{j=0}^{m-1} h_j$$

using $\min(m, i_H(\tau)) = \#\{j \mid 0 \leq j \leq m-1,\ \tau \in H_j\}$ (the groups H_j decrease, so $\tau \in H_j$ for $j < i_H(\tau)$ and no further). By lemma 3.6.5 and $e' = h_0$, and since $\varphi_{L/K'}(m-1) = \sum_{j=1}^{m-1} h_j/h_0$,

$$i_{\bar{G}}(\bar{\sigma}) = \frac{1}{h_0}\sum_{j=0}^{m-1} h_j = 1 + \frac{1}{h_0}\sum_{j=1}^{m-1} h_j = 1 + \varphi_{L/K'}(m-1)$$

Thus $i_{\bar{G}}(\bar{\sigma}) - 1 = \varphi_{L/K'}(m-1)$. Now, for $v = \varphi_{L/K'}(u)$,

$$\bar{\sigma} \in (G/H)_v \iff i_{\bar{G}}(\bar{\sigma}) - 1 \geq v \iff \varphi_{L/K'}(m-1) \geq v \iff m - 1 \geq u$$

since $\varphi_{L/K'}$ is increasing with $\varphi_{L/K'}(u) = v$. The last condition says some $s \in \sigma H$ has $i_G(s) \geq u+1$, that is, σH meets G_u, equivalently $\bar{\sigma} \in G_u H/H$. Therefore $(G/H)_{\varphi_{L/K'}(u)} = G_u H/H$, completing the proof.

The same reindexing makes the Herbrand function transitive, and with it the upper numbering descends to quotients cleanly.

Theorem 3.6.7: Transitivity of the Herbrand Function

For a tower $L/K'/K$ with K'/K Galois,

$$\varphi_{L/K} = \varphi_{K'/K} \circ \varphi_{L/K'}, \qquad \psi_{L/K} = \psi_{L/K'} \circ \psi_{K'/K}$$

***Proof*:**
Both $\varphi_{L/K}$ and $\varphi_{K'/K} \circ \varphi_{L/K'}$ vanish at 0 and are continuous and piecewise linear, so it suffices to compare right derivatives at a noninteger $u > 0$. Write $\bar{G} = G/H$ and $w = \varphi_{L/K'}(u)$. By the chain rule and proposition 3.6.3,

$$\left(\varphi_{K'/K} \circ \varphi_{L/K'}\right)'(u) = \frac{1}{(\bar{G}_0 : \bar{G}_w)} \cdot \frac{1}{(H_0 : H_u)}$$

Herbrand's theorem (theorem 3.6.6) gives $\bar{G}_w = G_u H/H$, so $|\bar{G}_w| = |G_u|/|G_u \cap H| = g_u/h_u$ and, at $u = 0$, $|\bar{G}_0| = g_0/h_0$. Hence $(\bar{G}_0 : \bar{G}_w) = (g_0 h_u)/(h_0 g_u)$. With $(H_0 : H_u) = h_0/h_u$, the product is

$$\frac{h_0 g_u}{g_0 h_u} \cdot \frac{h_u}{h_0} = \frac{g_u}{g_0} = \frac{1}{(G_0 : G_u)} = \varphi'_{L/K}(u)$$

So the two functions have equal derivatives and agree at 0, hence coincide. Inverting gives the identity for ψ, completing the proof.

Corollary 3.6.8: Herbrand's Theorem in the Upper Numbering

For $H \trianglelefteq G$ and $K' = L^H$, $(G/H)^v = G^v H/H$ for all $v \geq -1$.

***Proof*:**
By definition $(G/H)^v = (G/H)_{\psi_{K'/K}(v)}$. Taking $u = \psi_{L/K}(v)$ and using transitivity (theorem 3.6.7) gives $\varphi_{L/K'}(u) = \varphi_{L/K'}(\psi_{L/K'}(\psi_{K'/K}(v))) = \psi_{K'/K}(v)$, so Herbrand's theorem (theorem 3.6.6) yields $(G/H)_{\psi_{K'/K}(v)} = G_u H/H = G_{\psi_{L/K}(v)} H/H = G^v H/H$, as we sought to show.

The upper numbering is the natural home for one of the deepest facts about abelian ramification. Its jumps, a priori rational numbers, are in fact integers.

Theorem 3.6.9: Hasse-Arf

Let L/K be a finite *abelian* extension of local fields. Then every jump of the upper-numbering filtration (G^v) is an integer: if $G^v \neq G^{v+\epsilon}$ for all $\epsilon > 0$, then $v \in \mathbb{Z}$.

The proof, a reduction to totally ramified cyclic extensions of p-power degree followed by an analysis of how a generator moves a uniformizer, is given in full in appendix A. The integrality is what allows the upper numbering to be transported to a character of G: the conductor of an abelian character, defined as the least integer $v+1$ with G^v acting trivially, is well defined precisely because the jumps are integral, and it is this conductor that indexes the local factors of an L-function. Without Hasse-Arf the upper-numbering jumps could be fractional and the notion of conductor would collapse.

Example 3.6.10: Upper Numbering of the Cyclotomic Tower

For $L = \mathbb{Q}_p(\zeta_{p^n})/\mathbb{Q}_p$ the lower breaks lie at $i = p^k - 1$ for $k = 0, 1, \dots, n-1$ (example 3.4.6), where $G_i = \{\sigma_a \mid a \equiv 1 \bmod p^k\}$ has order $g_i = p^{n-k}$. Grouping the sum defining φ by the value of g_i,

$$\varphi(p^k - 1) = \frac{1}{g_0}\sum_{i=1}^{p^k-1} g_i = \frac{1}{p^{n-1}(p-1)}\sum_{k'=1}^{k}\left(p^{k'} - p^{k'-1}\right)p^{n-k'} = \frac{1}{p^{n-1}(p-1)}\sum_{k'=1}^{k}(p-1)p^{n-1} = k$$

So the upper breaks fall at the integers $v = 0, 1, \dots, n-1$, and

$$G^v = \left\{\sigma_a \;\middle|\; a \equiv 1 \bmod p^{\lceil v \rceil}\right\} \qquad (0 \le v \le n-1)$$

with $G^v = 1$ for $v > n-1$. The integrality of these breaks is Hasse-Arf made explicit, the extension being abelian.

Exercise 3.6.1

1. Compute $\varphi_{L/\mathbb{Q}_2}$ for $L = \mathbb{Q}_2(\sqrt{2})$ and determine the upper-numbering filtration. Verify that the unique jump is at an integer, as Hasse-Arf requires for this abelian extension.
2. Show that $G^{-1} = G$ and $G^0 = G_0$ directly from the definitions, and that for a totally *tamely* ramified extension of degree e one has $\varphi_{L/K}(u) = u/e$ for $u \ge 0$ and a single upper break at $v = 0$.
3. For the tower $\mathbb{Q}_p(\zeta_{p^2})/\mathbb{Q}_p(\zeta_p)/\mathbb{Q}_p$, compute $\varphi_{L/K'}$, $\varphi_{K'/K}$, and $\varphi_{L/K}$ explicitly and verify the transitivity $\varphi_{L/K} = \varphi_{K'/K} \circ \varphi_{L/K'}$.
4. Deduce from corollary 3.6.8 that if $G^v = 1$ then $(G/H)^v = 1$ for every $H \trianglelefteq G$. Interpret this as "the conductor can only drop in a quotient."

3.7 The Hilbert Symbol

The different and the ramification groups measure how an extension sits over its base. The Hilbert symbol measures something dual: whether an element is a norm. It packages the question "is a a norm from $K(\sqrt{b})$?" into a pairing (a, b) taking values ± 1, and the payoff is that this pairing is

bilinear, so norm conditions can be manipulated by linear algebra over $\mathbb{F}_2$. The symbol is the local atom of quadratic reciprocity, the invariant that classifies quaternion algebras over K, and the object that [7] reads from here to compute local Brauer groups. This section develops the quadratic symbol from scratch. The n-th symbol for general n, defined through the norm-residue map of local class field theory, and the explicit reciprocity laws belong to the sequel [10].

Throughout, K is a local field of characteristic $\neq 2$, with valuation ring $\mathcal{O}$, maximal ideal $\mathfrak{p}$, uniformizer π, and residue field $\mathbb{F}_q$.

Definition 3.7.1: The Hilbert Symbol

For $a, b \in K^\times$, the *Hilbert symbol* is

$$(a,b) = \begin{cases} 1 & \text{if } ax^2 + by^2 = z^2 \text{ has a solution } (x,y,z) \neq (0,0,0) \text{ in } K, \\ -1 & \text{otherwise.} \end{cases}$$

The symbol is unchanged if a or b is multiplied by a square, since substituting $x \mapsto cx$ turns ac^2x^2 back into ax^2; thus (a,b) depends only on the classes of a, b in $K^\times/(K^\times)^2$. It is also symmetric, $(a,b) = (b,a)$, by exchanging the roles of x and y. The elementary identities below hold over any field of characteristic $\neq 2$.

Proposition 3.7.2: Elementary Identities

For all $a, b \in K^\times$:

1. $(a,b) = 1$ if a or b is a square;
2. $(a,-a) = 1$ and $(a, 1-a) = 1$ (the latter when $a \neq 1$);
3. $(a,b) = 1$ if and only if a is a norm from $K(\sqrt{b})$, when b is not a square.

***Proof*:**

1. If $b = c^2$, then $(x,y,z) = (0,1,c)$ solves $ax^2 + by^2 = z^2$; symmetrically for a.
2. The projective point $(1,1,0)$ solves $ax^2 - ay^2 = z^2$, and $(1,1,1)$ solves $ax^2 + (1-a)y^2 = z^2$, since $a + (1-a) = 1$.
3. Suppose b is not a square, so $K(\sqrt{b})/K$ is a quadratic extension with norm $N(x + y\sqrt{b}) = x^2 - by^2$. If $(a,b) = 1$, take a nonzero solution of $as^2 + bt^2 = w^2$; the case $s = 0$ would force $b = (w/t)^2$ a square, so $s \neq 0$, and dividing by s^2 gives $a = (w/s)^2 - b(t/s)^2 = N\big((w/s) + (t/s)\sqrt{b}\big)$. Conversely, if $a = x^2 - by^2$ is a norm, then $(x,y,1)$ solves $z^2 = a \cdot 1 + by^2$ after rearranging $x^2 = a + by^2$, so $(1,y,x)$ is a nonzero solution and $(a,b) = 1$.

This exhausts the cases, as we sought to show.

The symbol can be evaluated in general because $K^\times$ modulo squares is small. Because K has odd residue characteristic, Hensel's lemma makes every principal unit a square: for $u \in 1 + \mathfrak{p}$, the polynomial $X^2 - u$ reduces to $X^2 - 1 = (X-1)(X+1)$ with simple roots, which lift. Combined

with the decomposition $K^\times = \pi^{\mathbb{Z}} \times \mu_{q-1} \times (1+\mathfrak{p})$ of proposition 1.3.5, this gives the following.

Lemma 3.7.3: Squares in a Local Field

If K has odd residue characteristic, then $K^\times/(K^\times)^2$ is a group of order 4, represented by $\{1, u_0, \pi, u_0\pi\}$ where $u_0 \in \mu_{q-1}$ is any non-square unit. The class of a unit u is trivial if and only if its residue $\bar{u}$ is a square in $\mathbb{F}_q^\times$.

***Proof*:**
By proposition 1.3.5 every element is $\pi^\alpha \zeta u$ with $\zeta \in \mu_{q-1}$ and $u \in 1+\mathfrak{p}$; the principal unit u is a square by the Hensel argument above, and π^α contributes α mod 2. The cyclic group μ_{q-1} has even order $q-1$, so its squares form the unique subgroup of index 2, and ζ is a square iff $\bar{\zeta}$ is a square in $\mathbb{F}_q^\times$. Hence $K^\times/(K^\times)^2$ is generated by the classes of π and of a non-square unit u_0, each of order 2, giving a group of order 4, as we sought to show.

The values on units and uniformizers now determine everything. Write $\left(\frac{u}{\mathfrak{p}}\right) = 1$ if $\bar{u}$ is a square in $\mathbb{F}_q^\times$ and -1 otherwise, the quadratic residue symbol of the residue field.

Theorem 3.7.4: The Tame Hilbert Symbol

Let K have odd residue characteristic. For units $u, w \in \mathcal{O}^\times$,

$$(u,w) = 1, \qquad (\pi, u) = \left(\frac{u}{\mathfrak{p}}\right), \qquad (\pi,\pi) = \left(\frac{-1}{\mathfrak{p}}\right)$$

Consequently, writing $a = \pi^\alpha u$ and $b = \pi^\beta w$ with $u, w \in \mathcal{O}^\times$,

$$(a,b) = \left(\frac{-1}{\mathfrak{p}}\right)^{\alpha\beta} \left(\frac{u}{\mathfrak{p}}\right)^{\beta} \left(\frac{w}{\mathfrak{p}}\right)^{\alpha}$$

***Proof*:**
Units. Reducing $uX^2 + wY^2 = Z^2$ modulo $\mathfrak{p}$ gives a nondegenerate conic over $\mathbb{F}_q$; as the two sets $\{\bar{u}\,x^2 \mid x \in \mathbb{F}_q\}$ and $\{\bar{z}^2 - \bar{w}\,y^2 \mid y, z \in \mathbb{F}_q\}$ each have more than $q/2$ elements, they meet, yielding a nonzero $(\bar{x}, \bar{y}, \bar{z})$ with $\bar{u}\bar{x}^2 + \bar{w}\bar{y}^2 = \bar{z}^2$ and, after relabeling, $\bar{z} \neq 0$. Then $f(Z) = ux^2 + wy^2 - Z^2$ has $f(z) \equiv 0$ and $f'(z) = -2z \not\equiv 0 \bmod \mathfrak{p}$, so Hensel lifts z to a root in $\mathcal{O}$; hence $(u,w) = 1$.

Uniformizer against a unit. If $\bar{u}$ is a square then u is a square (lemma 3.7.3) and $(\pi, u) = 1$ by proposition 3.7.2. If $\bar{u}$ is not a square, suppose $\pi x^2 + uy^2 = z^2$ had a nonzero solution, scaled so that $\min(\nu(x), \nu(y), \nu(z)) = 0$. Reducing modulo $\mathfrak{p}$ kills πx^2 (valuation ≥ 1) and leaves $\bar{z}^2 = \bar{u}\,\bar{y}^2$; were $\bar{y} \neq 0$ this would make $\bar{u}$ a square, so $\bar{y} = 0$ and then $\bar{z} = 0$, forcing $\nu(y), \nu(z) \geq 1$. But then $\nu(\pi x^2) = \nu(z^2 - uy^2) \geq 2$, so $\nu(x) \geq 1$, contradicting the normalization. Thus $(\pi, u) = -1 = \left(\frac{u}{\mathfrak{p}}\right)$.

Uniformizer against itself. By proposition 3.7.2, $(\pi, -\pi) = 1$, so multiplying $b = \pi$ by the square-free part reveals $(\pi,\pi) = (\pi,-\pi)\,(\pi,-1) = (\pi,-1) = \left(\frac{-1}{\mathfrak{p}}\right)$ by the previous case applied

to the unit -1.

The formula. Both sides depend only on a, b modulo squares, so by lemma 3.7.3 it suffices to check the four generators, which is exactly the three values above (with $(u, w) = 1$ handling $\alpha = \beta = 0$). Since the right-hand side is multiplicative in each of (α, u) and (β, w) and agrees with (a, b) on generators, it equals (a, b), completing the proof.

Corollary 3.7.5: Bilinearity and Nondegeneracy

For K of odd residue characteristic, the Hilbert symbol is a nondegenerate symmetric bilinear pairing

$$K^\times/(K^\times)^2 \times K^\times/(K^\times)^2 \to \{\pm 1\}$$

That is, $(aa', b) = (a, b)(a', b)$ for all a, a', b, and for each $a \notin (K^\times)^2$ there is a b with $(a, b) = -1$.

***Proof*:**
Bilinearity is visible from the explicit formula of theorem 3.7.4, whose right-hand side is a product of the multiplicative residue symbols and the exponents α, β combined additively. For nondegeneracy, view $K^\times/(K^\times)^2$ as an $\mathbb{F}_2$-vector space with basis the classes of π and u_0, and write the symbol additively by sending $1 \mapsto 0$ and $-1 \mapsto 1$ in $\mathbb{F}_2$. Then the pairing has Gram matrix $\left(\begin{smallmatrix} \epsilon & 1 \\ 1 & 0 \end{smallmatrix}\right)$, where $\epsilon = 0$ if $q \equiv 1 \bmod 4$ and $\epsilon = 1$ if $q \equiv 3 \bmod 4$, coming from $(\pi, \pi) = \left(\frac{-1}{\mathfrak{p}}\right)$, $(\pi, u_0) = (u_0, \pi) = -1$, and $(u_0, u_0) = 1$. This matrix has determinant $-1 = 1 \neq 0$ in $\mathbb{F}_2$, so the form is nondegenerate; in particular every nonzero class pairs nontrivially with some class, completing the proof.

3.7.6 The Dyadic Case and the n-th Symbol For $K = \mathbb{Q}_2$ the group $K^\times/(K^\times)^2$ has order 8, with representatives $\pm 1, \pm 5, \pm 2, \pm 10$ [14, Chapter II, corollary to Theorem 4], against 4 in the tame case (lemma 3.7.3). The explicit formula for (a, b) over $\mathbb{Q}_2$, proved in lemma 3.7.7 as in [14, Chapter III, Theorem 1], involves, besides the valuations of a and b, the classes modulo 2 of $\frac{u-1}{2}$ and $\frac{u^2-1}{8}$ for their unit parts u. The symbol of $\mathbb{Q}_2$ is still a nondegenerate symmetric bilinear pairing [14, Chapter III, Proposition 2 and Theorem 2]. For $n > 2$, and for a full account of the norm-residue symbol $(a, b) = \sigma_a(\sqrt[n]{b})/\sqrt[n]{b}$ defined through the local reciprocity map, see [10].

Lemma 3.7.7: The Dyadic Hilbert Symbol

Let $K = \mathbb{Q}_2$, and for $u \in \mathbb{Z}_2^\times$ let $\varepsilon(u)$ and $\omega(u)$ be the classes modulo 2 of $\frac{u-1}{2}$ and $\frac{u^2-1}{8}$.

1. A unit $u \in \mathbb{Z}_2^\times$ is a square in $\mathbb{Q}_2$ if and only if $u \equiv 1 \bmod 8$.
2. The maps $\varepsilon, \omega \colon \mathbb{Z}_2^\times \to \mathbb{Z}/2\mathbb{Z}$ are well defined homomorphisms.
3. If $a = 2^\alpha u$ and $b = 2^\beta w$ with $u, w \in \mathbb{Z}_2^\times$, then
$$(a,b) = (-1)^{\varepsilon(u)\varepsilon(w)+\alpha\omega(w)+\beta\omega(u)}$$
In particular, the symbol of $\mathbb{Q}_2$ is bilinear.

***Proof*:**
A unit u of $\mathbb{Z}_2$ is $u = 1 + 2t$ with $t \in \mathbb{Z}_2$, and $u^2 = 1 + 4t(t+1) \equiv 1 \bmod 8$, as t or $t+1$ lies in $2\mathbb{Z}_2$. So for $x \in \mathbb{Z}_2$, $x^2 \equiv 1 \bmod 8$ if x is a unit, $x^2 \equiv 4 \bmod 8$ if $\nu_2(x) = 1$, and $x^2 \equiv 0 \bmod 8$ if $\nu_2(x) \geq 2$.

1. If $u = c^2$ with $c \in \mathbb{Q}_2$, then $\nu_2(c) = 0$, so $u \equiv 1 \bmod 8$. Conversely, let $u \equiv 1 \bmod 8$, $f(X) = X^2 - u$ and $a_0 = 1$. Then $|f(a_0)| = |1-u| \leq |8| < |4| = |f'(a_0)|^2$, so corollary 2.1.4 gives a root of f in $\mathbb{Z}_2$.
2. With $u = 1 + 2t$ as above, $\frac{u-1}{2} = t$ and $\frac{u^2-1}{8} = \frac{t(t+1)}{2}$ lie in $\mathbb{Z}_2$. For units u and u',
$$\frac{uu'-1}{2} - \frac{u-1}{2} - \frac{u'-1}{2} = \frac{(u-1)(u'-1)}{2}$$
and
$$\frac{u^2u'^2-1}{8} - \frac{u^2-1}{8} - \frac{u'^2-1}{8} = \frac{(u^2-1)(u'^2-1)}{8}$$
where both right-hand sides lie in $2\mathbb{Z}_2$, as $u-1, u'-1 \in 2\mathbb{Z}_2$ and $u^2-1, u'^2-1 \in 8\mathbb{Z}_2$. So ε and ω are homomorphisms.
3. *Reductions.* Replacing a by ac^2 and x by x/c shows that (a,b) depends only on the classes of a and b modulo squares, and exchanging x and y shows $(a,b) = (b,a)$. The right-hand side has the same two properties: it is symmetric, and multiplying a by a square of $\mathbb{Q}_2^\times$, which is $4^k c^2$ with c a unit, changes α by $2k$ and, by (2), leaves $\varepsilon(u)$ and $\omega(u)$ unchanged. If units u and u' are congruent modulo 8, then $u/u' \equiv 1 \bmod 8$ is a square by (1). So it suffices to treat $\alpha, \beta \in \{0,1\}$, with u and w given modulo 8, and by symmetry the three cases below.

   (N) If $(c,b) = 1$, then $(cc', b) = (c', b)$ for every $c' \in \mathbb{Q}_2^\times$.

   Indeed, by proposition 3.7.2(1), if b is a square, both sides are 1. Otherwise, by proposition 3.7.2(3), $(c,b) = 1$ says that c lies in the group of norms from $\mathbb{Q}_2(\sqrt{b})^\times$, which then contains cc' exactly when it contains c'. Applying (N) to $(-a,a) = 1$ (proposition 3.7.2(2) and symmetry) gives $(-ab,a) = (b,a)$, that is, $(a,-ab) = (a,b)$.

   Case $\alpha = \beta = 0$. The claim is that $(u,w) = -1$ exactly when $u \equiv w \equiv 3 \bmod 4$. If

$u \equiv 1 \bmod 8$, then u is a square and $(u, w) = 1$ by proposition 3.7.2. If $u \equiv 5 \bmod 8$, then $u + 4w \equiv 1 \bmod 8$ is a square c^2 by (1), and $(x, y, z) = (1, 2, c)$ solves $ux^2 + wy^2 = z^2$, so $(u, w) = 1$. By symmetry, $(u, w) = 1$ whenever u or w is congruent to 1 modulo 4. Let $u \equiv w \equiv 3 \bmod 4$, and suppose $ux^2 + wy^2 = z^2$ with $x, y, z \in \mathbb{Z}_2$ not all in $2\mathbb{Z}_2$, which any nonzero solution becomes after scaling. Then $x^2 + y^2 + z^2 \equiv 0 \bmod 4$, and as squares are 0 or 1 modulo 4, all three lie in $2\mathbb{Z}_2$, a contradiction. So $(u, w) = -1$.

Case $\alpha = 1$, $\beta = 0$. First, $(2, w) = (-1)^{\omega(w)}$, that is, $(2, w) = 1$ exactly when $w \equiv \pm 1 \bmod 8$. If $w \equiv 1 \bmod 8$, then w is a square. If $w \equiv -1 \bmod 8$, then $-w$ is a square, so $(2, w) = (2, -1)$, and $(x, y, z) = (1, 1, 1)$ solves $2x^2 - y^2 = z^2$. So $(2, w) = 1$ in both cases. Let $w \equiv \pm 3 \bmod 8$, and suppose $2x^2 + wy^2 = z^2$ with $x, y, z \in \mathbb{Z}_2$ not all in $2\mathbb{Z}_2$. Then $y^2 \equiv z^2 \bmod 2$. If $y, z \in 2\mathbb{Z}_2$, then $2x^2 = z^2 - wy^2 \in 4\mathbb{Z}_2$ forces $x \in 2\mathbb{Z}_2$, which is excluded. So y and z are units, and $2x^2 \equiv 1 - w \bmod 8$. But $1 - w$ is 4 or 6 modulo 8, while $2x^2$ is 0 or 2 modulo 8, a contradiction. So $(2, w) = -1$.

Now if $(2, w) = 1$, then (N) gives $(2u, w) = (u, w)$, and if $(u, w) = 1$, then (N) gives $(2u, w) = (2, w)$. In both cases the formula follows from the previous case and the value of $(2, w)$. Otherwise $(2, w) = (u, w) = -1$, so $w \equiv 3 \bmod 8$ and $u \equiv 3$ or $7 \bmod 8$, and the formula asserts $(2u, w) = 1$. Up to squares, (u, w) is then $(-1, 3)$ or $(3, -5)$ (as $7 \equiv -1$ and $3 \equiv -5 \bmod 8$), and in both cases $(x, y, z) = (1, 1, 1)$ solves $2ux^2 + wy^2 = z^2$.

Case $\alpha = \beta = 1$. By $(a, -ab) = (a, b)$, $(2u, 2w) = (2u, -4uw) = (2u, -uw)$. The previous case, applied to the unit $-uw$, gives the exponent $\varepsilon(u)\varepsilon(-uw) + \omega(-uw)$. By (2), with $\varepsilon(-1) = 1$ and $\omega(-1) = 0$, this is $\varepsilon(u)\big(1 + \varepsilon(u) + \varepsilon(w)\big) + \omega(u) + \omega(w) = \varepsilon(u)\varepsilon(w) + \omega(u) + \omega(w)$, as $\varepsilon(u)\big(1 + \varepsilon(u)\big) = 0$ in $\mathbb{Z}/2\mathbb{Z}$.

Bilinearity. As $\nu_2(aa') = \nu_2(a) + \nu_2(a')$ and ε and ω are homomorphisms, the exponent is additive in a, and by symmetry it is additive in b.

This completes the proof.

Proposition 3.7.8: The Product Formula

For $a, b \in \mathbb{Q}^\times$, the local symbols $(a, b)_v$, one for each prime v (the symbol of $\mathbb{Q}_v$) and one for $v = \infty$ (the formula of definition 3.7.1 read over $\mathbb{R}$), satisfy $(a, b)_v = 1$ for all but finitely many v, and

$$\prod_v (a, b)_v = 1$$

***Proof*:**

Replacing a by ac^2 and x by x/c shows that each symbol depends only on the classes of a and b modulo squares, and exchanging x and y shows $(a, b)_v = (b, a)_v$. Both arguments work over $\mathbb{Q}_v$ and over $\mathbb{R}$. So we may take $a, b \in \mathbb{Z}$.

Step 1: the real symbol. If $a > 0$, then $(x, y, z) = (1, 0, \sqrt{a})$ solves $ax^2 + by^2 = z^2$ in $\mathbb{R}$, and if

$b > 0$, then $(0, 1, \sqrt{b})$ does. If $a < 0$ and $b < 0$, then $ax^2 + by^2 \leq 0 \leq z^2$, with equality only for $x = y = z = 0$. So $(a, b)_\infty = (-1)^{s(a)s(b)}$, where $s(a) = 0$ for $a > 0$ and $s(a) = 1$ for $a < 0$. As $s(aa') \equiv s(a) + s(a') \bmod 2$, the real symbol is bilinear.

Step 2: the dyadic symbol. By lemma 3.7.7, with ε and ω as there, if $a = 2^\alpha u$ and $b = 2^\beta w$ with $u, w \in \mathbb{Z}_2^\times$, then

$$(a, b)_2 = (-1)^{\varepsilon(u)\varepsilon(w)+\alpha\omega(w)+\beta\omega(u)}$$

and the dyadic symbol is bilinear.

Step 3: finiteness. For an odd prime $p \nmid ab$, both a and b are units of $\mathbb{Z}_p$, so $(a, b)_p = 1$ by theorem 3.7.4. Only finitely many primes divide $2ab$, so $(a, b)_v = 1$ for all but finitely many v.

Step 4: reduction to generators. By corollary 3.7.5 at the odd primes, and by Steps 1 and 2 at ∞ and 2, every symbol $(a, b)_v$ is bilinear. Write $a = \prod_g g^{m_g}$ and $b = \prod_h h^{n_h}$ with integers $m_g, n_h \geq 0$, where g and h run through -1 and the primes dividing ab. Then $(a, b)_v = \prod_{g,h} (g, h)_v^{m_g n_h}$ for every v. For each of the finitely many pairs, $(g, h)_v = 1$ for all but finitely many v, by Step 3, so

$$\prod_v (a, b)_v = \prod_{g,h} \Big(\prod_v (g, h)_v \Big)^{m_g n_h}$$

It therefore suffices to prove $\prod_v (g, h)_v = 1$ when each of g, h is -1 or a prime. By symmetry this leaves the seven pairs below, in which $p \neq q$ are odd primes.

Step 5: the generator pairs. By Step 1, $(g, h)_\infty = 1$ unless $g = h = -1$. At an odd prime ℓ, the residue symbol of theorem 3.7.4 for $K = \mathbb{Q}_\ell$, with $\pi = \ell$ and residue field $\mathbb{F}_\ell$, is the Legendre symbol, so for an integer w prime to ℓ the theorem gives $(\ell, w)_\ell = \left(\frac{w}{\ell}\right)$ and $(\ell, \ell)_\ell = \left(\frac{-1}{\ell}\right)$, while $(g, h)_\ell = 1$ when $\ell \nmid gh$, by Step 3. In Step 2, the elements -1, 2 and p have $(\alpha, u) = (0, -1)$, $(1, 1)$ and $(0, p)$, with $\varepsilon(1) = \omega(1) = \omega(-1) = 0$, $\varepsilon(-1) = 1$, and $\varepsilon(p)$, $\omega(p)$ the parities of $\frac{p-1}{2}$ and $\frac{p^2-1}{8}$.

- $(-1, -1)$: $(-1, -1)_\infty = -1$ and $(-1, -1)_2 = (-1)^{\varepsilon(-1)^2} = -1$, with product 1.
- $(-1, 2)$: $(-1, 2)_2 = (-1)^{\varepsilon(-1)\varepsilon(1)+\omega(-1)} = 1$.
- $(2, 2)$: $(2, 2)_2 = (-1)^{\varepsilon(1)^2+2\omega(1)} = 1$.
- $(-1, p)$: $(-1, p)_2 = (-1)^{\varepsilon(-1)\varepsilon(p)} = (-1)^{\frac{p-1}{2}}$ and $(-1, p)_p = \left(\frac{-1}{p}\right)$, with product 1 by [2, First Supplement To Quadratic Reciprocity].
- (p, p): $(p, p)_2 = (-1)^{\varepsilon(p)^2} = (-1)^{\frac{p-1}{2}}$ and $(p, p)_p = \left(\frac{-1}{p}\right)$, with product 1 by the same supplement.
- $(2, p)$: $(2, p)_2 = (-1)^{\omega(p)} = (-1)^{\frac{p^2-1}{8}}$ and $(2, p)_p = \left(\frac{2}{p}\right)$, with product 1 by [2, Second Supplement To Quadratic Reciprocity].

- (p,q): $(p,q)_2 = (-1)^{\varepsilon(p)\varepsilon(q)} = (-1)^{\frac{p-1}{2}\frac{q-1}{2}}$, $(p,q)_p = \left(\frac{q}{p}\right)$ and $(p,q)_q = \left(\frac{p}{q}\right)$, with product 1 by [2, Quadratic Reciprocity].

In each case every other symbol is 1, so $\prod_v (g,h)_v = 1$, completing the proof.

Example 3.7.9: Hilbert Symbols over $\mathbb{Q}_p$

Over $\mathbb{Q}_p$ with p odd, $(p,p) = \left(\frac{-1}{p}\right)$ by theorem 3.7.4, and $\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}$ by [2, First Supplement To Quadratic Reciprocity], so $(p,p) = 1$ for $p \equiv 1 \bmod 4$ and -1 for $p \equiv 3 \bmod 4$. For a unit u, $(p,u) = 1$ exactly when u is a quadratic residue mod p. Thus $(p,u) = -1$ detects that $\mathbb{Q}_p(\sqrt{u})$ is the unramified quadratic extension while $\mathbb{Q}_p(\sqrt{p})$ is ramified. Two units always pair trivially: $(u,w) = 1$, reflecting that a conic in two unit coefficients already has a smooth point over $\mathbb{F}_p$.

Exercise 3.7.1

1. Compute all Hilbert symbols (a,b) over $\mathbb{Q}_5$ for $a,b \in \{2,5,10\}$, using $\left(\frac{2}{5}\right) = -1$.
2. Using proposition 3.7.2 and the fact that the norms from a quadratic extension form a group, show that $(a,-ab) = (a,b)$ for all $a,b \in K^\times$.
3. Prove that for K of odd residue characteristic, $(a,b) = 1$ for all $b \in K^\times$ if and only if a is a square, giving a second proof of nondegeneracy.
4. Let u be a unit of K (odd residue characteristic). Show that $K(\sqrt{u})/K$ is unramified and that $(\pi,u) = -1$ if and only if this extension is nontrivial, linking the symbol to the ramification of the quadratic extension it cuts out.

3.8 *Witt Vectors and the Structure of Complete DVRs

The local fields of characteristic 0 are the finite extensions of $\mathbb{Q}_p$, and those of characteristic p are the fields $\mathbb{F}_q((t))$ (proposition 1.3.2). More generally, a complete DVR of characteristic p with perfect residue field k is isomorphic to $k[[t]]$. In mixed characteristic $(0,p)$ the role of $k[[t]]$ is played by the ring of *Witt vectors* $W(k)$, built from k by universal polynomial formulas: every complete DVR of characteristic 0 with residue field k is a totally ramified extension of $W(k)$ (theorem 3.8.4). For finite k this recovers $\mathbb{Z}_p$ and the valuation rings of the unramified extensions of $\mathbb{Q}_p$. Nothing later in the book depends on this section.

Fix a prime p. For $n \geq 0$ the *Witt polynomial* is

$$w_n(X_0,\dots,X_n) = \sum_{i=0}^{n} p^i X_i^{p^{n-i}} = X_0^{p^n} + pX_1^{p^{n-1}} + \cdots + p^n X_n$$

Given a commutative ring A, the *ghost map* $w\colon A^{\mathbb{N}} \to A^{\mathbb{N}}$ sends $(x_0,x_1,\dots)$ to its *ghost components* $(w_0(x),w_1(x),\dots)$. The idea is to make $A^{\mathbb{N}}$ a ring, the ring $W(A)$ of Witt vectors, so that the ghost map becomes a ring homomorphism to the product ring $A^{\mathbb{N}}$. Over a $\mathbb{Z}[1/p]$-algebra the ghost map is bijective, so this forces the operations, and the theorem is that they have integer coefficients. Over a

ring in which p is a nonzerodivisor the ghost map is injective, since $w_n(x) = \sum_{i<n} p^i x_i^{p^{n-i}} + p^n x_n$ determines x_n from $w_n(x)$ and $x_0, \dots, x_{n-1}$.

Lemma 3.8.1: Dwork's Congruence Lemma

Let A be a ring in which p is a nonzerodivisor, equipped with a ring endomorphism φ satisfying $\varphi(a) \equiv a^p \pmod{pA}$ for all $a \in A$. A sequence $(w_0, w_1, \dots) \in A^{\mathbb{N}}$ is the sequence of ghost components of some (necessarily unique) $x \in A^{\mathbb{N}}$ if and only if

$$w_n \equiv \varphi(w_{n-1}) \pmod{p^n A} \qquad (n \geq 1)$$

***Proof*:**

First note the elementary fact that $a \equiv b \pmod{p^m}$ with $m \geq 1$ implies $a^p \equiv b^p \pmod{p^{m+1}}$: writing $a = b + p^m c$, the binomial expansion $a^p = b^p + p\, b^{p-1}(p^m c) + \cdots$ has every term after b^p divisible by p^{m+1}. Iterating, $a \equiv b \pmod{p}$ gives $a^{p^k} \equiv b^{p^k} \pmod{p^{k+1}}$.

($\Rightarrow$) If $w_n = w_n(x)$ then, since $w_n(x) = \sum_{i=0}^{n-1} p^i x_i^{p^{n-i}} + p^n x_n$ and $\varphi(w_{n-1}(x)) = \sum_{i=0}^{n-1} p^i \varphi(x_i)^{p^{n-1-i}}$, the difference is $w_n - \varphi(w_{n-1}) = p^n x_n + \sum_{i=0}^{n-1} p^i \big(x_i^{p^{n-i}} - \varphi(x_i)^{p^{n-1-i}}\big)$. Each bracket vanishes modulo p^{n-i} by the elementary fact (as $\varphi(x_i) \equiv x_i^p \bmod p$ raised to p^{n-1-i}), so each term is divisible by p^n. Hence $w_n \equiv \varphi(w_{n-1}) \pmod{p^n}$.

($\Leftarrow$) Conversely, construct x_n recursively so that $w_n(x) = w_n$. Take $x_0 = w_0$. Given $x_0, \dots, x_{n-1}$ with $w_m(x) = w_m$ for $m < n$, solving $w_n(x) = w_n$ requires $p^n x_n = w_n - \sum_{i<n} p^i x_i^{p^{n-i}}$. By the computation above $\sum_{i<n} p^i x_i^{p^{n-i}} \equiv \varphi(w_{n-1}(x)) = \varphi(w_{n-1}) \pmod{p^n}$, and the hypothesis gives $w_n \equiv \varphi(w_{n-1}) \pmod{p^n}$, so the right-hand side lies in $p^n A$ and $x_n \in A$ exists, uniquely since p is a nonzerodivisor. This determines x, completing the proof.

Theorem 3.8.2: The Ring of Witt Vectors

There are unique polynomials $S_n, P_n \in \mathbb{Z}[X_0, \dots, X_n, Y_0, \dots, Y_n]$ with $w_n(S) = w_n(X) + w_n(Y)$ and $w_n(P) = w_n(X)\, w_n(Y)$. For every commutative ring A, the operations $(x) + (y) = (S_n(x, y))_n$ and $(x) \cdot (y) = (P_n(x, y))_n$ make $A^{\mathbb{N}}$ a commutative ring $W(A)$, functorial in A, whose zero and one are $(0, 0, \dots)$ and $(1, 0, 0, \dots)$, and for which the ghost map $w \colon W(A) \to A^{\mathbb{N}}$ is a ring homomorphism.

***Proof*:**

Work in the p-torsion-free ring $A_0 = \mathbb{Z}[X_i, Y_i]$ with the endomorphism φ raising every variable to its p-th power, so $\varphi(a) \equiv a^p \bmod p$ for every $a \in A_0$. By lemma 3.8.1, applied to the vectors X and Y, each of $w_n(X)$ and $w_n(Y)$ is congruent to the image under φ of its predecessor modulo p^n. As φ is a ring map, the sequences $\big(w_n(X) + w_n(Y)\big)_n$ and $\big(w_n(X)\, w_n(Y)\big)_n$ satisfy the same congruences. By lemma 3.8.1 each is the ghost sequence of a unique vector $S = (S_n)$, respectively $P = (P_n)$, with entries in A_0, and the recursion in the proof of the lemma shows that S_n and P_n involve only $X_0, \dots, X_n, Y_0, \dots, Y_n$. In the same way the sequence $(-w_n(X))_n$ is the ghost sequence of a vector of integer polynomials, which gives additive inverses.

Each ring axiom, for instance associativity of the addition, is an equality of Witt vectors

whose components are integer polynomials in the components of at most three vectors X, Y, Z. In the p-torsion-free ring $\mathbb{Z}[X_i, Y_i, Z_i]$ the ghost map is injective, and it carries the Witt operations to the componentwise operations of the product ring, where the axiom holds. The ghost components of $(0, 0, \dots)$ and $(1, 0, 0, \dots)$ are all 0 and all 1. So each axiom holds as an identity of integer polynomials, and specializing the variables gives it in $W(A)$ for every A. Functoriality holds because the operations are given by integer polynomials, and the ghost map is a ring homomorphism because $w_n(S) = w_n(X) + w_n(Y)$ and $w_n(P) = w_n(X)\, w_n(Y)$ are identities of polynomials, completing the proof.

The *Teichmüller lift* is $\tau\colon A \to W(A)$, $\tau(a) = (a, 0, 0, \dots)$, with ghost components $w_n(\tau(a)) = a^{p^n}$. The *Verschiebung* is $V(a_0, a_1, \dots) = (0, a_0, a_1, \dots)$, with $w_0(Vx) = 0$ and $w_n(Vx) = p\, w_{n-1}(x)$ for $n \geq 1$. In $A_1 = \mathbb{Z}[X_0, X_1, \dots]$ with $\varphi(X_i) = X_i^p$, the $(\Rightarrow)$ direction of lemma 3.8.1 shows that the sequence $(w_{n+1}(X))_{n\geq 0}$ satisfies Dwork's congruence, so there are integer polynomials F_n in $X_0, \dots, X_{n+1}$ with $w_n(F(X)) = w_{n+1}(X)$, and they define the *Frobenius* $F\colon W(A) \to W(A)$. Comparing ghost components in a p-torsion-free polynomial ring and specializing, as in the proof of theorem 3.8.2, shows that τ is multiplicative, V is additive, F is a ring endomorphism, and

$$FV = p \qquad \text{on } W(A)$$

since $w_n(FVx) = w_{n+1}(Vx) = p\, w_n(x)$ for every $n \geq 0$.

Now let $A = k$ have characteristic p. In A_1, the vector $\varphi(X) = (X_0^p, X_1^p, \dots)$ has $w_n(\varphi(X)) = \varphi(w_n(X)) \equiv w_{n+1}(X) = w_n(F(X)) \pmod{p^{n+1}}$ by lemma 3.8.1. Over a p-torsion-free ring, two vectors a, b with $w_m(a) \equiv w_m(b) \pmod{p^{m+1}}$ for all $m \leq n$ have $a_m \equiv b_m \pmod p$ for all $m \leq n$. By induction on m, $a_i \equiv b_i \pmod p$ for $i < m$, so $p^i a_i^{p^{m-i}} \equiv p^i b_i^{p^{m-i}} \pmod{p^{m+1}}$ by the elementary fact in the proof of lemma 3.8.1, which leaves $p^m a_m \equiv p^m b_m \pmod{p^{m+1}}$ and $a_m \equiv b_m \pmod p$. Hence $F_n(X) \equiv X_n^p \pmod p$, and over k the Frobenius is componentwise, $F(a_0, a_1, \dots) = (a_0^p, a_1^p, \dots)$. It commutes with the shift V, so

$$VF = FV = p \qquad \text{on } W(k)$$

and if k is perfect, F is an automorphism.

Theorem 3.8.3: Witt Vectors of a Perfect Field

Let k be a perfect field of characteristic p. Then $W(k)$ is a complete DVR with uniformizer p and residue field k, the reduction map being the first-component map $W(k) \to k$. Every element has a unique convergent expansion

$$x = \sum_{i=0}^{\infty} \tau(c_i)\, p^i \qquad c_i \in k$$

***Proof*:**
As S_m, P_m and the polynomials for additive inverses involve only components of index at most m, the truncation of a vector to its first n components is a ring homomorphism onto the ring $W_n(k)$ of truncated Witt vectors of length n. Its kernel is $V^n W(k)$, the set of vectors whose first n components vanish. By $VF = p$, with F bijective and commuting with V, $p^n W(k) = V^n F^n W(k) = V^n W(k)$. So $W(k)/p^n W(k) \cong W_n(k)$, and for $n = 1$ the first-component map is a surjective ring homomorphism $W(k) \to k$ with kernel $pW(k)$. Two vectors are congruent modulo p^n exactly when their first n components agree. A p-adic Cauchy sequence

therefore has eventually constant components in each index, and it converges to the vector of these components, so $W(k)$ is p-adically complete and separated. The element p is a nonzerodivisor, as V is injective and F is bijective.

If $x_0 \neq 0$, then x is a unit: $y = \tau(x_0^{-1})x$ has first component 1, so $y = 1 - z$ with $z \in pW(k)$, and $\sum_{m\geq 0} z^m$ converges to an inverse of y. Now let $x \neq 0$, and let v be the least index with $x_v \neq 0$. Then $x = V^v(x') = p^v F^{-v}(x')$ with $x' = (x_v, x_{v+1}, \dots)$, and $F^{-v}(x')$ has first component $x_v^{p^{-v}} \neq 0$, so it is a unit. Hence every nonzero element is p^v times a unit. As p is a nonzerodivisor, $W(k)$ is a domain, and its nonzero ideals are the $p^m W(k)$. So $W(k)$ is a complete DVR with uniformizer p and residue field k.

For the expansion, $x - \tau(x_0)$ has first component 0, so $x = \tau(c_0) + p\,x^{(1)}$ with $c_0 = x_0$ and $x^{(1)} \in W(k)$. Repeating this with $x^{(1)}$ gives $c_0, c_1, \dots \in k$ with $x - \sum_{i<n} \tau(c_i)p^i \in p^n W(k)$ for every n, so the series converges to x. If two expansions of x first differ at the index j, then cancelling p^j, a nonzerodivisor, and reducing modulo p shows that their j-th digits agree, which is impossible. Hence the expansion is unique, completing the proof.

A complete DVR of characteristic 0 whose residue field has characteristic p is *absolutely unramified* if p is a uniformizer.

Theorem 3.8.4: Cohen Structure Theorem

Let $\mathcal{O}$ be a complete DVR with maximal ideal $\mathfrak{p}$ and perfect residue field k of characteristic p, and let $\omega\colon k \to \mathcal{O}$ be the map of proposition 1.4.1.

1. If $\mathcal{O}$ has characteristic p, then $\mathcal{O} \cong k[[t]]$.
2. If $\mathcal{O}$ has characteristic 0, then the map

$$\theta\colon W(k) \to \mathcal{O} \qquad \theta(x) = \sum_{n\geq 0} p^n\, \omega\big(x_n^{p^{-n}}\big)$$

is the unique ring homomorphism whose composite with the reduction $\mathcal{O} \to k$ is the first-component map. It is injective, $\mathcal{O}$ is a free $\theta(W(k))$-module of rank $e = \nu_{\mathcal{O}}(p)$, and $\mathrm{Frac}(\mathcal{O})$ is a totally ramified extension of $\mathrm{Frac}(\theta(W(k)))$ of degree e. In particular, if $\mathcal{O}$ is absolutely unramified, then θ is an isomorphism.

***Proof*:**
Fix a uniformizer π of $\mathcal{O}$.

1. By proposition 1.4.1, ω is a ring homomorphism. Define $\Phi\colon k[[t]] \to \mathcal{O}$ by $\Phi\big(\sum_i c_i t^i\big) = \sum_i \omega(c_i)\pi^i$, which converges as $\omega(c_i)\pi^i \in \mathfrak{p}^i$. It is additive, and $\Phi(g) \equiv \Phi\big(\sum_{i<N} c_i t^i\big) \pmod{\mathfrak{p}^N}$ for every N. As ω is a ring homomorphism, Φ is multiplicative on polynomials, so $\Phi(gh) \equiv \Phi(g)\Phi(h) \pmod{\mathfrak{p}^N}$ for every N, whence $\Phi(gh) = \Phi(g)\Phi(h)$. It is injective: if $g \neq 0$ has first nonzero coefficient c_j, then $\Phi(g) = \pi^j(\omega(c_j) + \pi y)$ with $y \in \mathcal{O}$ and $\omega(c_j)$ a unit, so $\Phi(g) \neq 0$. It is surjective: for $x \in \mathcal{O}$ put $c_0 = x \bmod \mathfrak{p}$, write $x - \omega(c_0) = \pi x_1$, put $c_1 = x_1 \bmod \mathfrak{p}$, and continue, so that $x - \Phi\big(\sum_{i<N} c_i t^i\big) \in \mathfrak{p}^N$ for every N and $x = \Phi\big(\sum_i c_i t^i\big)$. Hence $\mathcal{O} \cong k[[t]]$.

2. By proposition 1.4.1, ω is multiplicative, and $p \in \mathfrak{p}$. If $a \equiv b \pmod{\mathfrak{p}^m}$ with $m \geq 1$, then $a^p \equiv b^p \pmod{\mathfrak{p}^{m+1}}$, because every term of $a^p - b^p = \sum_{i=1}^{p} \binom{p}{i} b^{p-i}(a-b)^i$ lies in $\mathfrak{p}^{m+1}$. Iterating, if $a_i \equiv b_i \pmod{\mathfrak{p}}$ for $0 \leq i \leq n$, then $p^i a_i^{p^{n-i}} \equiv p^i b_i^{p^{n-i}} \pmod{\mathfrak{p}^{n+1}}$, as $p^i \in \mathfrak{p}^i$, and so $w_n(a) \equiv w_n(b) \pmod{\mathfrak{p}^{n+1}}$.

   For $x \in W(k)$ and $n \geq 0$ let $\hat{x}^{(n)} = \big(\omega(x_0^{p^{-n}}), \ldots, \omega(x_n^{p^{-n}})\big)$. Multiplicativity of ω gives

$$w_n\big(\hat{x}^{(n)}\big) = \sum_{i=0}^{n} p^i\, \omega\big(x_i^{p^{-n}}\big)^{p^{n-i}} = \sum_{i=0}^{n} p^i\, \omega\big(x_i^{p^{-i}}\big)$$

   a partial sum of the series defining $\theta(x)$, so $\theta(x) \equiv w_n(\hat{x}^{(n)}) \pmod{\mathfrak{p}^{n+1}}$. Let $x, y \in W(k)$ and $s = x + y$, so that $s_i = S_i(x, y)$. The inverse of the Frobenius of k is a ring automorphism and S_i has integer coefficients, so $s_i^{p^{-n}}$ is S_i evaluated at the p^n-th roots of the components of x and y. Hence $\omega(s_i^{p^{-n}})$ and $S_i(\hat{x}^{(n)}, \hat{y}^{(n)})$ both reduce to $s_i^{p^{-n}}$. With $w_n(S(X,Y)) = w_n(X) + w_n(Y)$, this gives

$$\theta(s) \equiv w_n\big(\hat{s}^{(n)}\big) \equiv w_n\big(S(\hat{x}^{(n)}, \hat{y}^{(n)})\big) = w_n\big(\hat{x}^{(n)}\big) + w_n\big(\hat{y}^{(n)}\big) \equiv \theta(x) + \theta(y) \pmod{\mathfrak{p}^{n+1}}$$

   for every n, so $\theta(x+y) = \theta(x) + \theta(y)$. The same argument with P_i gives $\theta(xy) = \theta(x)\theta(y)$, and $\theta(1) = \omega(1) = 1$, so θ is a ring homomorphism. Its composite with reduction sends x to the residue of $\omega(x_0)$, which is x_0.

   For uniqueness, let $\psi\colon W(k) \to \mathcal{O}$ be a ring homomorphism whose composite with reduction is the first-component map. For $c \in k$, multiplicativity of τ gives $\tau(c) = \tau(c^{p^{-n}})^{p^n}$ for every n, so $\psi(\tau(c)) \in \bigcap_n \mathcal{O}^{p^n}$, and $\psi(\tau(c))$ reduces to c. By proposition 1.4.1, $\psi(\tau(c)) = \omega(c)$. As $\psi(p^n W(k)) \subseteq p^n \mathcal{O}$, the expansion of theorem 3.8.3 gives $\psi\big(\sum_i \tau(c_i) p^i\big) = \sum_i \omega(c_i) p^i$. The map θ satisfies the same hypotheses, so $\psi = \theta$.

   The ideals $p^m W(k)$ are the nonzero ideals of $W(k)$ (theorem 3.8.3), and $\theta(p^m) = p^m \neq 0$ as $\mathcal{O}$ has characteristic 0, so θ is injective. Write $A = \theta(W(k))$, a complete DVR with uniformizer p and residue field k, and let $e = \nu_{\mathcal{O}}(p)$, which is finite as $p \neq 0$. Every $y \in \mathcal{O}$ lies in $M + p\mathcal{O}$ with $M = \sum_{j<e} A\pi^j$: choosing digits $c_j \in k$ successively as in the first item gives $y - \sum_{j<e} \omega(c_j)\pi^j \in \mathfrak{p}^e = p\mathcal{O}$, and $\omega(c_j) = \theta(\tau(c_j)) \in A$. Applying this to y, then to the resulting element of $\mathcal{O}$, and so on, gives $m_i \in M$ with $y - \sum_{i<N} p^i m_i \in p^N \mathcal{O}$ for every N. Write $m_i = \sum_{j<e} a_{ij}\pi^j$ with $a_{ij} \in A$. The series $a_j = \sum_i p^i a_{ij}$ converge in A, since $W(k)$ is complete and $\theta(p^n W(k)) \subseteq p^n \mathcal{O}$, and $y = \sum_{j<e} a_j \pi^j$. So $M = \mathcal{O}$. A nonzero $a \in A$ is p^m times a unit, so $\nu_{\mathcal{O}}(a) \in e\mathbb{Z}$ and $\nu_{\mathcal{O}}(a\pi^j) \equiv j \pmod{e}$. In a relation $\sum_{j<e} a_j \pi^j = 0$ with not all a_j zero, the nonzero terms would have distinct valuations, which is impossible. Hence $1, \pi, \ldots, \pi^{e-1}$ is a basis of $\mathcal{O}$ over A, and of $\mathrm{Frac}(\mathcal{O})$ over $\mathrm{Frac}(A)$, which therefore has degree e. The valuation of $\mathcal{O}$ restricts to e times that of A, so by the uniqueness in theorem 2.1.9, $\mathcal{O}$ is the valuation ring of $\mathrm{Frac}(\mathcal{O})$ in the sense of definition 2.1.10. As $p\mathcal{O} = \mathfrak{p}^e$, the ramification index of $\mathrm{Frac}(\mathcal{O})/\mathrm{Frac}(A)$ is e, equal to the degree, and the extension is totally ramified. If $\mathcal{O}$ is absolutely unramified, then $e = 1$ and θ is an isomorphism, completing the proof.

Corollary 3.8.5: Witt Vectors of Finite Fields

Let $q = p^f$. Then $W(\mathbb{F}_q)$ is isomorphic to the valuation ring of the unramified extension of $\mathbb{Q}_p$ of degree f. In particular, $W(\mathbb{F}_p) \cong \mathbb{Z}_p$.

***Proof*:**
By theorem 3.3.3 there is an unramified extension U of $\mathbb{Q}_p$ with residue field $\mathbb{F}_q$, of degree $[\mathbb{F}_q : \mathbb{F}_p] = f$, and by proposition 3.3.7 every unramified extension of $\mathbb{Q}_p$ of degree f is isomorphic to it. Its valuation ring $\mathcal{O}_U$ is a complete DVR (theorem 2.1.9) with perfect residue field $\mathbb{F}_q$, and $e(U : \mathbb{Q}_p) = 1$ by corollary 3.1.5, so p is a uniformizer of $\mathcal{O}_U$. By theorem 3.8.4, $\theta\colon W(\mathbb{F}_q) \to \mathcal{O}_U$ is an isomorphism. For $f = 1$, $U = \mathbb{Q}_p$ and $\mathcal{O}_U = \mathbb{Z}_p$, completing the proof.

Example 3.8.6: Witt Vectors of $\overline{\mathbb{F}}_p$

The field $\overline{\mathbb{F}}_p$ is perfect, so $W(\overline{\mathbb{F}}_p)$ is a complete DVR with uniformizer p and residue field $\overline{\mathbb{F}}_p$ (theorem 3.8.3). The maximal unramified extension $\mathbb{Q}_p^{nr}$ has value group $\mathbb{Z}$ and residue field $\overline{\mathbb{F}}_p$ (example 3.3.13 and proposition 3.3.15), and its completion $\widehat{\mathbb{Q}_p^{nr}}$ has the same value group and residue field (proposition 1.2.4). So the valuation ring of $\widehat{\mathbb{Q}_p^{nr}}$ is an absolutely unramified complete DVR with residue field $\overline{\mathbb{F}}_p$, and theorem 3.8.4 identifies it with $W(\overline{\mathbb{F}}_p)$. The Frobenius $c \mapsto c^p$ of $\overline{\mathbb{F}}_p$ induces, through functoriality of W, an automorphism of $W(\overline{\mathbb{F}}_p)$. Under the identification it corresponds to the continuous extension of the Frobenius of $\mathbb{Q}_p^{nr}/\mathbb{Q}_p$, the topological generator of $\mathrm{Gal}_{\mathbb{Q}_p}(\mathbb{Q}_p^{nr}) \cong \widehat{\mathbb{Z}}$, since both are ring homomorphisms inducing $c \mapsto c^p$ on residue fields, and the uniqueness argument in the proof of theorem 3.8.4 applies to such maps.

Exercise 3.8.1

1. Using the Witt polynomials, compute the first two components of $\tau(a) + \tau(b)$ in $W(\mathbb{F}_p)$, and verify that the second is $-\frac{1}{p}\big((a+b)^p - a^p - b^p\big)$ reduced through the Witt addition.
2. Show that for k perfect of characteristic p, the ring $W(k)$ has characteristic 0, and contrast this with the truncation $W_1(k) = k$.
3. Show that on $W(\mathbb{F}_p)$ the Frobenius F is the identity and the Verschiebung V is multiplication by p.
4. Deduce from theorem 3.8.4 that two absolutely unramified complete DVRs with the same perfect residue field are isomorphic.

4

The Local-Global Correspondence

The local fields were introduced as the completions of a global field at its primes. This chapter closes the circle: the arithmetic of a global field is faithfully reflected in the family of its completions, and constructions carried out locally can be reassembled into global ones. The analytic engine is Krasner's lemma, which controls how the roots of a polynomial move under small perturbations of its coefficients; with it we prove the local-global correspondence, deduce the Kronecker-Weber theorem, and settle the Hasse principle for conics.

4.1 Krasner's Lemma and Continuity of Roots

If $f \in \mathbb{R}[x]$ or $\mathbb{C}[x]$ is separable, a small perturbation of its coefficients moves each root only slightly and keeps the polynomial separable. Over the fraction field K of a complete DVR more is true: every monic polynomial close enough to a monic irreducible separable $f \in K[x]$ has roots that generate the same extensions of K as the roots of f (theorem 4.1.8). The tool is Krasner's lemma, which compares $K(\alpha)$ with $K(\beta)$ for β close to α. Arguments that need a polynomial with prescribed roots and one further property use this to replace a given polynomial by a nearby one that has the property.

Lemma 4.1.1: Automorphism Preserves Absolute Value

Let K be a fraction field of a complete DVR with algebraic closure $\overline{K}$, and extend the absolute value $|-|$ to $\overline{K}$. Then for all $\alpha \in \overline{K}$ and $\sigma \in \mathrm{Aut}_K(\overline{K})$

$$|\sigma(\alpha)| = |\alpha|$$

***Proof*:**
First, α and $\sigma(\alpha)$ have the same minimal polynomial f, of degree n, since

$$f(\sigma(\alpha)) = \sigma(f(\alpha)) = \sigma(0) = 0$$

The norm of a generator of an extension of degree n is $(-1)^n$ times the constant term of its minimal polynomial, as multiplication by the generator has the companion matrix of that polynomial in the basis of its first n powers. Hence

$$N_{K(\alpha)/K}(\alpha) = (-1)^n f(0) = N_{K(\sigma(\alpha))/K}(\sigma(\alpha))$$

Then theorem 2.1.9 gives

$$|\sigma(\alpha)| = |N_{K(\sigma(\alpha))/K}(\sigma(\alpha))|^{1/n} = |N_{K(\alpha)/K}(\alpha)|^{1/n} = |\alpha|$$

completing the proof.

Definition 4.1.2: Belonging

Let K be the field of fraction of a complete DVR with absolute value $|-|$ extended to $\overline{K}$. Then for any $\alpha, \beta \in \overline{K}$, we say β *belongs* to α if

$$|\beta - \alpha| < |\beta - \sigma(\alpha)| \qquad \forall \sigma \in \mathrm{Aut}_K(\overline{K}) \setminus \{\sigma \mid \sigma(\alpha) = \alpha\}$$

In other words, β is strictly closer to α than all its conjugates.

As all nonarchimedean triangles are isosceles[1], it is equivalent to

$$|\beta - \alpha| < |\alpha - \sigma(\alpha)|$$

Theorem 4.1.3: Krasner's Lemma

Let K be a field of fraction of a complete DVR, $\alpha, \beta \in \overline{K}$, and α is separable over K. Then if β belongs to α, $K(\alpha) \subseteq K(\beta)$ (i.e. $\alpha \in K(\beta)$).

***Proof*:**
For the sake of contradiction, suppose β belongs to α but $\alpha \notin K(\beta)$. As α is separable over $K(\beta)$ and not in $K(\beta)$, its minimal polynomial over $K(\beta)$ has a root $\alpha' \neq \alpha$, and there is an automorphism $\sigma \in \mathrm{Aut}_{K(\beta)}(\overline{K}) \subseteq \mathrm{Aut}_K(\overline{K})$ with $\sigma(\alpha) = \alpha'$. Applying lemma 4.1.1 to $\beta - \alpha \in \overline{K}$, and using that σ fixes β,

$$|\beta - \alpha| = |\sigma(\beta - \alpha)| = |\sigma(\beta) - \sigma(\alpha)| = |\beta - \sigma(\alpha)|$$

which contradicts that β belongs to α, as $\sigma(\alpha) \neq \alpha$, completing the proof.

[1] If one side is shorter than another, it is shorter than both

4.1.4 Krasner's Lemma over Henselian Fields Like Hensel's lemma, Krasner's lemma holds over every henselian field (box 2.1.6). The proof above uses completeness only through the formula $|x| = |N_{K(x)/K}(x)|^{1/n}$, with $n = [K(x) : K]$, for the extended absolute value (lemma 4.1.1), and over a henselian field the absolute value extends uniquely to every algebraic extension, by the same formula [12, Chapter II, (6.2)]. Although the lemma is named after Krasner, Ostrowski proved it earlier, in 1917.

Example 4.1.5: Krasner's Lemma Fails Without Completeness

Let $K = \mathbb{Q}$ with the 5-adic absolute value, for which $\mathbb{Q}$ is not complete. The polynomial $x^2 + 1$ reduces modulo 5 to $(x - 2)(x - 3)$, with simple roots, so by corollary 2.1.2 it has a root $i \in \mathbb{Z}_5$ with $i \equiv 2 \bmod 5$, and then $-i \equiv 3 \bmod 5$. Give $\mathbb{Q}(i)$ the absolute value induced by the embedding $\mathbb{Q}(i) \to \mathbb{Q}_5$, and take $\alpha = i$ and $\beta = 2$. Then

$$|\beta - \alpha|_5 = |2 - i|_5 \leq 5^{-1} < 1 = |2 + i|_5$$

as $2 + i \equiv 4 \bmod 5$ is a unit. So β is strictly closer to α than to its only other conjugate $-i$, yet $\alpha \notin \mathbb{Q}(\beta) = \mathbb{Q}$. The proof of theorem 4.1.3 breaks down because the automorphism $i \mapsto -i$ of $\mathbb{Q}(i)$ does not preserve this absolute value: it sends $2 - i$ to $2 + i$.

Definition 4.1.6: L1 Norm

Let K be a field with absolute value $|-|$, and let $f = \sum_i f_i x^i \in K[x]$. Then

$$\|f\|_1 = \sum_i |f_i|$$

It is a norm on the K-vector space $K[x]$: it vanishes only at 0, $\|cf\|_1 = |c| \, \|f\|_1$, and the triangle inequality holds coefficientwise.

Lemma 4.1.7: Roots Smaller Than Polynomial

Let K be a field with absolute value $|-|$, let L/K be an extension with an absolute value extending $|-|$, and let $f = \prod_{i=1}^{n}(x - \alpha_i) \in K[x]$ be a monic polynomial with roots $\alpha_1, \ldots, \alpha_n \in L$. Then every root α_i of f satisfies

$$|\alpha_i| < \|f\|_1$$

***Proof*:**

Write $f = \sum_{i=0}^{n} f_i x^i$ with $f_n = 1$, so that $\|f\|_1 \geq 1$, and let α be a root. If $|\alpha| < 1$, then $|\alpha| < \|f\|_1$. If $|\alpha| = 1$, then $f \neq x^n$, as $\alpha \neq 0$, so some f_i with $i < n$ is nonzero and $\|f\|_1 > 1 = |\alpha|$. If $|\alpha| > 1$, then $\alpha^n = -\sum_{i<n} f_i \alpha^i$ and the triangle inequality give

$$|\alpha|^n \leq \sum_{i<n} |f_i| \, |\alpha|^i \leq |\alpha|^{n-1} \sum_{i<n} |f_i| = |\alpha|^{n-1}(\|f\|_1 - 1)$$

Dividing by $|\alpha|^{n-1}$ gives $|\alpha| \leq \|f\|_1 - 1 < \|f\|_1$, as we sought to show.

Theorem 4.1.8: Continuity of Roots

Let K be the fraction field of a complete DVR and $f \in K[x]$ a monic irreducible separable polynomial. Then there exists a $\delta \in \mathbb{R}_{>0}$ such that for every monic polynomial $g \in K[x]$ with $\|f - g\|_1 < \delta$,

$$\text{every root } \beta \text{ of } g \text{ belongs to a root } \alpha \text{ of } f \text{ for which } K(\alpha) = K(\beta)$$

In particular, every such g is irreducible, separable, and has the same splitting field as f.

***Proof*:**

Fix an algebraic closure $\overline{K}$ of K, with the absolute value extended to it, and let $\alpha_1, \ldots, \alpha_n$ be the roots of f in $\overline{K}$, which are distinct as f is separable, so that

$$f = \prod_{i=1}^{n}(x - \alpha_i) = \sum_{i=0}^{n} f_i x^i$$

Let ϵ be the least element of $\{1\} \cup \{|\alpha_i - \alpha_j| \mid i \neq j\}$, and put

$$\delta = \left(\frac{\epsilon}{2(\|f\|_1 + 1)}\right)^n$$

As $\epsilon \leq 1$ and $\|f\|_1 \geq 1$, $0 < \delta < 1$. Let $g = \sum_i g_i x^i \in K[x]$ be monic with $\|f - g\|_1 < \delta$. Then $\deg g = n$: otherwise the leading coefficient of whichever of f and g has larger degree contributes 1 to $\|f - g\|_1$. Also $\|g\|_1 \leq \|f\|_1 + \|f - g\|_1 < \|f\|_1 + 1$, and $\|g\|_1 \geq 1$.

Let $\beta \in \overline{K}$ be a root of g. By lemma 4.1.7, $|\beta| < \|g\|_1$, so $|\beta|^i \leq \|g\|_1^n$ for $0 \leq i \leq n$. As $f_n = g_n = 1$,

$$|f(\beta)| = |(f - g)(\beta)| \leq \sum_{i<n} |f_i - g_i|\,|\beta|^i \leq \|f - g\|_1\,\|g\|_1^n < \delta(\|f\|_1 + 1)^n = \left(\frac{\epsilon}{2}\right)^n$$

As $|f(\beta)| = \prod_{i=1}^{n} |\beta - \alpha_i|$, some root $\alpha = \alpha_i$ has $|\beta - \alpha| < \epsilon/2$. For every other root α_j, $|\alpha - \alpha_j| \geq \epsilon$, and the ultrametric inequality gives $|\beta - \alpha_j| = |\alpha - \alpha_j| \geq \epsilon > |\beta - \alpha|$. The conjugates of α different from α are among the α_j, so β belongs to α. By Krasner's lemma (theorem 4.1.3), $K(\alpha) \subseteq K(\beta)$, and as $n = [K(\alpha) : K] \leq [K(\beta) : K] \leq \deg g = n$, $K(\alpha) = K(\beta)$. Hence g, monic of degree $n = [K(\beta) : K]$ with β as a root, is the minimal polynomial of β, so it is irreducible, and it is separable, as β lies in the separable extension $K(\alpha)$ of K.

So every root of g generates the same field over K as some root of f. Conversely, let α' be a root of f, fix a root β of g and the root α of f it belongs to, and, as f is irreducible, choose $\tau \in \mathrm{Aut}_K(\overline{K})$ with $\tau(\alpha) = \alpha'$. The conjugates of α' different from α' are the $\tau(\sigma(\alpha))$ with $\sigma \in \mathrm{Aut}_K(\overline{K})$ and $\sigma(\alpha) \neq \alpha$, and lemma 4.1.1 gives

$$|\tau(\beta) - \alpha'| = |\beta - \alpha| < |\alpha - \sigma(\alpha)| = |\alpha' - \tau(\sigma(\alpha))|$$

So the root $\tau(\beta)$ of g belongs to α', and Krasner's lemma gives $K(\alpha') = K(\tau(\beta))$ as before. Hence the roots of f and the roots of g generate the same extension of K, which is the splitting field of both, completing the proof.

4.2 The Local-Global Correspondence

Let $\hat{L}$ be a local field. Then either by definition or by proposition 1.3.2 we know that $\hat{L}$ is a finite extension of $\mathbb{Q}_p$ or $\mathbb{F}_q((t))$ for some $p \leq \infty$ or some $q = p^r$. Furthermore, any completion of a global field with any nontrivial absolute value is a local field. It is thus reasonable to ask if we can complete the square; where $\hat{L}$ is the completion of a corresponding global field L that is a finite extension of $\mathbb{Q}$ or $\mathbb{F}_q(t)$. In the local number field case:

$$\begin{array}{ccc} \mathbb{Q}_p & \xrightarrow{\text{finite ext.}} & \hat{L} \\ {\scriptstyle |-|_p \text{ compl.}}\uparrow & & \uparrow{\scriptstyle |-|_{\mathfrak{p}} \text{ compl.}} \\ \mathbb{Q} & \overset{\text{finite ext.}}{\dashrightarrow} & L \end{array}$$

Generalizing the question, given a fixed global field K and a local field $\hat{K}$ that is the completion of K with respect to a nontrivial absolute value $|-|$, if it the case that every finite extension of local field $\hat{L}/\hat{K}$ corresponds to an extension of global field L/K where $\hat{L}$ is the completion of L with respect to to an absolute value whose restriction to K must be equivalent to $|-|$. This is indeed the case! We shall show this for the separable case as it is what we need and the inseparable case offers more complication

Theorem 4.2.1: Local to Global Correspondence

Let K be a global with with a non-trivial absolute value $|-|$ and $\hat{K}$ the completion of K with respect to $|-|$. Then every finite separable extension $\hat{L}/\hat{K}$ is the completion of a finite separable extension of L of K with respect to an absolute value that restricts to $|-|$. Furthermore, L can be chosen so that

$$[L : K] = [\hat{L} : \hat{K}]$$

in which case $\hat{L} = \hat{K} \cdot L$.

***Proof*:**

Let $\hat{L}/\hat{K}$ be a separable extension of degree n. Suppose first that $|-|$ is archimedean. Then $\hat{K}$ is complete with respect to an archimedean absolute value, so by [5, Ostrowski's Theorem For Complete Fields] there are an isomorphism σ from $\hat{K}$ onto $\mathbb{R}$ or $\mathbb{C}$ and an $s \in (0, 1]$ with $|a| = |\sigma(a)|_\infty^s$ for all $a \in \hat{K}$. If $\sigma(\hat{K}) = \mathbb{C}$, then $\hat{K}$ is algebraically closed by [5, Fundamental Theorem of Algebra], so $\hat{L} = \hat{K}$, and $L = K$ has every stated property. If $\sigma(\hat{K}) = \mathbb{R}$, write $\hat{L} = \hat{K}(\alpha)$ and let g be the minimal polynomial of α over $\hat{K}$. The irreducible polynomial $\sigma(g) \in \mathbb{R}[x]$ has a root $z \in \mathbb{C}$ by the same theorem, so $n = \deg g = [\mathbb{R}(z) : \mathbb{R}] \leq [\mathbb{C} : \mathbb{R}] = 2$. If $n = 1$, again $L = K$ works. If $n = 2$, then $z \notin \mathbb{R}$, so $\mathbb{R}(z) = \mathbb{C}$, and σ extends to an isomorphism $\tau\colon \hat{L} = \hat{K}(\alpha) \to \mathbb{R}(z) = \mathbb{C}$ with $\tau(\alpha) = z$. Give $\hat{L}$ the absolute value $|y| = |\tau(y)|_\infty^s$, which is an absolute value because $(u + v)^s \leq u^s + v^s$ for $u, v \geq 0$, which extends that of $\hat{K}$, and for which $\hat{L}$ is complete as $\mathbb{C}$ is. Let $j = \tau^{-1}(i)$, so $j^2 = -1$. As -1 is not a square in $\mathbb{R}$, $j \notin \hat{K}$, so $L := K(j)$ has $[L : K] = 2 = [\hat{L} : \hat{K}]$, and L/K is separable as K has characteristic 0. The absolute value of $\hat{L}$ restricts to one on L extending $|-|$. Every $y \in \hat{L}$ is $a + bj$ with $a, b \in \hat{K}$, and choosing $a_k, b_k \in K$ converging to a, b gives $a_k + b_k j \in L$ converging to y. So L is dense in the complete field $\hat{L}$, which is therefore the completion of L, and $\hat{L} = \hat{K} + \hat{K}j = \hat{K}L$.

Now suppose $|-|$ is nonarchimedean, so that the valuation ring of $\hat{K}$ is a complete DVR and $|-|$ is induced by its discrete valuation. As the extension is separable, by the primitive element theorem, we have a monic irreducible and separable polynomial $f \in \hat{K}[x]$ such that

$$\hat{L} = \frac{\hat{K}[x]}{f}$$

Let $\delta > 0$ be as in theorem 4.1.8 for f. As K is dense in its completion $\hat{K}$, approximating each non-leading coefficient of f gives a monic $g \in K[x]$ of the same degree with $\|g - f\|_1 < \delta$. By theorem 4.1.8, g is irreducible and separable over $\hat{K}$, and each root β of g in an algebraic closure of $\hat{K}$ has $\hat{K}(\beta) = \hat{K}(\alpha)$ for a root α of f, so

$$\hat{L} \cong \frac{\hat{K}[x]}{(f)} \cong \hat{K}(\alpha) = \hat{K}(\beta) \cong \frac{\hat{K}[x]}{(g)}$$

Then $\hat{L}$ is a finite separable extension of the field of fraction of a complete DVR, and so it is a field of fraction of a complete DVR and has a unique absolute value that extends the absolute value $|-|$ on $\hat{K}$.

Now, consider $L := K[x]/(g)$. The polynomial is irreducible in $\hat{K}[x]$, and hence in $K[x]$, and so

$$[L : K] = \deg g = [\hat{L} : \hat{K}]$$

Note the field $\hat{L}$ contains both $\hat{K}$ and L, and by construction is the smallest field that does this (note that g is irreducible in $\hat{K}[x]$), and so

$$\hat{L} = \hat{K}L$$

Finally, the absolute value on $\hat{L}$ restricts to an absolute value on L extending the absolute value $|-|$ on K , and hence $\hat{L}$ is complete and hence contains the completion of L with respect to $|-|$. On the other hand, the completion of L with respect to $|-|$ contains L and $\hat{K}$, and so must be $\hat{L}$, as we sought to show.

Let us now assume that $\hat{L}/\hat{K}$ is galois. Then is the corresponding global extension L/K galois, and are their galois groups isomorphic? As it turns out, it need not even be galois:

Example 4.2.2: Corresponding Global Extension Not Galois

Let $K = \mathbb{Q}$ and $\hat{K} = \mathbb{Q}_7$. Take $\hat{L} = \hat{K}[x]/(x^3 - 2)$. This extension is galois. The extension by definition contains the cube-root of 2, $\sqrt[3]{2}$. The other roots are this root times ζ_3, which is contains in $\mathbb{Q}_7$. This is true as $x^2 + x + 1 \in \mathbb{F}_7[x]$ has two roots which by Hensel's lemma both lift to roots in $\mathbb{Q}_7[x]$. However, $L = K[x]/(x^3 - 2)$ is *not* a galois extension, containing only one root.

However, if we replace K with $\mathbb{Q}(\zeta_3)$, this does not change the completion to $\hat{K}$ (take the completion of K with respect to the absolute value induced by a prime above 7), and certainly does not change $\hat{L}$, but now $L = K[x]/(x^3 - 2)$ *is* a galois extension.

Is it always possible to give a nice enough adjustment in the base global field K to get a correspondence of galois groups? This turns out to be the case.

Corollary 4.2.3: Local To Global Correspondence, Galois Groups

Let $\hat{L}/\hat{K}$ be a finite galois extension of local fields. Then there exists a finite galois extension of global fields L/K and an absolute value $|-|$ on L such that $\hat{L}$ is the completion of L with respect to $|-|$, $\hat{K}$ is the completion of K with respect to to the restriction of $|-|$ to K, and

$$\mathrm{Gal}_K(L) \cong \mathrm{Gal}_{\hat{K}}(\hat{L})$$

***Proof*:**

The absolute value $|-|$ of the local field $\hat{L}$ is nonarchimedean, and it restricts to that of $\hat{K}$. By proposition 1.3.2, $\hat{K}$ is a finite extension of $\mathbb{Q}_p$, separable as the characteristic is 0, or $\hat{K} \cong \mathbb{F}_q((t))$, which is spanned over $\mathbb{F}_p((t))$ by $\mathbb{F}_q$, so generated by a root of the separable polynomial $x^q - x$, hence separable over $\mathbb{F}_p((t))$. As $\mathbb{Q}_p$ and $\mathbb{F}_p((t))$ are the completions of the global fields $\mathbb{Q}$ and $\mathbb{F}_p(t)$, theorem 4.2.1 shows that $\hat{K}$ is the completion of a global field K with respect to the restriction of $|-|$. As in the proof of that theorem, take a monic separable irreducible polynomial $g \in K[x]$ that has all those properties in $\hat{K}[x]$ such that $\hat{L} = \hat{K}[x]/(g)$ and define again $L = K[x]/(g)$ so that $\hat{L} = \hat{K}L$.

This time, let M/K be the splitting field of g. As $\hat{L}/\hat{K}$ is galois, $\hat{L}$ contains the roots too, and so by the universal property of splitting fields $\hat{L}$ contains a subextension in of K isomorphic to M; we may identify this extension with M.

Now, note that $\hat{L}$ is equal to the completion of M with respect to to the restriction of $|-|$ to M. From this we have a group homomorphism induce by restriction:

$$\varphi : \mathrm{Gal}_{\hat{K}}(\hat{L}) \to \mathrm{Gal}_K(M)$$

This map is injective, as each $\sigma \in \mathrm{Gal}_{\hat{K}}(\hat{L})$ is determined by its action on the roots of g in M. Now, replace K by a fixed field of the image of φ, and replace L with M. The completion of K with respect to to the restriction of $|-|$ is still $\hat{K}$, and similarity for L and $\hat{L}$, and so:

$$\mathrm{Gal}_{\hat{K}}(\hat{L}) \cong \mathrm{Gal}_K(L)$$

as we sought to show.

Let us now put this all together and consider the $AKLB$ case and look a primes $\mathfrak{P}$ lying over $\mathfrak{p}$:

Theorem 4.2.4: Global To Local Correspondence

Assume AKLB[a], $\mathfrak{p} \subseteq A$, $\mathfrak{p}B = \prod_{\mathfrak{P}/\mathfrak{p}} \mathfrak{P}^{e_{\mathfrak{P}}}$. Let $K_{\mathfrak{p}}$ be the completion of K with respect to $|-|_{\mathfrak{p}}$ which has maximal ideal $\hat{\mathfrak{p}}$ in its valuation ring. Similarly, for each $\mathfrak{P}/\mathfrak{p}$, let $L_{\mathfrak{P}}$ denote the completion of L with respect to $|-|_{\mathfrak{P}}$ and let $\hat{\mathfrak{P}}$ denote the maximal ideal of the corresponding maximal ideal of the valuation ring. Then:

1. Each $L_{\mathfrak{P}}$ is a finite separable extension of $K_{\mathfrak{p}}$ with $[L_{\mathfrak{P}} : K_{\mathfrak{p}}] \leq [L : K]$.
2. Each $\hat{\mathfrak{P}}$ is the unique prime of $L_{\mathfrak{P}}$ lying over $\hat{\mathfrak{p}}$.
3. Each $\hat{\mathfrak{P}}$ has ramification index $e_{\hat{\mathfrak{P}}} = e_{\mathfrak{P}}$ and residue field degree $f_{\hat{\mathfrak{P}}} = f_{\mathfrak{P}}$.
4. $[L_{\mathfrak{P}} : K_{\mathfrak{p}}] = e_{\mathfrak{P}} f_{\mathfrak{P}}$
5. The map $L \otimes_K K_{\mathfrak{p}} \to \prod_{\mathfrak{P}/\mathfrak{p}} L_{\mathfrak{P}}$ given by $\ell \otimes x \mapsto (\ell x, ..., \ell x)$ is an isomorphism of finite étale $K_{\mathfrak{p}}$ algebras
6. If L/K is galois, then each $L_{\mathfrak{P}}/K_{\mathfrak{p}}$ is galois and we have isomorphisms of decomposition groups and inertia groups:

$$D_{\mathfrak{P}} \cong D_{\hat{\mathfrak{P}}} = \mathrm{Gal}_{K_{\mathfrak{p}}}(L_{\mathfrak{P}}) \qquad I_{\mathfrak{P}} \cong I_{\hat{\mathfrak{P}}}$$

[a] Recall L/K is finite and separable

***Proof*:**
(2) follows from theorem 3.1.3, and (4) from (2), (3) and the fundamental identity (corollary 3.1.5). We prove (1), (3), (5) and (6). By the primitive element theorem, $L = K(\alpha)$ for some $\alpha \in L$, whose minimal polynomial $f \in K[x]$ is separable of degree $n = [L : K]$.

1. For each $\mathfrak{P}/\mathfrak{p}$, the embedding $K \hookrightarrow L$ induces an embedding $K_{\mathfrak{p}} \to L_{\mathfrak{P}}$ by mapping the equivalence classes, which is well-defined as a sequence which is Cauchy in K with respect to $|-|_{\mathfrak{p}}$ is also Cauchy in L with respect to $|-|_{\mathfrak{P}}$ (the valuation is extended). Thus $K_{\mathfrak{p}}$ can be viewed as a topological subfield of $L_{\mathfrak{P}}$. The $K_{\mathfrak{p}}$-span W of $1, \alpha, \ldots, \alpha^{n-1}$ in $L_{\mathfrak{P}}$ is a finite-dimensional $K_{\mathfrak{p}}$-vector space, normed by the absolute value of $L_{\mathfrak{P}}$, so it is complete by lemma 2.1.8, hence closed in $L_{\mathfrak{P}}$. It contains $L = \sum_i K\alpha^i$, which is dense in $L_{\mathfrak{P}}$, so $W = L_{\mathfrak{P}}$. Hence $L_{\mathfrak{P}} = K_{\mathfrak{p}}(\alpha)$ has degree at most n over $K_{\mathfrak{p}}$, and it is separable over $K_{\mathfrak{p}}$, as the minimal polynomial of α over $K_{\mathfrak{p}}$ divides the separable polynomial f. In particular the $K_{\mathfrak{p}}$-algebra homomorphism
$$\varphi_{\mathfrak{P}} : L \otimes_K K_{\mathfrak{p}} \to L_{\mathfrak{P}} \qquad \ell \otimes x \mapsto \ell x$$
is surjective, as its image contains $K_{\mathfrak{p}}$ and α.

3. Write $\hat{\nu}_{\mathfrak{P}}$ and $\hat{\nu}_{\mathfrak{p}}$ for the extended valuations of $L_{\mathfrak{P}}$ and $K_{\mathfrak{p}}$. By proposition 1.2.4, they take the same values as $\nu_{\mathfrak{P}}$ and $\nu_{\mathfrak{p}}$, so $\hat{\nu}_{\mathfrak{P}}(L_{\mathfrak{P}}^{\times}) = \nu_{\mathfrak{P}}(L^{\times}) = \mathbb{Z}$ and $\hat{\nu}_{\mathfrak{p}}(K_{\mathfrak{p}}^{\times}) = \mathbb{Z}$, and the residue fields of $L_{\mathfrak{P}}$ and $K_{\mathfrak{p}}$ are those of $\nu_{\mathfrak{P}}$ on L and of $\nu_{\mathfrak{p}}$ on K, namely $B_{\mathfrak{P}}/\mathfrak{P}B_{\mathfrak{P}} \cong B/\mathfrak{P}$ and $A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} \cong A/\mathfrak{p}$, the inclusion between them being induced by $A \subseteq B$. By theorem 3.1.2, $\nu_{\mathfrak{P}} = e_{\mathfrak{P}}\nu_{\mathfrak{p}}$ on K, and passing to limits along Cauchy sequences gives $\hat{\nu}_{\mathfrak{P}} = e_{\mathfrak{P}}\hat{\nu}_{\mathfrak{p}}$ on the closure $K_{\mathfrak{p}}$ of K. Hence the ramification index of $L_{\mathfrak{P}}/K_{\mathfrak{p}}$ is $[\hat{\nu}_{\mathfrak{P}}(L_{\mathfrak{P}}^{\times}) : \hat{\nu}_{\mathfrak{P}}(K_{\mathfrak{p}}^{\times})] = [\mathbb{Z} : e_{\mathfrak{P}}\mathbb{Z}] = e_{\mathfrak{P}}$, and its residue degree is $[B/\mathfrak{P} : A/\mathfrak{p}] = f_{\mathfrak{P}}$.

5. Over $K_\mathfrak{p}$, $f = f_1 \cdots f_r$ with monic irreducible $f_j \in K_\mathfrak{p}[x]$, which are distinct as f is separable. The $K_\mathfrak{p}$-algebra $L \otimes_K K_\mathfrak{p}$ has basis $\alpha^i \otimes 1$ $(0 \leq i < n)$ and is generated by the root $\alpha \otimes 1$ of f, so $x \mapsto \alpha \otimes 1$ induces a surjection $K_\mathfrak{p}[x]/(f) \to L \otimes_K K_\mathfrak{p}$ between $K_\mathfrak{p}$-vector spaces of dimension n, which is an isomorphism. The ideals (f_j) of $K_\mathfrak{p}[x]$ are pairwise comaximal, so [5, Chinese Remainder Theorem] gives a $K_\mathfrak{p}$-algebra isomorphism

$$\psi \colon L \otimes_K K_\mathfrak{p} \to \prod_{j=1}^{r} M_j \qquad M_j = K_\mathfrak{p}[x]/(f_j)$$

sending $\alpha \otimes 1$ to $(\alpha_1, \ldots, \alpha_r)$, where α_j is the image of x in the field M_j. Composing $\ell \mapsto \ell \otimes 1$ with ψ and the projection to M_j gives a K-embedding $\iota_j \colon L \to M_j$ with $\iota_j(\alpha) = \alpha_j$.

The field M_j is a finite extension of $K_\mathfrak{p}$, so it is complete for the unique absolute value extending that of $K_\mathfrak{p}$ (theorem 2.1.9). Restricted along ι_j, this absolute value comes from a discrete valuation on L extending $\nu_\mathfrak{p}$, so by theorem 3.1.2 it is equivalent to $|-|_{\mathfrak{P}_j}$ for exactly one prime $\mathfrak{P}_j$ of B over $\mathfrak{p}$. As $M_j = \sum_i K_\mathfrak{p} \alpha_j^i$ and K is dense in $K_\mathfrak{p}$, the subfield $\iota_j(L) \supseteq \sum_i K \alpha_j^i$ is dense in M_j, by the norm comparison of lemma 2.1.8. So ι_j extends by continuity to a field embedding $\theta_j \colon L_{\mathfrak{P}_j} \to M_j$, which restricts to the inclusion on $K_\mathfrak{p}$, and whose image is complete, hence closed, and dense. Thus θ_j is an isomorphism with $\theta_j(\alpha) = \alpha_j$, and $\theta_j \circ \varphi_{\mathfrak{P}_j}$ is the j-th component of ψ, both being $K_\mathfrak{p}$-algebra homomorphisms that agree on the elements $\ell \otimes 1$.

The map $j \mapsto \mathfrak{P}_j$ is injective: if $\mathfrak{P}_j = \mathfrak{P}_k$, then $\theta_k \circ \theta_j^{-1} \colon M_j \to M_k$ is a $K_\mathfrak{p}$-isomorphism sending α_j to α_k, so $f_j = f_k$ and $j = k$. It is surjective: for a prime $\mathfrak{P}$ over $\mathfrak{p}$, the surjection $\varphi_\mathfrak{P} \circ \psi^{-1}$ of (1) from $\prod_j M_j$ onto the field $L_\mathfrak{P}$ has a maximal ideal as kernel, which is the kernel of the projection to some factor M_k. So it induces a $K_\mathfrak{p}$-isomorphism $\chi \colon M_k \to L_\mathfrak{P}$ with $\chi(\iota_k(\ell)) = \ell$ for $\ell \in L$. The absolute value $|\chi(-)|_\mathfrak{P}$ on M_k extends a power of that of $K_\mathfrak{p}$, so by uniqueness it is equivalent to the absolute value of M_k, and restricting to L gives $\mathfrak{P}_k = \mathfrak{P}$. Hence $\varphi = \prod_\mathfrak{P} \varphi_\mathfrak{P}$ is ψ followed by the product of the isomorphisms θ_j^{-1}, an isomorphism of $K_\mathfrak{p}$-algebras.

6. Assume L/K is galois, with group $G = \mathrm{Gal}_K(L)$, and let $\hat{B}_\mathfrak{P}$ be the valuation ring of $L_\mathfrak{P}$. Let $\sigma \in D_\mathfrak{P} = \{\sigma \in G \mid \sigma(\mathfrak{P}) = \mathfrak{P}\}$. Then σ maps $B_\mathfrak{P}$ onto itself and $\mathfrak{P}B_\mathfrak{P}$ onto itself, so $\nu_\mathfrak{P}(\sigma(x)) = \nu_\mathfrak{P}(x)$ for $x \in L^\times$, and σ is an isometry of L for $|-|_\mathfrak{P}$. It therefore extends by continuity to a field automorphism $\hat{\sigma}$ of $L_\mathfrak{P}$, the extension of σ^{-1} being its inverse, and $\hat{\sigma}$ fixes the closure $K_\mathfrak{p}$ of K pointwise. By uniqueness of continuous extensions, $\varphi \colon \sigma \mapsto \hat{\sigma}$ is a homomorphism $D_\mathfrak{P} \to \mathrm{Aut}_{K_\mathfrak{p}}(L_\mathfrak{P})$, injective as $\hat{\sigma}$ restricts to σ on L.

By (1), $L_\mathfrak{P} = K_\mathfrak{p}(\alpha)$. As L/K is normal, all roots of f lie in $L \subseteq L_\mathfrak{P}$, so $L_\mathfrak{P}$ is a splitting field over $K_\mathfrak{p}$ of the separable polynomial f, and $L_\mathfrak{P}/K_\mathfrak{p}$ is galois [5, Fundamental Theorem of Galois Theory I]. Let $\tau \in \mathrm{Gal}_{K_\mathfrak{p}}(L_\mathfrak{P})$. The function $x \mapsto |\tau(x)|$ is an absolute value on $L_\mathfrak{P}$ extending that of $K_\mathfrak{p}$, so it is the absolute value of $L_\mathfrak{P}$ by theorem 2.1.9, and τ is an isometry. As $\tau(\alpha)$ is a root of f, it lies in L, so τ maps $L = K(\alpha)$ into L and restricts to some $\sigma \in G$. Then σ preserves B and the absolute value $|-|_\mathfrak{P}$, hence maps $\mathfrak{P} = \{x \in B \mid |x|_\mathfrak{P} < 1\}$ onto itself, so $\sigma \in D_\mathfrak{P}$. As $\hat{\sigma}$ and τ are continuous and agree on the dense subfield L, $\varphi(\sigma) = \tau$. So φ is an isomorphism $D_\mathfrak{P} \cong \mathrm{Gal}_{K_\mathfrak{p}}(L_\mathfrak{P})$, and

$D_{\hat{\mathfrak{P}}} = \mathrm{Gal}_{K_\mathfrak{p}}(L_\mathfrak{P})$, as $\hat{\mathfrak{P}}$ is the only prime of $\hat{B}_\mathfrak{P}$. In particular $|D_\mathfrak{P}| = [L_\mathfrak{P} : K_\mathfrak{p}] = e_\mathfrak{P} f_\mathfrak{P}$ by (4).

For the inertia groups, recall from (3) that the inclusion $B \subseteq \hat{B}_\mathfrak{P}$ induces an isomorphism $B/\mathfrak{P} \cong \hat{B}_\mathfrak{P}/\hat{\mathfrak{P}}$, so every residue class of $\hat{B}_\mathfrak{P}$ contains an element of B. For $\sigma \in D_\mathfrak{P}$ and $x \in B$ we have $\hat{\sigma}(x) = \sigma(x)$, so the automorphisms of $B/\mathfrak{P}$ and of $\hat{B}_\mathfrak{P}/\hat{\mathfrak{P}}$ induced by σ and by $\hat{\sigma}$ correspond under this isomorphism. Hence σ acts trivially on $B/\mathfrak{P}$ exactly when $\hat{\sigma}$ acts trivially on $\hat{B}_\mathfrak{P}/\hat{\mathfrak{P}}$, that is, $\sigma \in I_\mathfrak{P}$ exactly when $\varphi(\sigma) \in I_{\hat{\mathfrak{P}}}$, and φ restricts to an isomorphism $I_\mathfrak{P} \cong I_{\hat{\mathfrak{P}}}$.

This completes the proof.

Corollary 4.2.5: Global To Local Correspondence, Norm and Trace

Assume AKLB and let $\mathfrak{p}$ be a prime of A. Then for every $\alpha \in L$, we have:

$$N_{L/K}(\alpha) = \prod_{\mathfrak{P}/\mathfrak{p}} N_{L_\mathfrak{P}/K_\mathfrak{p}}(\alpha) \qquad \mathrm{tr}_{L/K}(\alpha) = \sum_{\mathfrak{P}/\mathfrak{p}} \mathrm{tr}_{L_\mathfrak{P}/K_\mathfrak{p}}(\alpha)$$

***Proof*:**

The norm and trace are the determinant and trace of the K-linear map $L \xrightarrow{\times\alpha} L$. A K-basis of L is a $K_\mathfrak{p}$-basis of $L \otimes_K K_\mathfrak{p}$ in which multiplication by $\alpha \otimes 1$ has the same matrix, and under the isomorphism of theorem 4.2.4(5) this map becomes multiplication by α on each factor $L_\mathfrak{P}$. Determinants and traces are multiplicative and additive over products, which gives the two identities, as we sought to show.

Corollary 4.2.6: Global To Local Correspondence, Decomposition

Assume AKLB with A a Dedekind domain, let $\mathfrak{p}$ be a nonzero prime of A with $\mathfrak{p}B = \prod_{\mathfrak{P}/\mathfrak{p}} \mathfrak{P}^{e_\mathfrak{P}}$, let $\hat{A}_\mathfrak{p}$ be the completion of A with respect to $|-|_\mathfrak{p}$ and for each $\mathfrak{P}/\mathfrak{p}$ let $\hat{B}_\mathfrak{P}$ denote the completion of B with respect to $|-|_\mathfrak{P}$. Then

$$B \otimes_A \hat{A}_\mathfrak{p} \cong_{\hat{A}_\mathfrak{p}} \prod_{\mathfrak{P}/\mathfrak{p}} \hat{B}_\mathfrak{P}$$

***Proof*:**

Write $\hat{A} = \hat{A}_\mathfrak{p}$ and $n = [L : K]$. By [2, Generation of Ring of Integers], applied to the Noetherian normal domain A and the separable extension L/K, B is a finitely generated A-module. It is torsion-free, so $B_\mathfrak{p} = B \otimes_A A_\mathfrak{p}$ is a finitely generated torsion-free module over the DVR $A_\mathfrak{p}$, hence free [5, Invariant Factor Form of Modules over PID's], with a basis $b_1, \ldots, b_n \subseteq B_\mathfrak{p} \subseteq L$ of n elements, as $B_\mathfrak{p}$ spans L over K.

Step 1: the left side. An element $s \in A \setminus \mathfrak{p}$ is a unit modulo $\mathfrak{p}^k$, as $A/\mathfrak{p}^k$ is local with maximal ideal $\mathfrak{p}/\mathfrak{p}^k$, so multiplication by s is bijective on $B/\mathfrak{p}^k B$, and $B/\mathfrak{p}^k B \cong B_\mathfrak{p}/\mathfrak{p}^k B_\mathfrak{p} = \bigoplus_i (A/\mathfrak{p}^k)\, b_i$.

By propositions 1.2.3 and 1.2.4, $\hat{A} = \varprojlim_k A/\mathfrak{p}^k$. Hence $B \otimes_A \hat{A} = B_\mathfrak{p} \otimes_{A_\mathfrak{p}} \hat{A} = \bigoplus_i \hat{A}\, b_i = \varprojlim_k B/\mathfrak{p}^k B$.

Step 2: the right side. As $\mathfrak{p}B = \prod_{\mathfrak{P}/\mathfrak{p}} \mathfrak{P}^{e_\mathfrak{P}}$, $\mathfrak{p}^k B = \prod_{\mathfrak{P}/\mathfrak{p}} \mathfrak{P}^{ke_\mathfrak{P}}$, and the powers of the distinct maximal ideals $\mathfrak{P}$ are pairwise comaximal. So [5, Chinese Remainder Theorem] gives $B/\mathfrak{p}^k B \cong \prod_{\mathfrak{P}/\mathfrak{p}} B/\mathfrak{P}^{ke_\mathfrak{P}}$. For each $\mathfrak{P}$, proposition 1.2.4 gives $B/\mathfrak{P}^m \cong B_\mathfrak{P}/\mathfrak{P}^m B_\mathfrak{P} \cong \hat{B}_\mathfrak{P}/\hat{\mathfrak{P}}^m$, and $\hat{B}_\mathfrak{P} \cong \varprojlim_m \hat{B}_\mathfrak{P}/\hat{\mathfrak{P}}^m$ by proposition 1.2.3. Passing to the limit over k therefore gives $\varprojlim_k B/\mathfrak{p}^k B \cong \prod_{\mathfrak{P}/\mathfrak{p}} \hat{B}_\mathfrak{P}$.

All these isomorphisms are ring isomorphisms compatible with the actions of $\hat{A}$, induced by $b \otimes a \mapsto (ba)_\mathfrak{P}$ on $B \otimes_A \hat{A}$. Combining the two steps gives the stated isomorphism, completing the proof.

4.3 The Kronecker-Weber Theorem

The Kronecker-Weber theorem states that every finite abelian extension of $\mathbb{Q}$ lies in a cyclotomic field $\mathbb{Q}(\zeta_m)$. Throughout, extensions of $\mathbb{Q}_p$ lie in a fixed algebraic closure, in which $\mathbb{Q}_p(\zeta_m)$ is the field generated by the m-th roots of unity. The roots of unity of order dividing a or b generate those of order dividing $\mathrm{lcm}(a,b)$, so $\mathbb{Q}_p(\zeta_a)\mathbb{Q}_p(\zeta_b) = \mathbb{Q}_p(\zeta_{\mathrm{lcm}(a,b)})$.

Lemma 4.3.1: Kronecker Weber Theorem, Lemma

For any prime p,

$$\mathbb{Q}_p\left((-p)^{1/(p-1)}\right) = \mathbb{Q}_p(\zeta_p)$$

***Proof*:**
Let $\zeta = \zeta_p$ and $\pi = \zeta - 1$. By lemma 2.2.6 with $n = 1$, the field $F = \mathbb{Q}_p(\zeta)$ is totally ramified of degree $p-1$ over $\mathbb{Q}_p$, with uniformizer π and residue field $\mathbb{F}_p$, and π is a root of

$$\frac{(x+1)^p - 1}{x} = x^{p-1} + \sum_{k=1}^{p-1} \binom{p}{k} x^{k-1}$$

Solving for π^{p-1} and dividing by $-p$ gives

$$u := -\frac{\pi^{p-1}}{p} = 1 + \sum_{k=2}^{p-1} \frac{1}{p}\binom{p}{k} \pi^{k-1}$$

where each $\frac{1}{p}\binom{p}{k}$ with $2 \le k \le p-1$ is an integer. So $u \equiv 1 \bmod \pi$ is a unit of $\mathcal{O}_F$. The polynomial $x^{p-1} - u \in \mathcal{O}_F[x]$ is monic, and its reduction $x^{p-1} - 1$ has 1 as a simple root, the derivative there being $p - 1 \neq 0$ in $\mathbb{F}_p$. By corollary 2.1.2 there is a $\beta \in \mathcal{O}_F$ with $\beta^{p-1} = u$, and then

$$\left(\frac{\pi}{\beta}\right)^{p-1} = \frac{\pi^{p-1}}{u} = -p$$

So $\pi/\beta \in F$ is a root of $x^{p-1}+p$, which is irreducible of degree $p-1$ by corollary 2.2.4. As F is the splitting field of x^p-1, it is a normal extension of $\mathbb{Q}_p$, and since it contains one root of the irreducible polynomial $x^{p-1}+p$, it contains all of them. For any root α, the field $\mathbb{Q}_p(\alpha) \subseteq F$ has degree $p-1=[F:\mathbb{Q}_p]$, so $\mathbb{Q}_p(\alpha)=F$, as we sought to show.

Lemma 4.3.2: No $(\mathbb{Z}/p\mathbb{Z})^3$-Extension of $\mathbb{Q}_p$ for Odd p

Let p be an odd prime. There is no Galois extension $M/\mathbb{Q}_p$ with $\mathrm{Gal}_{\mathbb{Q}_p}(M) \cong (\mathbb{Z}/p\mathbb{Z})^3$.

***Proof*:**
Let ζ be a primitive p-th root of unity, $F=\mathbb{Q}_p(\zeta)$ and $\Delta=\mathrm{Gal}_{\mathbb{Q}_p}(F)$. By lemma 2.2.6, $F/\mathbb{Q}_p$ is totally ramified of degree $p-1$ with uniformizer $\pi=\zeta-1$ and residue field $\mathbb{F}_p$, and $\omega\colon \Delta \to (\mathbb{Z}/p\mathbb{Z})^\times$, defined by $\delta(\zeta)=\zeta^{\omega(\delta)}$, is an isomorphism. Let ν be the normalized valuation of F, so $\nu(\pi)=1$ and $\nu(p)=p-1$, and let $U^{(i)}$ be the higher unit groups of F. Every $\delta \in \Delta$ preserves ν, since the absolute value of F is $|N_{F/\mathbb{Q}_p}(-)|_p^{1/(p-1)}$ (theorem 2.1.9) and $N_{F/\mathbb{Q}_p}(\delta(x))=N_{F/\mathbb{Q}_p}(x)$. Let $V=F^\times/(F^\times)^p$, a group in which every element has order dividing p, on which Δ acts by $\delta\big(x(F^\times)^p\big)=\delta(x)(F^\times)^p$, and put

$$V_\omega = \left\{ v \in V \;\middle|\; \delta(v)=v^{\omega(\delta)} \text{ for all } \delta \in \Delta \right\}$$

a subgroup of V, the exponent $\omega(\delta) \in \mathbb{Z}/p\mathbb{Z}$ being meaningful as $v^p=1$.

Step 1: V_ω has at most p^2 elements. Since p is odd, $(\mathbb{Z}/p\mathbb{Z})^\times$ is nontrivial and there is a $\delta_0 \in \Delta$ with $k_0 := \omega(\delta_0) \neq 1$. If $x(F^\times)^p \in V_\omega$, then $\delta_0(x)=x^{k_0}y^p$ for some $y \in F^\times$, and applying ν gives $(k_0-1)\nu(x) \equiv 0 \bmod p$, so $p \big| \nu(x)$. Then $x\pi^{-\nu(x)} \in \mathcal{O}_F^\times$ represents the same class. As $\mathcal{O}_F^\times = \mu_{p-1} \times U^{(1)}$ (proposition 1.3.5) and $\mu_{p-1} \subseteq (F^\times)^p$, raising to the p-th power being a bijection of the group μ_{p-1} of order prime to p, every element of V_ω is the class of a principal unit. For $i \geq 1$ let $V^{(i)} \subseteq V$ be the image of $U^{(i)}$, so $V_\omega \subseteq V^{(1)} \supseteq V^{(2)} \supseteq \cdots$.

1. $V^{(p+1)}=1$. As $2 > \nu(p)/(p-1)=1$, theorem 1.4.2 makes $\log\colon U^{(2)} \to \mathfrak{p}^2$ and exp mutually inverse isomorphisms. For $w \in U^{(p+1)}$, $\log w \in \mathfrak{p}^{p+1}=p\mathfrak{p}^2$, say $\log w = pz$ with $z \in \mathfrak{p}^2$, and then $w=\exp(pz)=\exp(z)^p \in (F^\times)^p$.
2. Each quotient $V^{(i)}/V^{(i+1)}$ has at most p elements, being a quotient of $U^{(i)}/U^{(i+1)} \cong \mathcal{O}_F/\mathfrak{p} \cong \mathbb{F}_p$ (proposition 1.2.2).
3. Let $\delta \in \Delta$, let $k \geq 1$ be an integer representing $\omega(\delta)$, and let $w \in U^{(i)}$ with $i \geq 1$. Then $\delta(w) \equiv w^{k^i} \bmod U^{(i+1)}$. First, $\delta(\pi)=\zeta^k-1=\pi(1+\zeta+\cdots+\zeta^{k-1})$ with each $\zeta^j \equiv 1 \bmod \mathfrak{p}$, so $\delta(\pi)=\pi t$ with $t \equiv k \bmod \mathfrak{p}$. Write $w=1+c\pi^i$ with $c \in \mathcal{O}_F$. As the residue field is $\mathbb{F}_p$, $c \equiv c_0 \bmod \mathfrak{p}$ for an integer c_0, and $\delta(c)-c_0=\delta(c-c_0) \in \mathfrak{p}$, so $\delta(c) \equiv c \bmod \mathfrak{p}$. Therefore $\delta(w)=1+\delta(c)t^i\pi^i \equiv 1+k^ic\pi^i \equiv (1+c\pi^i)^{k^i} \bmod \mathfrak{p}^{i+1}$, and as w^{k^i} is a unit, $\delta(w)w^{-k^i} \in U^{(i+1)}$.

Now let $v \in V_\omega$ with $v \neq 1$, and let i be the largest index with $v \in V^{(i)}$, so $1 \leq i \leq p$ by the first item. The image $\bar{v}$ of v in $V^{(i)}/V^{(i+1)}$ is nontrivial, so this quotient has order exactly p by the second item. For $\delta \in \Delta$ and k as in the third item, $\delta(\bar{v})=\bar{v}^{k^i}$, while $v \in V_\omega$ gives $\delta(\bar{v})=\bar{v}^k$.

Hence $p \mid k^i - k$ for every k prime to p, as ω is surjective. Taking k to generate the cyclic group $(\mathbb{Z}/p\mathbb{Z})^\times$ of order $p-1$ gives $p-1 \mid i-1$, so $i = 1$ or $i = p$. Consequently every nontrivial element of $V_\omega \cap V^{(2)}$ lies in $V^{(p)}$, which has at most p elements by the first two items, and $V_\omega/(V_\omega \cap V^{(2)})$ embeds in $V^{(1)}/V^{(2)}$, which has at most p elements. So V_ω has at most p^2 elements.

Step 2: Kummer generators. Suppose $M/\mathbb{Q}_p$ is Galois with group $(\mathbb{Z}/p\mathbb{Z})^3$, and let $E \subseteq M$ be a subfield with $[E : \mathbb{Q}_p] = p$, so $E/\mathbb{Q}_p$ is cyclic of degree p. The degree $[EF : \mathbb{Q}_p]$ is divisible by p and by $p-1$, while $[EF : F] \leq [E : \mathbb{Q}_p] = p$, so $[EF : F] = p$. By [5, Galois Composite Fields II], EF is Galois over $\mathbb{Q}_p$ with group embedding in $\mathrm{Gal}_{\mathbb{Q}_p}(E) \times \Delta$, hence abelian, and EF/F is Galois of order p, hence cyclic. As F has characteristic 0 and contains the p-th roots of unity, [5, Primitive Element For Cyclic Kummer Extension] gives $\alpha_E \in EF$ with $EF = F(\alpha_E)$ and $a_E := \alpha_E^p \in F^\times$. The class v_E of a_E in V is nontrivial, since $a_E = b^p$ with $b \in F$ would give $\alpha_E = \zeta^j b \in F$ for some j, both being roots of $x^p - a_E$.

The class v_E lies in V_ω. Let $\delta \in \Delta$, let k represent $\omega(\delta)$, extend δ to $\tilde{\delta} \in \mathrm{Gal}_{\mathbb{Q}_p}(EF)$, and let σ generate $\mathrm{Gal}_F(EF)$, say $\sigma(\alpha_E) = \zeta^t \alpha_E$. Since $\mathrm{Gal}_{\mathbb{Q}_p}(EF)$ is abelian,

$$\sigma(\tilde{\delta}(\alpha_E)) = \tilde{\delta}(\zeta^t \alpha_E) = \zeta^{tk} \tilde{\delta}(\alpha_E) \qquad \sigma(\alpha_E^k) = \zeta^{tk} \alpha_E^k$$

so $c := \tilde{\delta}(\alpha_E)/\alpha_E^k$ is fixed by σ, that is, $c \in F$. Then $\delta(a_E) = \tilde{\delta}(\alpha_E)^p = c^p a_E^k$, and so $\delta(v_E) = v_E^k$.

Distinct E give distinct subgroups $\langle v_E \rangle$. If $\langle v_E \rangle = \langle v_{E'} \rangle$, then $a_{E'} = a_E^j c^p$ for some integer j and some $c \in F^\times$. Then $\alpha_E^j c$ is a p-th root of $a_{E'}$, so $\alpha_{E'} = \zeta^t \alpha_E^j c \in EF$ for some t, and $E'F \subseteq EF$, whence $E'F = EF$ as both have degree p over F. The field $EF \cap M$ contains E, and its degree over $\mathbb{Q}_p$ divides both $[EF : \mathbb{Q}_p] = p(p-1)$ and $[M : \mathbb{Q}_p] = p^3$, so $EF \cap M = E$. Likewise $E'F \cap M = E'$, and so $E = E'$.

Step 3: counting. The group $(\mathbb{Z}/p\mathbb{Z})^3$ has $(p^3-1)/(p-1) = p^2 + p + 1$ subgroups of index p, so M has $p^2 + p + 1$ subfields of degree p over $\mathbb{Q}_p$, and by Step 2 they give $p^2 + p + 1$ distinct subgroups of order p in V_ω. Each of these contains $p-1$ nontrivial elements and any two meet only in 1, so V_ω has at least $(p^2 + p + 1)(p-1) + 1 = p^3$ elements, contradicting Step 1. Hence no such M exists, as we sought to show.

For $p = 2$ the conclusion of lemma 4.3.2 fails: $\mathbb{Q}_2(\zeta_{24}) = \mathbb{Q}_2(\zeta_3)\mathbb{Q}_2(\zeta_8)$ is the composite of the unramified quadratic extension and the totally ramified extension with group $(\mathbb{Z}/8\mathbb{Z})^\times \cong (\mathbb{Z}/2\mathbb{Z})^2$, so its group is $(\mathbb{Z}/2\mathbb{Z})^3$.

Lemma 4.3.3: Dyadic Galois Groups Excluded

There is no Galois extension $M/\mathbb{Q}_2$ with $\mathrm{Gal}_{\mathbb{Q}_2}(M)$ isomorphic to $(\mathbb{Z}/2\mathbb{Z})^4$ or to $(\mathbb{Z}/4\mathbb{Z})^3$.

***Proof*:**
Step 1: $\mathbb{Q}_2$ has exactly seven quadratic extensions. By proposition 1.3.5, with residue field $\mathbb{F}_2$, and theorem 1.4.3, $\mathbb{Q}_2^\times = 2^\mathbb{Z} \times U^{(1)}$ with $U^{(1)} \cong \mu_{2^\infty}(\mathbb{Q}_2) \times \mathbb{Z}_2$. The group $\mu_{2^\infty}(\mathbb{Q}_2)$ is $\{\pm 1\}$. It is cyclic, being a finite subgroup of $\mathbb{Q}_2^\times$, so if it had order at least 4 it would contain a square root of -1. Such an element would lie in $\mathbb{Z}_2$, as its absolute value is 1, while squares of 2-adic integers

are 0 or 1 modulo 4 and $-1 \equiv 3 \bmod 4$. Hence $\mathbb{Q}_2^\times \cong \mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}_2$, and since $\mathbb{Z}_2/2\mathbb{Z}_2 \cong \mathbb{Z}/2\mathbb{Z}$ (proposition 1.1.6), the group $\mathbb{Q}_2^\times/(\mathbb{Q}_2^\times)^2$ has 8 elements. Completing the square shows that every quadratic extension of $\mathbb{Q}_2$ is $\mathbb{Q}_2(\sqrt{a})$ for some $a \in \mathbb{Q}_2^\times \setminus (\mathbb{Q}_2^\times)^2$. If $\mathbb{Q}_2(\sqrt{a}) = \mathbb{Q}_2(\sqrt{b})$, write $\sqrt{b} = x + y\sqrt{a}$ with $x, y \in \mathbb{Q}_2$. Then $b = x^2 + ay^2 + 2xy\sqrt{a}$ forces $xy = 0$, and $y = 0$ would make b a square, so $x = 0$ and $b = ay^2$. Conversely a and ay^2 give the same field. So the quadratic extensions of $\mathbb{Q}_2$ correspond to the 7 nontrivial elements of $\mathbb{Q}_2^\times/(\mathbb{Q}_2^\times)^2$.

Step 2: the group $(\mathbb{Z}/2\mathbb{Z})^4$. A Galois extension of $\mathbb{Q}_2$ with group $(\mathbb{Z}/2\mathbb{Z})^4$ would have $2^4 - 1 = 15$ subgroups of index 2, hence 15 distinct quadratic subextensions, contradicting Step 1.

Step 3: no cyclic quartic extension contains $k = \mathbb{Q}_2(i)$*, where* $i^2 = -1$. The field k is quadratic over $\mathbb{Q}_2$, as -1 is not a square. Suppose $N/\mathbb{Q}_2$ is cyclic of degree 4 with $k \subseteq N$, and let σ generate $\mathrm{Gal}_{\mathbb{Q}_2}(N)$. Then σ^2 generates $\mathrm{Gal}_k(N)$, the unique subgroup of order 2, and σ restricts to the nontrivial automorphism $i \mapsto -i$ of k. Completing the square, $N = k(s)$ with $s^2 = \beta \in k$. Since σ^2 moves s and $\sigma^2(s)^2 = \beta$, it follows that $\sigma^2(s) = -s$. Put $\gamma = s\,\sigma(s) \neq 0$. Then $\sigma(\gamma) = \sigma(s)\sigma^2(s) = -\gamma$, so $\gamma \notin \mathbb{Q}_2$, while $\sigma^2(\gamma) = \gamma$ gives $\gamma \in k$, and $\gamma^2 = \beta\,\sigma(\beta)$ is fixed by σ, so lies in $\mathbb{Q}_2$. Writing $\gamma = u + wi$ with $u, w \in \mathbb{Q}_2$, the condition $\gamma^2 = u^2 - w^2 + 2uwi \in \mathbb{Q}_2$ forces $uw = 0$, and $\gamma \notin \mathbb{Q}_2$ forces $w \neq 0$, so $u = 0$ and $\gamma^2 = -w^2$. Writing $\beta = x + yi$ with $x, y \in \mathbb{Q}_2$, also $\gamma^2 = \beta\,\sigma(\beta) = (x + yi)(x - yi) = x^2 + y^2$. Hence $-1 = (x/w)^2 + (y/w)^2$ would be a sum of two squares in $\mathbb{Q}_2$. This is impossible. Given $X^2 + Y^2 = -1$ with $X, Y \in \mathbb{Q}_2$, choose $m \geq 0$ so that $A = 2^m X$ and $B = 2^m Y$ lie in $\mathbb{Z}_2$, with at least one of them a unit when $m > 0$. Squares of 2-adic integers are 0 or 1 modulo 4, so $A^2 + B^2$ is 0, 1 or 2 modulo 4, and it is 1 or 2 when A or B is a unit. But $A^2 + B^2 = -4^m$ is 3 modulo 4 when $m = 0$ and 0 modulo 4 when $m > 0$.

Step 4: the group $(\mathbb{Z}/4\mathbb{Z})^3$. Suppose $M/\mathbb{Q}_2$ is Galois with $G = \mathrm{Gal}_{\mathbb{Q}_2}(M) \cong (\mathbb{Z}/4\mathbb{Z})^3$. The fixed field of $2G$ has group $G/2G \cong (\mathbb{Z}/2\mathbb{Z})^3$, so it has 7 distinct quadratic subextensions, which by Step 1 are all the quadratic extensions of $\mathbb{Q}_2$. In particular $k = M^H$ for a subgroup $H \supseteq 2G$ of index 2. Identify $G = (\mathbb{Z}/4\mathbb{Z})^3$. Then $H/2G$ is the kernel of a nonzero homomorphism $(\mathbb{Z}/2\mathbb{Z})^3 \to \mathbb{Z}/2\mathbb{Z}$, $(y_1, y_2, y_3) \mapsto \sum_i c_i y_i$, and with $\tilde{c}_i \in \{0, 1\}$ lifting c_i, the homomorphism $\psi\colon G \to \mathbb{Z}/4\mathbb{Z}$, $\psi(x) = \sum_i \tilde{c}_i x_i$, is surjective and has $\ker\psi \subseteq H$, as $\psi(x) = 0$ forces $\sum_i c_i \bar{x}_i = 0$ for the reduction $\bar{x}$ of x modulo 2. The fixed field $N = M^{\ker\psi}$ is then cyclic of degree 4 over $\mathbb{Q}_2$ and contains $M^H = k$, contradicting Step 3. So neither group occurs, as we sought to show.

Theorem 4.3.4: Local Kronecker-Weber Theorem

Every finite abelian extension of $\mathbb{Q}_p$ lies in a Cyclotomic field $\mathbb{Q}_p(\zeta_m)$

***Proof*:**

Step 1: reduction to cyclic extensions of prime-power degree. Let $K/\mathbb{Q}_p$ be finite abelian with group G. By [5, The Fundamental Theorem of f.g. Abelian Groups] and [5, Chinese Remainder Theorem For Integers], $G = C_1 \times \cdots \times C_n$ with each C_i cyclic of prime-power order. Let $H_i \subseteq G$ be the product of the C_j with $j \neq i$, and let $K_i = K^{H_i}$. Then $\mathrm{Gal}_{\mathbb{Q}_p}(K_i) \cong G/H_i \cong C_i$, and the composite $K_1 \cdots K_n$ is the fixed field of $H_1 \cap \cdots \cap H_n = 1$, that is, K. If $K_i \subseteq \mathbb{Q}_p(\zeta_{m_i})$ for each i, then $K \subseteq \mathbb{Q}_p(\zeta_{m_1 \cdots m_n})$. So it suffices to treat $K/\mathbb{Q}_p$ cyclic of degree ℓ^r for a prime ℓ and $r \geq 1$.

Step 2: the case $\ell \neq p$. By theorem 3.3.16 there is an intermediate field F with $[F : \mathbb{Q}_p] = f(K/\mathbb{Q}_p)$ and K/F totally ramified of degree $e = [K : F]$, a power of ℓ, so $p \nmid e$. As residue degrees multiply in towers (proposition 3.3.21), $f(F/\mathbb{Q}_p) = f(K/\mathbb{Q}_p) = [F : \mathbb{Q}_p]$, so $F/\mathbb{Q}_p$ is unramified and p is a uniformizer of $\mathcal{O}_F$. Applied to K/F over the complete DVR $\mathcal{O}_F$, theorem 3.3.22 gives $K = F(\gamma)$ with γ^e a uniformizer of $\mathcal{O}_F$, say $\gamma^e = -pu$ with $u \in \mathcal{O}_F^\times$. Let α be a root of $x^e + p$ and $\beta = \gamma/\alpha$, so that $\beta^e = u$ and $K = F(\gamma) \subseteq F(\alpha, \beta)$.

The element β is a root of $x^e - u \in \mathcal{O}_F[x]$, whose reduction $x^e - \bar{u}$ is separable over the residue field, as its derivative ex^{e-1} vanishes only at 0, which is not a root. So $F(\beta)/F$ is unramified by lemma 3.3.2, and $F(\beta)/\mathbb{Q}_p$ is unramified by proposition 3.3.21. By proposition 3.3.7, $F(\beta)$ is cyclic over $\mathbb{Q}_p$ and isomorphic over $\mathbb{Q}_p$ to $\mathbb{Q}_p(\zeta_{p^{f'}-1})$, where $f' = [F(\beta) : \mathbb{Q}_p]$. The field $\mathbb{Q}_p(\zeta_{p^{f'}-1})$ is the splitting field of $x^{p^{f'}-1} - 1$, so it is the only subfield of the algebraic closure isomorphic to it over $\mathbb{Q}_p$ [5, Finite Normal Extension Equivalences], and $F(\beta) = \mathbb{Q}_p(\zeta_{p^{f'}-1})$.

The extension $\mathbb{Q}_p(\alpha)/\mathbb{Q}_p$ is Galois, as $\alpha = \gamma/\beta$ lies in $K \cdot F(\beta)$, a composite of two abelian extensions of $\mathbb{Q}_p$, which is abelian by [5, Galois Composite Fields II], and every subextension of an abelian extension is Galois. By corollary 2.2.4, $x^e + p$ is irreducible and $\mathbb{Q}_p(\alpha)/\mathbb{Q}_p$ is totally ramified of degree e. Being normal, $\mathbb{Q}_p(\alpha)$ contains all e roots of $x^e + p$, hence their ratios, which are all the e-th roots of unity. The field $\mathbb{Q}_p(\zeta_e)$ is unramified over $\mathbb{Q}_p$ by lemma 3.3.6 and lies in the totally ramified $\mathbb{Q}_p(\alpha)$, so its residue degree divides $f(\mathbb{Q}_p(\alpha)/\mathbb{Q}_p) = 1$, and $\mathbb{Q}_p(\zeta_e) = \mathbb{Q}_p$. The e-th roots of unity then lie in $\mathbb{Z}_p^\times$, and reduction to $\mathbb{F}_p^\times$ is injective on them: if $\xi^e = 1$ and $\xi = 1 + z$ with $\nu_p(z) \geq 1$, then $0 = \xi^e - 1 = ez + \sum_{k \geq 2} \binom{e}{k} z^k$, where for $z \neq 0$ the term ez has valuation $\nu_p(z)$, smaller than that of every other term, so $z = 0$. Hence the e distinct e-th roots of unity embed in $\mathbb{F}_p^\times$, and $e \big| p - 1$. Let λ be a root of $x^{p-1} + p$. By lemma 4.3.1, $\lambda \in \mathbb{Q}_p(\zeta_p)$, and $\lambda^{(p-1)/e}$ is a root of $x^e + p$, so $\alpha = \xi\lambda^{(p-1)/e}$ for some e-th root of unity $\xi \in \mathbb{Q}_p$. Hence $\alpha \in \mathbb{Q}_p(\zeta_p)$, and

$$K \subseteq F(\beta)(\alpha) \subseteq \mathbb{Q}_p(\zeta_{p^{f'}-1})\mathbb{Q}_p(\zeta_p) = \mathbb{Q}_p(\zeta_{(p^{f'}-1)p})$$

Step 3: the case $\ell = p$, set-up. Let $K/\mathbb{Q}_p$ be cyclic of degree p^r with $r \geq 1$. Let U be an unramified extension of $\mathbb{Q}_p$ of degree p^r in the algebraic closure, which exists by theorem 3.3.3. As in Step 2, U is cyclic over $\mathbb{Q}_p$, and $U = \mathbb{Q}_p(\zeta_{p^{p^r}-1})$ by proposition 3.3.7 and [5, Finite Normal Extension Equivalences]. Put $T = \mathbb{Q}_p(\zeta_{p^t})$, where $t = r + 1$ for odd p and $t = r + 2$ for $p = 2$, totally ramified of degree $\varphi(p^t)$ with group $(\mathbb{Z}/p^t\mathbb{Z})^\times$ by lemma 2.2.6. A subfield of both U and T has ramification index and residue degree 1, so $U \cap T = \mathbb{Q}_p$. With $m = (p^{p^r} - 1)p^t$, this gives $\mathbb{Q}_p(\zeta_m) = UT$ and, by [5, Galois Composite Fields II],

$$\mathrm{Gal}_{\mathbb{Q}_p}(\mathbb{Q}_p(\zeta_m)) \cong \mathbb{Z}/p^r\mathbb{Z} \times (\mathbb{Z}/p^t\mathbb{Z})^\times$$

Suppose $K \not\subseteq \mathbb{Q}_p(\zeta_m)$, and let $G = \mathrm{Gal}_{\mathbb{Q}_p}(K(\zeta_m))$, abelian as in Step 2. Restriction to K embeds $\mathrm{Gal}_{\mathbb{Q}_p(\zeta_m)}(K(\zeta_m))$ into the cyclic group $\mathrm{Gal}_{\mathbb{Q}_p}(K)$ of order p^r, and it is nontrivial, so it has order p^s with $1 \leq s \leq r$. Hence $|G| = p^s \cdot p^r \cdot \varphi(p^t)$, and restriction embeds G into $\mathrm{Gal}_{\mathbb{Q}_p}(K) \times \mathbb{Z}/p^r\mathbb{Z} \times (\mathbb{Z}/p^t\mathbb{Z})^\times$.

Step 4: the case $\ell = p$ odd. Here $t = r + 1$ and $\varphi(p^t) = p^r(p - 1)$. Let P be the subgroup of elements of p-power order in G, of order p^{2r+s}. The image of an element of P in each of the three factors above has p-power order dividing that factor's order, which is p^r, p^r and $p^r(p - 1)$

respectively, so every element of P has order dividing p^r. By [5, The Fundamental Theorem of f.g. Abelian Groups], $P \cong \mathbb{Z}/p^{a_1}\mathbb{Z} \times \cdots \times \mathbb{Z}/p^{a_k}\mathbb{Z}$ with $1 \leq a_j \leq r$, so $p^{2r+s} = |P| \leq p^{kr}$ and $k \geq 3$. The projection $G \to P$, followed by $P \to P/pP \cong (\mathbb{Z}/p\mathbb{Z})^k$ and a projection onto three factors, is a surjection $G \to (\mathbb{Z}/p\mathbb{Z})^3$. The fixed field of its kernel is Galois over $\mathbb{Q}_p$ with group $(\mathbb{Z}/p\mathbb{Z})^3$, contradicting lemma 4.3.2. So $K \subseteq \mathbb{Q}_p(\zeta_m)$.

Step 5: the case $\ell = p = 2$. Here $t = r+2$ and $\varphi(2^t) = 2^{r+1}$, so $|G| = 2^{2r+1+s}$. Every odd integer x satisfies $x^{2^r} \equiv 1 \bmod 2^{r+2}$: for $r = 1$, $x^2 - 1 = (x-1)(x+1)$ is a product of two consecutive even integers, hence divisible by 8, and if $x^{2^r} = 1 + 2^{r+2}y$, then $x^{2^{r+1}} = 1 + 2^{r+3}y + 2^{2r+4}y^2$. So $(\mathbb{Z}/2^{r+2}\mathbb{Z})^\times$ has exponent dividing 2^r, every element of G has order dividing 2^r, and [5, The Fundamental Theorem of f.g. Abelian Groups] gives $G \cong \mathbb{Z}/2^{a_1}\mathbb{Z} \times \cdots \times \mathbb{Z}/2^{a_k}\mathbb{Z}$ with $r \geq a_1 \geq \cdots \geq a_k \geq 1$ and $a_1 + \cdots + a_k = 2r+1+s \geq 2r+2$. If $k \geq 4$, then $G \to G/2G \cong (\mathbb{Z}/2\mathbb{Z})^k$ surjects onto $(\mathbb{Z}/2\mathbb{Z})^4$. If $k \leq 3$, then $k = 3$ and $a_3 \geq 2$, since otherwise $a_1 + \cdots + a_k \leq 2r+1$, so G surjects onto $G/4G \cong (\mathbb{Z}/4\mathbb{Z})^3$. Either way $K(\zeta_m)$ contains a Galois extension of $\mathbb{Q}_2$ with group $(\mathbb{Z}/2\mathbb{Z})^4$ or $(\mathbb{Z}/4\mathbb{Z})^3$, contradicting lemma 4.3.3. So $K \subseteq \mathbb{Q}_2(\zeta_m)$.

Steps 2, 4 and 5 place every cyclic extension of $\mathbb{Q}_p$ of prime-power degree in a cyclotomic field, and Step 1 extends this to every finite abelian extension, completing the proof.

Proposition 4.3.5: Local Implies Global Kronecker-Weber

Suppose that for every prime p, every finite abelian extension of $\mathbb{Q}_p$ lies in a cyclotomic field $\mathbb{Q}_p(\zeta_m)$. Then every finite abelian extension of $\mathbb{Q}$ lies in a cyclotomic field $\mathbb{Q}(\zeta_m)$.

***Proof*:**
Let $K/\mathbb{Q}$ be a finite abelian extension. Only finitely many primes ramify in K, namely the primes dividing its discriminant ([2, Discriminant And Ramified Primes]). For each ramified prime p pick a prime $\mathfrak{p}|p$ of K. By theorem 4.2.4, $K_\mathfrak{p}/\mathbb{Q}_p$ is Galois with $\mathrm{Gal}_{\mathbb{Q}_p}(K_\mathfrak{p}) \cong D_\mathfrak{p} \subseteq \mathrm{Gal}_\mathbb{Q}(K)$, so it is abelian, and by hypothesis $K_\mathfrak{p} \subseteq \mathbb{Q}_p(\zeta_{m_p})$ for some $m_p \geq 1$. This containment holds in every algebraic closure of $K_\mathfrak{p}$, since any two are isomorphic over $K_\mathfrak{p}$ by an isomorphism carrying the m_p-th roots of unity of one onto those of the other. Put $n_p = \nu_p(m_p)$,

$$m = \prod_{p \text{ ramified in } K} p^{n_p}$$

and $L = K(\zeta_m) = K \cdot \mathbb{Q}(\zeta_m)$. By [5, Galois Composite Fields II], $L/\mathbb{Q}$ is Galois with group embedding in $\mathrm{Gal}_\mathbb{Q}(K) \times \mathrm{Gal}_\mathbb{Q}(\mathbb{Q}(\zeta_m))$, hence abelian.

Fix a prime p and a prime $\mathfrak{P}|p$ of L, taken above the chosen $\mathfrak{p}$ when p is ramified in K, and let $I_p = I_\mathfrak{P} \subseteq \mathrm{Gal}_\mathbb{Q}(L)$ be its inertia group. We claim that $|I_p| \leq \varphi(p^{\nu_p(m)})$. By [2, Inertia Group Properties] and theorem 4.2.4, $|I_p| = e_\mathfrak{P} = e(L_\mathfrak{P}/\mathbb{Q}_p)$. The closure of K in $L_\mathfrak{P}$ is the completion $K_{\mathfrak{p}'}$ at $\mathfrak{p}' = \mathfrak{P} \cap K$, and $K_{\mathfrak{p}'}(\zeta_m) \subseteq L_\mathfrak{P}$ is a finite extension of $\mathbb{Q}_p$, hence complete (theorem 2.1.9), containing the dense subfield $L = K(\zeta_m)$, so $L_\mathfrak{P} = K_{\mathfrak{p}'}(\zeta_m)$. Work in an algebraic closure of $L_\mathfrak{P}$.

- If p is ramified in K, then $\mathfrak{p}' = \mathfrak{p}$ and $\nu_p(m) = n_p$, so $L_\mathfrak{P} \subseteq \mathbb{Q}_p(\zeta_{m_p})\mathbb{Q}_p(\zeta_m) = \mathbb{Q}_p(\zeta_{\mathrm{lcm}(m_p,m)})$ with $\nu_p(\mathrm{lcm}(m_p,m)) = n_p$. Write $\mathrm{lcm}(m_p,m) = p^{n_p}c$ with $p \nmid c$, and

let $W = \mathbb{Q}_p(\zeta_c)$, unramified over $\mathbb{Q}_p$ by lemma 3.3.6. Then $L_{\mathfrak{P}} \subseteq W(\zeta_{p^{n_p}})$, and as ramification indices multiply in towers (proposition 3.3.21),

$$e(L_{\mathfrak{P}}/\mathbb{Q}_p) \leq e(W(\zeta_{p^{n_p}})/\mathbb{Q}_p) = e(W(\zeta_{p^{n_p}})/W) \leq [W(\zeta_{p^{n_p}}) : W] \leq \varphi(p^{n_p})$$

the last inequality because $\zeta_{p^{n_p}}$ is a root of the p^{n_p}-th cyclotomic polynomial, of degree $\varphi(p^{n_p})$.

- If p is unramified in K, then $p \nmid m$, and $e(K_{\mathfrak{p}'}/\mathbb{Q}_p) = e_{\mathfrak{p}'} = 1$ by theorem 4.2.4. As $\mathbb{Q}_p(\zeta_m)/\mathbb{Q}_p$ is unramified by lemma 3.3.6, the composite $L_{\mathfrak{P}} = K_{\mathfrak{p}'}\mathbb{Q}_p(\zeta_m)$ is unramified over $\mathbb{Q}_p$ by corollary 3.3.11, and $|I_p| = 1 = \varphi(p^0)$.

Let $I \subseteq \mathrm{Gal}_{\mathbb{Q}}(L)$ be the subgroup generated by all the I_p. Only those with $p \mid m$ can be nontrivial, and as $\mathrm{Gal}_{\mathbb{Q}}(L)$ is abelian, I is the image of their product under multiplication, so

$$|I| \leq \prod_{p \mid m} |I_p| \leq \prod_{p \mid m} \varphi(p^{\nu_p(m)}) = \varphi(m) = [\mathbb{Q}(\zeta_m) : \mathbb{Q}]$$

the last equality by [5, Irreducibility Of Cyclotomic Polynomials]. The fixed field L^I is unramified at every prime p. With $L_I = L^{I_{\mathfrak{P}}}$ and $\mathfrak{P}_I = \mathfrak{P} \cap L_I$, [2, Inertia Group Properties] gives $e(\mathfrak{P}/\mathfrak{P}_I) = e_{\mathfrak{P}}$, so $e(\mathfrak{P}_I/p) = 1$ by [2, Multiplicity of Ramification And Inertia]. As $L^I \subseteq L_I$, the same multiplicativity gives $e(\mathfrak{P} \cap L^I/p) = 1$, and as $L^I/\mathbb{Q}$ is Galois, every prime of L^I above p has this ramification index ([2, Galois Extension, Inertia Degree And Ramification Index Independent]). By [2, Discriminant And Ramified Primes] the discriminant of L^I is ± 1, and by [2, Discriminant Nonzero] this forces $L^I = \mathbb{Q}$. Hence $I = \mathrm{Gal}_{\mathbb{Q}}(L)$, and

$$[L : \mathbb{Q}] = |I| \leq [\mathbb{Q}(\zeta_m) : \mathbb{Q}]$$

As $\mathbb{Q}(\zeta_m) \subseteq L$, this gives $L = \mathbb{Q}(\zeta_m)$, and so $K \subseteq L = \mathbb{Q}(\zeta_m)$, as we sought to show.

Theorem 4.3.6: Global Kronecker-Weber Theorem

Every finite abelian extension of $\mathbb{Q}$ lies in a Cyclotomic field $\mathbb{Q}(\zeta_m)$

***Proof*:**
By theorem 4.3.4, for every prime p every finite abelian extension of $\mathbb{Q}_p$ lies in a cyclotomic field $\mathbb{Q}_p(\zeta_m)$. This is the hypothesis of proposition 4.3.5, which then places every finite abelian extension of $\mathbb{Q}$ in a cyclotomic field $\mathbb{Q}(\zeta_m)$, as we sought to show.

4.4 *Application: The Hasse Principle for Conics

This section is not necessary for the rest of this book, but it demonstrates some of the usefulness of local fields. It skims the necessary algebraic curve theory, and *Classical Algebraic Geometry* [9] has the details.

Using the rational root test ([5, Rational Roots Test]) finding rational solutions to any degree

polynomial with 1 variable is simple, and there are necessarily finitely many. Once 2 variables are introduced, the complexity of finding rational solutions increases greatly, and the degree plays an important role in compartmentalizing the complexity. Degree 1 polynomials in two variables are linear, and hence are immediately solvable using linear algebra. A plane cubic whose projective closure is smooth and has a rational point is an *elliptic curve*, and the complexity of finding rational solutions jumps greatly (*Elliptic Curves* [6]). A plane curve of degree $d \geq 4$ whose projective closure is smooth has genus $(d-1)(d-2)/2 \geq 3$, so it has only finitely many rational points by *Faltings's Theorem*, but as of writing this book there is no *constructive method* of finding them.

In the case of degree 2 polynomials with two variables, we can find all rational solutions using the *Hasse Principle*.

Definition 4.4.1: Conic

Let

$$C : ax^2 + by^2 + cxy + dx + ey + f = 0$$

be an arbitrary equation of degree 2. Then if the projective closure of C is smooth, C is called a *conic*.

Naturally, projectivising creates a homogeneous polynomial in 3 variables of degree 2, and hence we can also think of a conic as:

$$C : F(x_1, x_2, x_3) = \sum f_{ij} x_i x_j = 0$$

where $f_{ij} = f_{ji}$ and $\det f_{ij} \neq 0$. By [5, Every Symmetric Form Has an Orthogonal Basis], over any field of characteristic $\neq 2$ a linear change of variables takes any conic to a curve of the form:

$$f_1 x_1^2 + f_2 x_2^2 + f_3 x_3^2 = 0$$

for $f_i \neq 0$.

Lemma 4.4.2: Rational Points on a Conic

Let C be a conic over $\mathbb{Q}$. Then C has either no rational points or infinitely many.

***Proof*:**
Write $q(x, y) = ax^2 + by^2 + cxy + dx + ey + f$, so that C is $q = 0$, and suppose $P = (x_0, y_0)$ is a rational point of C. For $t \in \mathbb{Q}$ the line $L_t \colon y = y_0 + t(x - x_0)$ meets C where

$$g_t(x) = q\big(x, y_0 + t(x - x_0)\big) = 0,$$

and $g_t \in \mathbb{Q}[x]$ has degree at most 2 and the root $x = x_0$. If $g_t = 0$ for some t, then C contains L_t, which has infinitely many rational points. Otherwise, the coefficient of x^2 in g_t is $a + ct + bt^2$, and a, b, c are not all zero, so g_t has degree 2 for all but at most two values of t. For such t the two roots of g_t sum to a rational number, so the second root x_t is rational and $P_t = (x_t, y_0 + t(x_t - x_0))$ is a rational point of C on L_t. Two distinct lines through P meet only at P, so the points $P_t \neq P$ are pairwise distinct. Finally, $P_t = P$ means that x_0 is a double root of g_t, that is, $q_x(P) + t\, q_y(P) = 0$. Since C is smooth at P (its projective closure is smooth at $(x_0 : y_0 : 1)$), the partials $q_x(P)$ and $q_y(P)$ do not both vanish, so this holds for at most one t. Hence infinitely many t give distinct rational points P_t.

Theorem 4.4.3: Local-to-global For Conics

Let C be a conic over $\mathbb{Q}$, with homogeneous equation $F(x_1, x_2, x_3) = 0$. For a field $k \supseteq \mathbb{Q}$, call a nonzero solution of $F = 0$ in k^3 a k-point of C. Then C has a $\mathbb{Q}$-point if and only if C has a $\mathbb{Q}_p$-point for each prime p and an $\mathbb{R}$-point.

Lemma 4.4.4: Minkowski's Lattice Point Theorem

Let Γ be a discrete complete lattice in $\mathbb{R}^n$, with its standard inner product, and let X be a centrally symmetric convex subset of $\mathbb{R}^n$. Then if

$$\mathrm{vol}(X) > 2^n \mathrm{vol}(\Gamma)$$

then X contains at least one nonzero lattice point $\gamma \in \Gamma$

***Proof*:**
The statement coincides with [2, Minkowski's Lattice Point Theorem], which proves it, completing the proof.

***Proof of theorem 4.4.3*:**
Write $F = \sum_{i,j} f_{ij} x_i x_j$ with $f_{ij} = f_{ji} \in \mathbb{Q}$ and $\det(f_{ij}) \neq 0$, as after definition 4.4.1. As $\mathbb{Q} \subseteq \mathbb{Q}_p$ and $\mathbb{Q} \subseteq \mathbb{R}$, a rational point of C is a $\mathbb{Q}_p$-point for every p and an $\mathbb{R}$-point. Conversely, suppose that C has a $\mathbb{Q}_p$-point for every prime p and an $\mathbb{R}$-point.

Step 1: reduction. An invertible linear change of variables over $\mathbb{Q}$, or multiplication of F by a nonzero rational, matches the nonzero zeros of the old and the new form in k^3 for every field $k \supseteq \mathbb{Q}$, so it affects neither the hypothesis nor the conclusion. By [5, Every Symmetric Form Has an Orthogonal Basis], such a change of variables makes $F = f_1 x_1^2 + f_2 x_2^2 + f_3 x_3^2$, with $f_1 f_2 f_3 \neq 0$ as $\det(f_{ij}) \neq 0$, and clearing denominators makes each f_i an integer. If $p^2 \big| f_i$ for a prime p, the substitution $x_i = y_i/p$ replaces f_i by f_i/p^2, so we may take each f_i square-free. Let $N = f_1 f_2 f_3$. If a prime p divides all three f_i, replace F by F/p. If p divides exactly two of them, say f_i and f_j, replace F by pF and substitute $x_i = y_i/p$ and $x_j = y_j/p$, which turns the three coefficients into f_i/p, f_j/p and pf_k, again square-free. Each move lowers $|N|$, so after finitely many moves no prime divides two of the f_i, and N is square-free.

Step 2: a normalized local zero. Let p be a prime and $a \in \mathbb{Q}_p^3$ a nonzero zero of F. Multiplying a by a power of p, we may take $a \in \mathbb{Z}_p^3$ with at least one coordinate in $\mathbb{Z}_p^\times$. Suppose that $p \big| N$, let i be the index with $p \big| f_i$, and let j and k be the other two, so that $p \nmid f_j f_k$ and $\nu_p(f_i) = 1$. If $a_j \in p\mathbb{Z}_p$, then $f_k a_k^2 = -f_i a_i^2 - f_j a_j^2 \in p\mathbb{Z}_p$ gives $a_k \in p\mathbb{Z}_p$, then $f_i a_i^2 = -f_j a_j^2 - f_k a_k^2 \in p^2\mathbb{Z}_p$ gives $a_i \in p\mathbb{Z}_p$, and no coordinate is a unit. Exchanging the roles of j and k, the coordinates a_j and a_k are units.

Step 3: a congruence condition at each prime dividing $2N$. For each prime $p \big| 2N$ there is a surjective homomorphism φ_p from $\mathbb{Z}^3$ onto a finite group A_p such that $|A_p|$ divides $F(x)$ for every $x \in \ker \varphi_p$. To construct it, fix a as in Step 2. Recall that a unit $u = 1 + 2t$ of $\mathbb{Z}_2$ has

$u^2 = 1 + 4t(t+1) \in 1 + 8\mathbb{Z}_2$, as t or $t+1$ lies in $2\mathbb{Z}_2$, while $u^2 \in 4\mathbb{Z}_2$ for $u \in 2\mathbb{Z}_2$.

- *p odd.* Then $p\big|N$. With i, j, k as in Step 2, $f_j a_j^2 + f_k a_k^2 = -f_i a_i^2 \in p\mathbb{Z}_p$. Choose $r \in \mathbb{Z}$ with $r \equiv a_k a_j^{-1} \bmod p\mathbb{Z}_p$. Then $f_j + f_k r^2 \equiv a_j^{-2}(f_j a_j^2 + f_k a_k^2) \equiv 0 \bmod p\mathbb{Z}_p$, so $p\big|f_j + f_k r^2$. Let $\varphi_p \colon \mathbb{Z}^3 \to \mathbb{Z}/p\mathbb{Z}$ send x to the class of $x_k - r x_j$, which is surjective. For $x \in \ker \varphi_p$, $F(x) \equiv (f_j + f_k r^2) x_j^2 \equiv 0 \bmod p$, as $p\big|f_i$.
- *$p = 2$ and N odd.* All f_i are odd, so $0 = F(a) \equiv a_1^2 + a_2^2 + a_3^2 \bmod 2\mathbb{Z}_2$, and an even number of the a_i are units. By Step 2 one of them is, so exactly two are, say a_j and a_k, and $a_i \in 2\mathbb{Z}_2$, where $\{i,j,k\} = \{1,2,3\}$. Then $0 = F(a) \equiv f_j + f_k \bmod 4\mathbb{Z}_2$, so $4\big|f_j + f_k$. Let $\varphi_2 \colon \mathbb{Z}^3 \to (\mathbb{Z}/2\mathbb{Z})^2$ send x to the classes of x_i and $x_j - x_k$, which is surjective. For $x \in \ker \varphi_2$, $4\big|f_i x_i^2$, and x_j, x_k are either both even, so that $4\big|f_j x_j^2 + f_k x_k^2$, or both odd, so that $f_j x_j^2 + f_k x_k^2 \equiv f_j + f_k \equiv 0 \bmod 4$. So $4\big|F(x)$.
- *$p = 2$ and N even.* With i, j, k as in Step 2, $\nu_2(f_i) = 1$ and $a_j^2 \equiv a_k^2 \equiv 1 \bmod 8\mathbb{Z}_2$. If $a_i \in 2\mathbb{Z}_2$, then $f_i a_i^2 \in 8\mathbb{Z}_2$ and $8\big|f_j + f_k$, and we put $s = 0$. Otherwise $a_i^2 \equiv 1 \bmod 8\mathbb{Z}_2$ and $8\big|f_i + f_j + f_k$, and we put $s = 1$. Let $\varphi_2 \colon \mathbb{Z}^3 \to \mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ send x to the classes of $x_j - x_k$ and $x_i - s x_k$, which is surjective. Let $x \in \ker \varphi_2$. Then $x_j^2 - x_k^2 = (x_j - x_k)(x_j + x_k) \in 8\mathbb{Z}$, as $x_j - x_k \in 4\mathbb{Z}$ and $x_j + x_k = 2x_k + (x_j - x_k)$ is even. If x_k is even, then so is x_i, so $8\big|f_i x_i^2$, and $f_j x_j^2 + f_k x_k^2 \equiv (f_j + f_k) x_k^2 \equiv 0 \bmod 8$, as $f_j + f_k$ is even and $4\big|x_k^2$. If x_k is odd, then x_j is odd, $x_j^2 \equiv x_k^2 \equiv 1 \bmod 8$ and $x_i \equiv s \bmod 2$, so $f_i x_i^2 \equiv s f_i \bmod 8$ and $F(x) \equiv s f_i + f_j + f_k \equiv 0 \bmod 8$. So $8\big|F(x)$.

Step 4: the lattice. Let $\varphi = (\varphi_p)_{p|2N} \colon \mathbb{Z}^3 \to \prod_{p|2N} A_p$ and $\Lambda = \ker \varphi = \bigcap_{p|2N} \ker \varphi_p$. Each φ_p factors through reduction modulo d_p, where $d_p = p$ for odd p and $d_2 = 4$, and these moduli are pairwise coprime. Given $(t_p)_p$, pick $w_p \in \mathbb{Z}^3$ with $\varphi_p(w_p) = t_p$. By [5, Chinese Remainder Theorem], applied in each coordinate to the pairwise comaximal ideals $d_p\mathbb{Z}$, some $w \in \mathbb{Z}^3$ has $w \equiv w_p \bmod d_p$ for every p, so that $\varphi(w) = (t_p)_p$. Hence φ is surjective, and by [5, First Isomorphism Theorem]

$$[\mathbb{Z}^3 : \Lambda] = \prod_{p|2N} |A_p| = 4|N|$$

as N is square-free and $|A_2|$ is 4 for odd N and 8 for even N. Put $m = 4|N|$. For $x \in \Lambda$, Step 3 gives $p\big|F(x)$ for each odd $p\big|N$ and $|A_2|\big|F(x)$. These divisors are pairwise coprime with product m, so $m\big|F(x)$.

Step 5: the volume of Λ. By [5, Lagrange Theorem] in $\mathbb{Z}^3/\Lambda$, every $x \in \mathbb{Z}^3$ has $mx \in \Lambda$, so $m\mathbb{Z}^3 \subseteq \Lambda$ and Λ spans $\mathbb{R}^3$. It is discrete, as $\mathbb{Z}^3$ meets the open unit ball only in 0. So Λ is a lattice in $\mathbb{R}^3$, free of rank 3 by [2, Equivalence Of Conditions], and lemma 4.4.4 applies to it. A $\mathbb{Z}$-basis v_1, v_2, v_3 of Λ spans $\mathbb{R}^3$, so it is an $\mathbb{R}$-basis, and its fundamental mesh $\Phi = \{\sum_l t_l v_l \mid 0 \le t_l < 1\}$ has Lebesgue volume $\mathrm{vol}(\Lambda)$ by [2, Volume of Lattice]. Splitting each coordinate of $y = \sum_l y_l v_l$ into its integer and fractional parts shows that every $y \in \mathbb{R}^3$ is uniquely $\eta + \mu$ with $\eta \in \Phi$ and $\mu \in \Lambda$. The same splitting in the standard basis writes every $y \in \mathbb{R}^3$ uniquely as $\kappa + z$ with $\kappa \in B = [0,1)^3$ and $z \in \mathbb{Z}^3$. Let $c_1, \ldots, c_m$ represent the cosets of Λ in $\mathbb{Z}^3$. As Lebesgue measure

is countably additive and translation invariant,

$$\mathrm{vol}(\Lambda) = \sum_{z\in\mathbb{Z}^3} \mathrm{vol}\big(\Phi \cap (B+z)\big) = \sum_{l=1}^{m}\sum_{\mu\in\Lambda} \mathrm{vol}\big((\Phi-\mu-c_l)\cap B\big) = \sum_{l=1}^{m} \mathrm{vol}(B) = m$$

The first equality holds as the sets $B+z$ partition $\mathbb{R}^3$. The second writes $z = c_l + \mu$ and translates by $-z$. The third holds as, for each l, the sets $\Phi - \mu - c_l$ with $\mu \in \Lambda$ partition $\mathbb{R}^3$.

Step 6: a rational point. Let $q(x) = |f_1|x_1^2 + |f_2|x_2^2 + |f_3|x_3^2$ and $S = \big\{x \in \mathbb{R}^3 \;\big|\; q(x) < m\big\}$. Then S is centrally symmetric, and convex, since for $x, x' \in \mathbb{R}^3$ and $t \in [0,1]$

$$q(tx + (1-t)x') = tq(x) + (1-t)q(x') - t(1-t)q(x-x') \le tq(x) + (1-t)q(x')$$

It is the image of the open ball of radius $\sqrt{m}$, of volume $\frac{4\pi}{3}m^{3/2}$, under the diagonal linear map with diagonal entries $|f_1|^{-1/2}$, $|f_2|^{-1/2}$ and $|f_3|^{-1/2}$, which multiplies Lebesgue measure by the absolute value $|N|^{-1/2}$ of its determinant. As $m^{3/2} = 8|N|^{3/2}$,

$$\mathrm{vol}(S) = \frac{4\pi}{3}\frac{m^{3/2}}{\sqrt{|N|}} = \frac{32\pi}{3}|N| = \frac{8\pi}{3}m > 2^3 m = 2^3\mathrm{vol}(\Lambda)$$

By lemma 4.4.4 with $n = 3$, some nonzero $\lambda \in \Lambda$ lies in S. Then $|F(\lambda)| \le q(\lambda) < m$, and $m \big| F(\lambda)$ by Step 4, so $F(\lambda) = 0$. Thus $\lambda \in \mathbb{Z}^3 \setminus \{0\}$ is a rational point for the form of Step 1, and undoing the changes of Step 1 gives a rational point of C. Only the $\mathbb{Q}_p$-points for $p \big| 2N$ entered the argument, completing the proof.

5

Places: Generalizing Primes

As we saw, primes uniquely correspond to discrete valuations (or equivalence classes of absolute values). From this perspective, the Archimedean prime could be thought of as the "prime at infinity". This can in fact be made very precise, and it must be taken into account when studying *ray class fields* in the sequel [10].

5.1 Places and Completions

Definition 5.1.1: Places

Let K be a field. Then a *place* of K is an equivalence class of nontrivial absolute values on K. The set of places of K shall be denoted M_K

Recall that the completions of K are in correspondence with absolute values and are invariant under equivalent absolute values, and hence completions of K are in correspondence with places. If $|-|_\nu$ is some representative absolute value of a place, then K_ν shall denote its completion. If a place is Archimedean, the field K_ν shall be called Archimedean, and nonarchimedean otherwise. If K is a global field, then the completion at a place gives a local field, which if ν is Archimedean we know that $K_\nu \cong \mathbb{R}$ or $K_\nu \cong \mathbb{C}$, and it is nonarchimedean then the absolute value of K_ν is induced by a discrete valuation, which we shall denote ν (this motivated the symbol for general fields, as the discrete valuation is independent of the choice of absolute value representative).

Example 5.1.2: Places

1. When $K = \mathbb{Q}$, this gives the familiar places given by the primes as well as ∞.
2. When $K = \mathbb{F}_q(t)$ for $q = p^r$, the places correspond to the monic irreducible polynomials of

$\mathbb{F}_q[t]$ along with a *nonarchimedean* place at infinity $|-|_\infty = q^{\deg(-)}$.

The other places that shall be considered are extensions of these fields.

Definition 5.1.3: Types of Places

Let K be a global field and ν a place. Then:

1. $K_\nu \cong \mathbb{R}$, ν is said to be a *real place*
2. $K_\nu \cong \mathbb{C}$, ν is said to be a *complex place*
3. If $|-|_\nu$ is induced by a discrete valuation $\nu_\mathfrak{p}$ for a prime $\mathfrak{p} \subseteq K$, then ν is a *finite place*. Otherwise it is an *infinite place*.

Every finite place must be nonarchimedean, every infinite place is Archimedean in characteristic 0 and nonarchimedean otherwise, every Archimedean place is an infinite place, but nonarchimedean may be finite or infinite (as demonstrated in example 5.1.2).

Definition 5.1.4: Extending Places

Lt L/K be an extension of global fields. Then for every ω of L, any absolute value $|-|_w$ that represents the equivalence class w restricts to an absolute value on K that represents a place ν of K independent of choice of $|-|_w$. Then we write $w|\nu$ and say that w *extends* ν or that w *lies above* ν

Here now comes the most important results about places that generalize the Archimedean embedding and theorem 4.2.4(5):

Theorem 5.1.5: Extending Global Ring Tensoring Local Ring

let L/K be a finite separable extension of global fields and let ν be a place of K. Then:

$$L \otimes_K K_\nu \cong \prod_{w|\nu} L_w$$

given by

$$\ell \otimes x \mapsto (\ell x, ..., \ell x)$$

Note that this was already proven for the nonarchimedean places in theorem 4.2.4.

***Proof*:**
Let ν be a place on K. First off, the separable extension L/K is a finite étale K-algebra, so the base change $L \otimes_K K_v$ is a finite étale K_v-algebra, and is therefore isomorphic to a finite product $\prod_{i\in I} L_i$ of finite separable extensions L_i of K_v, each of which is a local field (any finite extension of a local field is a local field). Thus, it suffices to show that there is a bijective correspondence between the sets of local fields and localizations:

$$\{L_i : i \in I\} \qquad \leftrightarrow \qquad \{L_w : w|v\}.$$

To that end, fix an absolute value $|-|_v$ on K_v. As L_i is a local field extending K_v, when ν is nonarchimedean by theorem 2.1.9 there is a unique absolute value $|-|_w$ that restricts to $|-|_v$; when v is Archimedean it is immediate since then either $K_v \cong L_w$ or $K_v \cong \mathbb{R} \subseteq \mathbb{C} \cong L_w$ and the Euclidean absolute value on $\mathbb{R}$ is the restriction of the Euclidean absolute value on $\mathbb{C}$. Then the map $L \hookrightarrow L \otimes_K K_v \cong \prod_i L_i \twoheadrightarrow L_i$ allows us to view L as a subfield of each L_i ($L \subseteq L_i$), so $|-|_w$ on L_i restricts to L that uniquely determines a place $w|v$. There is thus a well-defined map

$$\varphi : \{L_i : i \in I\} \to \{L_w : w|v\}$$

Then φ must be shown to be bijective and $\varphi(L_i) \cong L_i$.

For $\varphi(L_i) \cong L_i$, note that $L \otimes_K K_v \cong \prod_i L_i$ as an isomorphism of topological rings as both sides are finite dimensional vector spaces over the complete field K_v and thus have a unique topology induced by the sup-norm (which by uniqueness of norm-topologies on f.d. vector spaces, is the same as the topology on$\prod_i L_i$). Next, the image of the canonical embedding $L \hookrightarrow L \otimes_K K_v$ defined by $\ell \mapsto \ell \otimes 1$ is dense because $K \subseteq L$ is dense in K_v: given any $\ell \otimes x$ in $L \otimes_K K_v$ with $\ell \in L$ and $x \in K_v$, choose $y \in K^\times$ arbitrarily close to x so that $\ell y \otimes 1 = \ell \otimes y$ is an element of the image of L arbitrarily close to $\ell \otimes x$ (and similarly for sums of pure tensors). The image of L is therefore also dense in $\prod_i L_i$ and has dense image under the projections $\prod_i L_i \twoheadrightarrow L_i$ and $\prod_i L_i \to L_i \times L_j$ ($i \neq j$). Now, if $\varphi(L_i) = L_w$ then $L_i \cong L_w$. Both are complete fields in which L is dense, and their absolute values restrict to the same absolute value on L. The identity of L is therefore an isometry between dense subsets, so it extends through Cauchy sequences to an isometry $L_w \to L_i$. The extension is a ring homomorphism by continuity, and it is onto because its image is complete, hence closed, and contains the dense subfield L.

For injectivity of φ, note that if $\varphi(L_i) = \varphi(L_j) = L_w$ for some $i \neq j$ then image of the diagonal embedding $L \to L_w \times L_w$ is not dense in $L_w \times L_w$ (its closure is isomorphic to L_w!), but the image of L is dense in $L_i \times L_j$; a contradiction.

For surjectivity of φ, for each $w|v$ define a continuous homomorphism of finite étale K_v-algebras and topological rings:

$$\varphi_w : L \otimes_K K_v \to L_w \qquad \ell \otimes x \mapsto \ell x.$$

The map φ_w is surjective because its image contains L and is complete, and L_w is the completion of L. It follows that φ_w factors through the projection of $L \otimes_K K_v \cong \prod_i L_i$ on to one of its factors L_i and induces a homeomorphism from L_i to L_w. But then $L_i \cong L_w$ as topological fields, so $\varphi(L_i) = L_w$ and φ is surjective.

Corollary 5.1.6: Places and Irreducible Factors

Let L/K be a finite separable extension of global fields, ν a place of K, and $f \in K[x]$ an irreducible polynomial such that $L \cong K[x]/(f(x))$. Then there is a bijective correspondence between the irreducible factors of f in $K_\nu[x]$ and the places of L lying above ν. In particular, if $f = f_1 \cdots f_r$ are the factors of f in $K_\nu[x]$, then:

$$L_{w_i} \cong \frac{K_\nu[x]}{(f_i(x))} \qquad 1 \leq i \leq r$$

for appropriate w_i's.

***Proof*:**
Let $\alpha \in L$ be the image of x under the isomorphism $L \cong K[x]/(f(x))$, so that $L = K(\alpha)$ and f is the minimal polynomial of α. Since L/K is separable, f is a separable polynomial, so its irreducible factors $f_1, \dots, f_r$ in $K_\nu[x]$ are distinct, and in particular pairwise coprime. The Chinese Remainder Theorem thus gives isomorphisms of K_ν-algebras:

$$L \otimes_K K_\nu \cong \frac{K[x]}{(f(x))} \otimes_K K_\nu \cong \frac{K_\nu[x]}{(f(x))} \cong \prod_{i=1}^{r} \frac{K_\nu[x]}{(f_i(x))}$$

where each $K_\nu[x]/(f_i(x))$ is a finite field extension of K_ν. By theorem 5.1.5 there is also an isomorphism:

$$L \otimes_K K_\nu \cong \prod_{w|\nu} L_w$$

Now, in a finite product $R = R_1 \times \cdots \times R_m$ of fields, every ideal is a product of ideals of the factors, so the maximal ideals of R are exactly the m kernels of the projections $R \to R_i$, and the quotient of R by the i-th of them is R_i. Applying this to the two decompositions of $L \otimes_K K_\nu$ above, each maximal ideal $\mathfrak{m}$ of $L \otimes_K K_\nu$ picks out exactly one index i with $(L \otimes_K K_\nu)/\mathfrak{m} \cong K_\nu[x]/(f_i(x))$ and exactly one place $w|\nu$ with $(L \otimes_K K_\nu)/\mathfrak{m} \cong L_w$. Sending the one to the other over each maximal ideal is thus a bijection between $\{f_1, \dots, f_r\}$ and $\{w \mid w|\nu\}$ under which:

$$L_{w_i} \cong \frac{K_\nu[x]}{(f_i(x))}$$

as we sought to show.

Corollary 5.1.7: Classifying Embeddings

Let L/K be a finite separable extension of global fields and ν a place of K. Then given a fixed algebraic closure $\overline{K}_\nu$, there is a bijection

$$\frac{\mathrm{Hom}_K(L, \overline{K}_\nu)}{\mathrm{Gal}_{K_\nu}(\overline{K}_\nu)} \leftrightarrow \{w|\nu\}$$

where the left hand side is the $\mathrm{Gal}_{K_\nu}(\overline{K}_\nu)$-orbits of K-embeddings of L into $\overline{K}_\nu$, and the right hand side is the places of L above ν.

***Proof*:**
Let $L \cong K(\alpha) = k[x]/(f)$ for some $\alpha \in L$ with minimal polynomial $f \in K[x]$. Then from field theory, there is a bijection between $\mathrm{Hom}_K(L, \overline{K}_\nu)$ and the roots of α_i of f in $\overline{K}_\nu$ that is compatible with the action of $\mathrm{Gal}_{K_\nu}(\overline{K}_\nu)$ on both sets.

Now, if $f = f_1 \cdots f_r$ is the factorization of f in $K_\nu[x]$, each f_i corresponds to an orbit of the galois action on the roots of f, which by corollary corollary 5.1.6 we get a bijective correspondence with the places of L above ν, as we sought to show.

In the special case of $K = \mathbb{Q}$ and $v = \infty$, we have that

$$\frac{\mathrm{Hom}_{\mathbb{Q}}(L, \mathbb{C})}{\mathrm{Gal}_{\mathbb{R}}(\mathbb{C})} \leftrightarrow \{w|\infty\}$$

which recovers the real embeddings and the conjugate pairs of complex embeddings that index the coordinates of the Archimedean embedding [2, Archimedean Embedding].

Definition 5.1.8: Real and Complex Place

Let K be a number field. Then the elements of $\mathrm{Hom}_{\mathbb{Q}}(K, \mathbb{R})$ are the *real embeddings* and the elements of $\mathrm{Hom}_{\mathbb{Q}}(K, \mathbb{C})$ whose image does not lie strictly in $\mathbb{R}$ are called the *complex embeddings*

Corollary 5.1.9: Index Given Real and Complex Embeddings

Let K be a number field with r real places and s complex places. Then:

$$[K : \mathbb{Q}] = r + 2s$$

***Proof*:**
This is computation: take $K \cong \mathbb{Q}[x]/(f)$ for some irreducible separable $f \in \mathbb{Q}[x]$ so that:

$$[K : \mathbb{Q}] = \deg f = \#\mathrm{Hom}_{\mathbb{Q}}(K, \mathbb{C})$$

Then the action of $\mathrm{Gal}_{\mathbb{R}}(\mathbb{C})$ on $\mathrm{Hom}_{\mathbb{Q}}(K, \mathbb{C})$ as r orbits of size 1 and 2 orbits of size 2, and the rest follows.

Example 5.1.10: Real and Complex Places in Non-trivial Field
Take $K = \mathbb{Q}[x]/(x^3 - 2)$. Then there are three embeddigns $K \hookrightarrow \mathbb{C}$ for each root given by:

$$x \mapsto \sqrt[3]{2} \qquad x \mapsto \zeta_3 \sqrt[3]{2} \qquad x \mapsto \zeta_3^2 \sqrt[3]{2}$$

The first is a real embedding, the other two complex and conjugate to each other. Thus, K has a real place and a complex place, and we get that $[K : \mathbb{Q}] = 1 \cdot 1 + 2 \cdot 1 = 3$.

5.2 Extending Prime Factorization to Places

In this section, we shall continue the analogy between primes and places by showing that they respect many of the similar identities and properties that prime ideals do. We want to show an analogy for residue field, norm, inertia degree and ramification index, the fundamental identity, picard group, and divisors. In this section, we shall use the words prime and place interchangeably, distinguishing them as finite and infinite primes.

Let us first define the residue field for infinite primes. We shall see that

$$\kappa(\mathfrak{p}) = K_{\mathfrak{p}} \qquad \mathfrak{p} \big| \infty$$

will be the right notion. Now, for each prime $\mathfrak{p}$ we define the (canonical) homomorphism:

$$\nu_{\mathfrak{p}} : K^{\times} \to \mathbb{R}$$

where if $\mathfrak{p}$ is finite, it is the usual $\mathfrak{p}$-adic exponential valuation and $\nu_{\mathfrak{p}}(K^\times) = \mathbb{Z}$, and if $\mathfrak{p}$ is infinite then:

$$\nu_{\mathfrak{p}}(a) = -\log(|\tau a|)$$

where $\tau : K \to \mathbb{C}$ is the embedding which defined $\mathfrak{p}$. Now, for any $\mathfrak{p}/p$ (whether p is finite or infinite, define:

$$f_{\mathfrak{p}} = [\kappa(\mathfrak{p}) : \kappa(p)]$$

Where if $\mathfrak{p}\big|\infty$ we have $f_{\mathfrak{p}} = [K_{\mathfrak{p}} : \mathbb{R}]$ and

$$\mathfrak{N}(\mathfrak{p}) = \begin{cases} p^{f_{\mathfrak{p}}} & \mathfrak{p} \nmid \infty \\ e^{f_{\mathfrak{p}}} & \mathfrak{p}\big|\infty \end{cases}$$

where e here is Euler's number, not the ramification. In fact, we may consider e as an infinite prime number, and the extension $\mathbb{C}/\mathbb{R}$ can be thought of as an unramified extension with inertia degree 2. Next, define the absolute value $|-|_{\mathfrak{p}} : K \to \mathbb{R}$ via:

$$|a|_{\mathfrak{p}} = \mathfrak{N}(\mathfrak{p})^{-\nu_{\mathfrak{p}}(a)}$$

and $|0|_{\mathfrak{p}} = 0$. If $\mathfrak{p}$ is an infinite prime with associated embedding $\tau : K \to \mathbb{C}$, notice that:

$$|a|_{\mathfrak{p}} = |\tau a| \qquad |a|_{\mathfrak{p}} = |\tau a|^2$$

where $\mathfrak{p}$ is real and complex respectively.

Next, if L/K is a finite separable extension of K, denote the primes $\mathfrak{P}/\mathfrak{p}$ of L the valuations in the class $\mathfrak{P}$ which, when restricted to K, give $\mathfrak{p}$. In the case of an infinite prime $\mathfrak{P}$, we get the inertia degree and ramification index to be:

$$f_{\mathfrak{P}/\mathfrak{p}} = [L_{\mathfrak{P}} : K_{\mathfrak{p}}] \qquad e_{\mathfrak{P}/\mathfrak{p}} = 1$$

Thus, we may general the fundamental identity and many other results

Proposition 5.2.1: Fundamental Identity For Places

Let $\mathfrak{P}/\mathfrak{p}$ be an extension of primes (finite or infinite). Then:

1. $\sum_{\mathfrak{P}/\mathfrak{p}} e_{\mathfrak{P}/\mathfrak{p}} f_{\mathfrak{P}/\mathfrak{p}} = \sum_{\mathfrak{P}/\mathfrak{p}} [L_{\mathfrak{P}} : K_{\mathfrak{p}}] = [L : K]$
2. $\mathfrak{N}(\mathfrak{P}) = \mathfrak{N}(\mathfrak{p})^{f_{\mathfrak{P}/\mathfrak{p}}}$
3. $\nu_{\mathfrak{P}}(a) = e_{\mathfrak{P}/\mathfrak{p}} \nu_{\mathfrak{p}}(a)$ (for $a \in K^\times$)
4. $\nu_{\mathfrak{p}}(N_{L_{\mathfrak{P}}/K_{\mathfrak{p}}}(a)) = f_{\mathfrak{P}/\mathfrak{p}} \nu_{\mathfrak{P}}(a)$ (for $a \in L^\times$)
5. $|a|_{\mathfrak{P}} = |N_{L_{\mathfrak{P}}/K_{\mathfrak{p}}}(a)|_{\mathfrak{p}}$ (for $a \in L$)

***Proof*:**
Fix $\mathfrak{P}/\mathfrak{p}$ and write $e = e_{\mathfrak{P}/\mathfrak{p}}$ and $f = f_{\mathfrak{P}/\mathfrak{p}}$. When applied to elements of the completions, $\nu_{\mathfrak{p}}$ and $\nu_{\mathfrak{P}}$ denote the canonical extensions of these valuations to $K_{\mathfrak{p}}^\times$ and $L_{\mathfrak{P}}^\times$, given by the same formulas.

(1) By theorem 5.1.5 we have $L \otimes_K K_\mathfrak{p} \cong \prod_{\mathfrak{P}/\mathfrak{p}} L_\mathfrak{P}$, and comparing $K_\mathfrak{p}$-dimensions gives $\sum_{\mathfrak{P}/\mathfrak{p}} [L_\mathfrak{P} : K_\mathfrak{p}] = \dim_{K_\mathfrak{p}}(L \otimes_K K_\mathfrak{p}) = [L : K]$. For the remaining equality, if $\mathfrak{p}$ is finite then $[L_\mathfrak{P} : K_\mathfrak{p}] = ef$ by theorem 4.2.4(3)(4), and if $\mathfrak{p}$ is infinite then $e = 1$ and $f = [L_\mathfrak{P} : K_\mathfrak{p}]$ by definition.

(2) Suppose first that $\mathfrak{p} \nmid \infty$, and let p be the rational prime below $\mathfrak{p}$. The residue field tower $\kappa(\mathfrak{P})/\kappa(\mathfrak{p})/\kappa(p)$ gives $f_\mathfrak{P} = [\kappa(\mathfrak{P}) : \kappa(p)] = [\kappa(\mathfrak{P}) : \kappa(\mathfrak{p})][\kappa(\mathfrak{p}) : \kappa(p)] = f f_\mathfrak{p}$, hence:

$$\mathfrak{N}(\mathfrak{P}) = p^{f_\mathfrak{P}} = \left(p^{f_\mathfrak{p}}\right)^f = \mathfrak{N}(\mathfrak{p})^f$$

If $\mathfrak{p} \mid \infty$ then $f_\mathfrak{P} = [L_\mathfrak{P} : \mathbb{R}] = [L_\mathfrak{P} : K_\mathfrak{p}][K_\mathfrak{p} : \mathbb{R}] = f f_\mathfrak{p}$, and the same computation with the base of the natural logarithm in place of p gives $\mathfrak{N}(\mathfrak{P}) = \mathfrak{N}(\mathfrak{p})^f$.

(3) If $\mathfrak{p}$ is finite, this is the statement that $\nu_\mathfrak{P}$ extends $\nu_\mathfrak{p}$ with index e (theorem 3.1.2). If $\mathfrak{p}$ is infinite, then $e = 1$. Let $\tau' : L \to \mathbb{C}$ be an embedding defining $\mathfrak{P}$. The restriction to K of the absolute value $|\tau'(-)|$ representing $\mathfrak{P}$ is $|\tau(-)|$ for $\tau = \tau'|_K$, so τ is an embedding defining $\mathfrak{p}$, and for $a \in K^\times$:

$$\nu_\mathfrak{P}(a) = -\log|\tau' a| = -\log|\tau a| = \nu_\mathfrak{p}(a)$$

(4) If $\mathfrak{p}$ is finite, then $L_\mathfrak{P}/K_\mathfrak{p}$ is a finite separable extension of fraction fields of complete DVRs whose inertia degree is f by theorem 4.2.4(1)(3), and so corollary 3.1.6 gives $\nu_\mathfrak{p}(N_{L_\mathfrak{P}/K_\mathfrak{p}}(a)) = f\nu_\mathfrak{P}(a)$ for $a \in L^\times$. If $\mathfrak{p}$ is infinite there are two cases. If $L_\mathfrak{P} = K_\mathfrak{p}$ then $f = 1$ and the norm is the identity. Otherwise $K_\mathfrak{p} \cong \mathbb{R}$ and $L_\mathfrak{P} \cong \mathbb{C}$, so $f = 2$, $N_{L_\mathfrak{P}/K_\mathfrak{p}}(a) = a\overline{a}$, and:

$$\nu_\mathfrak{p}(N_{L_\mathfrak{P}/K_\mathfrak{p}}(a)) = -\log|a\overline{a}| = -2\log|a| = 2\nu_\mathfrak{P}(a)$$

(5) For $a = 0$ both sides are 0. For $a \neq 0$, combining (2) and (4):

$$|a|_\mathfrak{P} = \mathfrak{N}(\mathfrak{P})^{-\nu_\mathfrak{P}(a)} = \mathfrak{N}(\mathfrak{p})^{-f\nu_\mathfrak{P}(a)} = \mathfrak{N}(\mathfrak{p})^{-\nu_\mathfrak{p}(N_{L_\mathfrak{P}/K_\mathfrak{p}}(a))} = |N_{L_\mathfrak{P}/K_\mathfrak{p}}(a)|_\mathfrak{p}$$

completing the proof.

Proposition 5.2.2: Product Formula for Number Fields

Let K be a number field and $a \in K^\times$. Then $|a|_\mathfrak{p} = 1$ for all but finitely many primes $\mathfrak{p}$ of K, and

$$\prod_\mathfrak{p} |a|_\mathfrak{p} = 1$$

where the product runs over all primes of K, finite and infinite.

***Proof*:**
Write $a = b/c$ with $b, c \in \mathcal{O}_K \setminus \{0\}$. A finite prime $\mathfrak{p}$ with $\nu_\mathfrak{p}(a) \neq 0$ occurs in the factorization of $b\mathcal{O}_K$ or of $c\mathcal{O}_K$ [2, Unique Factorization Of Ideals In Dedekind Domains], so there are finitely many such $\mathfrak{p}$. Every prime $\mathfrak{p}$ of K lies over exactly one place p of $\mathbb{Q}$ (definition 5.1.4), and by proposition 5.2.1(1) only finitely many primes of K lie over ∞. So $|a|_\mathfrak{p} = 1$ for all but finitely

many $\mathfrak{p}$, and

$$\prod_{\mathfrak{p}} |a|_{\mathfrak{p}} = \prod_{p} \prod_{\mathfrak{p}|p} |a|_{\mathfrak{p}}$$

the outer product running over the places p of $\mathbb{Q}$.

Fix a place p of $\mathbb{Q}$. By theorem 5.1.5, $\ell \otimes x \mapsto (\ell x)_{\mathfrak{p}}$ is an isomorphism of $\mathbb{Q}_p$-algebras $K \otimes_{\mathbb{Q}} \mathbb{Q}_p \to \prod_{\mathfrak{p}|p} K_{\mathfrak{p}}$. The norm $N_{K/\mathbb{Q}}(a)$ is the determinant of multiplication by a on K, and a $\mathbb{Q}$-basis of K is a $\mathbb{Q}_p$-basis of $K \otimes_{\mathbb{Q}} \mathbb{Q}_p$ in which multiplication by $a \otimes 1$ has the same matrix. Under the isomorphism, multiplication by $a \otimes 1$ becomes multiplication by a on each factor $K_{\mathfrak{p}}$, so its determinant is also $\prod_{\mathfrak{p}|p} N_{K_{\mathfrak{p}}/\mathbb{Q}_p}(a)$. Hence $N_{K/\mathbb{Q}}(a) = \prod_{\mathfrak{p}|p} N_{K_{\mathfrak{p}}/\mathbb{Q}_p}(a)$, and proposition 5.2.1(5) gives

$$\prod_{\mathfrak{p}|p} |a|_{\mathfrak{p}} = \prod_{\mathfrak{p}|p} |N_{K_{\mathfrak{p}}/\mathbb{Q}_p}(a)|_p = |N_{K/\mathbb{Q}}(a)|_p$$

Taking the product over all places p of $\mathbb{Q}$, proposition 1.1.10 applied to $N_{K/\mathbb{Q}}(a) \in \mathbb{Q}^\times$ gives $\prod_{\mathfrak{p}} |a|_{\mathfrak{p}} = \prod_p |N_{K/\mathbb{Q}}(a)|_p = 1$, completing the proof.

5.2.1 Galois Theory of Global Fields

Next, let us take a moment to extend the theory of places when L/K is a galois extension of global field. Then we have a galois group $\mathrm{Gal}_K(L)$ acting on the set of places ω of L via $\omega \mapsto \sigma(w)$ where $\sigma(w)$ is the equivalence classes of absolute valued given by

$$\|\alpha\|_{\sigma(w)} = \|\sigma(\alpha)\|_w$$

This generalizes the case from the finite prime case. We can generalize the decomposition group:

Definition 5.2.3: Decomposition Group For Places

Let L/K be a galois extension of global fields and let w be a place of L. Then the *decomposition group* of w is the stabilizer:

$$D_w = \{\sigma \in \mathrm{Gal}_K(L) \mid \sigma(w) = w\}$$

We shall see that in this case we can get infinite primes that ramify. By letting $L \cong \mathbb{Q}[x]/(f)$, we get a one-to-one correspondence between the embeddings of L into $\mathbb{C}$ with the root of $f \in \mathbb{C}$. Then each embedding of L into $\mathbb{C}$ restricts to an embedding of K into $\mathbb{C}$. This map induces a map that sends each infinite place w of L to an infinite place v of K that extends to w. Now, this map may send a complex place to a real place, which happens when a pair of distinct complex conjugate embeddings of L restrict to the same embedding of K (implying it must be a real embedding). In this case, we say that v (and w) is *ramified* in the extension L/K, and the ramification index is $e_\nu = 2$ (and $e_\nu = 1$ otherwise). In this case, we shall also have $f_\nu = 1$ for $\nu|\infty$, and $g_\nu = \#\{w|\nu\}$. With all of this, we may generalize the fundamental identity to:

$$e_\nu f_\nu g_\nu = [L : K]$$

and more generally:

Definition 5.2.4: Ramification Index For Global Infinite Primes

Let L/K be a galois extension of number fields L/K. Then:

$$e_0(L/K) = \prod_{\nu \nmid \infty} e_\nu \qquad e_\infty(L/K) = \prod_{\nu \mid \infty} e_\nu$$

$$e(L/K) = e_0(L/K)e_\infty(L/K)$$

Now as L/K is galois, it induces a transitive action on the embeddings and their corresponding places. As $L \cong k[x]/(g)$, each embedding of K into $\mathbb{C}$ gives rise to $[L:K]$ distinct embeddings of L into $\mathbb{C}$ that extend it, one for each root of g in $\mathbb{C}$. Thus ,for each infinite place ν of K, the galois group acts transitively on $\{w|\nu\}$, and so either all of the places $w|\nu$ are ramified or none are, which is determined by the decomposition group for places (namely, whether the decomposition group is order 1 or 2).

Let us conclude by generalizing Dirichlet's unit theorem. The following refinement is due to Herbrand, who also introduced the Herbrand quotient used in the sequel [10].

Theorem 5.2.5: Herbrand Unit Theorem

Let L/K be a galois extension of number field. Let $w_1, ..., w_r$ be the real places of L and $w_{r+1}, ..., w_{r+s}$ be the complex places of L. Then there exists $\epsilon_1, ..., \epsilon_{r+s} \in \mathcal{O}_L^\times$ such that

1. $\sigma(\epsilon_i) = \epsilon_j$ if and only if $\sigma(w_j) = w_i$ for all $\sigma \in \mathrm{Gal}_K(L)$
2. $\epsilon_1, ..., \epsilon_{r+s}$ generate a finite index subgroup of $\mathcal{O}_L^\times$
3. $\epsilon_1\epsilon_2 \cdots \epsilon_{r+s} = 1$ and every relation among the ϵ_i is generated by this one.

***Proof*:**

Let $G = \mathrm{Gal}_K(L)$ and $n = r + s$. If $n = 1$, then $\mathcal{O}_L^\times = \mu(L)$ is finite [2, Dirichlet Unit Theorem], and $\epsilon_1 = 1$ has the three properties. Let $n \geq 2$. For an archimedean place w write $\|x\|_w = |x|_w$ for its normalized absolute value (section 5.2), so that $\|\sigma(x)\|_w = \|x\|_{\sigma(w)}$ by the definition of the action, and $(\sigma\tau)(w) = \tau(\sigma(w))$, as $\|x\|_{(\sigma\tau)(w)} = \|\tau(x)\|_{\sigma(w)}$. The map $\lambda(x) = (\log\|x\|_{w_1}, \ldots, \log\|x\|_{w_n})$ is the logarithmic embedding of the unit group, with one coordinate for each archimedean place. The logarithmic embedding $x \mapsto (\log|\tau(x)|)_\tau$, with one coordinate for each embedding $\tau\colon L \to \mathbb{C}$, takes values in the vectors (x_τ) with $x_{\overline{\tau}} = x_\tau$, and the linear isomorphism sending such a vector to the vector with entry x_τ at a real place and $x_\tau + x_{\overline{\tau}}$ at the complex place of a pair $\tau, \overline{\tau}$ carries it to λ, and the trace-zero hyperplane onto the hyperplane $H \subseteq \mathbb{R}^n$ of vectors with coordinate sum 0. So the kernel of λ on $\mathcal{O}_L^\times$ is the finite group $\mu(L)$ and its image Γ lies in H [2, Roots Of Unity In Group Of Units], Γ is discrete [2, Multiplicative Group Forms Lattice], and Γ is free of rank $n - 1 = \dim H$ [2, Dirichlet Unit Theorem]. A discrete subgroup of H that is free of rank $\dim H$ spans H [2, Equivalence Of Conditions], so a $\mathbb{Z}$-basis $g_1, \ldots, g_{n-1}$ of Γ is an $\mathbb{R}$-basis of H, and every $y = \sum_k t_k g_k \in H$ lies within sup-distance $C = \sum_k \|g_k\|_\infty > 0$ of $\sum_k \lfloor t_k \rfloor g_k \in \Gamma$.

Step 1: units large at one place. Let w_o be an archimedean place of L. Let $y \in H$ have coordinate $-2C$ at each place other than w_o and $2C(n-1)$ at w_o, and let $\epsilon \in \mathcal{O}_L^\times$ have $\lambda(\epsilon)$ within sup-distance C of y. Then $\log \|\epsilon\|_w \leq -C < 0$ for every archimedean $w \neq w_o$, and $\|\epsilon\|_{w_o} > 1$ because $\prod_w \|\epsilon\|_w = |N_{L/\mathbb{Q}}(\epsilon)| = 1$ [2, Trace and Norm Via Embeddings].

Step 2: equivariant units. Choose one place w_o in each G-orbit of archimedean places and a unit ϵ_o for it as in Step 1. For a place w_i in the orbit of w_o, the set $S_i = \{\sigma \in G \mid \sigma(w_i) = w_o\}$ is nonempty, and we put $\beta_i = \prod_{\sigma \in S_i} \sigma(\epsilon_o)$. Then $\|\beta_i\|_{w_j} = \prod_{\sigma \in S_i} \|\epsilon_o\|_{\sigma(w_j)}$, and for $\sigma \in S_i$ we have $\sigma(w_j) = w_o$ exactly when $j = i$, as σ permutes the places. Hence $\|\beta_i\|_{w_i} > 1$ and $\|\beta_i\|_{w_j} < 1$ for $j \neq i$, so the β_i are pairwise distinct. If $\tau \in G$ and $\tau(w_j) = w_i$, then $(\tau\sigma)(w_j) = \sigma(w_i)$, so $\sigma \mapsto \tau\sigma$ is a bijection from S_i onto S_j, and $\tau(\beta_i) = \prod_{\sigma \in S_i} (\tau\sigma)(\epsilon_o) = \beta_j$.

Step 3: independence. Let M be the $n \times n$ matrix with rows $\lambda(\beta_i)$, that is, $M_{ij} = \log \|\beta_i\|_{w_j}$. Then $M_{ii} > 0$, $M_{ij} < 0$ for $i \neq j$, and every row sums to 0. Fix k, and let N be M with row k and column k deleted. For each row $i \neq k$ of N, $N_{ii} = \sum_{j \neq i} |M_{ij}| > \sum_{j \neq i,k} |N_{ij}|$, as $M_{ik} \neq 0$. If $Nx = 0$ with $x \neq 0$, then for i with $|x_i|$ maximal, $N_{ii}|x_i| \leq \sum_{j \neq i,k} |N_{ij}|\,|x_i| < N_{ii}|x_i|$, which is impossible. So N is invertible, and the rows $\lambda(\beta_i)$ with $i \neq k$ are linearly independent.

Step 4: the units ϵ_i. Let B be the subgroup generated by the β_i. By Step 3, $\lambda(B) \subseteq \Gamma \cong \mathbb{Z}^{n-1}$ has rank $n-1$, so $\Gamma/\lambda(B)$ is finite [5, The Fundamental Theorem of f.g. Abelian Groups], and as $\ker \lambda = \mu(L)$ is finite, B has finite index in $\mathcal{O}_L^\times$. Let $R = \{m \in \mathbb{Z}^n \mid \prod_i \beta_i^{m_i} = 1\}$. The n elements $\lambda(\beta_i)$ of the free group Γ of rank $n-1$ satisfy $\sum_i m_i \lambda(\beta_i) = 0$ for some nonzero $m \in \mathbb{Z}^n$, so $\prod_i \beta_i^{m_i} \in \mu(L)$, and a power of this relation gives a nonzero element of R. Every nonzero $m \in R$ has all $m_i \neq 0$, by Step 3. So $R \cong \mathbb{Z}$: for $m, m' \in R$, the element $m'_1 m - m_1 m'$ of R has first coordinate 0, hence is 0. Let m generate R. Its entries have one sign: otherwise, with $P = \{i \mid m_i > 0\}$ and $Q = \{i \mid m_i < 0\}$ both nonempty, summing $\sum_i m_i M_{ij} = 0$ over $j \in P$ gives $\sum_i m_i \sum_{j \in P} M_{ij} = 0$, while $\sum_{j \in P} M_{ij} = -\sum_{j \notin P} M_{ij} > 0$ for $i \in P$ and $\sum_{j \in P} M_{ij} < 0$ for $i \in Q$, so every term of the sum is positive. Replacing m by $-m$, all $m_i > 0$. For $\tau \in G$, let $\pi_\tau(i)$ be the index with $\tau(w_{\pi_\tau(i)}) = w_i$. By Step 2, $\tau(\beta_i) = \beta_{\pi_\tau(i)}$, so applying τ to $\prod_i \beta_i^{m_i} = 1$ shows that the vector m' with $m'_{\pi_\tau(i)} = m_i$ lies in R. Then $m' = cm$ with $c \in \mathbb{Z}$, and $c = 1$ as m' and m have the same positive entries in another order. So $m_{\pi_\tau(i)} = m_i$ for all τ and i.

Put $\epsilon_i = \beta_i^{m_i}$. If $\tau(w_j) = w_i$, then $\tau(\epsilon_i) = \beta_j^{m_i} = \beta_j^{m_j} = \epsilon_j$. Conversely $\|\epsilon_i\|_{w_i} > 1 > \|\epsilon_i\|_{w_j}$ for $j \neq i$, so the ϵ_i are pairwise distinct and $\tau(\epsilon_i) = \epsilon_j$ forces $\tau(w_j) = w_i$. This is (1). The image of each β_i in $B/\langle \epsilon_1, \ldots, \epsilon_n \rangle$ has order dividing m_i, so this quotient is finite and the ϵ_i generate a subgroup of finite index in $\mathcal{O}_L^\times$, which is (2). Finally $\prod_i \epsilon_i = \prod_i \beta_i^{m_i} = 1$, and if $\prod_i \epsilon_i^{a_i} = 1$ then $(a_i m_i)_i \in R = \mathbb{Z}m$, so $a_i m_i = c m_i$ and $a_i = c$ for all i, which is (3), completing the proof.

5.2.2 Picard Group for Places

Let us extend fractional ideals of K by taking into account infinite primes

Definition 5.2.6: Replete Ideal

Let K be a number field with ring of integers $\mathcal{O}$ and group of fractional ideals $J(\mathcal{O})$. A *replete ideal* of K is an element of the group

$$J(\overline{\mathcal{O}}) = J(\mathcal{O}) \times \prod_{\mathfrak{p}\mid\infty} \mathbb{R}_+^\times$$

where $\mathbb{R}_+^\times$ is the multiplicative group of positive real numbers.

For notational consistency, for any infinite prime $\mathfrak{p}$ and any real number $\nu \in \mathbb{R}$, let:

$$\mathfrak{p}^\nu := e^\nu \in \mathbb{R}_+^\times$$

Next, given a system of real numbers $\nu_\mathfrak{p}, \mathfrak{p}\mid\infty$, let $\prod_{\mathfrak{p}\mid\infty} \mathfrak{p}^{\nu_\mathfrak{p}}$ denote the vector:

$$\prod_{\mathfrak{p}\mid\infty} \mathfrak{p}^{\nu_\mathfrak{p}} = (..., e^{\nu_\mathfrak{p}}, ...) \in \prod_{\mathfrak{p}\mid\infty} \mathbb{R}_+^\times$$

and *not* the product of the quantities $e^{\nu_\mathfrak{p}} \in \mathbb{R}$. Then every replete ideal $\mathfrak{a} \in J(\overline{\mathcal{O}})$ admits a unique factorization:

$$\mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{\nu_\mathfrak{p}} = \prod_{\mathfrak{p}\nmid\infty} \mathfrak{p}^{\nu_\mathfrak{p}} \times \prod_{\mathfrak{p}\mid\infty} \mathfrak{p}^{\nu_\mathfrak{p}}$$

with $\nu_\mathfrak{p} \in \mathbb{Z}$, almost all 0, for $\mathfrak{p} \nmid \infty$, and $\nu_\mathfrak{p} \in \mathbb{R}$ for $\mathfrak{p}\mid\infty$. we shall denote the finite portion $\mathfrak{a}_f$ and the infinite portion $\mathfrak{a}_\infty$ so that

$$\mathfrak{a} = \mathfrak{a}_f \times \mathfrak{a}_\infty$$

Then $\mathfrak{a}_f$ is a fractional ideal and $\mathfrak{a}_\infty$ is an element of $\prod_{\mathfrak{p}\mid\infty} \mathbb{R}_+^\times$. Viewing $\mathfrak{a}_f$ and $\mathfrak{a}_\infty$ as elements of

$$\mathfrak{a}_f \times \prod_{\mathfrak{p}\mid\infty} 1 \qquad (1) \times \mathfrak{a}_\infty$$

we can think of all the elements of $J(\overline{\mathcal{O}})$ as

$$\mathfrak{a} = \mathfrak{a}_f \cdot \mathfrak{a}_\infty$$

To each $a \in K^\times$ we associate the *replete principal ideal*

$$[a] = \prod_{\mathfrak{p}} \mathfrak{p}^{\nu_\mathfrak{p}(a)}$$

These form a subgroup $P(\overline{\mathcal{O}}) \subseteq J(\overline{\mathcal{O}})$, as $a \mapsto [a]$ is a homomorphism, and hence we can define the *replete Picard group* or *replete class group*:

Definition 5.2.7: Replete Picard Group

$$\mathrm{Pic}\,(\overline{\mathcal{O}}) = \frac{J(\overline{\mathcal{O}})}{P(\overline{\mathcal{O}})}$$

Next,

Definition 5.2.8: Absolute Norm For Replete Ideals

The *absolute norm* of a replete ideal $\mathfrak{a}$ is defined to be the positive real number:

$$\mathfrak{N}(\mathfrak{a}) = \prod_{\mathfrak{p}} \mathfrak{N}(\mathfrak{p})^{\nu_{\mathfrak{p}}}$$

The absolute norm is multiplicative and induces a surjective homomorphism:

$$\mathfrak{N} : J(\overline{\mathcal{O}}) \to \mathbb{R}_+^\times$$

The absolute norm of a replete principal ideal is 1: as $|a|_{\mathfrak{p}} = \mathfrak{N}(\mathfrak{p})^{-\nu_{\mathfrak{p}}(a)}$ for every prime $\mathfrak{p}$, proposition 5.2.2 gives

$$\mathfrak{N}([a]) = \prod_{\mathfrak{p}} \mathfrak{N}(\mathfrak{p})^{\nu_{\mathfrak{p}}(a)} = \prod_{\mathfrak{p}} |a|_{\mathfrak{p}}^{-1} = 1$$

Hence, we in fact get a surjective homomorphism on the replete Picard group

$$\mathfrak{N} : \mathrm{Pic}\,(\overline{\mathcal{O}}) \to \mathbb{R}_+^\times$$

5.3 *Haar Measure and the Product Formula

The real and complex numbers carry a *translation invariant Lebesgue measure*, unique up to a positive scalar and determined by giving the unit interval or the unit square measure 1. Each completion K_ν of a global field at a place is a locally compact abelian group, and every such group carries a translation invariant measure, its *Haar measure*, again unique up to a positive scalar. The construction belongs to analysis [11].

Definition 5.3.1: Radon Measure and Haar Measure

Let X be a locally compact Hausdorff space with Borel measure μ. Then μ is a *Radon measure* if for every Borel set S:

1. $\mu(S) < \infty$ if S is compact
2. $\mu(S) = \inf\{\mu(U) \mid S \subseteq U, U \text{ is open}\}$
3. $\mu(S) = \sup\{\mu(C) \mid C \subseteq S, C \text{ is compact}\}$

If X is furthermore a group, a Radon measure is a (left/right) *Haar* measure if it is *(left/right) translation invariant*:

$$\mu(S) = \mu(x + S) \qquad [\text{for right } \mu(S + x)]$$

If X is abelian, the left/right measures are identical and the measure is simply called a *Haar measure*.

Theorem 5.3.2: Existence and Uniqueness of Haar Measure

Let G be a locally compact group. Then there exists a Haar measure μ on G. If μ' is another Haar measure, then there exists a $\lambda \in \mathbb{R}_{>0}$ such that $\mu'(S) = \lambda\mu(S)$.

***Proof*:**
see [11, Existence of Haar Measure, Uniqueness of Haar Measure]

Example 5.3.3: Haar Measure On Real Space
The standard Haar measure on $\mathbb{R}^n$ is Lebesgue measure, determined on open boxes by

$$\mu\left(\prod_i (a_i, b_i)\right) = \prod_i |b_i - a_i|$$

and it is the unique Haar measure giving the unit cube measure 1.

By proposition 1.3.3, every local field K is locally compact, and its valuation ring A, the closed unit ball, is compact and open.

Proposition 5.3.4: Haar Measure On Local Fields

Let K be a local field with normalized discrete valuation v, residue field k, and absolute value

$$|-|_v := (\#k)^{-v(-)}$$

and let μ be a Haar measure on K. For every $x \in K$ and measurable set $S \subseteq K$ we have

$$\mu(xS) = |x|_v \mu(S)$$

Moreover, the absolute value $|-|_v$ is the unique absolute value compatible with the topology on K for which this is true.

***Proof*:**
Let A be the valuation ring of K, $\mathfrak{p} = \pi A$ its maximal ideal, and $q = \#k$. The set $A = \{x \mid |x|_v < q\}$ is open, and it is compact by proposition 1.3.3, so $\mu(A) < \infty$. Each $\pi^{-m}A$ is a finite union of translates of A, as $\pi^{-m}A/A \cong A/\mathfrak{p}^m$ is finite, and $K = \bigcup_m \pi^{-m}A$. So $\mu(A) = 0$ would force $\mu = 0$, and hence $\mu(A) > 0$. For $x = 0$ and S nonempty, the identity reads $\mu(\{0\}) = 0$, which holds because the infinitely many points of A all have the measure of $\{0\}$, while $\mu(A) < \infty$.

Let $x \neq 0$. Multiplication by x is an automorphism of the topological group $(K, +)$, which is locally compact by proposition 1.3.3, so $S \mapsto \mu(xS)$ is again a Haar measure, and theorem 5.3.2 gives $\chi(x) > 0$ with $\mu(xS) = \chi(x)\mu(S)$ for all S. For $x, y \neq 0$, $\chi(xy)\mu(A) = \mu(xyA) = \chi(x)\mu(yA) = \chi(x)\chi(y)\mu(A)$, so χ is multiplicative. Every $x \in K^\times$ has $x \in A$ or $x^{-1} \in A$, and both χ and $|-|_v$ are multiplicative, so it suffices to show $\chi(x) = |x|_v$ for $x \in A \setminus \{0\}$. For such x, $xA = \mathfrak{p}^{v(x)}$, and $A/\mathfrak{p}^{v(x)}$ has $q^{v(x)}$ elements, its filtration by the $\mathfrak{p}^i/\mathfrak{p}^{v(x)}$ having successive

quotients $\mathfrak{p}^i/\mathfrak{p}^{i+1} \cong A/\mathfrak{p}$ of q elements (multiplication by π^i). So A is a disjoint union of $q^{v(x)}$ translates of xA, and

$$\mu(A) = q^{v(x)}\mu(xA) = q^{v(x)}\chi(x)\mu(A)$$

which gives $\chi(x) = q^{-v(x)} = |x|_v$. Hence $\mu(xS) = |x|_v\mu(S)$ for all x and S.

For uniqueness, let $|-|$ be an absolute value defining the topology of K with $\mu(xS) = |x|\mu(S)$ for all x and S. By [5, Equivalent Topoloies], $|x| = |x|_v^c$ for some $c > 0$ and all x. For $x = \pi$, comparing $\mu(\pi A) = |\pi|\mu(A)$ with $\mu(\pi A) = |\pi|_v\mu(A)$ gives $|\pi|_v^c = |\pi|_v = q^{-1} \neq 1$, so $c = 1$ and $|-| = |-|_v$, as we sought to show.

The first paragraph of the proof shows $0 < \mu(A) < \infty$ for every Haar measure μ on K, so by theorem 5.3.2 exactly one Haar measure on K gives the valuation ring measure 1, the *normalized* Haar measure.

Exercise 5.3.1

1. Let K be a number field with s complex places. Show that the sign of the absolute discriminant Δ_K is $(-1)^s$.

5.4 *Arakelov Compactification

On a smooth projective curve over a field every principal divisor has degree 0, while on the affine line the function t has the single zero $t = 0$, its pole at infinity being invisible. For a number field K, the scheme $\operatorname{Spec}\mathcal{O}_K$ plays the role of an affine curve, and the infinite places of K play the role of the missing points at infinity. The object obtained by adding them is written $\overline{\operatorname{Spec}\mathcal{O}_K}$.

At a finite prime $\mathfrak{p}$ the unit ball $\{x \in K_\mathfrak{p} \mid |x|_\mathfrak{p} \leq 1\}$ is the valuation ring, but at an infinite place ν the unit ball $\{x \in K_\nu \mid |x|_\nu \leq 1\}$ is not closed under addition: for a real place it is the interval $[-1, 1]$, which contains 1 but not $1 + 1 = 2$. So no local ring is available at ν, and Arakelov geometry instead adds analytic data at the infinite places to divisors and line bundles. Throughout, $|-|_\mathfrak{p}$ and $|-|_\nu$ are the normalized absolute values of section 5.2, so that $|a|_\mathfrak{p} = \mathfrak{N}(\mathfrak{p})^{-\nu_\mathfrak{p}(a)}$ at a finite prime and $|a|_\nu = |\tau a|^{f_\nu}$ at an infinite place ν defined by an embedding $\tau\colon K \to \mathbb{C}$, where $f_\nu = [K_\nu : \mathbb{R}]$.

Definition 5.4.1: Arakelov Divisor

Let K be a number field. An *Arakelov divisor* of K is a formal sum

$$\widehat{D} = \sum_{\mathfrak{p}\nmid\infty} n_\mathfrak{p}(\mathfrak{p}) + \sum_{\nu\mid\infty} x_\nu(\nu)$$

over the finite primes $\mathfrak{p}$ and the infinite places ν of K, where $n_\mathfrak{p} \in \mathbb{Z}$, with $n_\mathfrak{p} = 0$ for all but finitely many $\mathfrak{p}$, and $x_\nu \in \mathbb{R}$. The Arakelov divisors form a group under addition, denoted $\widehat{\operatorname{Div}}(K)$.

Definition 5.4.2: Principal Arakelov Divisor

For $a \in K^\times$, the *principal Arakelov divisor* of a is

$$\widehat{(a)} = \sum_{\mathfrak{p} \nmid \infty} \nu_{\mathfrak{p}}(a)(\mathfrak{p}) - \sum_{\nu \mid \infty} \log |a|_\nu (\nu)$$

The sign of the infinite coefficients matches the finite ones: a large value of $|a|_\nu$ plays the role of a pole at ν, as $\nu_{\mathfrak{p}}(a) < 0$ does at $\mathfrak{p}$.

Definition 5.4.3: Arakelov Degree

The *degree* of an Arakelov divisor is the homomorphism $\deg\colon \widehat{\mathrm{Div}}(K) \to \mathbb{R}$ given by

$$\deg(\widehat{D}) = \sum_{\mathfrak{p} \nmid \infty} n_{\mathfrak{p}} \log \mathfrak{N}(\mathfrak{p}) + \sum_{\nu \mid \infty} x_\nu$$

Arakelov divisors are the replete ideals of section 5.2 written additively. With $\mathcal{O} = \mathcal{O}_K$, the map sending a replete ideal $\prod_{\mathfrak{p}} \mathfrak{p}^{m_{\mathfrak{p}}}$ to $\sum_{\mathfrak{p} \nmid \infty} m_{\mathfrak{p}}(\mathfrak{p}) + \sum_{\nu \mid \infty} f_\nu m_\nu(\nu)$ is an isomorphism $J(\overline{\mathcal{O}}) \to \widehat{\mathrm{Div}}(K)$.

It carries the replete principal ideal $[a]$ to $\widehat{(a)}$, because the exponent of $[a]$ at an infinite place ν is $-\log|\tau a|$ and $-f_\nu \log|\tau a| = -\log|a|_\nu$. It carries the logarithm of the absolute norm to the degree, because $\log \mathfrak{N}(\nu) = f_\nu$.

Theorem 5.4.4: Arakelov Degree of Principal Divisors

Let K be a number field and $a \in K^\times$. Then $\deg(\widehat{(a)}) = 0$.

***Proof*:**
At a finite prime, $\log|a|_{\mathfrak{p}} = -\nu_{\mathfrak{p}}(a) \log \mathfrak{N}(\mathfrak{p})$. Hence

$$\deg(\widehat{(a)}) = \sum_{\mathfrak{p} \nmid \infty} \nu_{\mathfrak{p}}(a) \log \mathfrak{N}(\mathfrak{p}) - \sum_{\nu \mid \infty} \log |a|_\nu = -\sum_{\mathfrak{p}} \log |a|_{\mathfrak{p}}$$

where the last sum runs over all primes of K, finite and infinite, and has finitely many nonzero terms. By proposition 5.2.2 it equals $-\log 1 = 0$, completing the proof.

This is the analogue of the fact that a nonzero meromorphic function on a compact Riemann surface has as many zeros as poles, counted with multiplicity. At the level of line bundles, the compactification equips an invertible module with metrics at the infinite places.

Definition 5.4.5: Hermitian (Arakelov) Line Bundle

An *Arakelov line bundle* on $\overline{\operatorname{Spec}\mathcal{O}_K}$ is a pair $\overline{\mathcal{L}} = (\mathcal{L}, (\|-\|_\nu)_{\nu\mid\infty})$, where $\mathcal{L}$ is an invertible $\mathcal{O}_K$-module, equivalently an invertible sheaf on $\operatorname{Spec}\mathcal{O}_K$, and for each infinite place ν, defined by an embedding $\tau_\nu\colon K \to \mathbb{C}$, $\|-\|_\nu$ is a hermitian metric on the one-dimensional complex vector space $\mathcal{L}\otimes_{\mathcal{O}_K,\tau_\nu}\mathbb{C}$.

At a complex place, replacing τ_ν by its complex conjugate gives an equivalent datum, by transport along $x\otimes z \mapsto x\otimes\bar{z}$. At a finite prime $\mathfrak{p}$, the module $\mathcal{L}$ gives the lattice $\mathcal{L}\otimes_{\mathcal{O}_K}\mathcal{O}_{K_\mathfrak{p}}$ in the one-dimensional $K_\mathfrak{p}$-vector space $\mathcal{L}\otimes_{\mathcal{O}_K}K_\mathfrak{p}$, and this lattice is the unit ball of a norm on that space. The metrics are the corresponding norms at the infinite places, where no lattice is available.

Definition 5.4.6: Arakelov Class Group

The *Arakelov class group* of K is the quotient

$$\widehat{\operatorname{Pic}}(\mathcal{O}_K) = \widehat{\operatorname{Div}}(K)/\left\{\widehat{(a)} \;\middle|\; a\in K^\times\right\}$$

by the subgroup of principal Arakelov divisors.

The principal Arakelov divisors form a subgroup because $a\mapsto\widehat{(a)}$ is a homomorphism, and the degree is well defined on $\widehat{\operatorname{Pic}}(\mathcal{O}_K)$ by theorem 5.4.4. The group $\widehat{\operatorname{Pic}}(\mathcal{O}_K)$ is the analogue of the divisor class group of a smooth projective curve, which is isomorphic to its Picard group. Forgetting the infinite coefficients sends an Arakelov divisor to the fractional ideal $\prod_\mathfrak{p}\mathfrak{p}^{n_\mathfrak{p}}$, and a principal Arakelov divisor to a principal ideal, so it induces a surjective homomorphism $\widehat{\operatorname{Pic}}(\mathcal{O}_K)\to\operatorname{Cl}_K$ onto the ideal class group. Its kernel consists of the classes of divisors $\sum_{\nu\mid\infty}x_\nu(\nu)$ supported at the infinite places. Such a divisor is principal exactly when there is a unit $u\in\mathcal{O}_K^\times$ with $x_\nu = -\log|u|_\nu$ for every ν, because an element $a\in K^\times$ has $\nu_\mathfrak{p}(a)=0$ at every finite prime exactly when $a\in\mathcal{O}_K^\times$. So the class group records the finite part of $\widehat{\operatorname{Pic}}(\mathcal{O}_K)$, and the unit group determines which divisors at infinity are principal. Neukirch develops the Riemann-Roch theory of these objects [12, Chapter III].

The unit ball at an infinite place is closed under multiplication but not under addition, and schemes are built from commutative rings, so the infinite places are not points of any scheme. Proposals for a geometry over the "field with one element" $\mathbb{F}_1$ replace commutative rings by more general algebraic objects, in some of which such unit balls are admissible. One of their aims is a compactification of $\operatorname{Spec}\mathbb{Z}$ in which the infinite place is a point on the same footing as the finite primes. None of these proposals has become standard.

What Next?

This book has built the local fields and the ramification theory of their extensions, stopping just before the theorem that organizes all abelian extensions of a local field. Several roads lead onward.

Local class field theory. The immediate sequel classifies the finite abelian extensions of a local field K by the finite-index subgroups of $K^\times$, through the local reciprocity map $K^\times \to \mathrm{Gal}_K(K^{\mathrm{ab}})$; the Hilbert symbol of section 3.7 and the norm groups met throughout are its raw material. *Class Field Theory* [10] develops it cohomologically, the route of Serre's *Local Fields* [15]. Lubin-Tate formal groups give it an explicit shape, and Serre develops them in his chapter of the Cassels-Fröhlich *Algebraic Number Theory* [1, Chapter VI, §3].

Galois cohomology and local duality. The Brauer group of a local field, the local duality theorems, and the cohomological reformulation of class field theory are the subject of *Galois Cohomology* [7], following Serre's *Galois Cohomology*.

The local Langlands program. Beyond abelian extensions, the smooth representation theory of GL_n over a local field encodes the whole Galois group; the case of GL_2 is *Langlands for GL2(F)* [8]. Bushnell and Henniart's *The Local Langlands Conjecture for GL(2)* is the definitive treatment.

p-adic Hodge theory. The Witt vectors of section 3.8 are the entry point to the study of p-adic Galois representations and the comparison theorems relating p-adic étale and de Rham cohomology. Fontaine's period rings and the CMI notes of Brinon and Conrad are where to begin; this lies beyond the present series.

A

The Hasse-Arf Theorem

This appendix proves the Hasse-Arf theorem stated in theorem 3.6.9: for a finite *abelian* extension L/K of local fields, the jumps of the upper-numbering ramification filtration are integers. The result is separated out because its proof, while elementary, is longer than the development in section 3.6 and would interrupt the thread there. It also cannot be routed through class field theory, which lies downstream of this book. The argument is self-contained, resting on the ramification theory of sections 3.4 and 3.6.

The general abelian case reduces to totally ramified cyclic extensions of p-power degree, where integrality of the jumps becomes a congruence between consecutive lower breaks. That congruence is a theorem of Sen [13] on the action of a wildly ramified automorphism on a uniformizer.

A.1 Reduction to Cyclic Extensions

Proposition A.1.1: Reduction to the Cyclic Case

If every finite cyclic extension of local fields has integral upper-numbering jumps, then so does every finite abelian extension.

***Proof*:**
Let L/K be abelian with group G, and let v be an upper break, so $G^v \neq G^{v+\epsilon}$ for all $\epsilon > 0$. As the filtration takes finitely many values, there is an $\epsilon_0 > 0$ such that $G' = G^{v+\epsilon}$ is the same group for all $0 < \epsilon \leq \epsilon_0$, and $G' \neq G^v$. The quotient G^v/G' is a nontrivial finite abelian group, so it carries a character not identically 1. It extends to a character of G/G': for a finite abelian group A the group of characters $A \to \mathbb{C}^\times$ has order $|A|$, so restriction from the characters of G/G', whose kernel is the group of characters of $(G/G')/(G^v/G')$, has image of

order $|G/G'|/|G/G^v| = |G^v/G'|$, which is all characters of G^v/G'. Composing with $G \to G/G'$ gives a character $\chi\colon G \to \mathbb{C}^\times$ with $\chi(G^v) \neq 1$ and $\chi(G') = 1$. Set $H = \ker\chi$, so that $G/H \cong \operatorname{im}\chi$ is cyclic.

By Herbrand's theorem in the upper numbering (corollary 3.6.8), $(G/H)^w = G^wH/H$ for all w. If $G^vH/H = G'H/H$, then any $g \in G^v$ would equal $g'h$ with $g' \in G'$ and $h \in H$, and then $\chi(g) = \chi(g')\chi(h) = 1$, contradicting the choice of a $g \in G^v$ with $\chi(g) \neq 1$. Hence $(G/H)^v \neq (G/H)^{v+\epsilon}$ for $0 < \epsilon \leq \epsilon_0$, and so for all $\epsilon > 0$, as the filtration decreases. So v is also an upper break of the cyclic quotient G/H, the Galois group of L^H/K, and by hypothesis $v \in \mathbb{Z}$. As v was an arbitrary upper break of G, the abelian case follows, completing the proof.

Lemma A.1.2: Reduction to Totally Ramified p-Extensions

If every cyclic, totally ramified extension of local fields whose degree is a power of the residue characteristic has integral upper-numbering jumps, then so does every finite cyclic extension of local fields.

***Proof*:**
Let L/K be cyclic with group G, let p be the residue characteristic, and let v be an upper break of G. For $-1 < w \leq 0$ we have $\psi_{L/K}(w) = w$ (proposition 3.6.3), so $G^w = G_w = G_0$ (definitions 3.6.2 and 3.6.4), and a break $v \leq 0$ is -1 or 0. Assume $v > 0$. Let $E = L^{G_0}$ be the inertia field, so that L/E is totally ramified with group G_0 (theorem 3.3.16). Its ramification groups are the G_u with $u \geq 0$ (proposition 3.6.1), so its Herbrand function is $\varphi_{L/K}$ (definition 3.6.2) and its upper-numbering groups are the G^w with $w \geq 0$. The cyclic group G_0 is $G_1 \times C$, where G_1 is its Sylow p-subgroup (corollary 3.4.5) and C is the subgroup of order $(G_0 : G_1)$, prime to p. Let $M = L^C$. By corollary 3.6.8, applied to L/E, the upper-numbering groups of $\mathrm{Gal}_E(M) = G_0/C$ are the G^wC/C. For $w > 0$ we have $G^w = G_{\psi_{L/K}(w)} \subseteq G_1$, and $G_1 \cap C = 1$, so $G^vC/C \neq G^{v+\epsilon}C/C$ for every $\epsilon > 0$, and v is an upper break of M/E. At $w = 0$ the same theorem gives $(G_0/C)^0 = G_0/C$, so the inertia group of M/E is the whole group and M/E is totally ramified. It is cyclic of degree $|G_1|$, a power of p, so $v \in \mathbb{Z}$ by hypothesis, completing the proof.

A.2 The Break Congruence

Let L/K be cyclic and totally ramified of degree p^n, where p is the residue characteristic, let G be its group, generated by τ, and write $A_k = \langle \tau^{p^k} \rangle$ for the subgroup of index p^k. As $G = G_0$ is a p-group and G_0/G_1 has order prime to p (theorem 3.4.4), $G = G_1$. Each G_i is a subgroup of the cyclic group G, hence one of the A_k, and for $i \geq 1$ the quotient G_i/G_{i+1} is elementary abelian (theorem 3.4.4), while A_k/A_{k+2} is cyclic of order p^2. So the filtration passes through every A_k, and as $G_i = 1$ for large i (proposition 3.4.3), there are integers $1 \leq \lambda_1 < \cdots < \lambda_n$, the lower breaks, with $G_i = A_{k-1}$ for $\lambda_{k-1} < i \leq \lambda_k$, where $\lambda_0 = 0$, and $G_i = 1$ for $i > \lambda_n$.

Proposition A.2.1: Cyclic Hasse-Arf as a Congruence

With the notation above, the upper-numbering jumps of L/K are the numbers $\varphi_{L/K}(\lambda_k)$, and

$$\varphi_{L/K}(\lambda_k) = \lambda_1 + \frac{\lambda_2 - \lambda_1}{p} + \cdots + \frac{\lambda_k - \lambda_{k-1}}{p^{k-1}}$$

Consequently all upper jumps are integers if and only if $\lambda_k \equiv \lambda_{k-1} \pmod{p^{k-1}}$ for $2 \le k \le n$.

***Proof*:**
On the interval $(\lambda_{k-1}, \lambda_k]$ the ramification group is $G_i = A_{k-1}$, of index $[G_0 : A_{k-1}] = p^{k-1}$, so $\varphi_{L/K}$ has slope $p^{-(k-1)}$ there (proposition 3.6.3). Integrating from 0, and using $\varphi_{L/K}(\lambda_1) = \lambda_1$ (slope 1 up to the first break), gives the displayed formula. The upper jumps are the images $\varphi_{L/K}(\lambda_k)$ of the lower breaks by definition of the upper numbering. Finally, arguing by induction on k: given $\varphi_{L/K}(\lambda_{k-1}) \in \mathbb{Z}$, the value $\varphi_{L/K}(\lambda_k) = \varphi_{L/K}(\lambda_{k-1}) + (\lambda_k - \lambda_{k-1})/p^{k-1}$ is an integer if and only if $p^{k-1} \big| \lambda_k - \lambda_{k-1}$, which is the stated congruence, completing the proof.

Fix a uniformizer π of L, let ν be the normalized valuation of L, and for $\sigma \in G$ and $k \in \mathbb{Z}$ put

$$c_k(\sigma) = \nu\Big(\frac{\sigma^k(\pi)}{\pi} - 1\Big) \in \mathbb{Z} \cup \{\infty\}$$

which is ∞ exactly when $\sigma^k = 1$, as $L = K(\pi)$ (theorem 3.3.18).

Lemma A.2.2: The Numbers $c_k(\sigma)$

Let $\sigma \in G$, let $L' = L^{\langle\sigma\rangle}$, and let $k \in \mathbb{Z}$.

1. If $\sigma^k \neq 1$, then $c_k(\sigma) = i_G(\sigma^k) - 1$, which depends only on the subgroup $\langle\sigma^k\rangle$. Consequently $c_k(\sigma)$ depends only on $v_p(k)$, and $j \mapsto c_{p^j}(\sigma)$ is nondecreasing, with value ∞ once $\sigma^{p^j} = 1$.
2. There is an $\alpha_k \in L^\times$ with $\nu(\alpha_k) = k$ and $\nu\big(\sigma(\alpha_k)/\alpha_k - 1\big) = c_k(\sigma)$.
3. Every nonzero $\gamma \in L$ is a convergent sum $\gamma = \sum_{k \ge \nu(\gamma)} \theta_k \alpha_k$, with the α_k of the previous item and each θ_k either 0 or a unit of the valuation ring of L'.

***Proof*:**

1. As L/K is totally ramified, the valuation ring of L is generated over that of K by π (theorem 3.3.18), so lemma 3.4.2 applies with $\alpha = \pi$: for $s \in G$ and $i \ge -1$, $s \in G_i$ if and only if $i_G(s) = \nu(s(\pi) - \pi) \ge i + 1$. As $\nu(\pi) = 1$, $c_k(\sigma) = \nu(\sigma^k(\pi) - \pi) - 1 = i_G(\sigma^k) - 1$. The G_i are subgroups (proposition 3.4.3), so $s \in G_i$ if and only if $\langle s\rangle \subseteq G_i$. Hence $i_G(s)$ depends only on $\langle s\rangle$, and $\langle s'\rangle \subseteq \langle s\rangle$ implies $i_G(s') \ge i_G(s)$. In the cyclic p-group $\langle\sigma\rangle$, the subgroup $\langle\sigma^k\rangle$ equals $\langle\sigma^{p^{v_p(k)}}\rangle$, and the subgroups $\langle\sigma^{p^j}\rangle$ decrease with j, which gives the remaining assertions.
2. For $k > 0$ put $\alpha_k = \prod_{i=0}^{k-1} \sigma^i(\pi)$, a product of k uniformizers, as $\nu \circ \sigma = \nu$. Then $\sigma(\alpha_k)/\alpha_k = \sigma^k(\pi)/\pi$. For $k < 0$ put $\alpha_k = \alpha_{-k}^{-1}$, so that $\sigma(\alpha_k)/\alpha_k = w^{-1}$ with $w =$

$\sigma^{-k}(\pi)/\pi$ a unit, and $\nu(w^{-1}-1) = \nu(w-1) = c_{-k}(\sigma) = c_k(\sigma)$ by the previous item. For $k = 0$ put $\alpha_0 = 1$.

3. As L/K is totally ramified, so is L/L', and the residue fields of L and L' coincide. Given $\gamma \neq 0$ with $\nu(\gamma) = k$, the unit γ/α_k is congruent modulo the maximal ideal of L to a unit θ_k of the valuation ring of L', so $\gamma - \theta_k\alpha_k$ is 0 or has valuation greater than k. Repeating this with $\gamma - \theta_k\alpha_k$ in place of γ, and setting $\theta_j = 0$ at the skipped indices, gives partial sums differing from γ by elements of valuation tending to infinity, completing the proof.

Proposition A.2.3: Sen's Congruence

For every integer $a \geq 0$, every $\sigma \in G$ and all integers k_1, k_2 with $\sigma^{k_1} \neq 1 \neq \sigma^{k_2}$ and $v_p(k_1 - k_2) \geq a$, we have $v_p\big(c_{k_1}(\sigma) - c_{k_2}(\sigma)\big) \geq a + 1$.

***Proof*:**
We argue by induction on a, assuming the claim for all smaller values of a and all elements of G.

Step 1: reduction to consecutive powers of p. It suffices to show that $v_p\big(c_{p^a}(\tau) - c_{p^{a+1}}(\tau)\big) \geq a+1$ for every $\tau \in G$ with $\tau^{p^{a+1}} \neq 1$. Granting this, let σ, k_1, k_2 be as in the claim. If $v_p(k_1) = v_p(k_2)$, then $c_{k_1}(\sigma) = c_{k_2}(\sigma)$ by lemma A.2.2. Otherwise, say $j_1 = v_p(k_1) < j_2 = v_p(k_2)$, both finite as $k_1, k_2 \neq 0$. Then $j_1 = v_p(k_1 - k_2) \geq a$ and $\sigma^{p^{j_2}} \neq 1$, as $\langle\sigma^{k_2}\rangle = \langle\sigma^{p^{j_2}}\rangle$. By lemma A.2.2,

$$c_{k_1}(\sigma) - c_{k_2}(\sigma) = \sum_{j=j_1}^{j_2-1} \big(c_{p^j}(\sigma) - c_{p^{j+1}}(\sigma)\big)$$

For each such j, the element $\tau = \sigma^{p^{j-a}}$ has $c_{p^a}(\tau) = c_{p^j}(\sigma)$, $c_{p^{a+1}}(\tau) = c_{p^{j+1}}(\sigma)$ and $\tau^{p^{a+1}} = \sigma^{p^{j+1}} \neq 1$, so every term of the sum is divisible by p^{a+1}.

Step 2: distinct values. Fix $\tau \in G$ with $\tau^{p^{a+1}} \neq 1$. Then the integers $k + c_k(\tau)$, for k with $v_p(k) \leq a$, are pairwise distinct and different from $c_{p^a}(\tau)$. All these $c_k(\tau)$ are finite. If $v_p(k_1) = v_p(k_2)$ and $k_1 \neq k_2$, then $c_{k_1}(\tau) = c_{k_2}(\tau)$ and $k_1 + c_{k_1}(\tau) \neq k_2 + c_{k_2}(\tau)$. If $v_p(k_1) \neq v_p(k_2)$, then $b = v_p(k_1 - k_2) = \min(v_p(k_1), v_p(k_2)) \leq a - 1$, and the induction hypothesis gives $v_p\big(c_{k_2}(\tau) - c_{k_1}(\tau)\big) \geq b + 1$, so $k_1 - k_2 \neq c_{k_2}(\tau) - c_{k_1}(\tau)$. Finally, if $v_p(k) = a$, then $c_k(\tau) = c_{p^a}(\tau)$ and $k + c_k(\tau) \neq c_{p^a}(\tau)$ as $k \neq 0$, while if $v_p(k) < a$, then $v_p(p^a - k) = v_p(k)$ and the induction hypothesis gives $v_p\big(c_{p^a}(\tau) - c_k(\tau)\big) \geq v_p(k) + 1$, so $c_{p^a}(\tau) - c_k(\tau) \neq k$.

Step 3: the congruence. Let τ be as in Step 2, put $d = c_{p^a}(\tau) - c_{p^{a+1}}(\tau)$, and suppose $v_p(d) \leq a$. For $a \geq 1$, the induction hypothesis applied to τ^p, p^{a-1} and p^a gives $v_p\big(c_{p^{a-1}}(\tau^p) - c_{p^a}(\tau^p)\big) \geq a$, that is, $v_p(d) \geq a$. Hence $v_p(d) = a$, the case $a = 0$ following from $v_p(d) \geq 0$, and in particular $d \neq 0$. By lemma A.2.2 applied to τ^p, there is an $\alpha \in L^\times$ with $\nu(\alpha) = d$ and $\nu\big(\tau^p(\alpha)/\alpha - 1\big) = c_d(\tau^p) = c_{p^a}(\tau^p) = c_{p^{a+1}}(\tau)$, using $v_p(d) = a$ and $\tau^{p^{a+1}} \neq 1$. So

$$\nu\big(\tau^p(\alpha) - \alpha\big) = d + c_{p^{a+1}}(\tau) = c_{p^a}(\tau)$$

Put $\beta = \sum_{i=0}^{p-1} \tau^i(\alpha)$, so that $\tau(\beta) - \beta = \tau^p(\alpha) - \alpha$ has valuation $c_{p^a}(\tau)$, and in particular $\beta \neq 0$. Since $G = G_1$, every $s \in G$ and $x \in L^\times$ satisfy $\nu\big(s(x)/x - 1\big) \geq 1$: for $x = \pi^j u$ with

u a unit, $s(\pi)/\pi$ and $s(u)/u$ lie in the group $U_L^{(1)}$, as $\nu(s(y)-y)\geq 2$ for y in the valuation ring. So $\beta = p\alpha + \sum_{i=0}^{p-1}\big(\tau^i(\alpha)-\alpha\big)$ has $\nu(\beta) > d$, as $\nu(p\alpha) > \nu(\alpha)$. By lemma A.2.2 for τ, write $\beta = \sum_{k\geq\nu(\beta)}\beta_k$ with $\beta_k = \theta_k\alpha_k$, so that $\tau(\beta_k)-\beta_k = \theta_k\big(\tau(\alpha_k)-\alpha_k\big)$ is 0 or has valuation $k + c_k(\tau)$, and $\tau(\beta)-\beta = S_1 + S_2$, where S_1 collects the terms $\tau(\beta_k)-\beta_k$ with $v_p(k)\leq a$ and S_2 those with $v_p(k)\geq a+1$. Every term of S_2 is 0 or has valuation at least $k + c_{p^{a+1}}(\tau) > d + c_{p^{a+1}}(\tau) = c_{p^a}(\tau)$, using the monotonicity in lemma A.2.2, so $\nu(S_2) > c_{p^a}(\tau)$ and therefore $\nu(S_1) = c_{p^a}(\tau)$. But by Step 2 the nonzero terms of S_1 have pairwise distinct valuations, tending to infinity, so $\nu(S_1)$ is the least of them, which is not $c_{p^a}(\tau)$, or $S_1 = 0$. This contradiction shows $v_p(d)\geq a+1$, which by Step 1 proves the claim for a, completing the proof.

Lemma A.2.4: The Break Congruence

Let L/K be cyclic and totally ramified of degree p^n, where p is the residue characteristic, with lower breaks $\lambda_1 < \cdots < \lambda_n$. Then $\lambda_k \equiv \lambda_{k-1} \pmod{p^{k-1}}$ for $2\leq k\leq n$.

***Proof*:**

With τ a generator of G, the element $\tau^{p^{k-1}}$ generates A_{k-1}, which lies in G_i exactly for $i\leq\lambda_k$. So $i_G(\tau^{p^{k-1}}) = \lambda_k + 1$ by lemma 3.4.2, and $c_{p^{k-1}}(\tau) = \lambda_k$ by lemma A.2.2. For $2\leq k\leq n$, proposition A.2.3 with $a = k-2$, $k_1 = p^{k-1}$ and $k_2 = p^{k-2}$, where $v_p(k_1-k_2) = k-2$ and $\tau^{k_1}, \tau^{k_2}\neq 1$, gives $p^{k-1}\big|\lambda_k - \lambda_{k-1}$, as we sought to show.

As a check that the congruence is the right statement, the cyclotomic tower realizes it exactly: for $L = \mathbb{Q}_p(\zeta_{p^n})$ the lower breaks lie at $\lambda_k = p^k - 1$ (example 3.4.6), and $\lambda_k - \lambda_{k-1} = p^{k-1}(p-1)\equiv 0 \pmod{p^{k-1}}$, so the upper jumps $\varphi(\lambda_k) = k$ are integers, consistent with example 3.6.10.

Theorem A.2.5: Hasse-Arf, restated

Every finite abelian extension of local fields has integral upper-numbering jumps.

***Proof*:**

By proposition A.1.1 it suffices to treat cyclic L/K, and by lemma A.1.2 cyclic and totally ramified of degree p^n with p the residue characteristic. For these, proposition A.2.1 shows that the jumps are integral exactly when $\lambda_k \equiv \lambda_{k-1} \pmod{p^{k-1}}$ for $2\leq k\leq n$, which is lemma A.2.4, completing the proof.

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